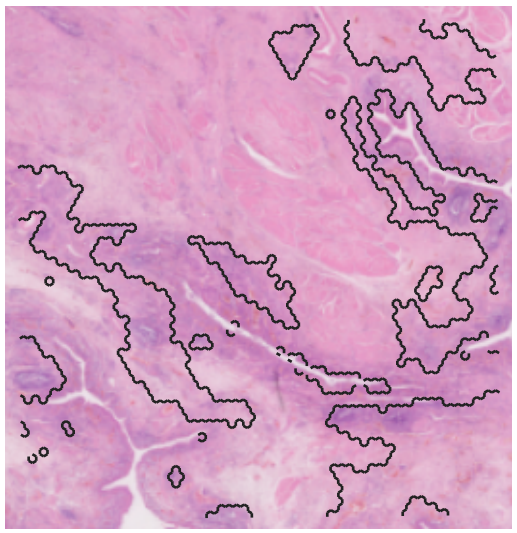
Inhousedata_T4384654 | Tissue context

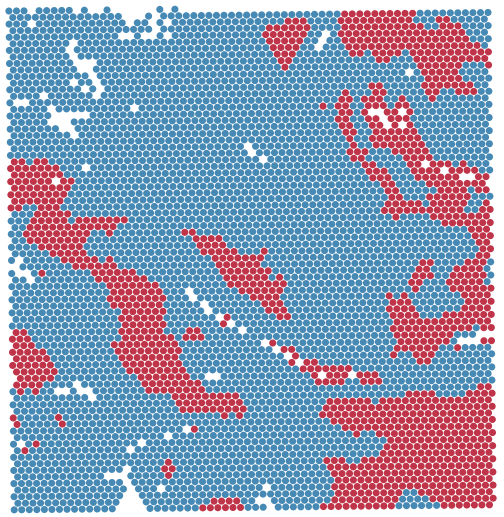
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



Black line: observed candidate outer edge

Final candidate location

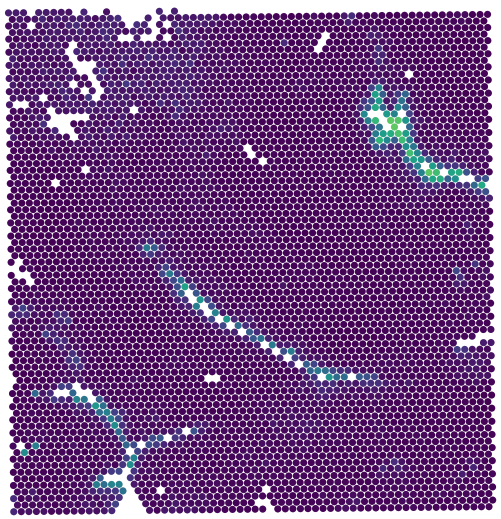


inside outside

Accepted R12 (original 0–8)

0 2

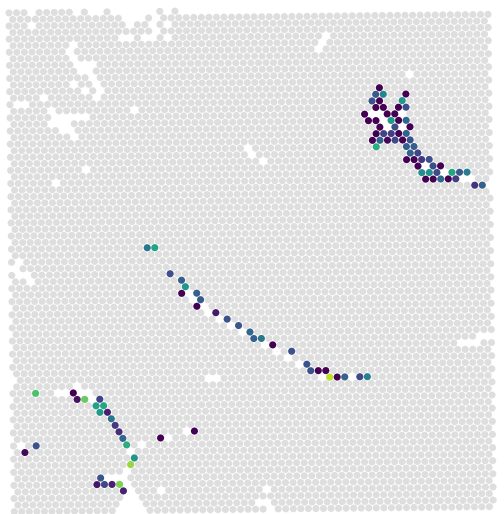
Epithelial support



0 1

Fraction of all 30

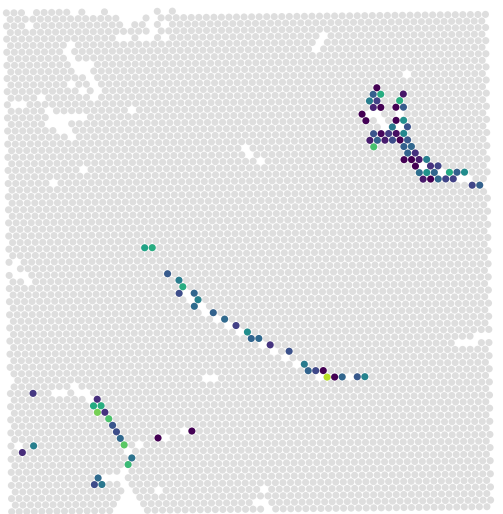
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



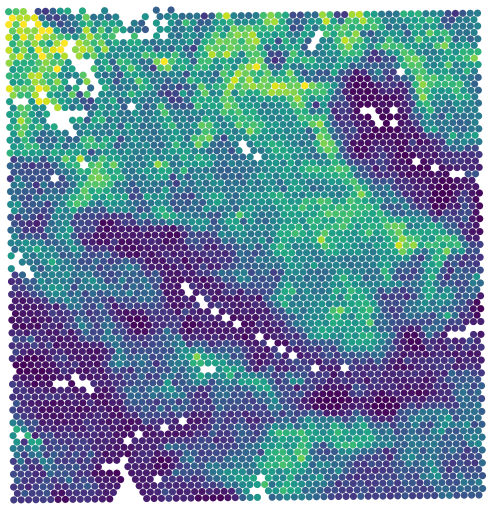
0 0.14

Matched-model AUCell

Inhousedata_T4384654 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



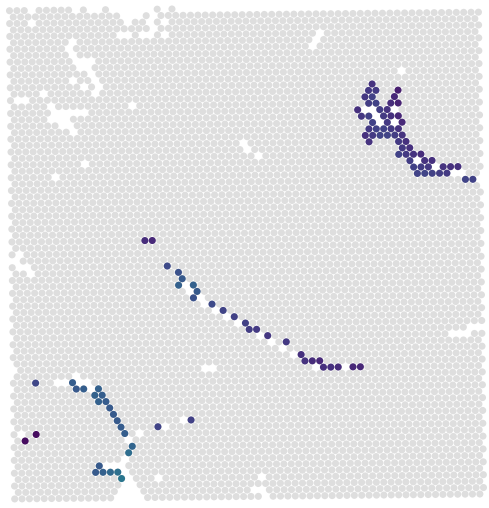
0 0.418

Fraction of all 30

CAF P80 targets

False True

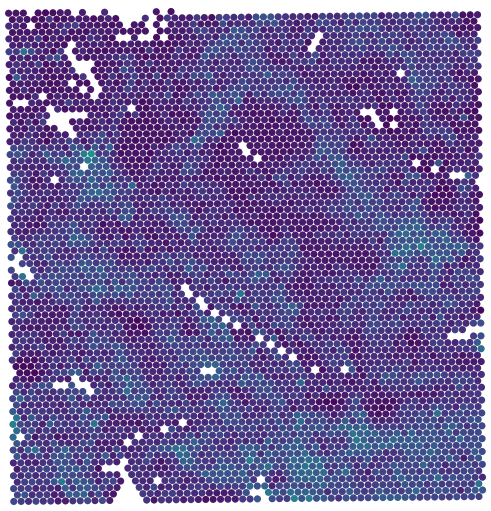
CAF distance



0 3.1e+03

Nearest spot center (μm)

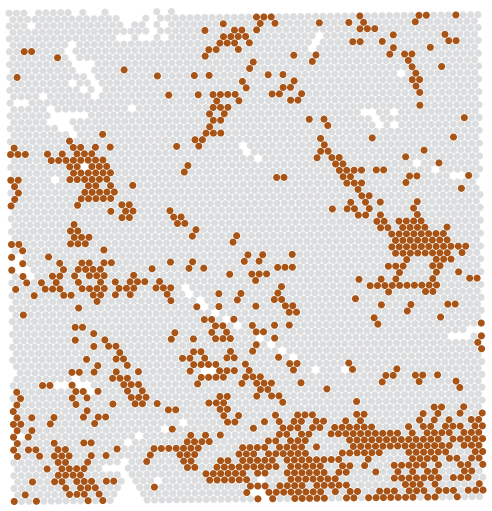
TAM composition



0 0.291

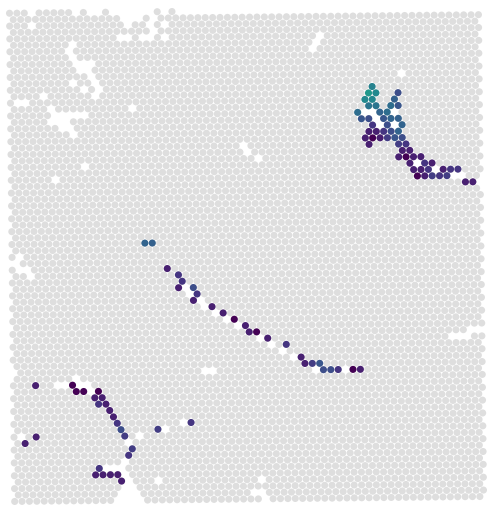
Fraction of all 30

TAM P80 targets



False True

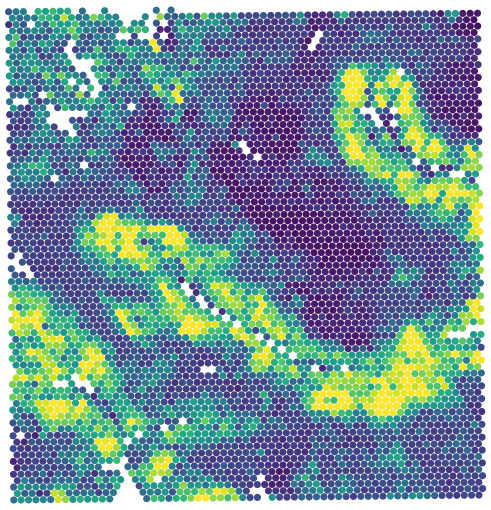
TAM distance



0 1.13e+03

Nearest spot center (μm)

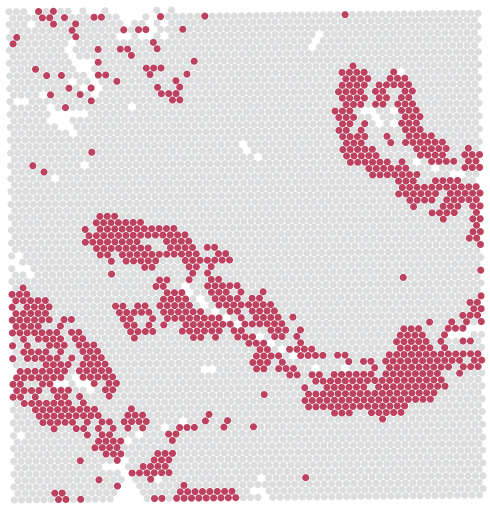
Treg composition



0 0.102

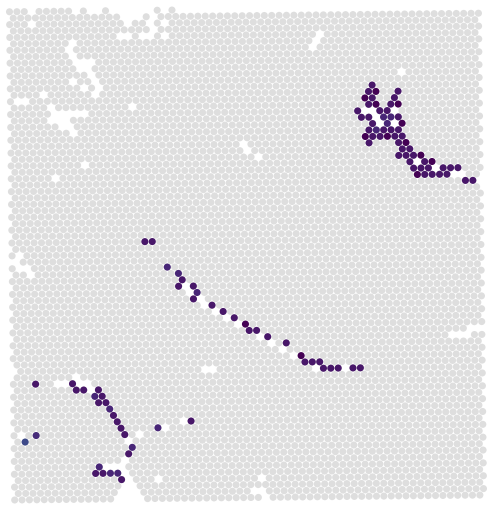
Fraction of all 30

Treg P80 targets



False True

Treg distance

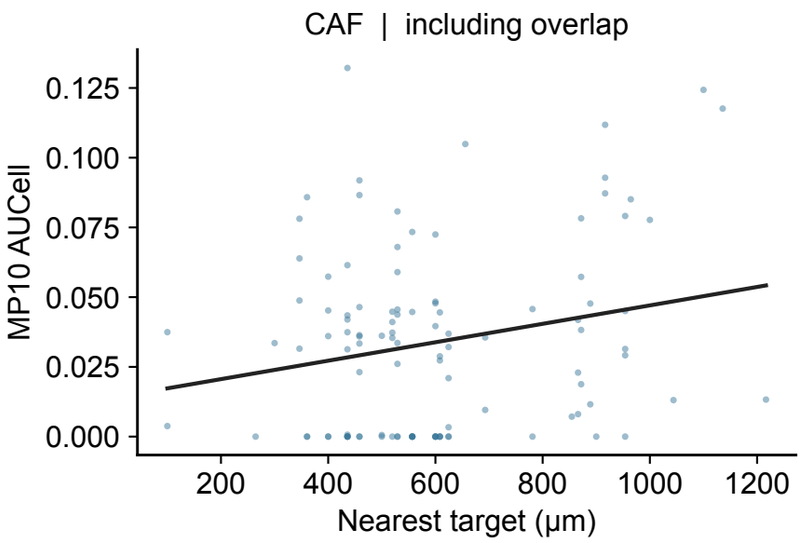


0 1.61e+03

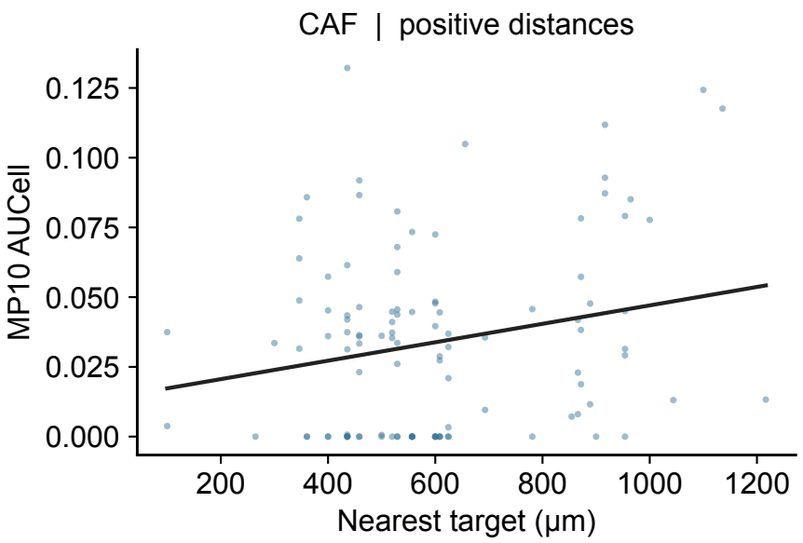
Nearest spot center (μm)

Inhousedata_T4384654 | MP10 versus nearest target

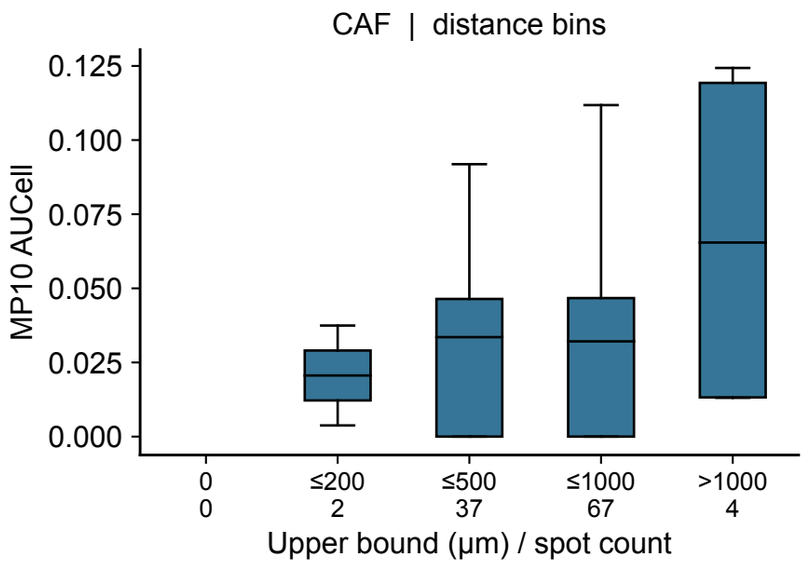
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



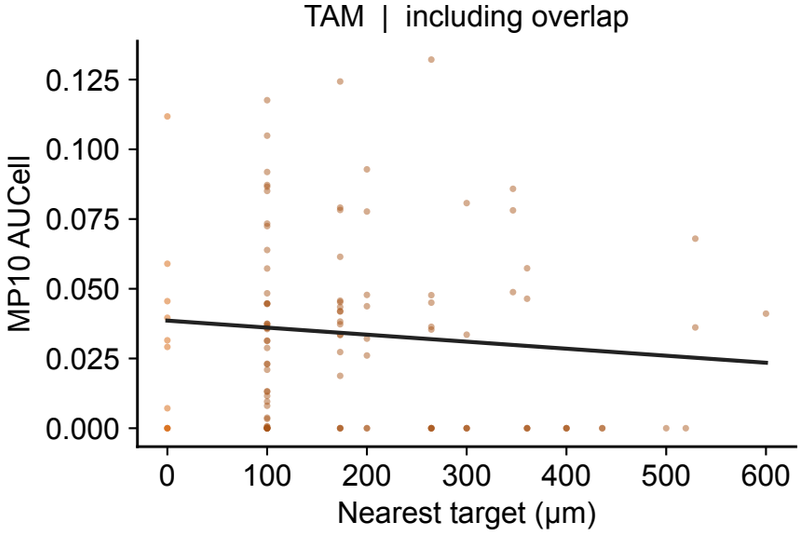
r +0.22 ρ +0.11
n 110/110 queries • 951 targets



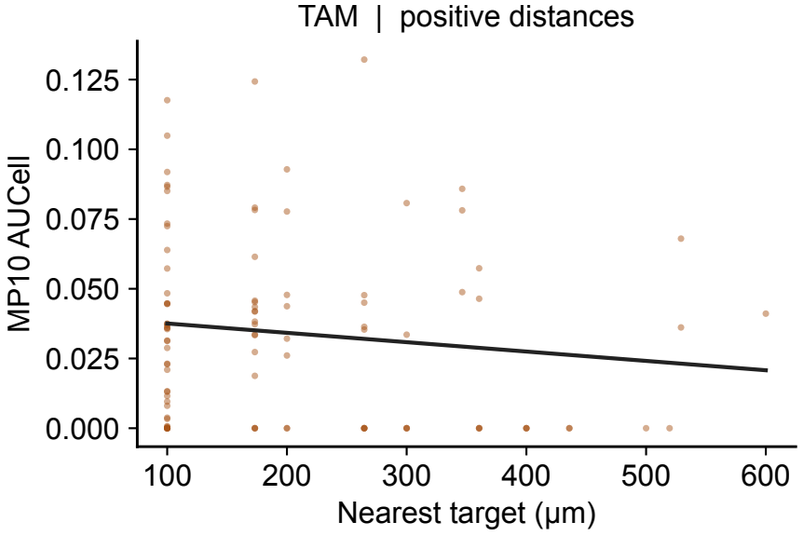
r +0.22 ρ +0.11
n 110/110 queries • 951 targets



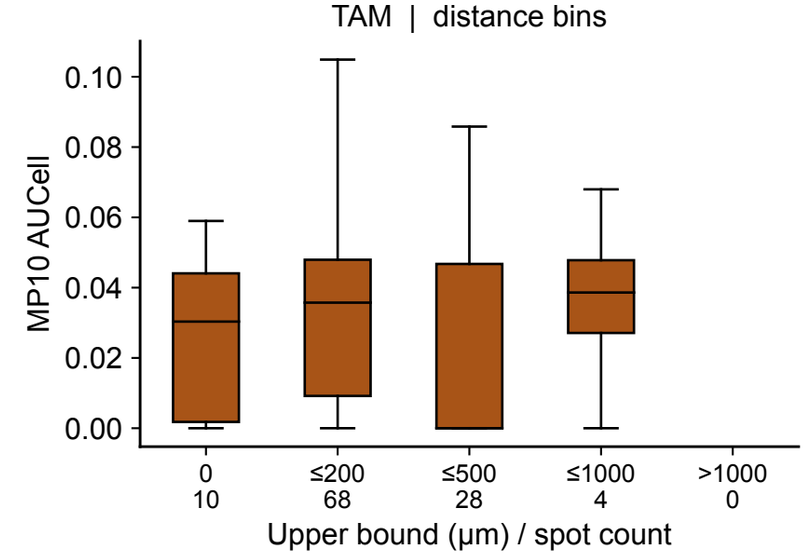
Orange = zero distance
Line = within-section linear fit



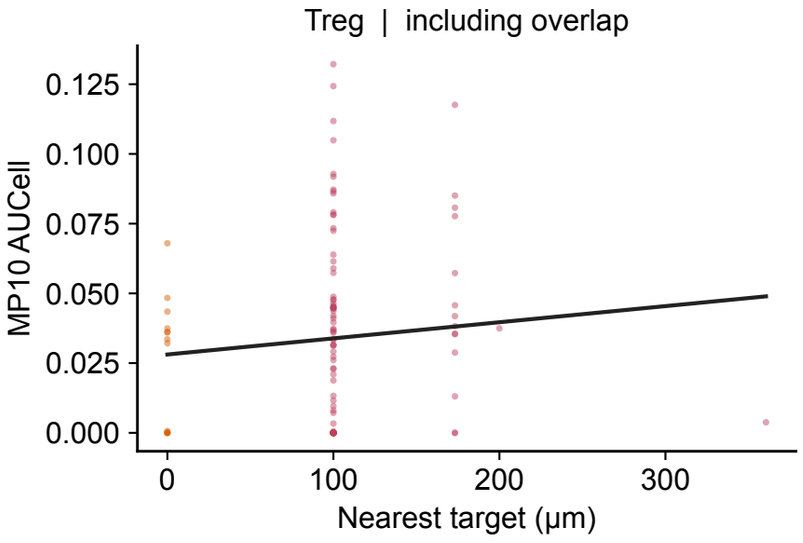
r -0.10 ρ -0.09
n 110/110 queries • 951 targets



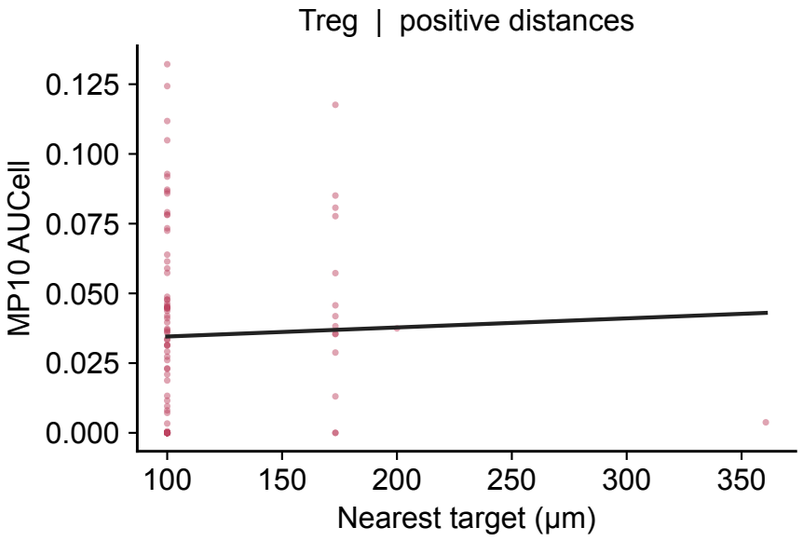
r -0.13 ρ -0.12
n 100/110 queries • 951 targets



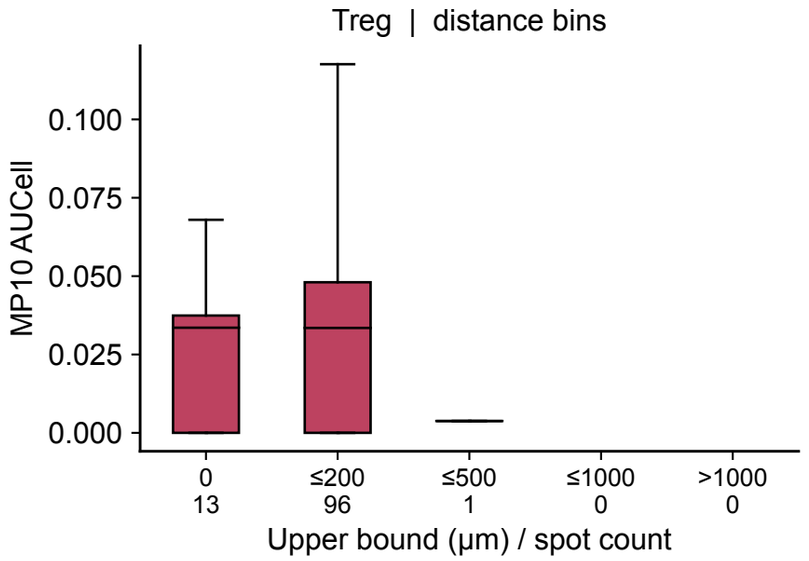
Orange = zero distance
Line = within-section linear fit



r +0.09 ρ +0.11
n 110/110 queries • 951 targets



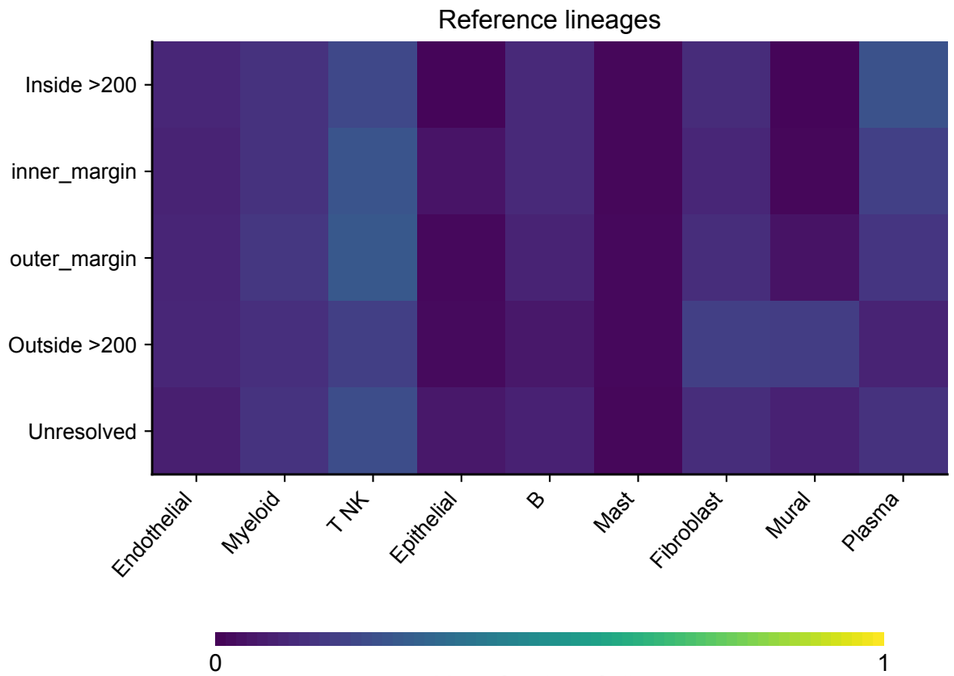
r +0.04 ρ +0.08
n 97/110 queries • 951 targets

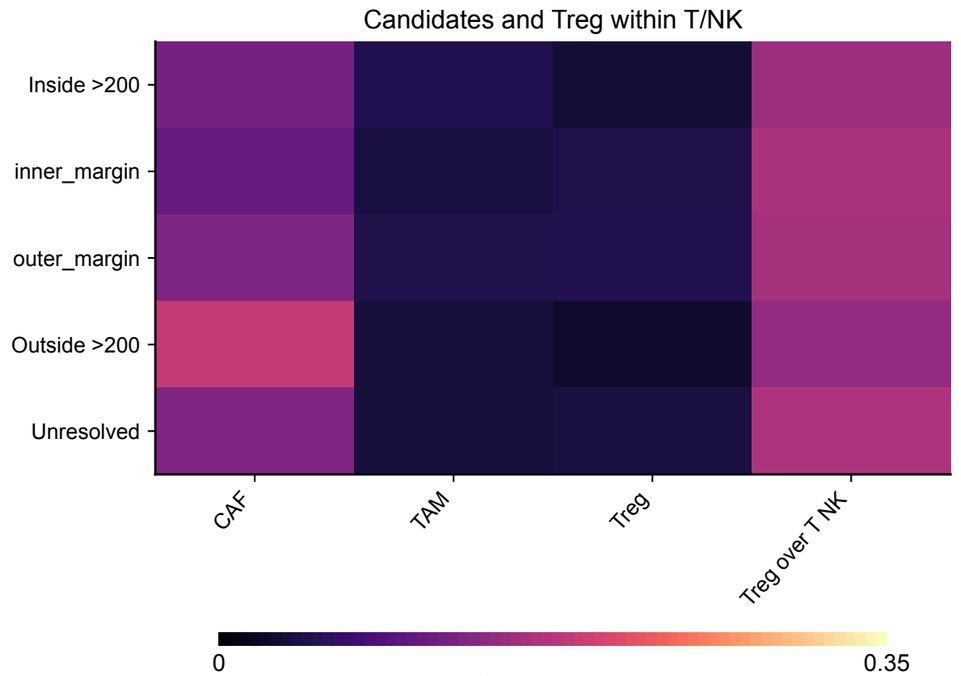


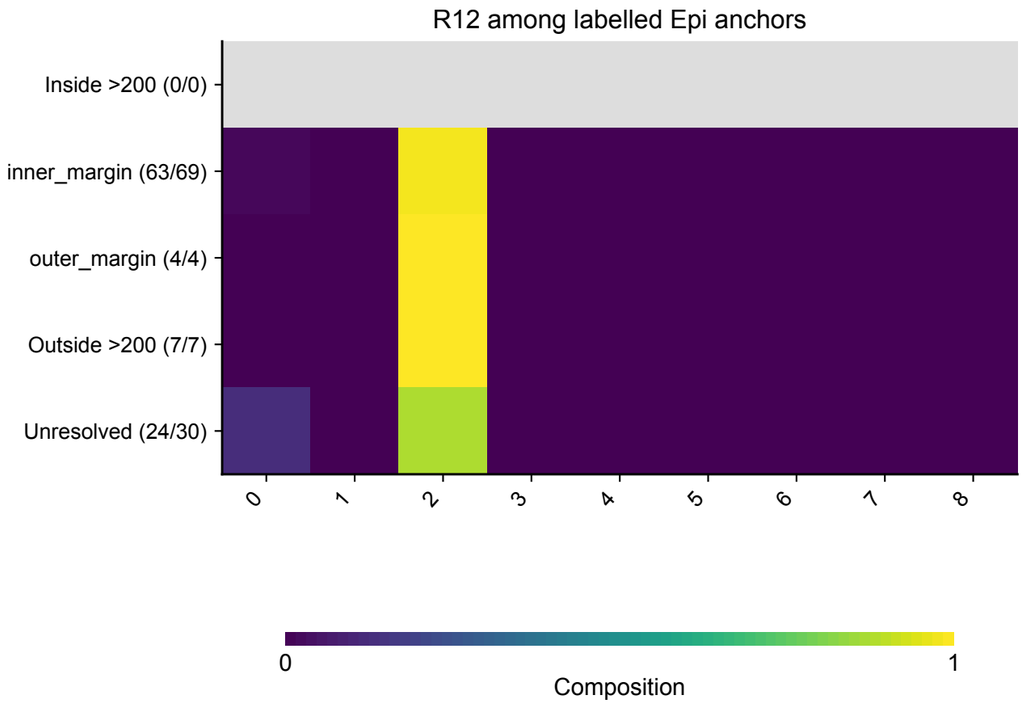
Orange = zero distance
Line = within-section linear fit

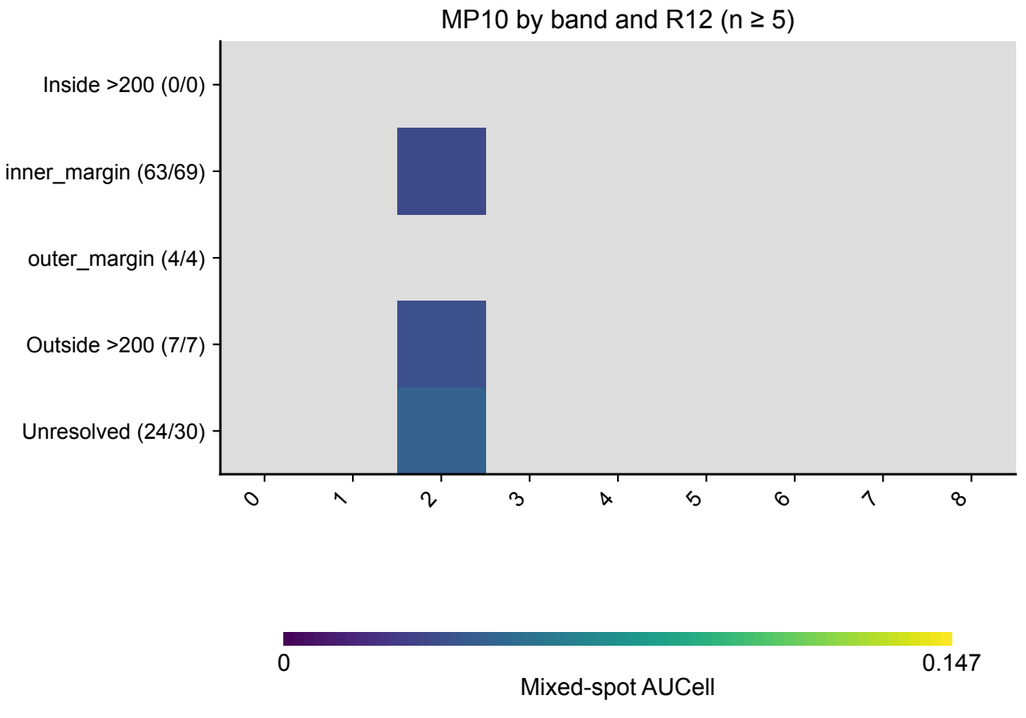
Inhousedata_T4384654 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





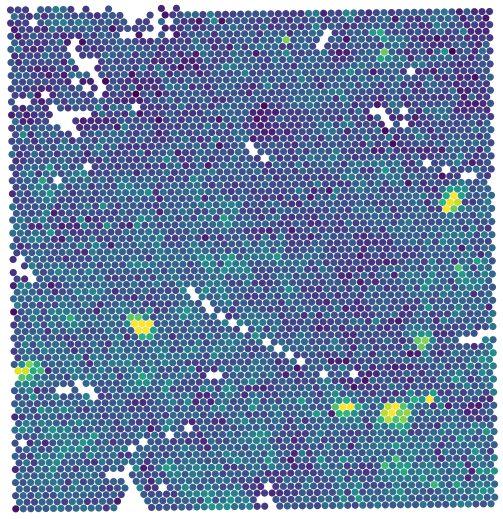




Inhousedata_T4384654 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

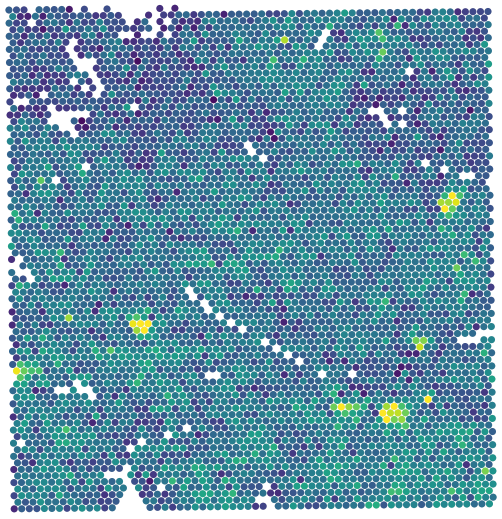
E2F Targets



0 0.152

UCell score

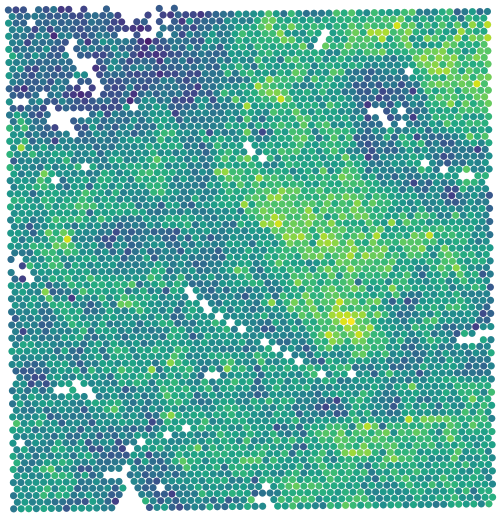
G2M Checkpoint



0 0.134

UCell score

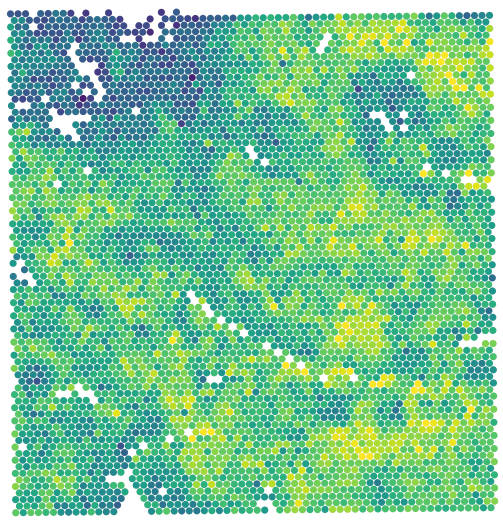
Hypoxia



0 0.211

UCell score

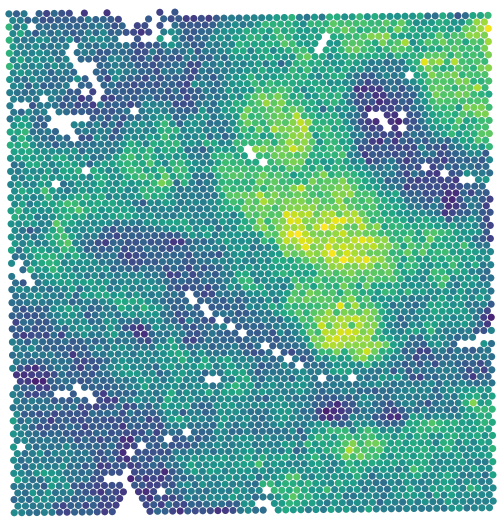
Tnfa Signaling Via Nfkb



0 0.275

UCell score

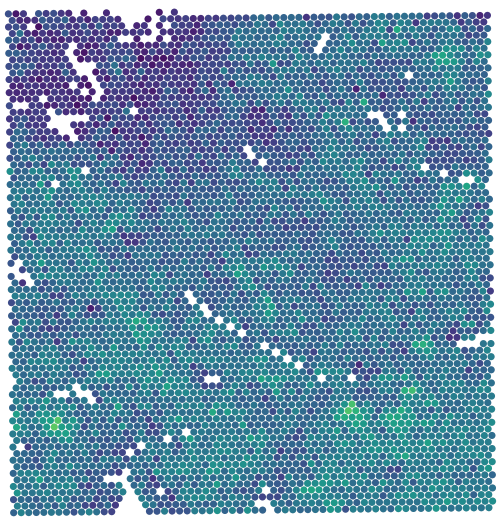
Emt



0 0.386

UCell score

Interferon Gamma Response



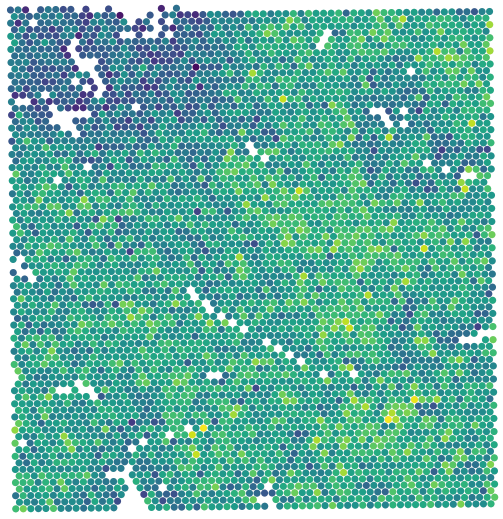
0 0.275

UCell score

Inhousedata_T4384654 | Immune, lipid and matrix programs

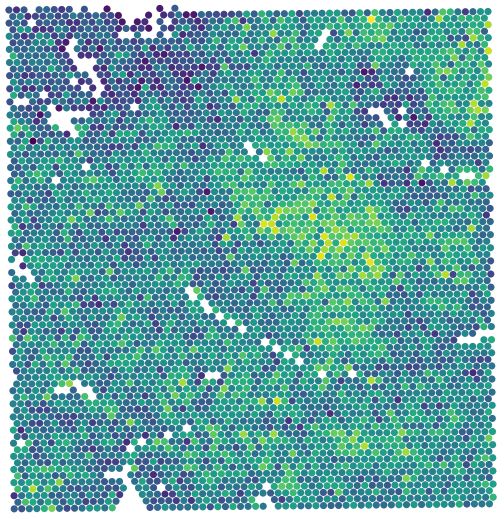
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling



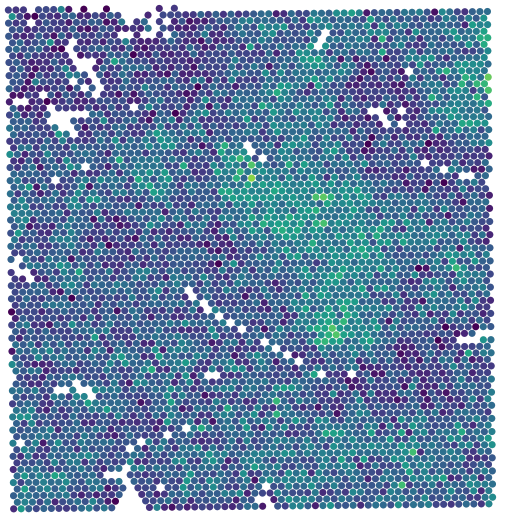
0 0.144 UCell score

Tgf Beta Signaling



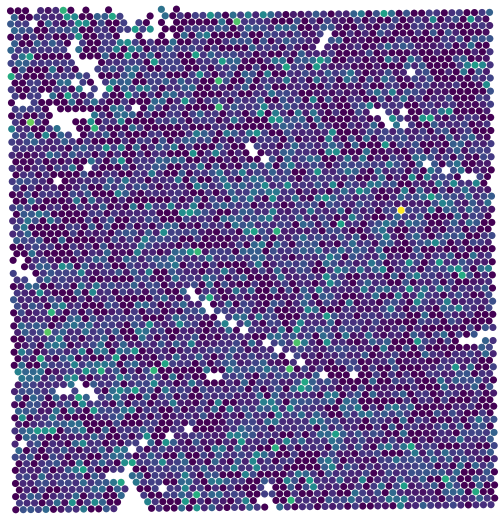
0 0.232 UCell score

Cholesterol Homeostasis



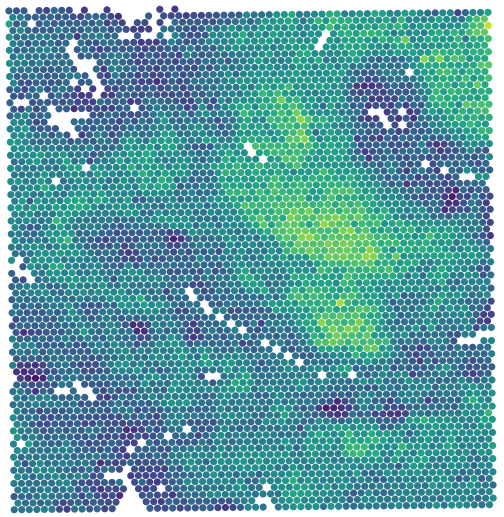
0 0.21 UCell score

Cholesterol Biosynthesis



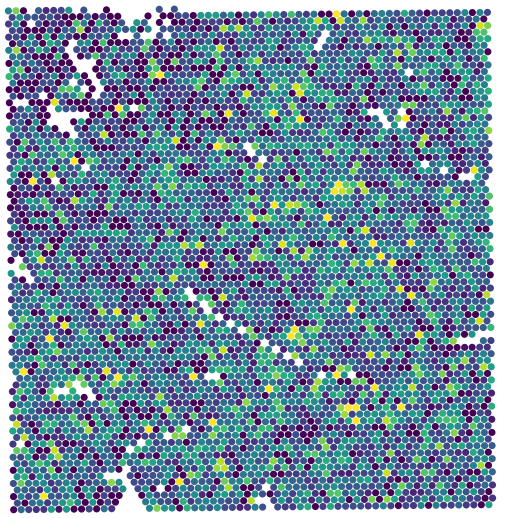
0 0.166 UCell score

Extracellular Matrix Organization



0 0.247 UCell score

Co Inhibition By Ctla4

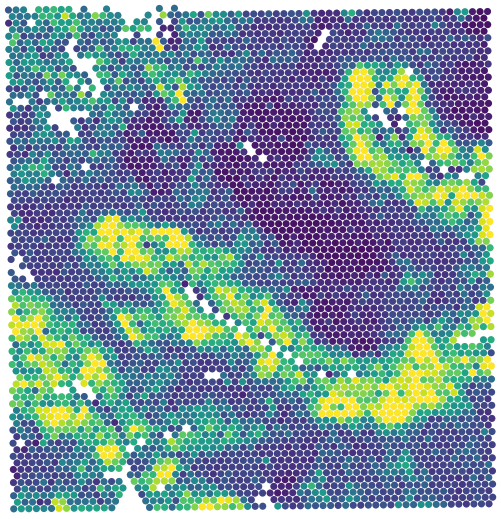


0 0.151 UCell score

Inhousedata_T4384654 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

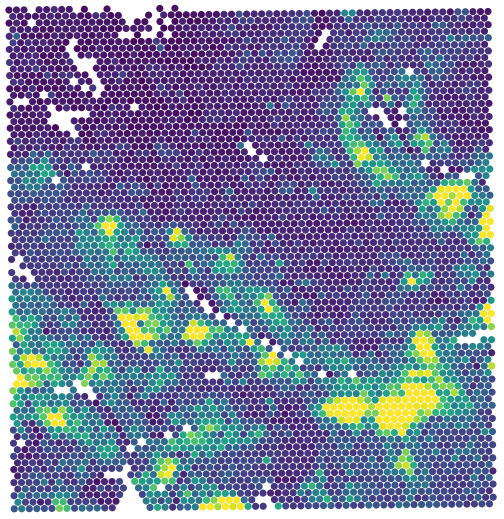
Treg composition



0 0.102

Fraction of all 30

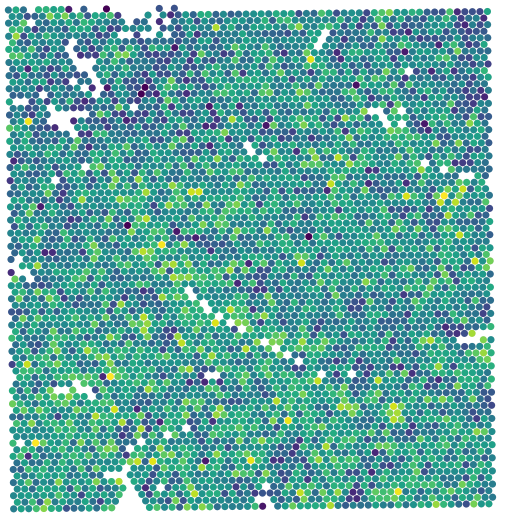
Treg estimated abundance



0 2.85

Posterior mean estimate

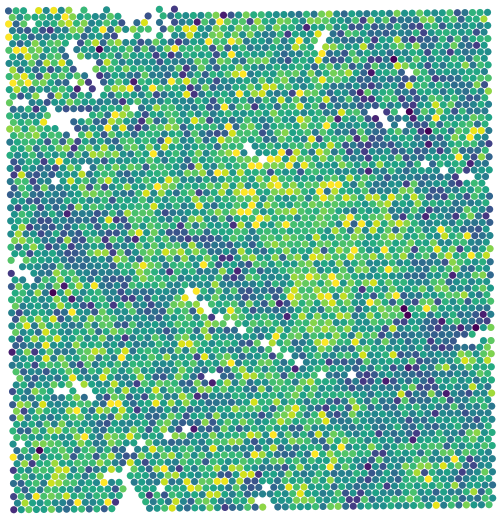
Treg Resting Vs Activated Dn



0 0.0778

UCell score

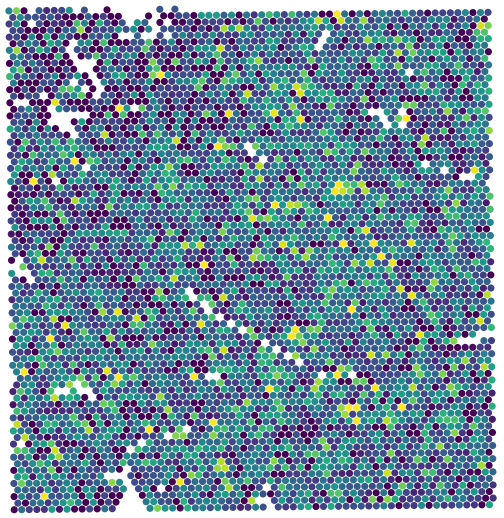
Treg Resting Vs Activated Up



0 0.0609

UCell score

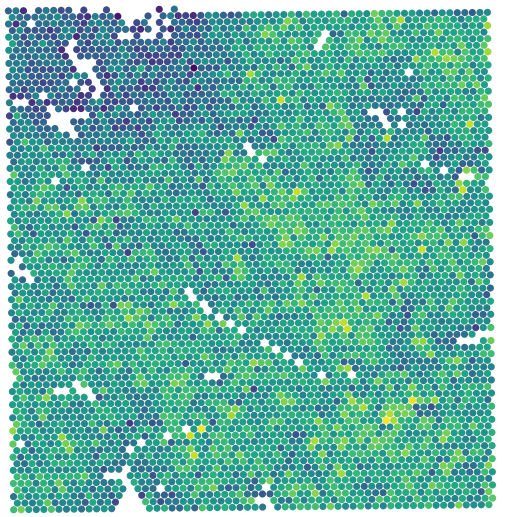
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

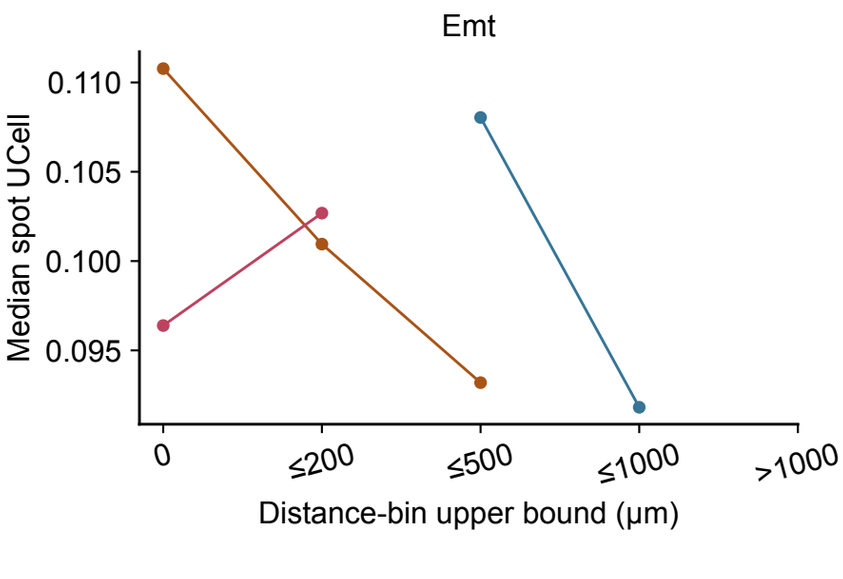


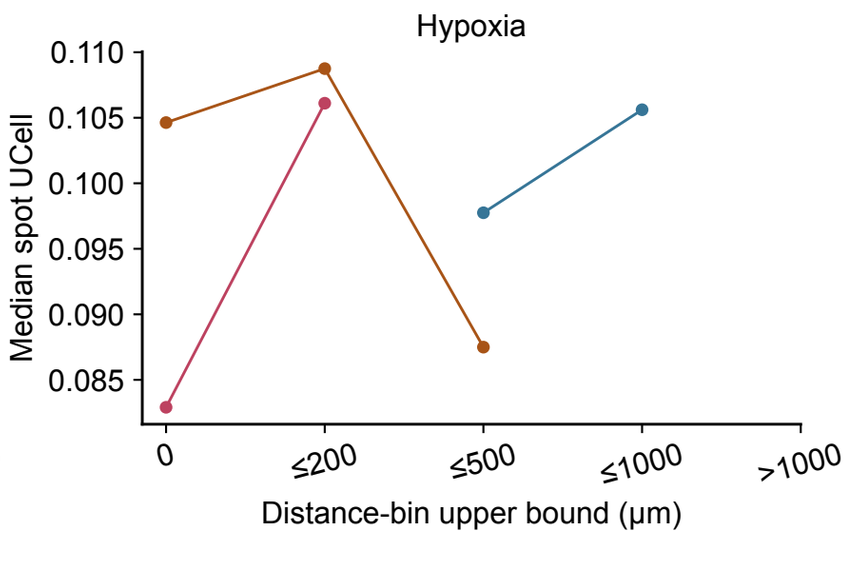
0 0.144

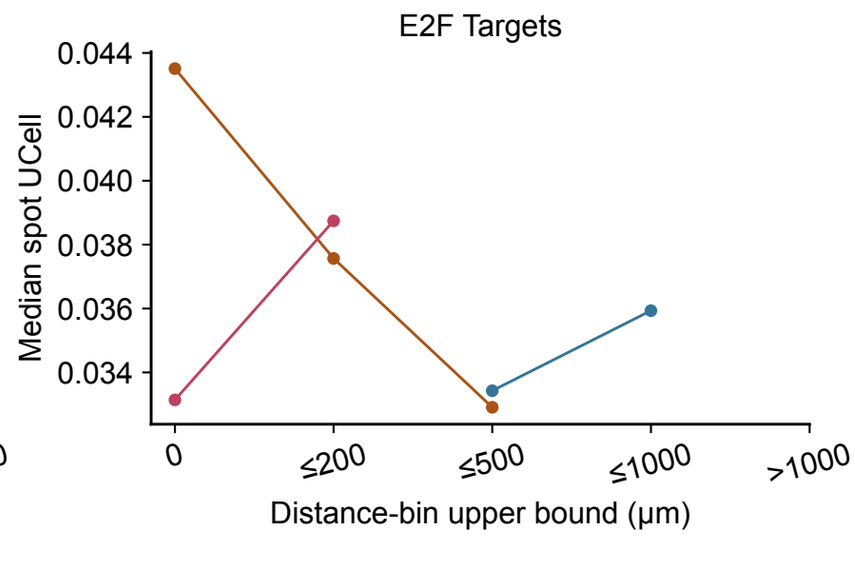
UCell score

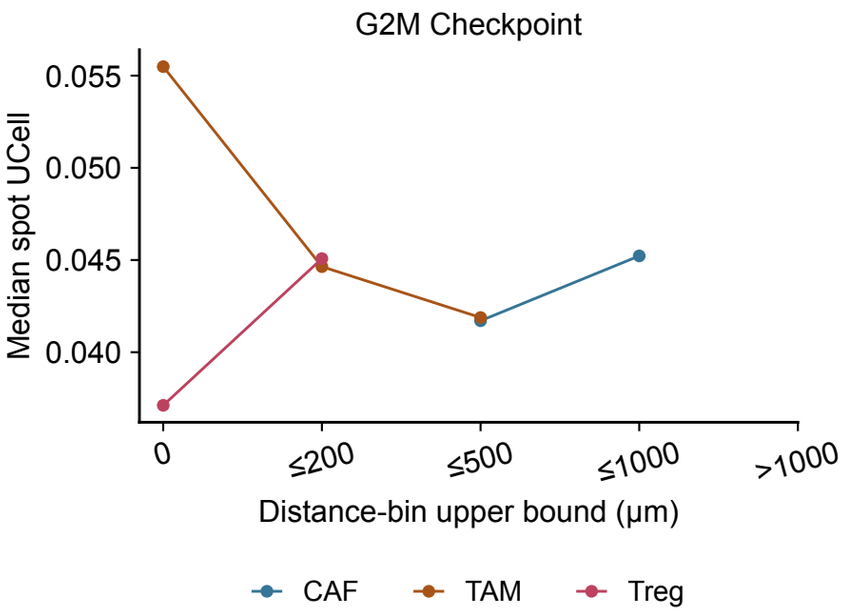
Inhousedata_T4384654 | Programs by target distance

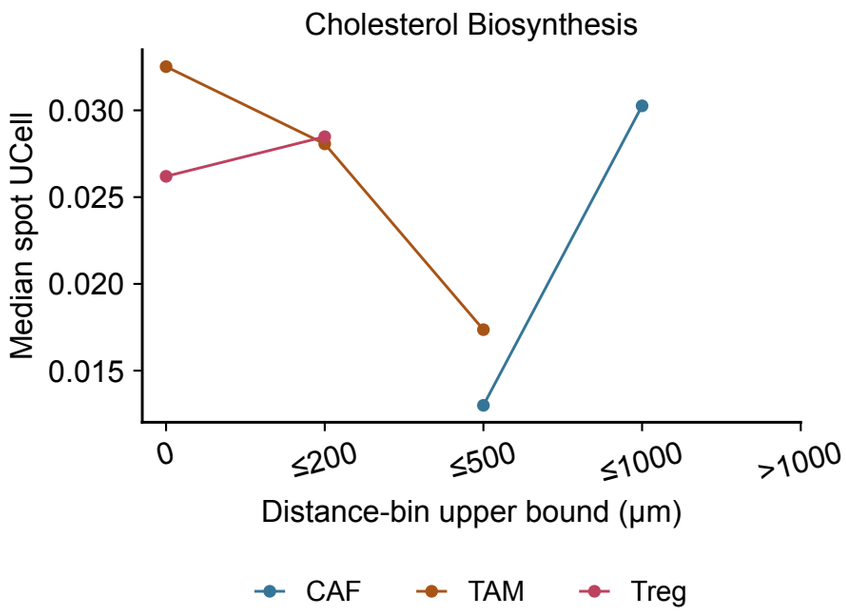
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

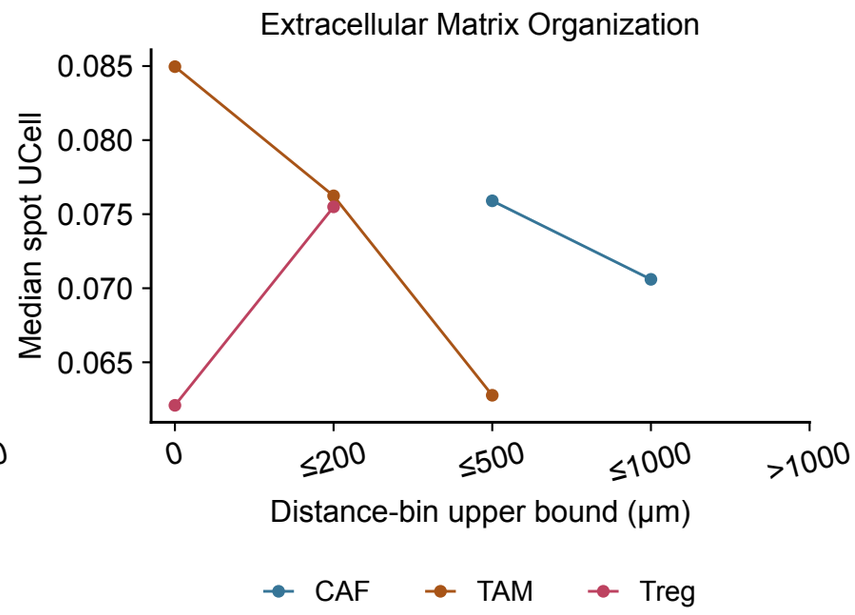






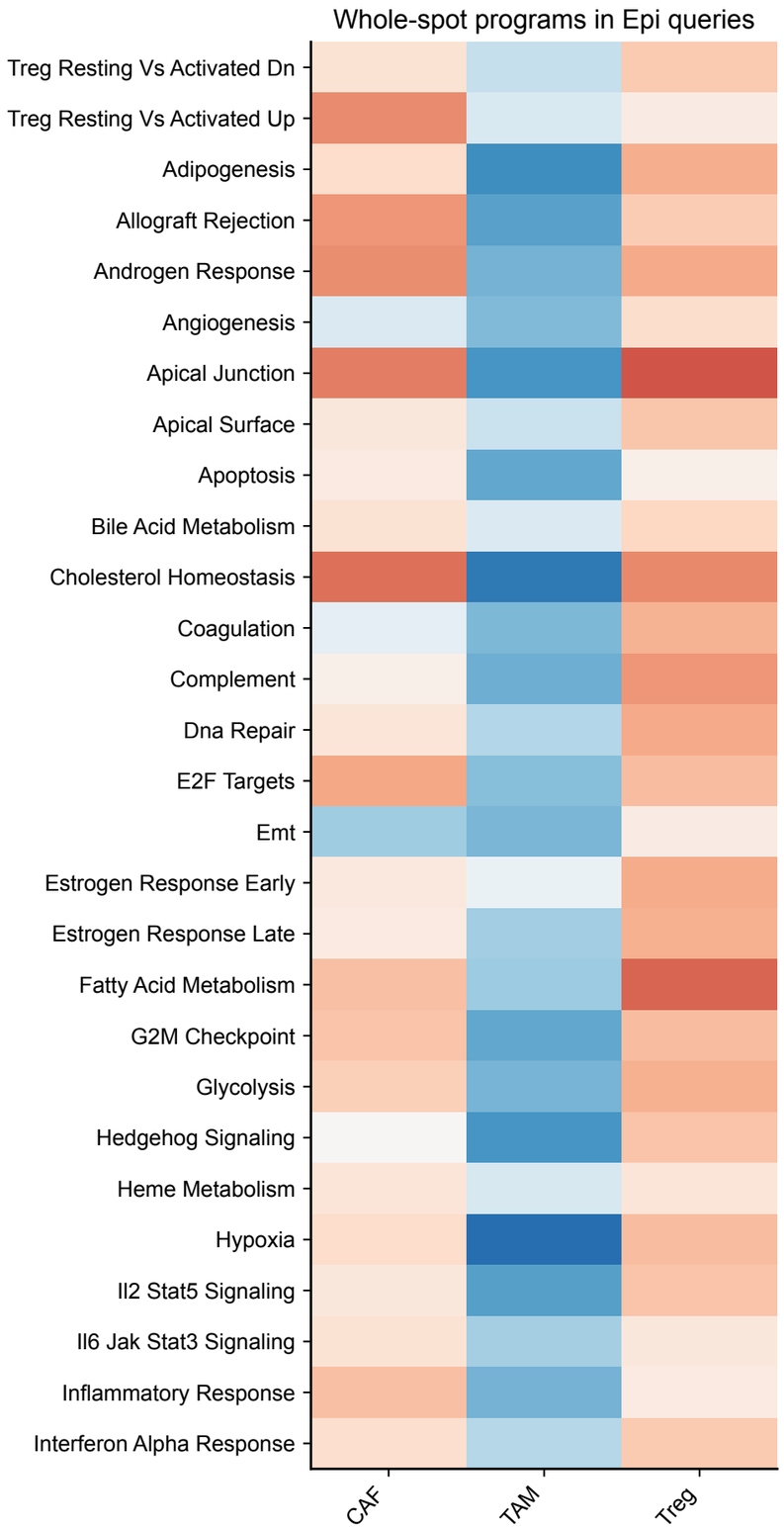


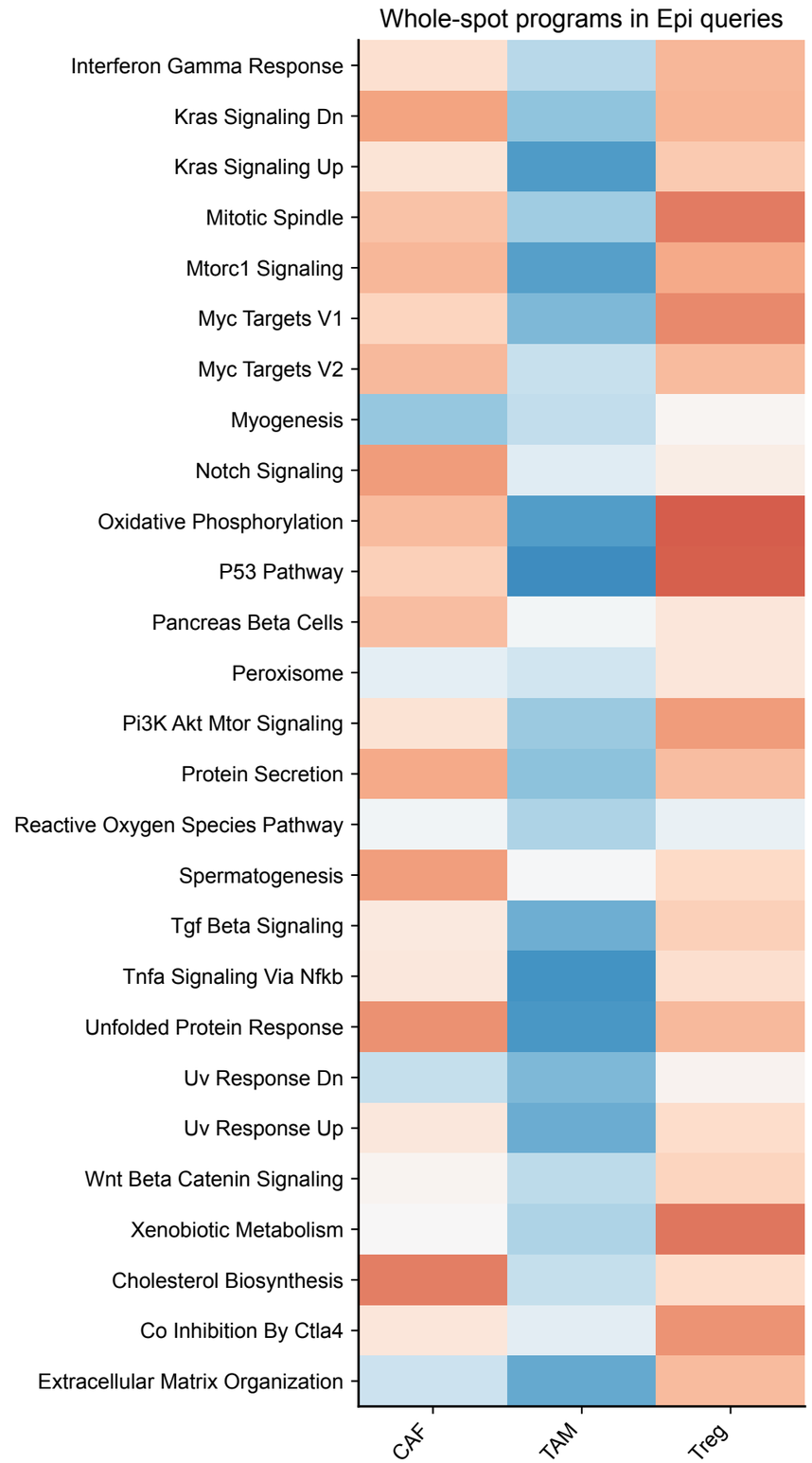




Inhousedata_T4384654 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

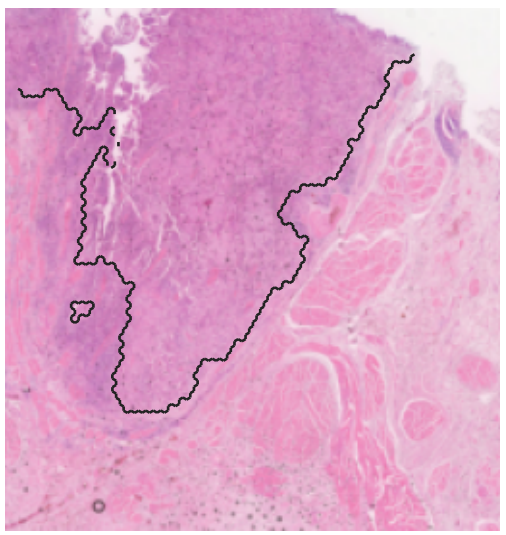




Inhousedata_T4398343 | Tissue context

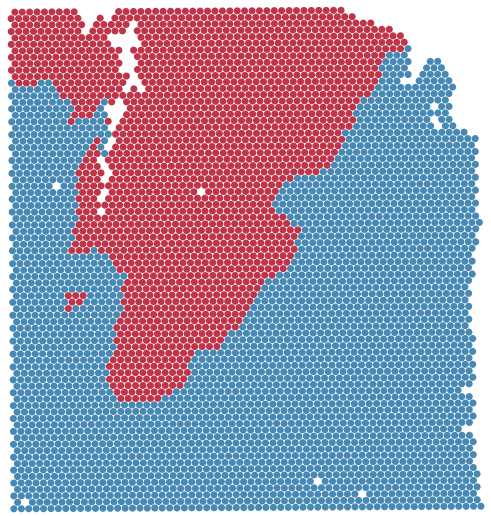
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



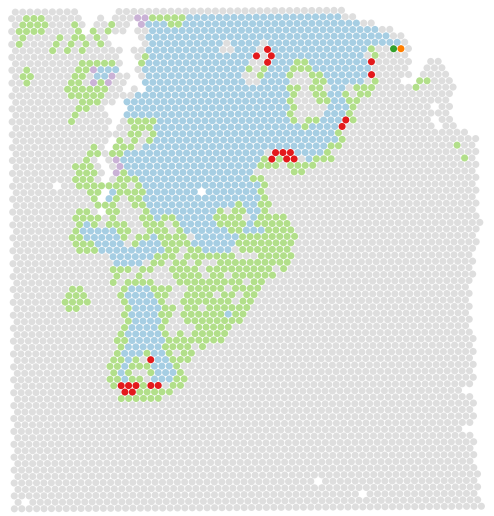
Black line: observed candidate outer edge

Final candidate location



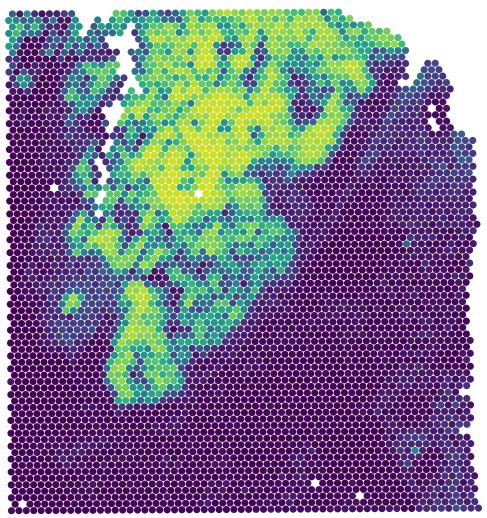
inside outside

Accepted R12 (original 0–8)



0 3 5 7 8 2

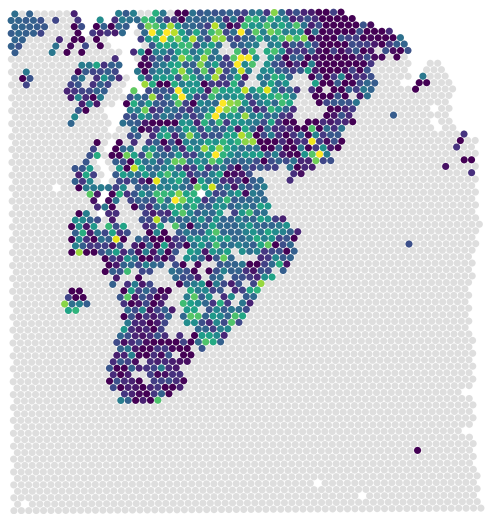
Epithelial support



0 1

Fraction of all 30

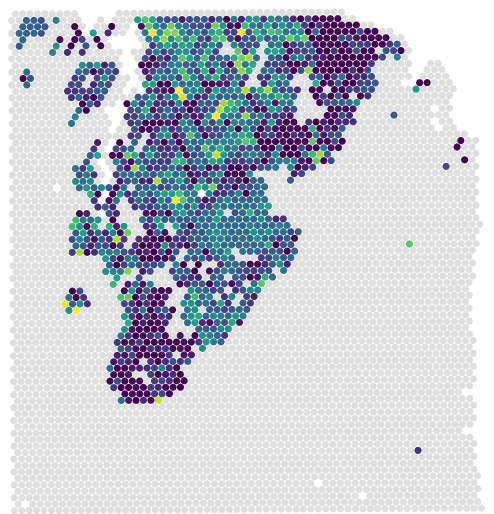
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



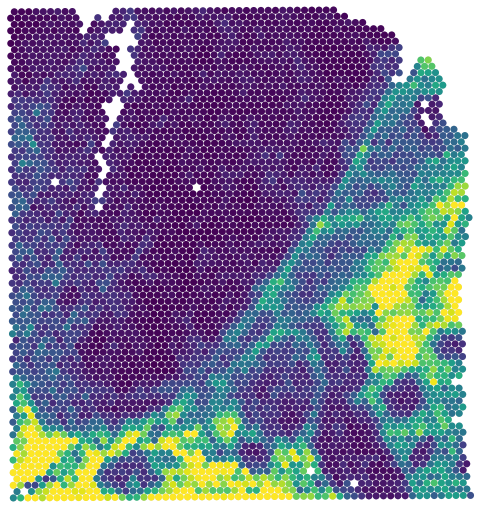
0 0.14

Matched-model AUCell

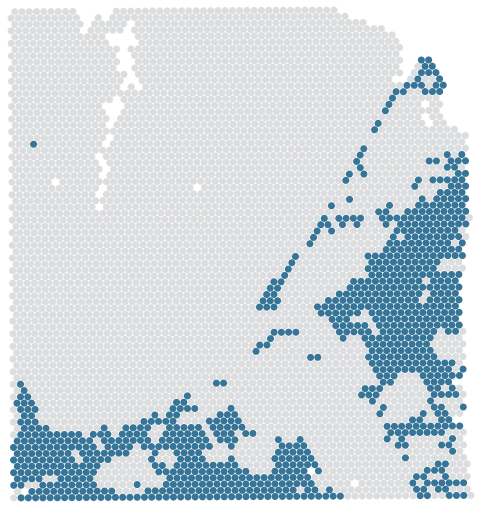
Inhousedata_T4398343 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

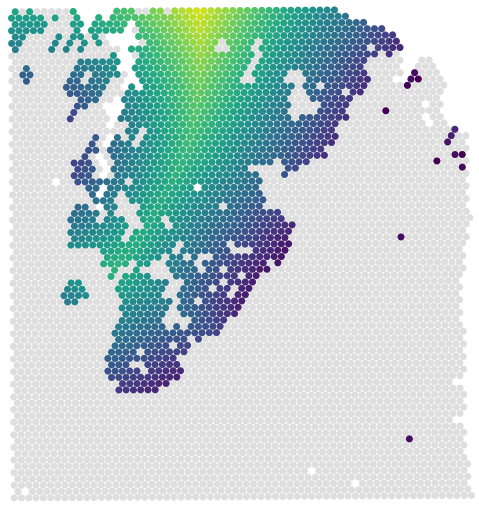
CAF composition



CAF P80 targets



CAF distance

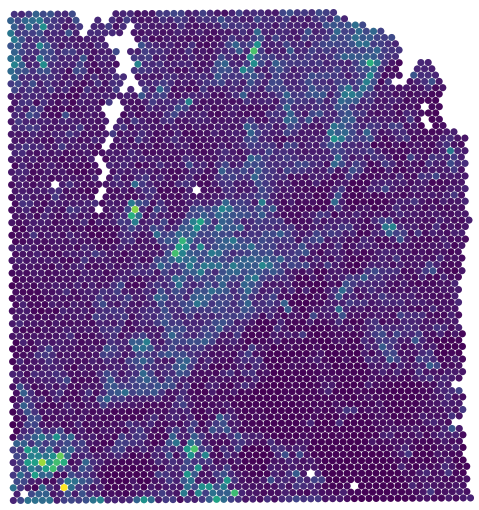


0 0.418
Fraction of all 30

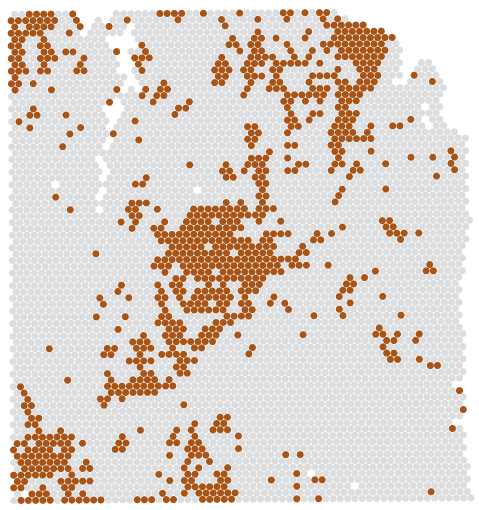
False True

0 3.1e+03
Nearest spot center (μm)

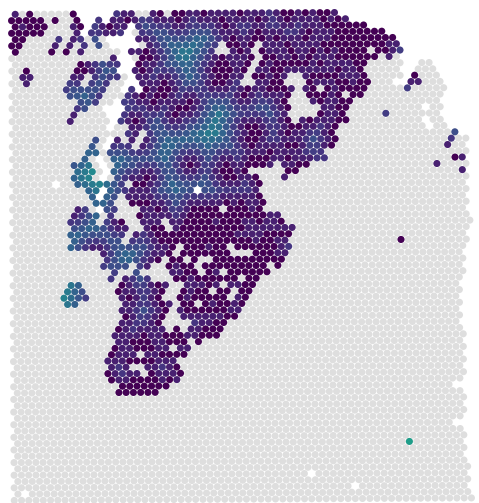
TAM composition



TAM P80 targets



TAM distance

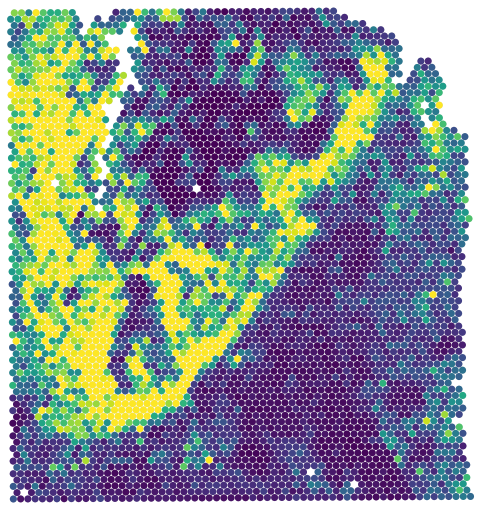


0 0.291
Fraction of all 30

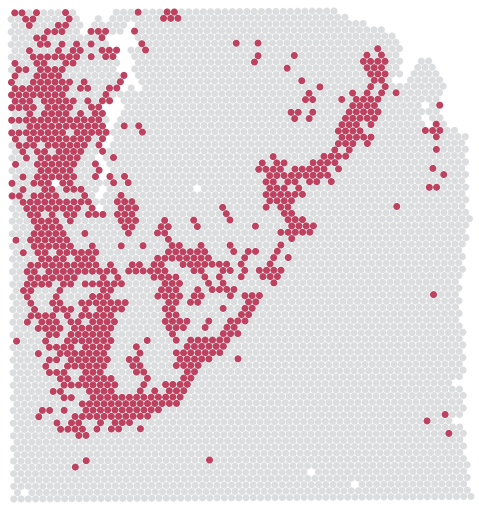
False True

0 1.13e+03
Nearest spot center (μm)

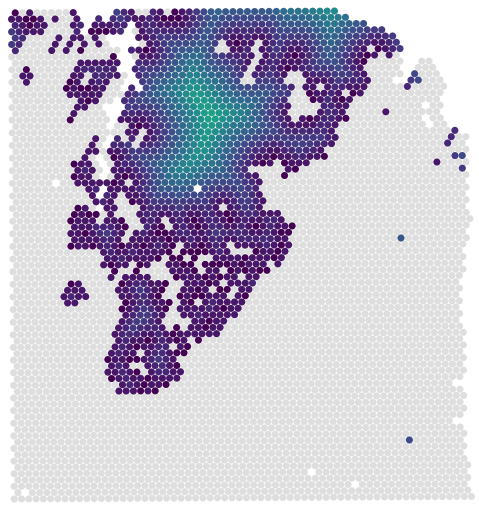
Treg composition



Treg P80 targets



Treg distance



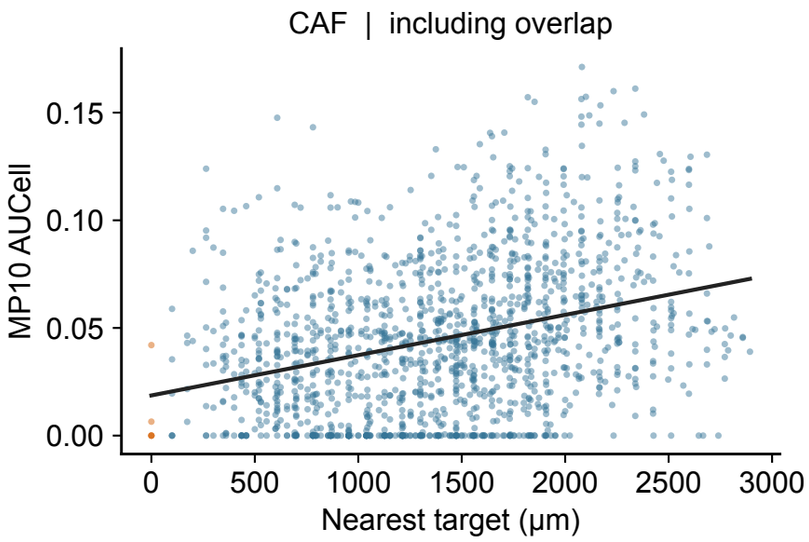
0 0.102
Fraction of all 30

False True

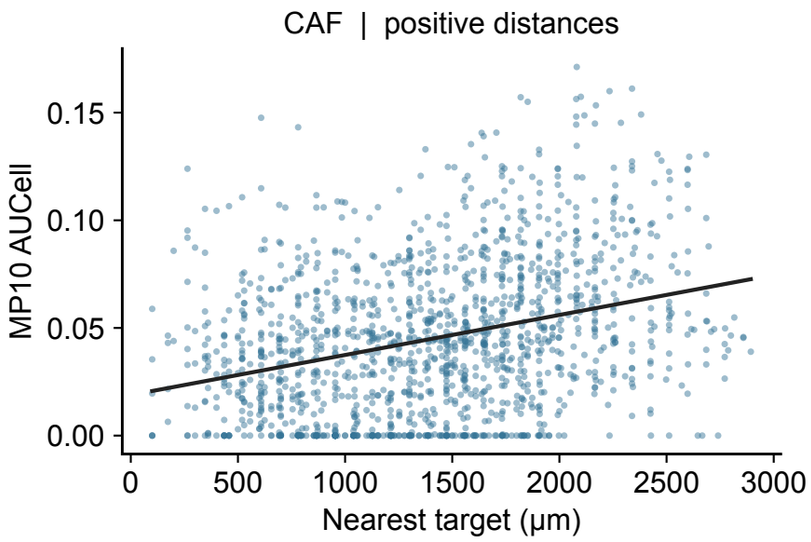
0 1.61e+03
Nearest spot center (μm)

Inhousedata_T4398343 | MP10 versus nearest target

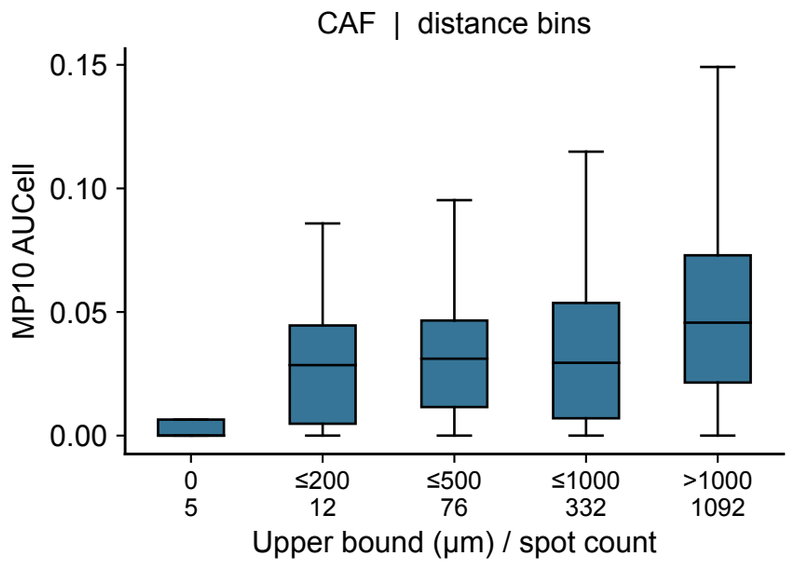
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



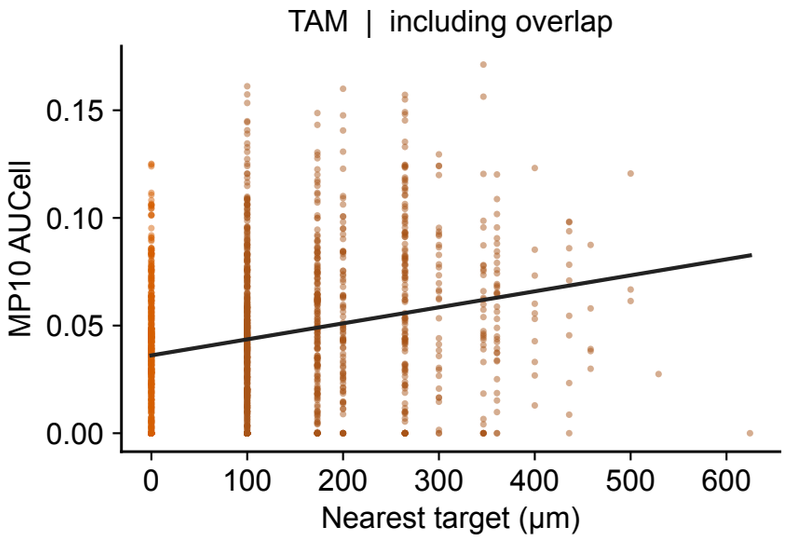
r +0.32 ρ +0.32
n 1517/1517 queries • 939 targets



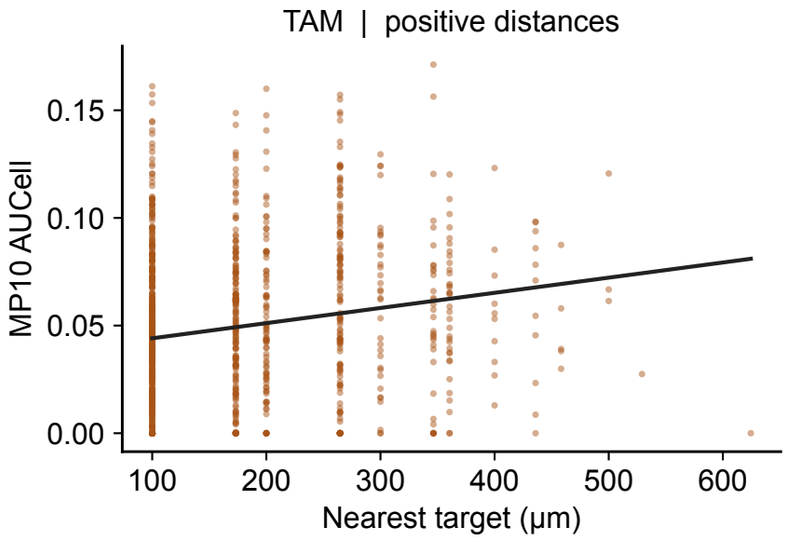
r +0.32 ρ +0.32
n 1512/1517 queries • 939 targets



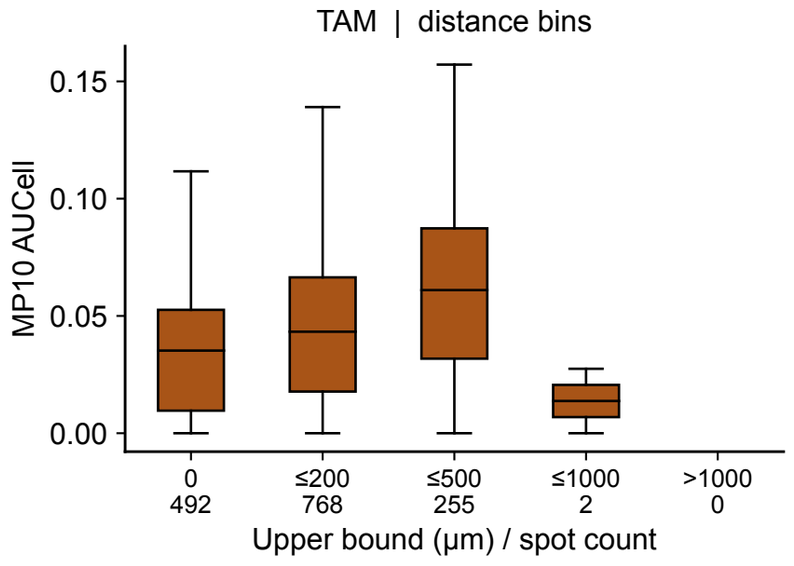
Orange = zero distance
Line = within-section linear fit



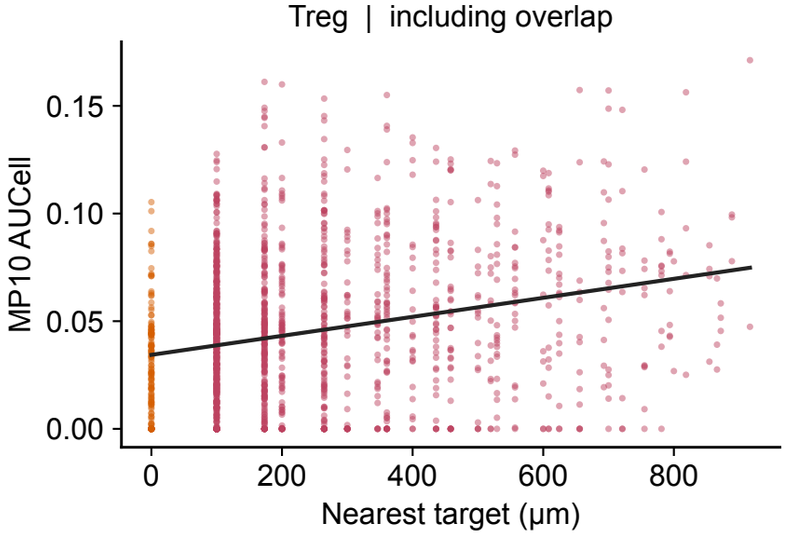
r +0.24 ρ +0.23
n 1517/1517 queries • 939 targets



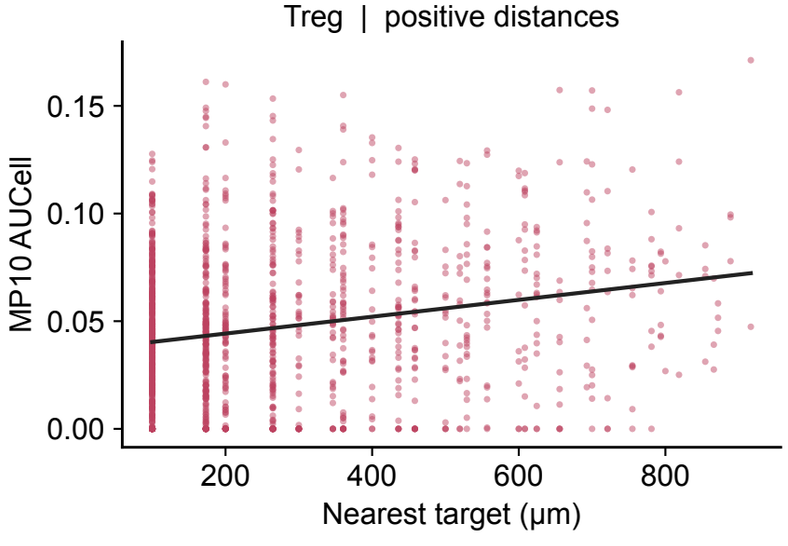
r +0.18 ρ +0.20
n 1025/1517 queries • 939 targets



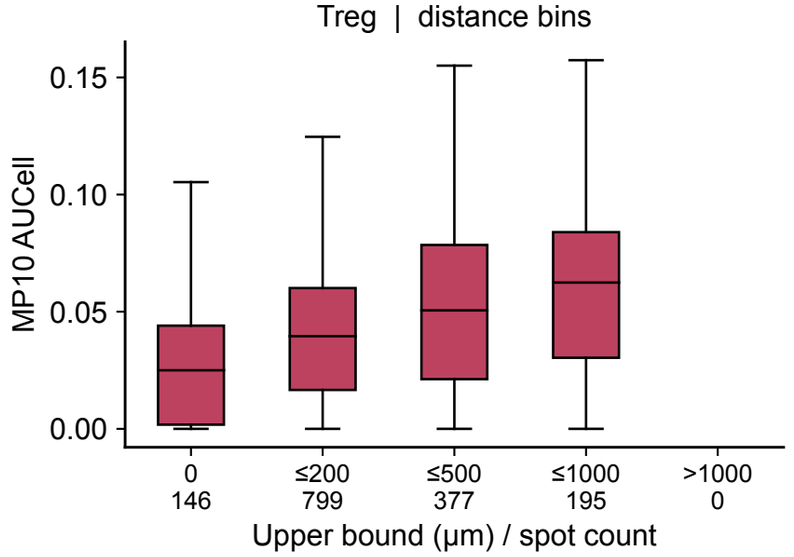
Orange = zero distance
Line = within-section linear fit



r +0.26 ρ +0.24
n 1517/1517 queries • 939 targets



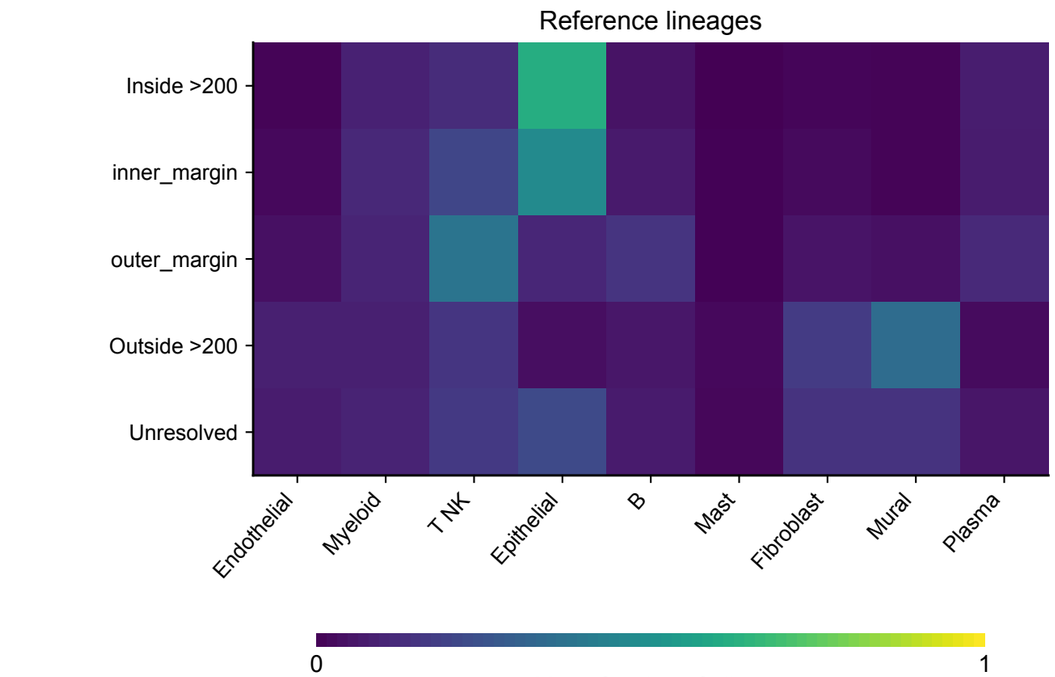
r +0.22 ρ +0.20
n 1371/1517 queries • 939 targets

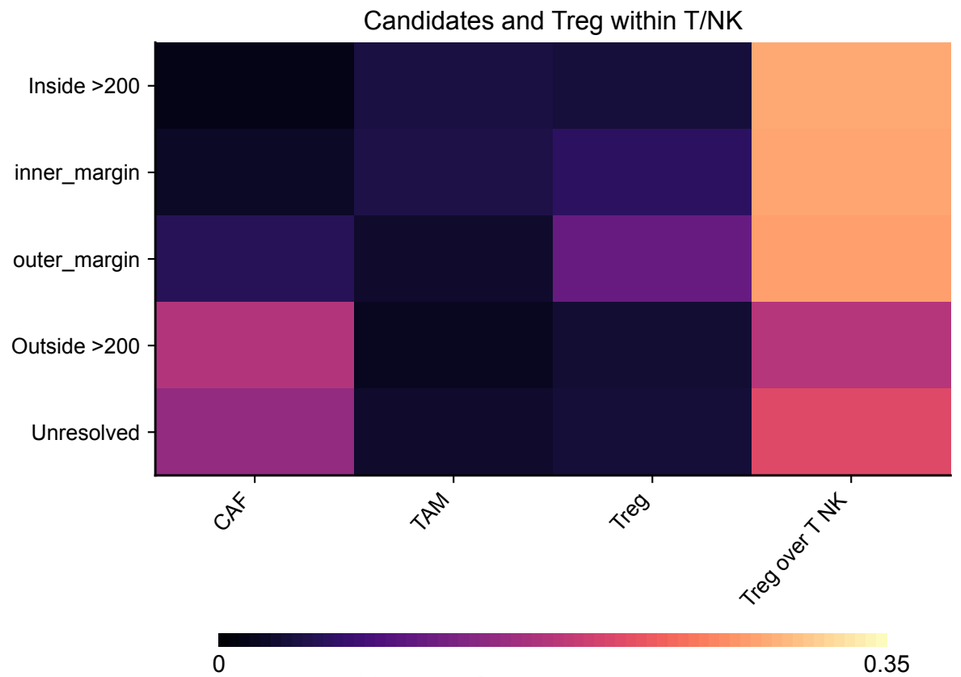


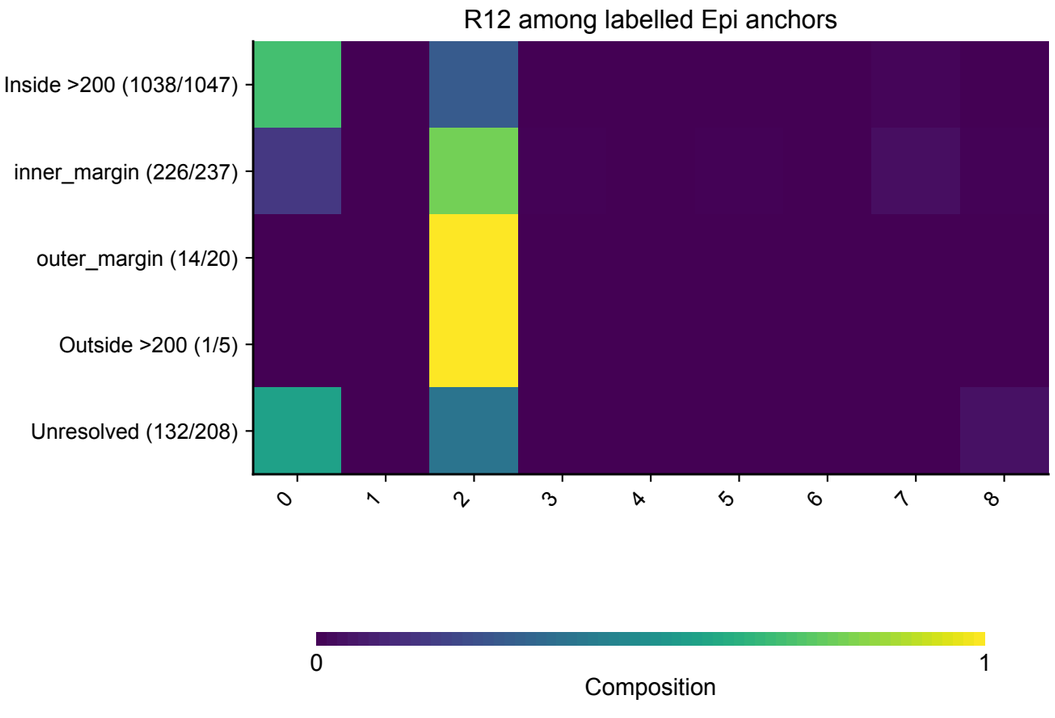
Orange = zero distance
Line = within-section linear fit

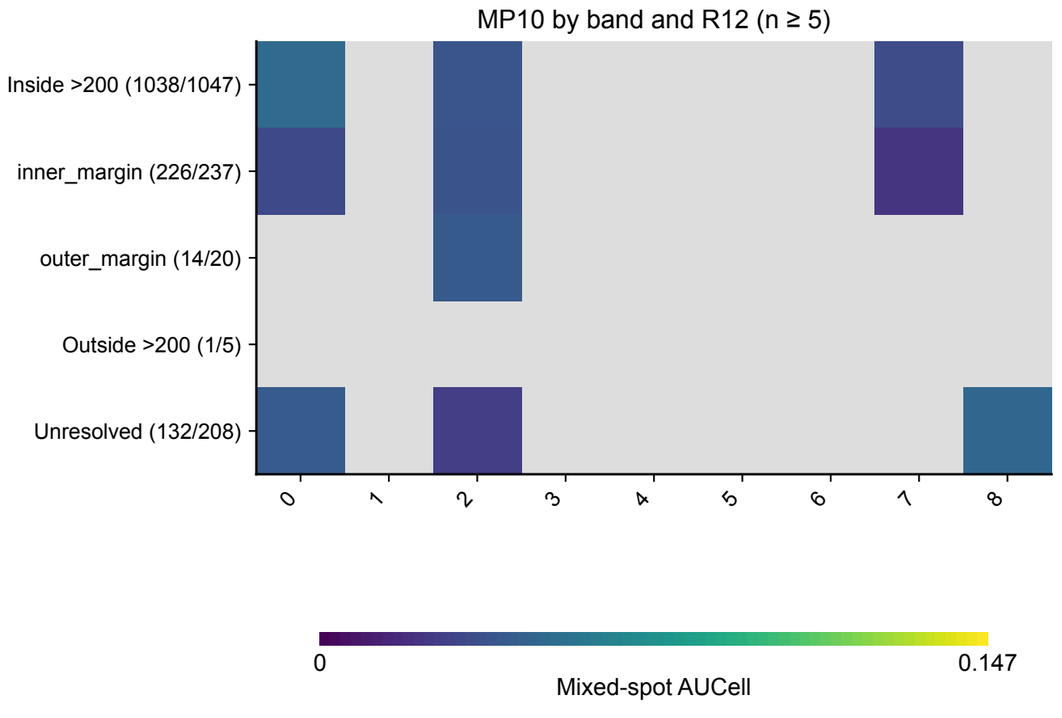
Inhousedata_T4398343 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





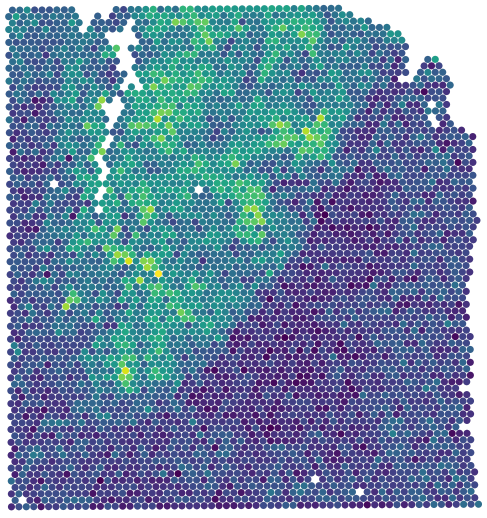




Inhousedata_T4398343 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

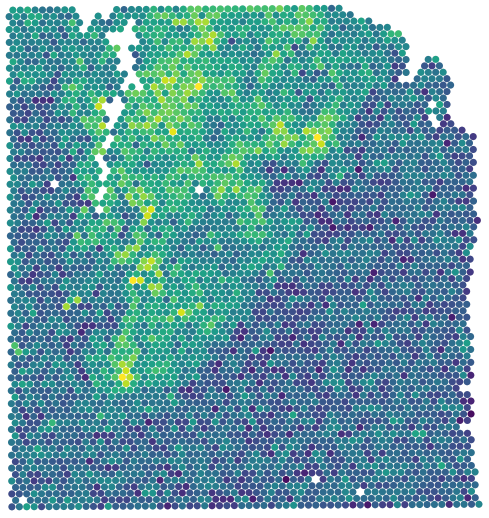
E2F Targets



0 0.152

UCell score

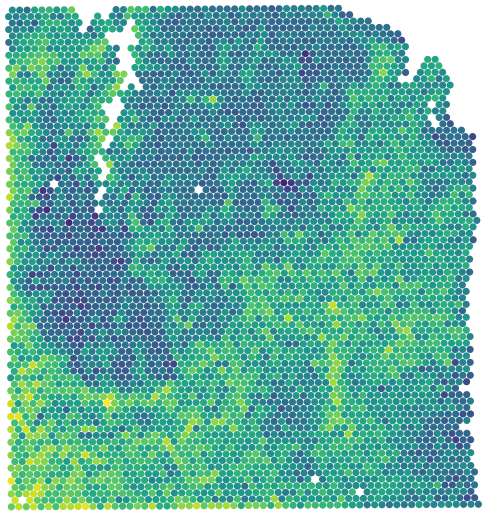
G2M Checkpoint



0 0.134

UCell score

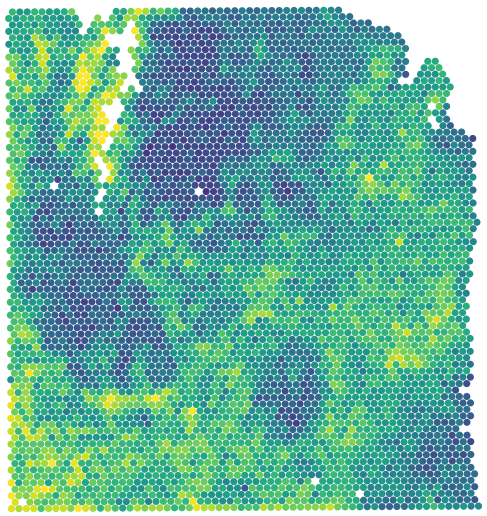
Hypoxia



0 0.211

UCell score

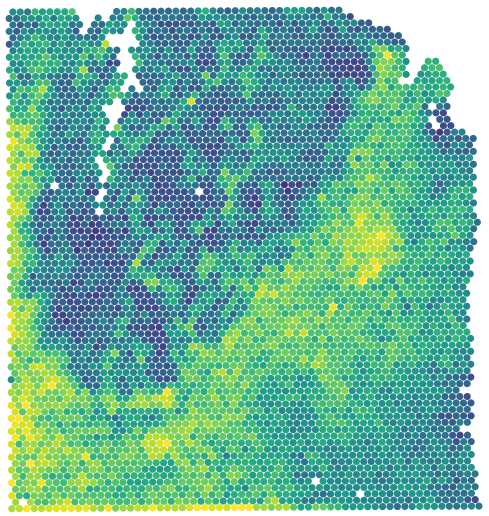
Tnfa Signaling Via Nfkb



0 0.275

UCell score

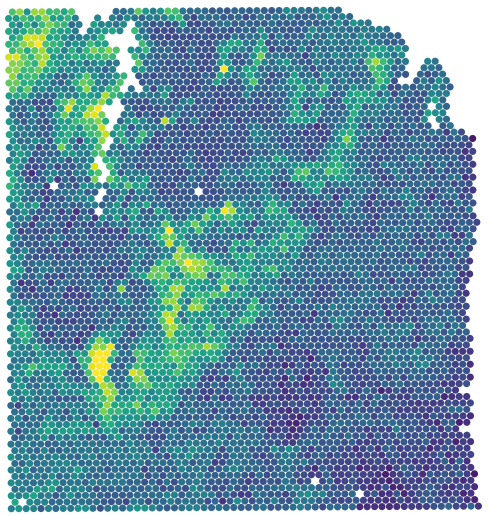
Emt



0 0.386

UCell score

Interferon Gamma Response



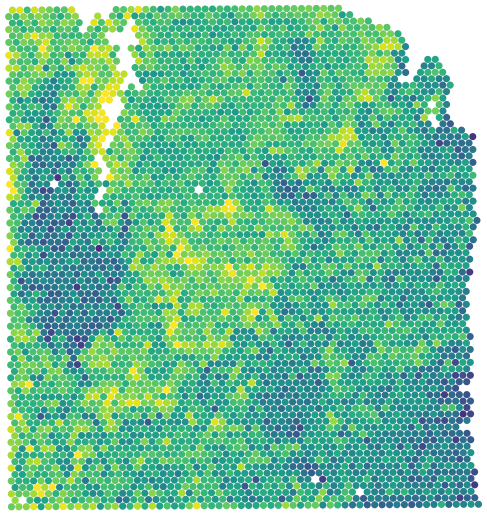
0 0.275

UCell score

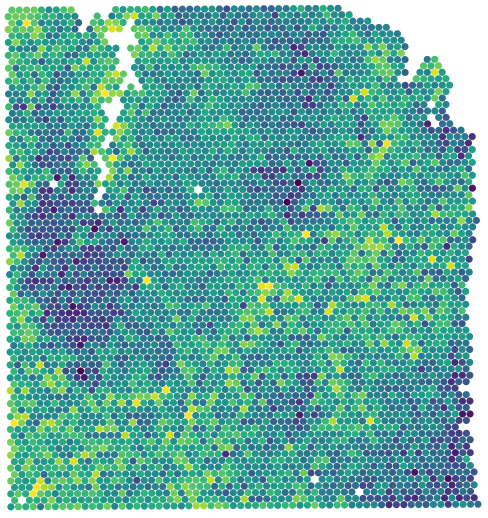
Inhousedata_T4398343 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

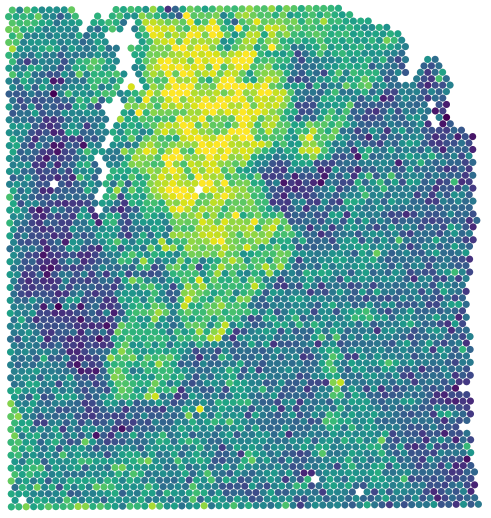
Il2 Stat5 Signaling



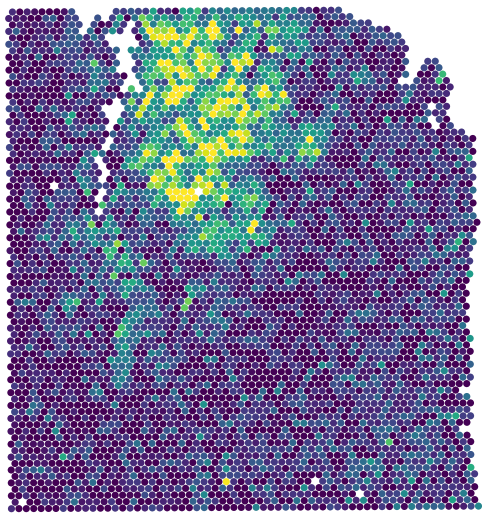
Tgf Beta Signaling

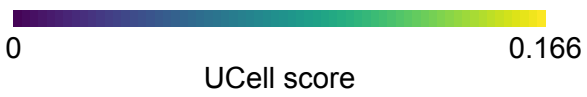


Cholesterol Homeostasis

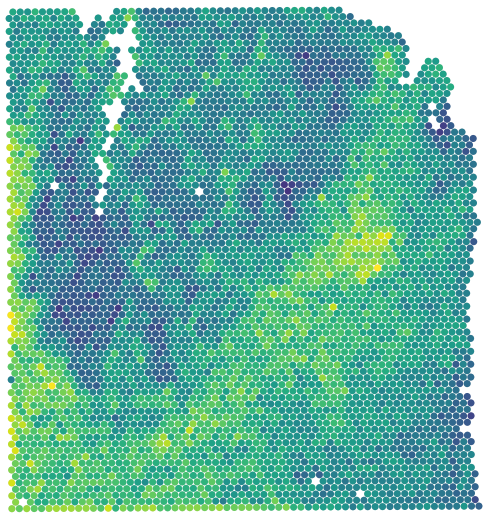


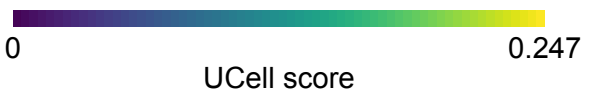
Cholesterol Biosynthesis



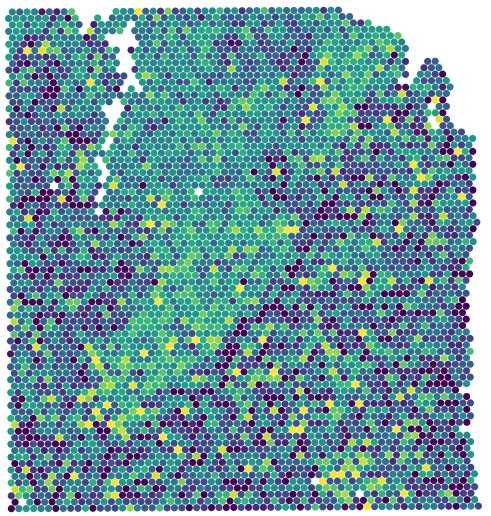


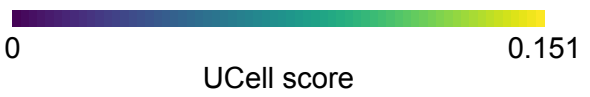
Extracellular Matrix Organization





Co Inhibition By Ctla4

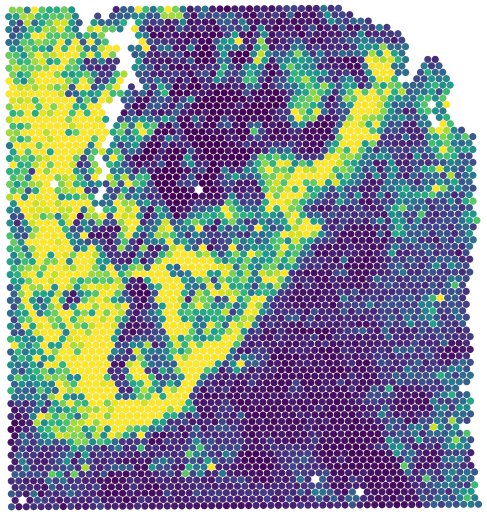




Inhousedata_T4398343 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

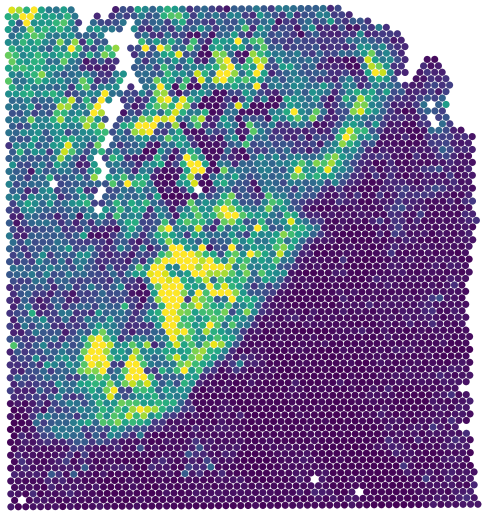
Treg composition



0 0.102

Fraction of all 30

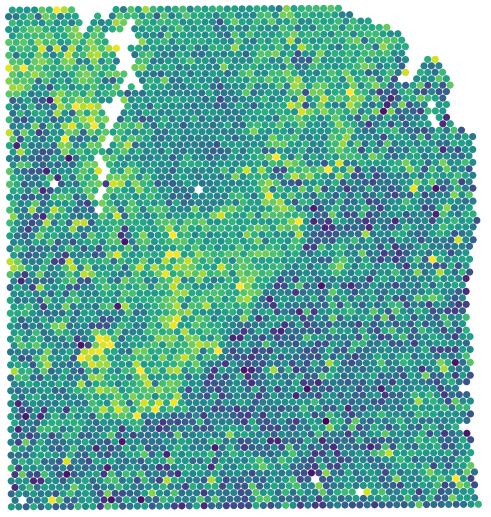
Treg estimated abundance



0 2.85

Posterior mean estimate

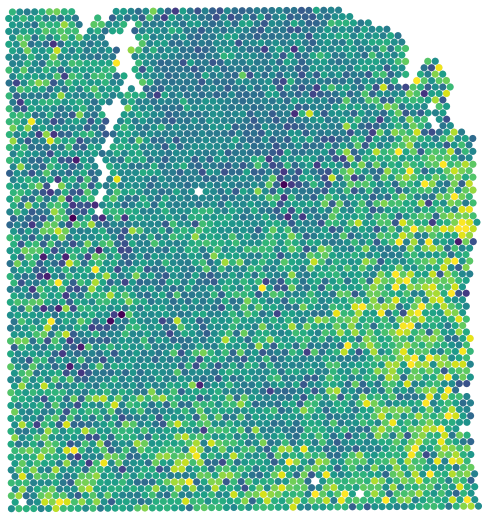
Treg Resting Vs Activated Dn



0 0.0778

UCell score

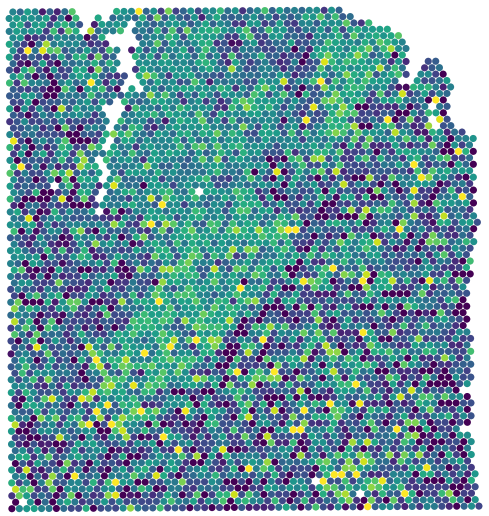
Treg Resting Vs Activated Up



0 0.0609

UCell score

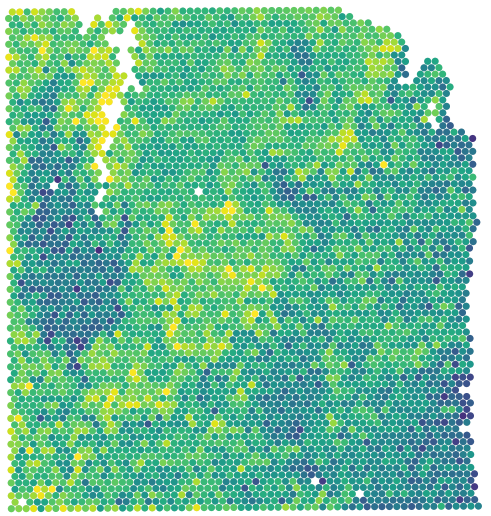
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

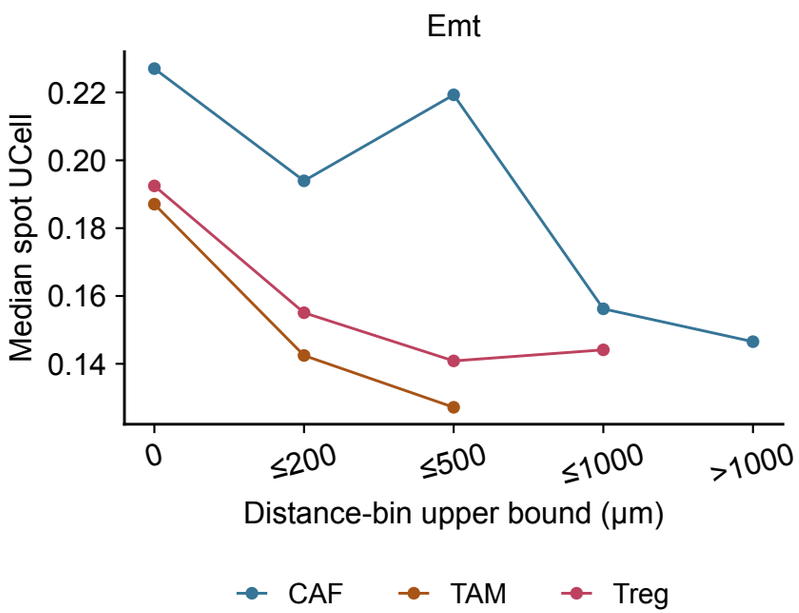


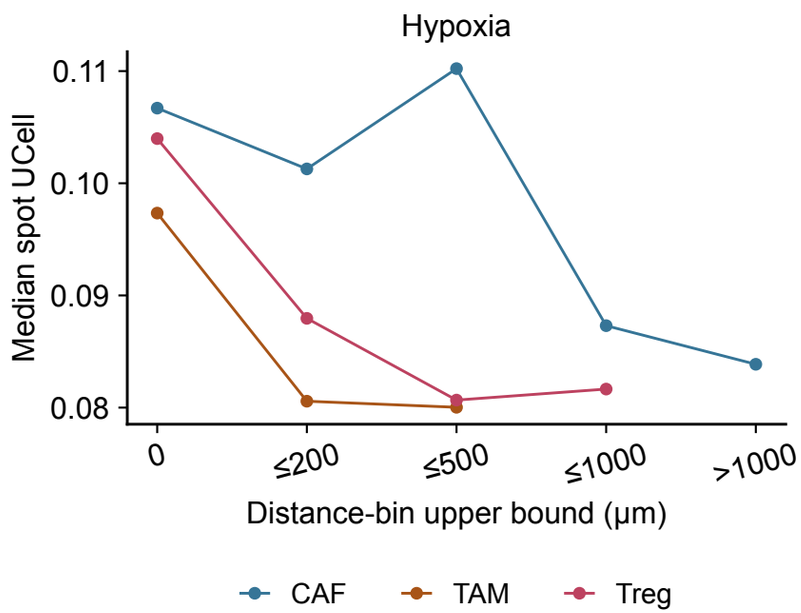
0 0.144

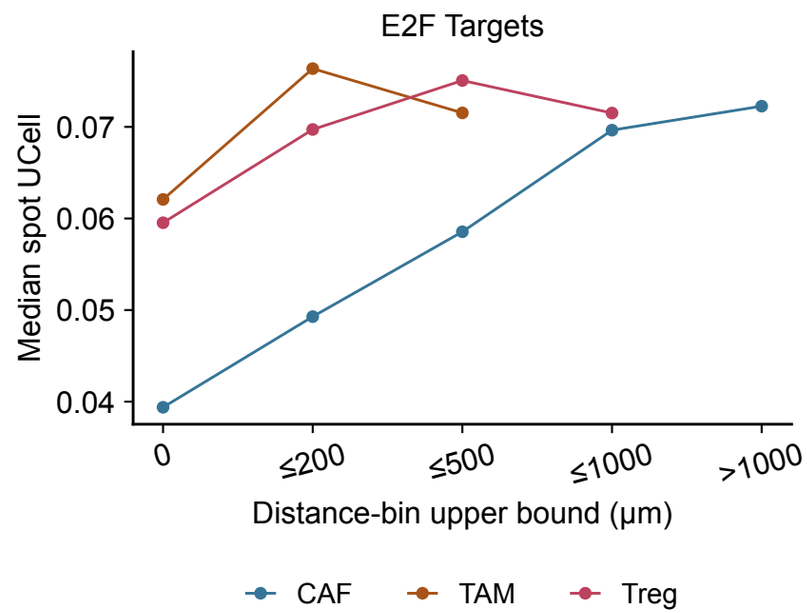
UCell score

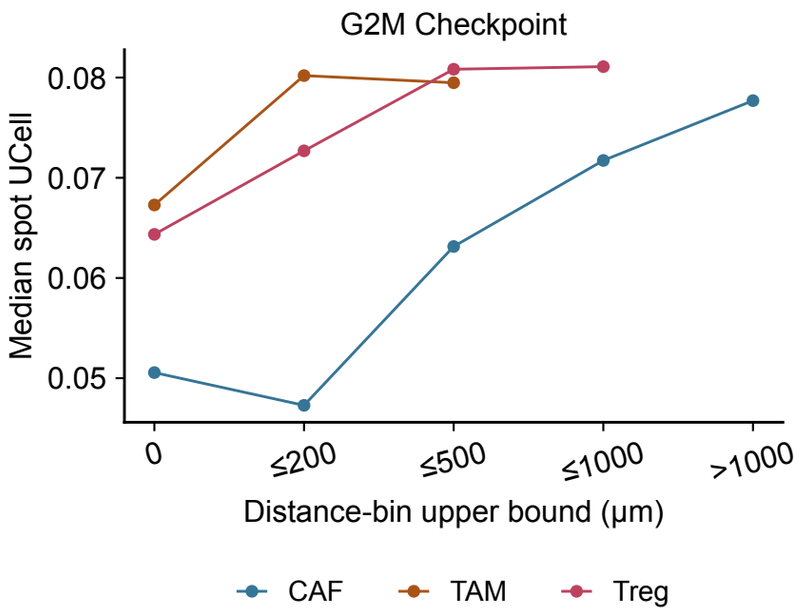
Inhousedata_T4398343 | Programs by target distance

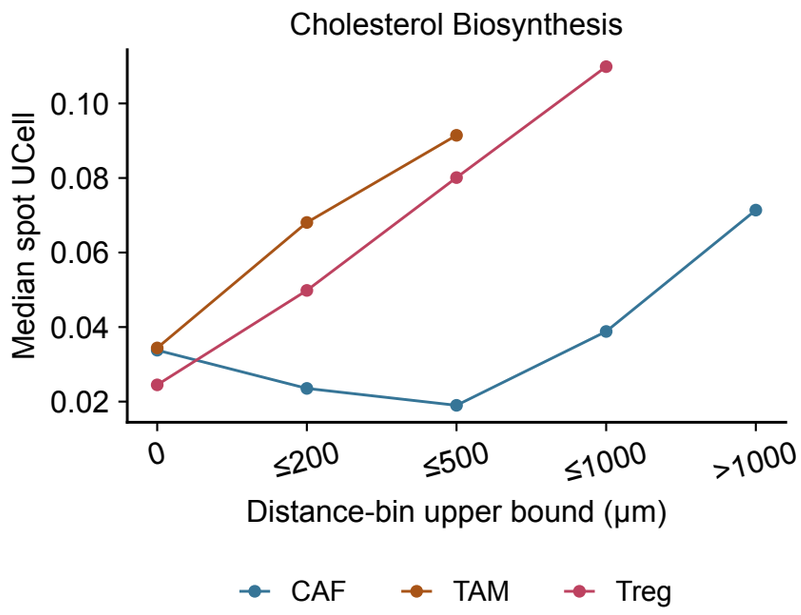
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

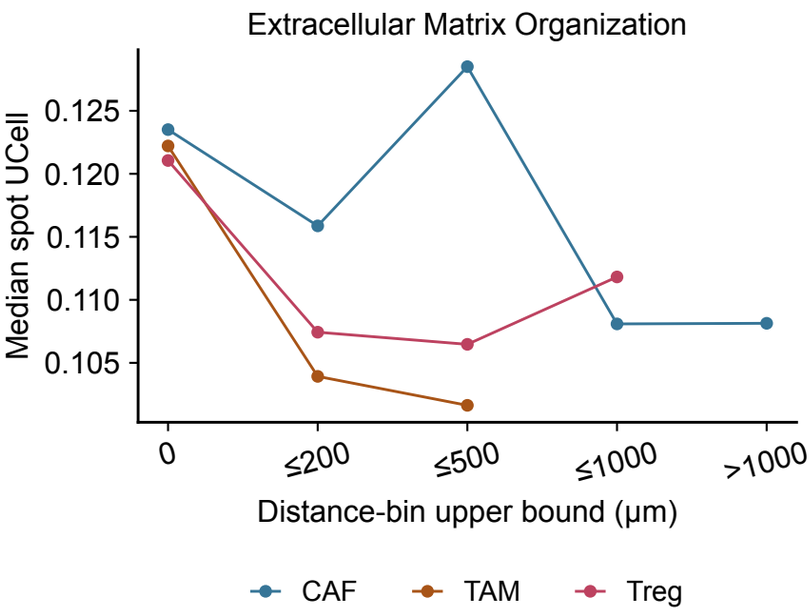






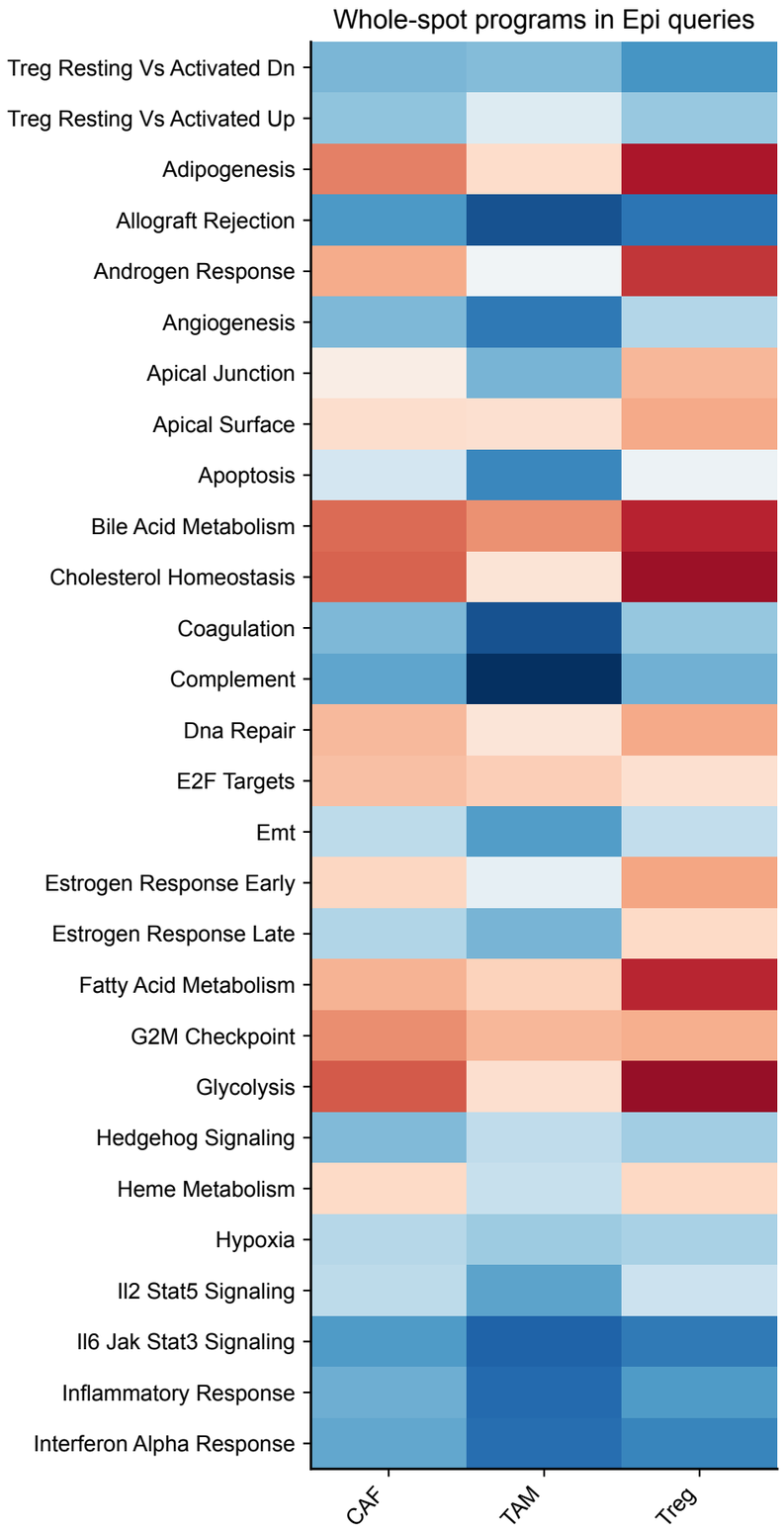


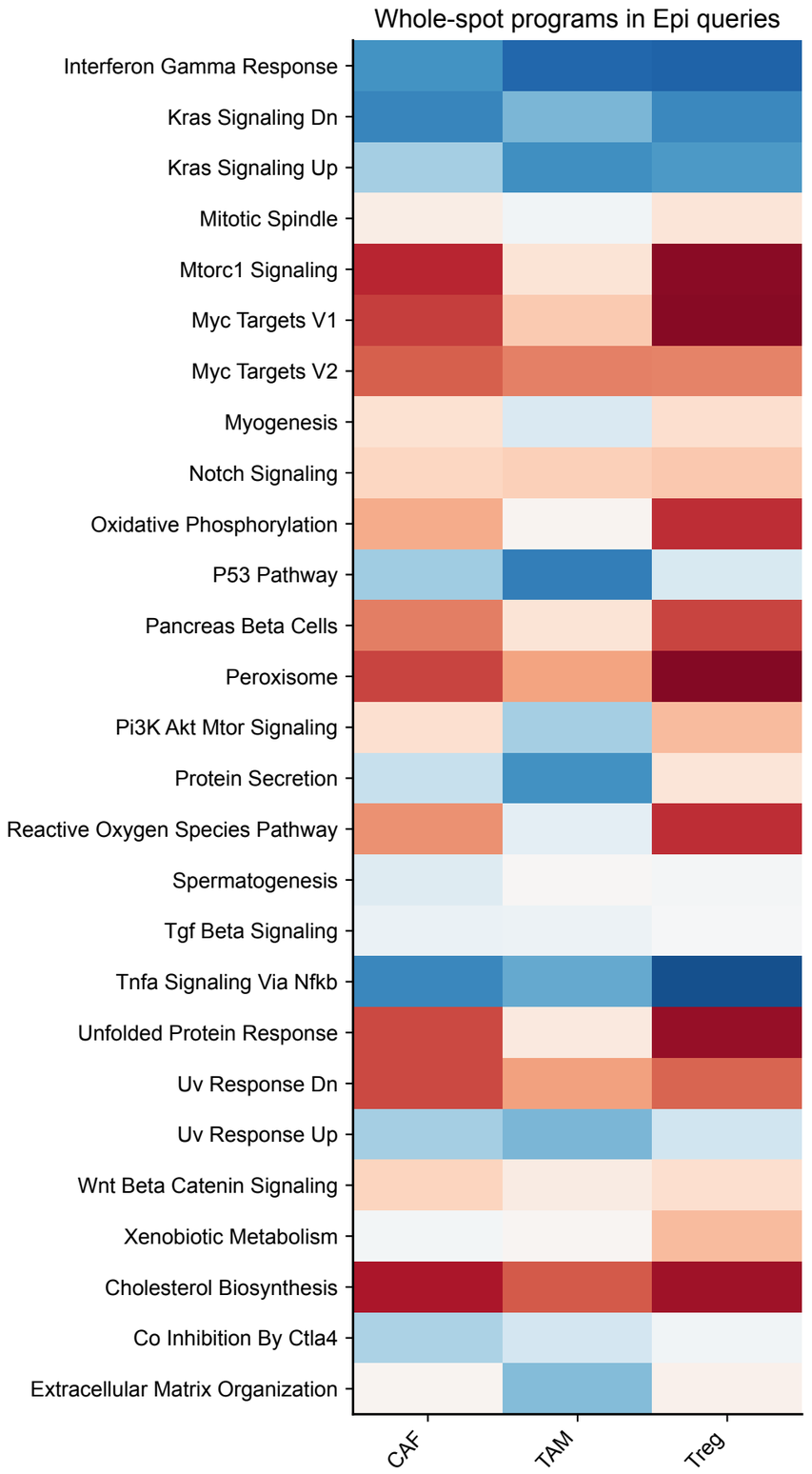




Inhousedata_T4398343 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

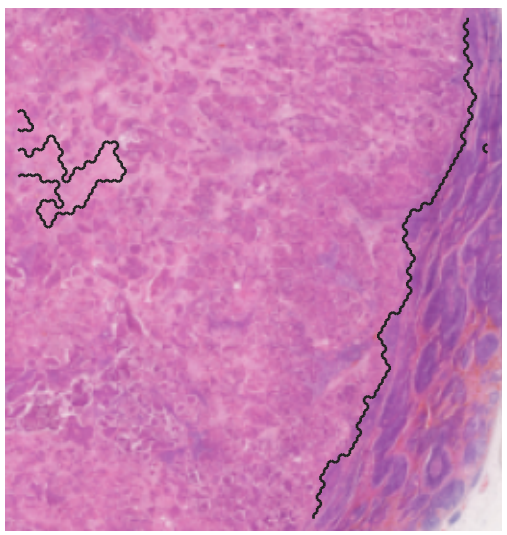




Inhousedata_T4413900 | Tissue context

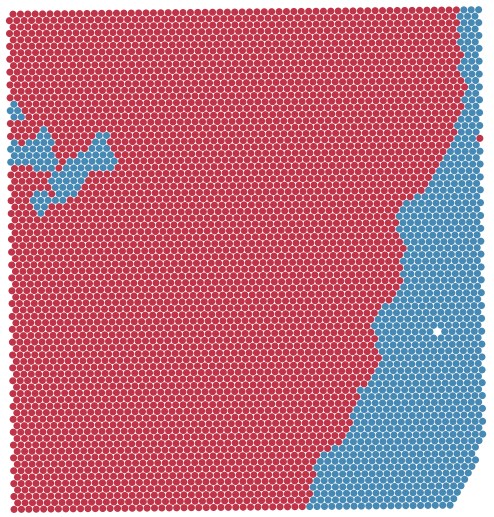
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



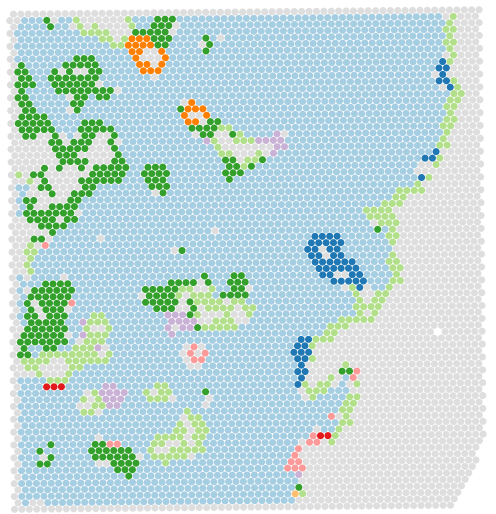
Black line: observed candidate outer edge

Final candidate location



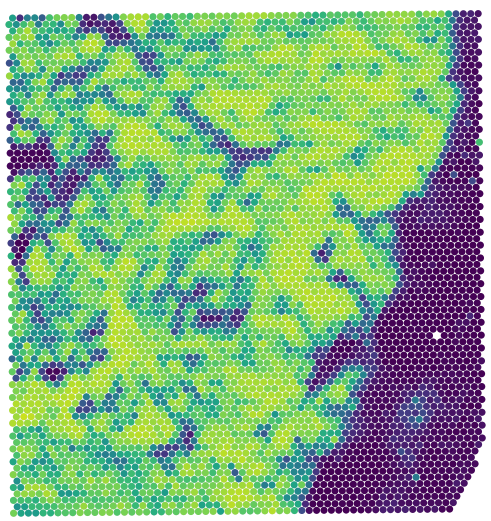
inside outside

Accepted R12 (original 0–8)



0 2 4 6 8
1 3 5 7

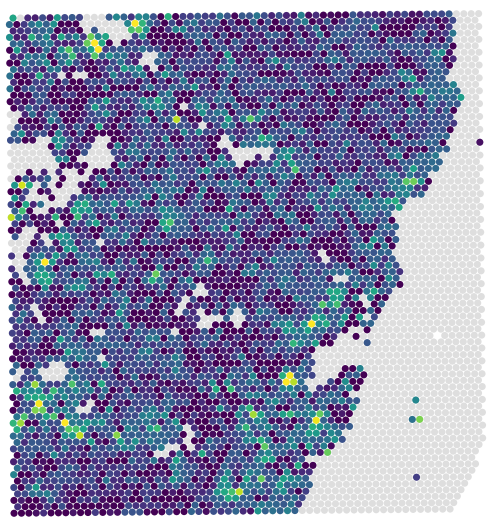
Epithelial support



0 1

Fraction of all 30

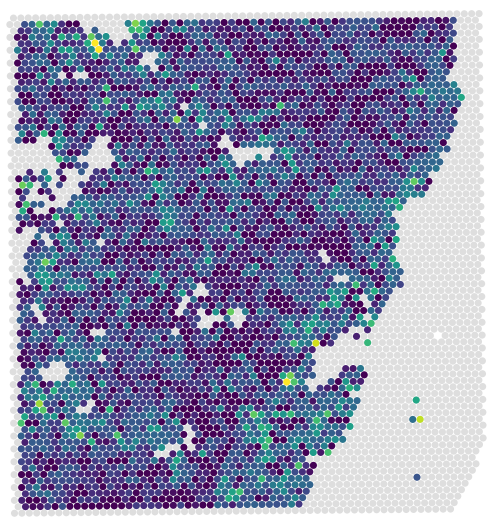
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



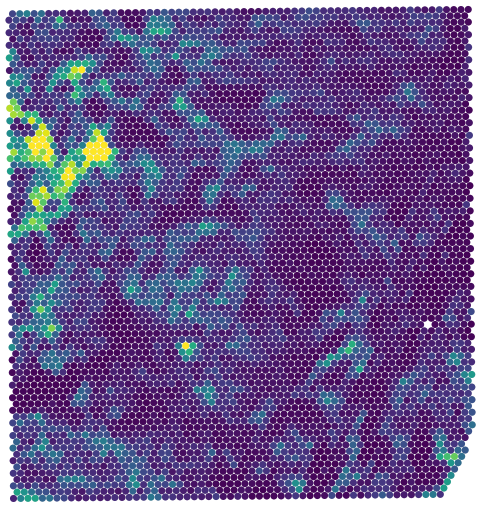
0 0.14

Matched-model AUCell

Inhousedata_T4413900 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



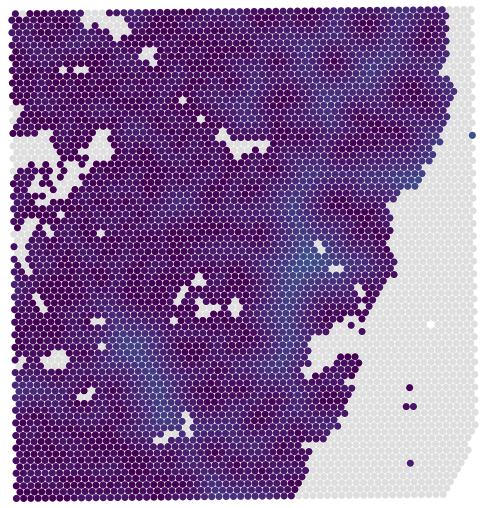
0 0.418

Fraction of all 30

CAF P80 targets

False True

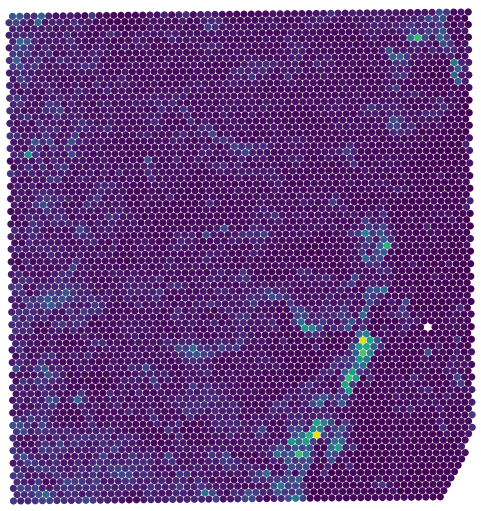
CAF distance



0 3.1e+03

Nearest spot center (μm)

TAM composition



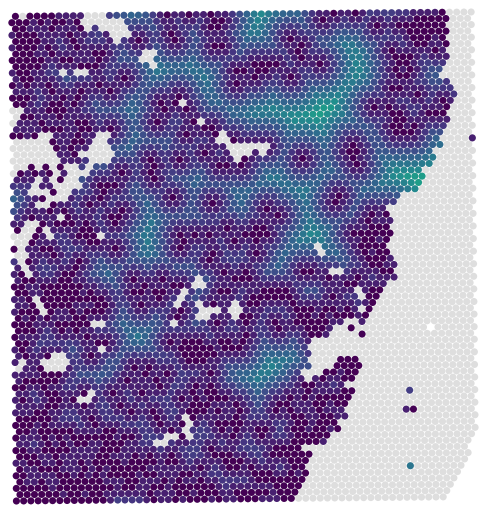
0 0.291

Fraction of all 30

TAM P80 targets

False True

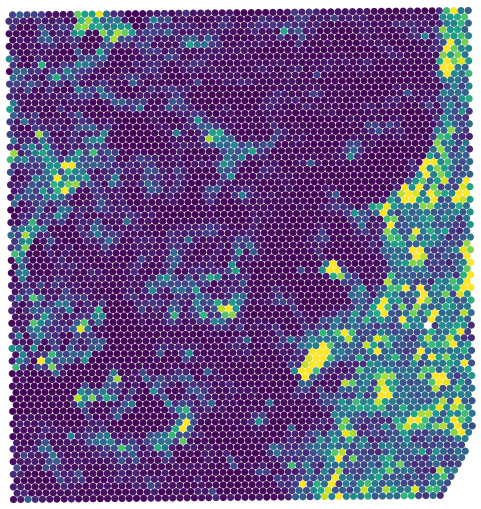
TAM distance



0 1.13e+03

Nearest spot center (μm)

Treg composition



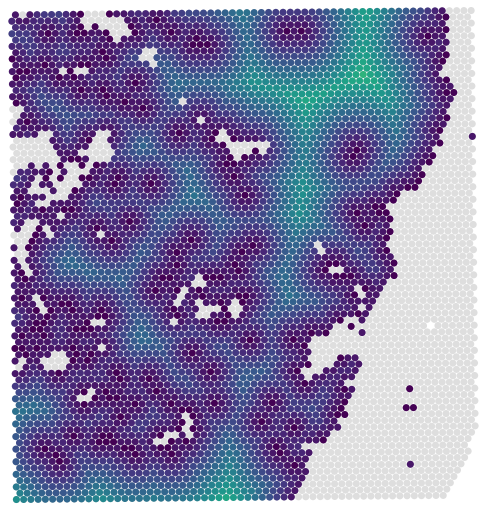
0 0.102

Fraction of all 30

Treg P80 targets

False True

Treg distance

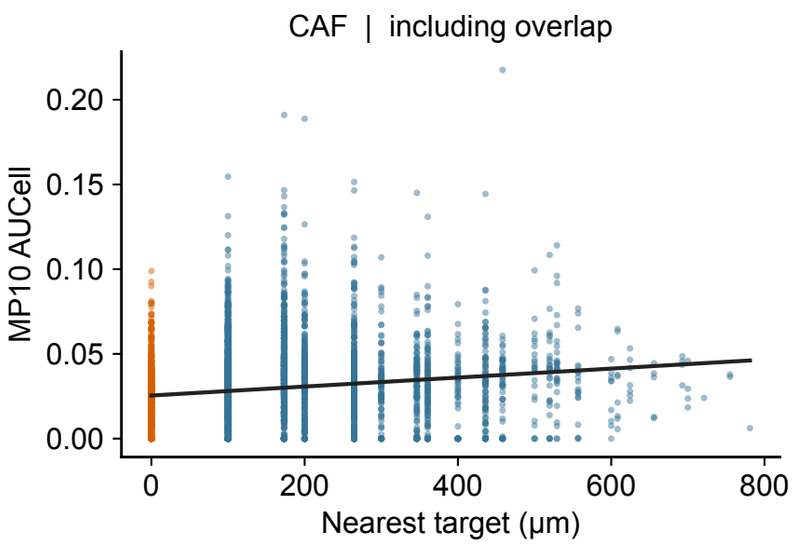


0 1.61e+03

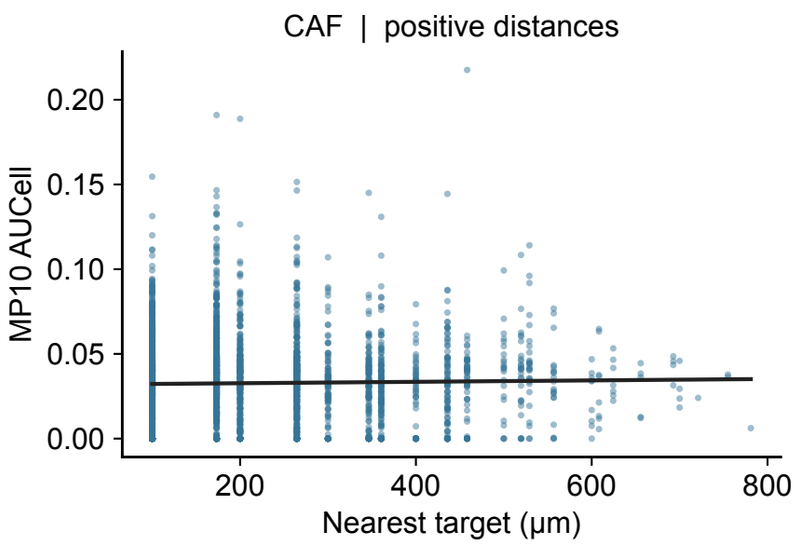
Nearest spot center (μm)

Inhousedata_T4413900 | MP10 versus nearest target

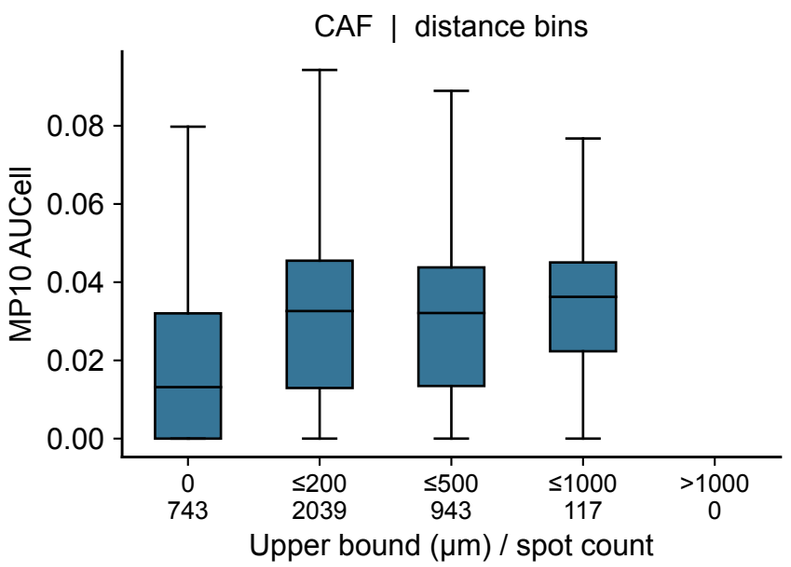
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



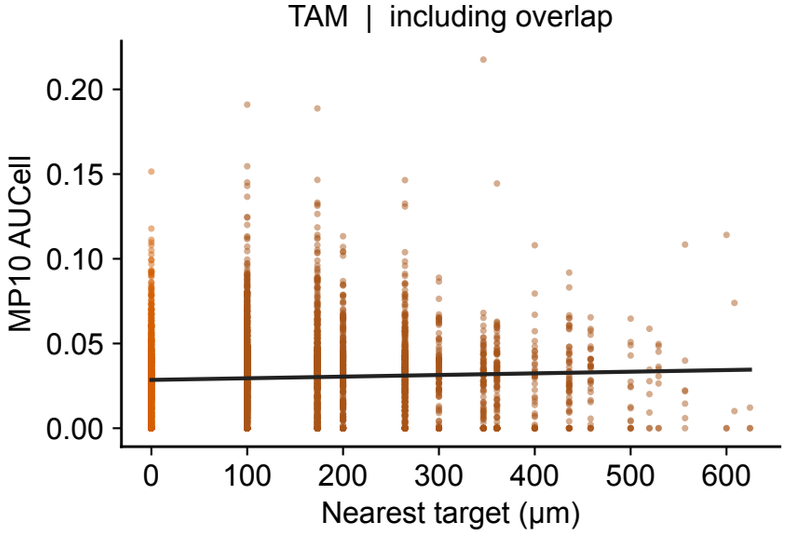
r +0.15 ρ +0.20
n 3842/3842 queries • 994 targets



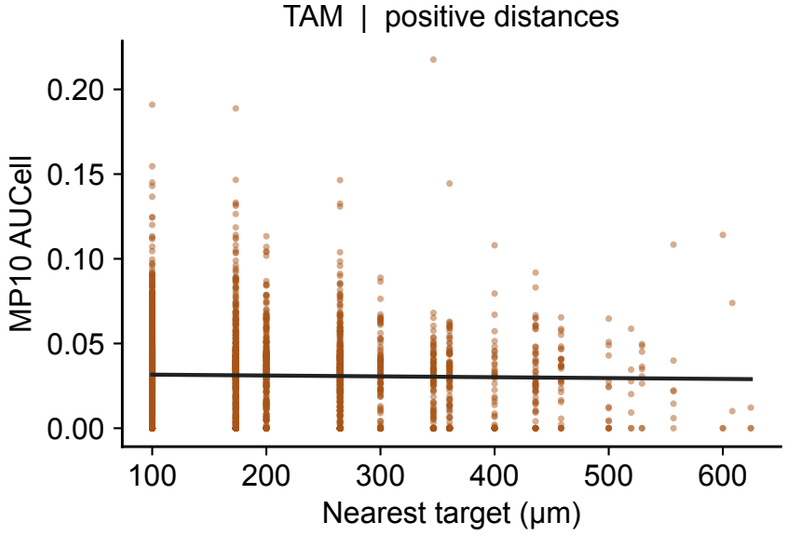
r +0.02 ρ +0.04
n 3099/3842 queries • 994 targets



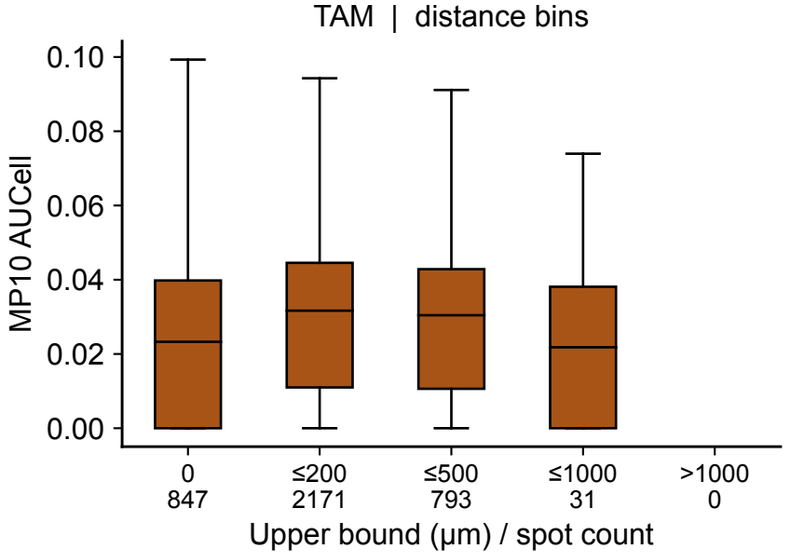
Orange = zero distance
Line = within-section linear fit



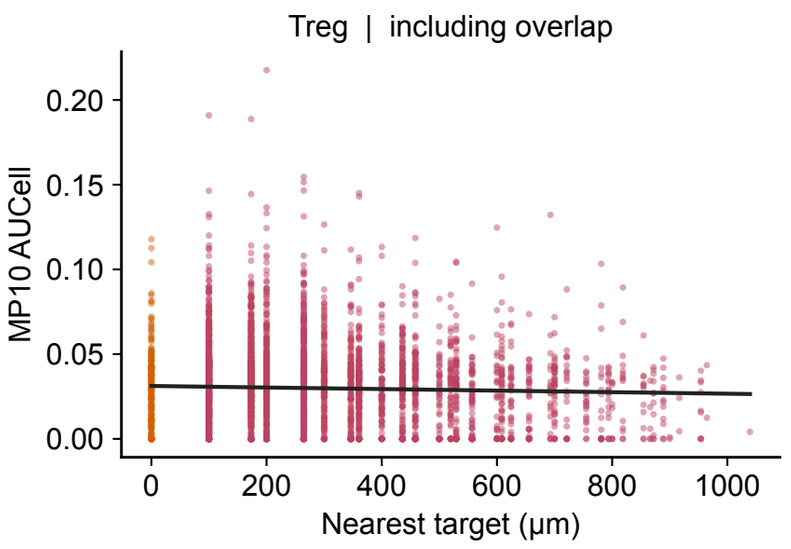
r +0.05 ρ +0.08
n 3842/3842 queries • 994 targets



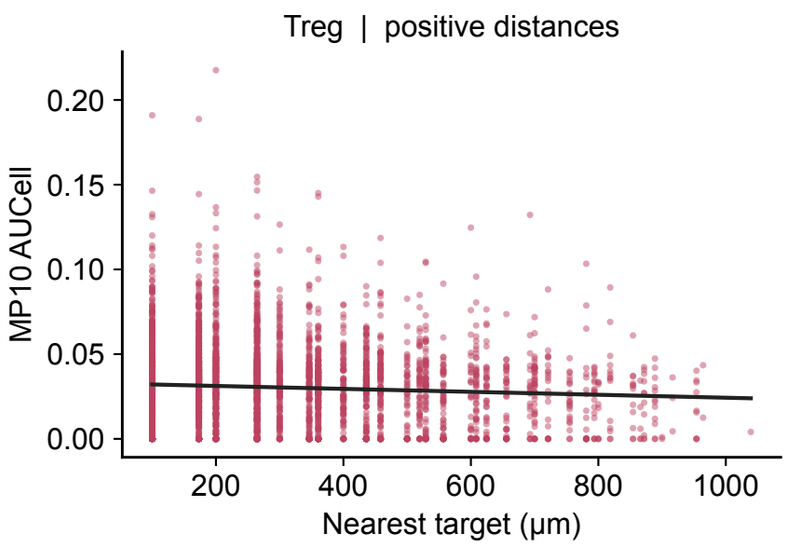
r -0.02 ρ +0.01
n 2995/3842 queries • 994 targets



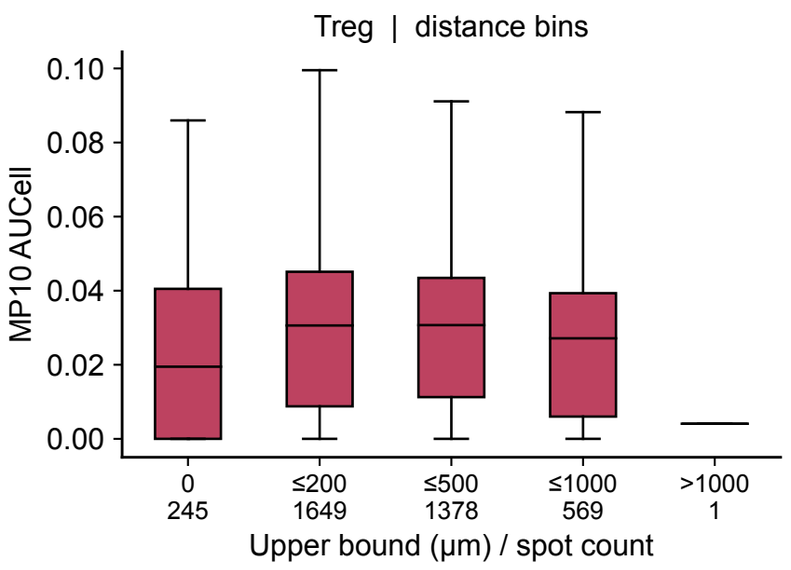
Orange = zero distance
Line = within-section linear fit



r -0.04 ρ +0.00
n 3842/3842 queries • 994 targets



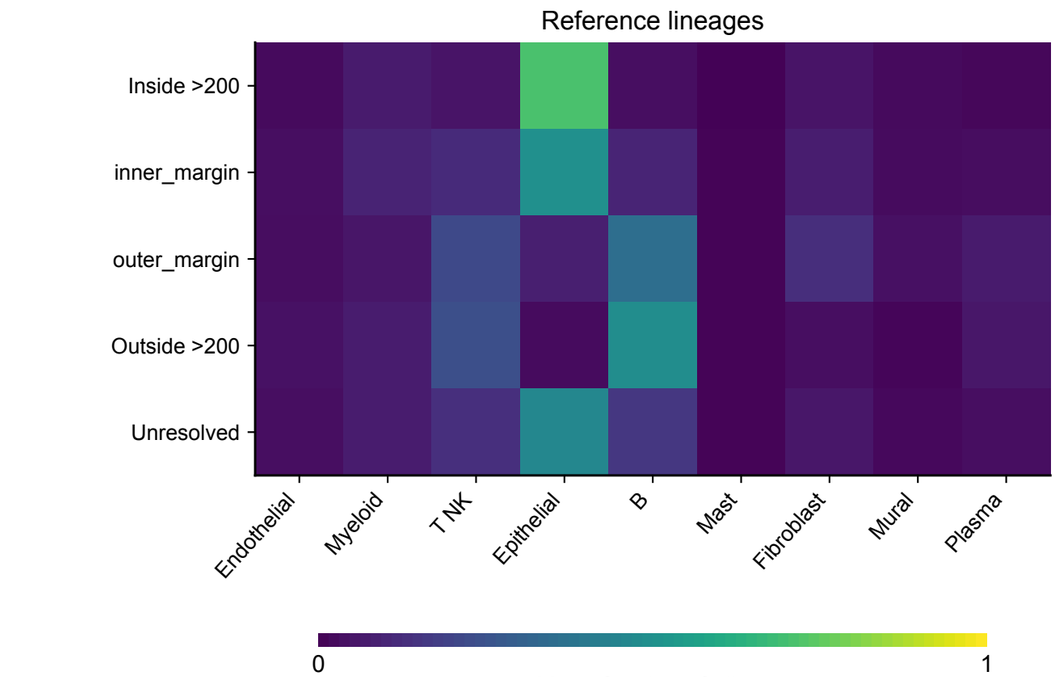
r -0.07 ρ -0.03
n 3597/3842 queries • 994 targets

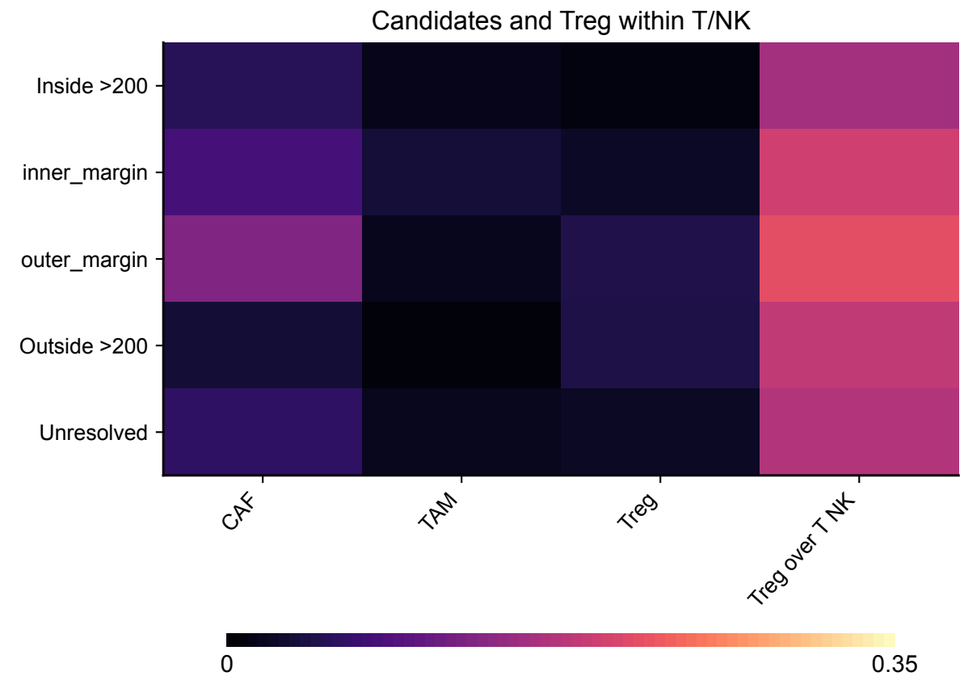


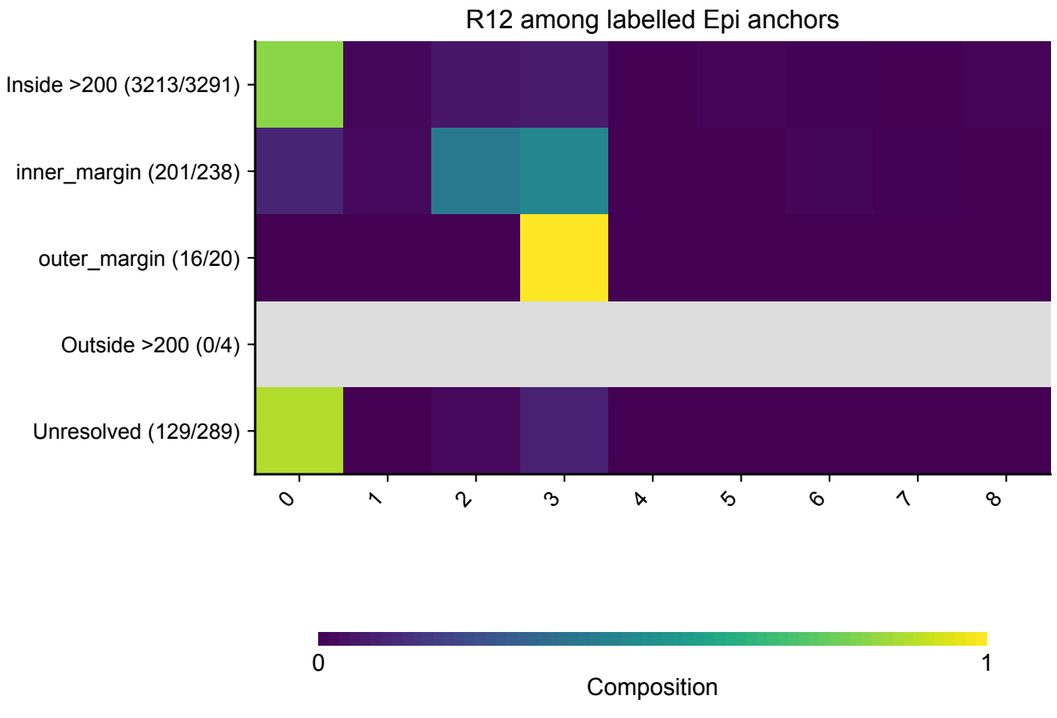
Orange = zero distance
Line = within-section linear fit

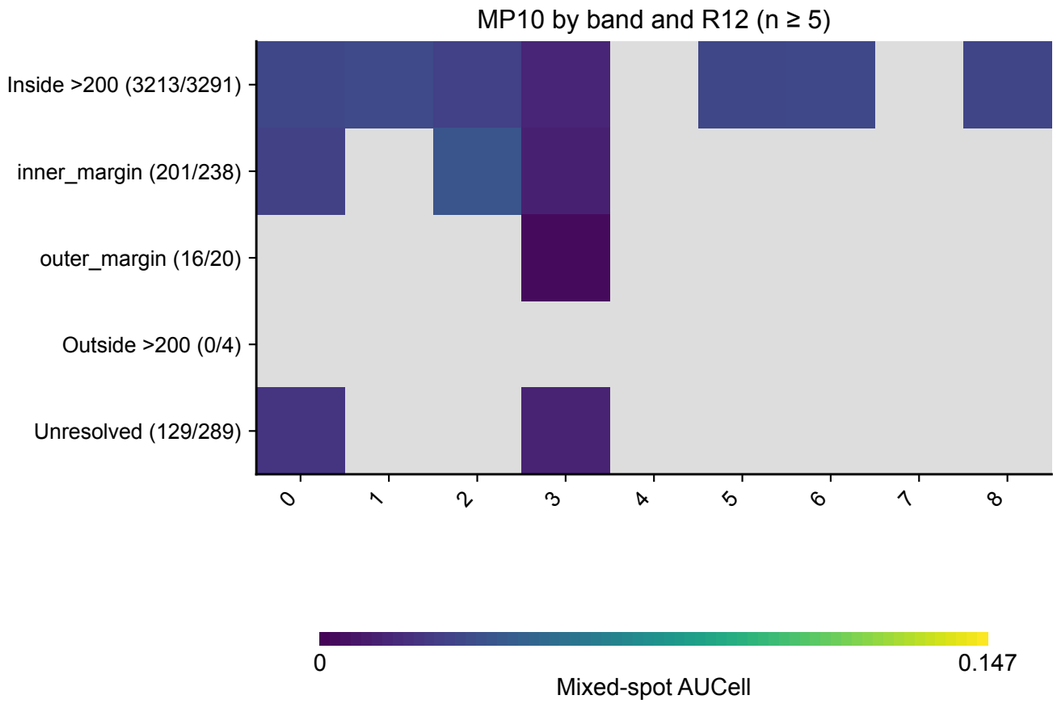
Inhousedata_T4413900 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





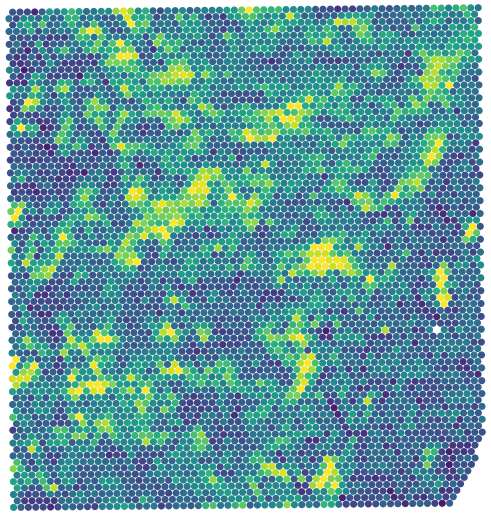




Inhousedata_T4413900 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

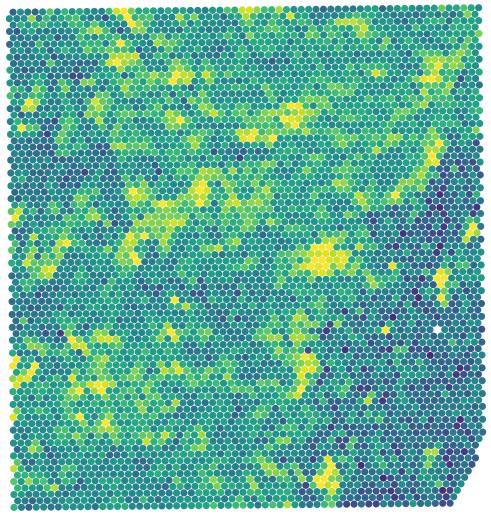
E2F Targets



0 0.152

UCell score

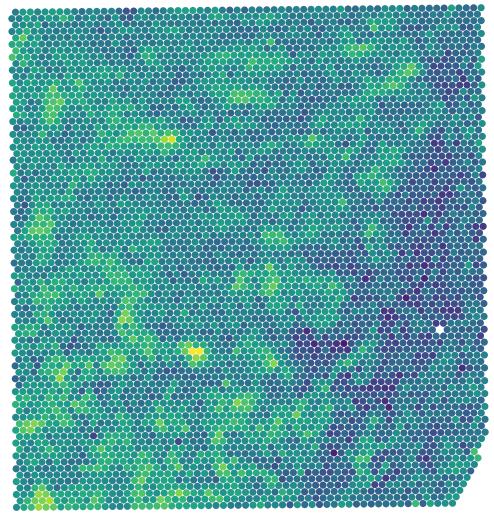
G2M Checkpoint



0 0.134

UCell score

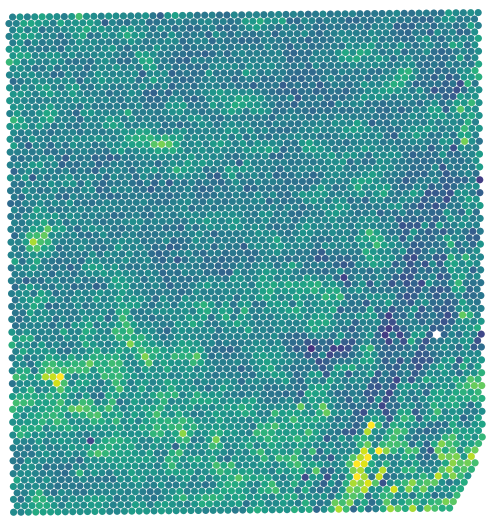
Hypoxia



0 0.211

UCell score

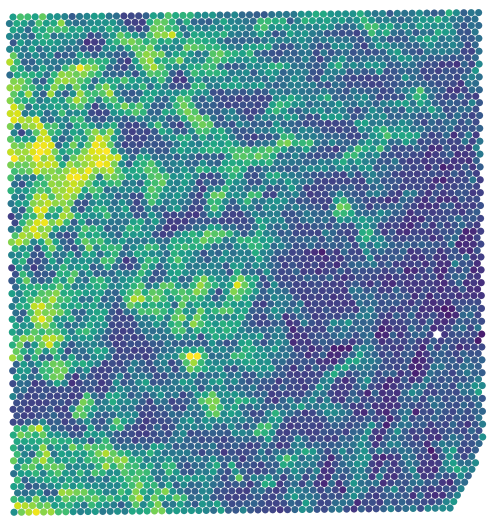
Tnfa Signaling Via Nfkb



0 0.275

UCell score

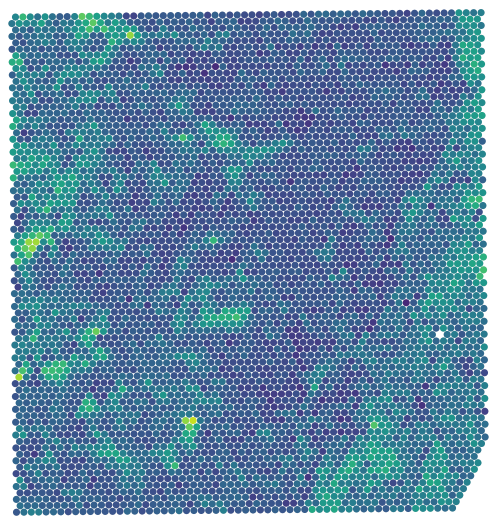
Emt



0 0.386

UCell score

Interferon Gamma Response



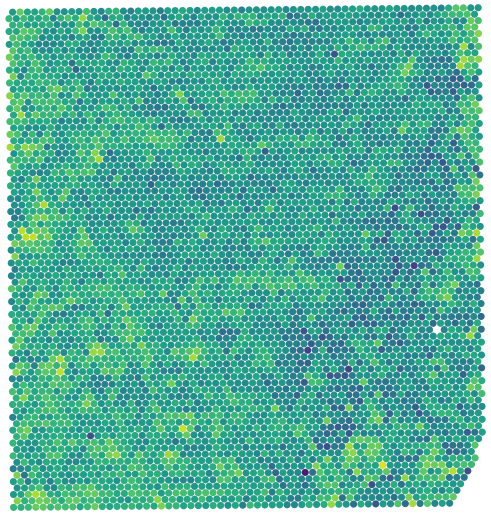
0 0.275

UCell score

Inhousedata_T4413900 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

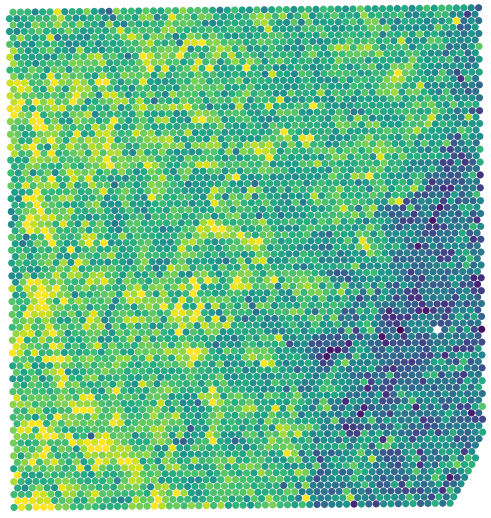
Il2 Stat5 Signaling



0 0.144

UCell score

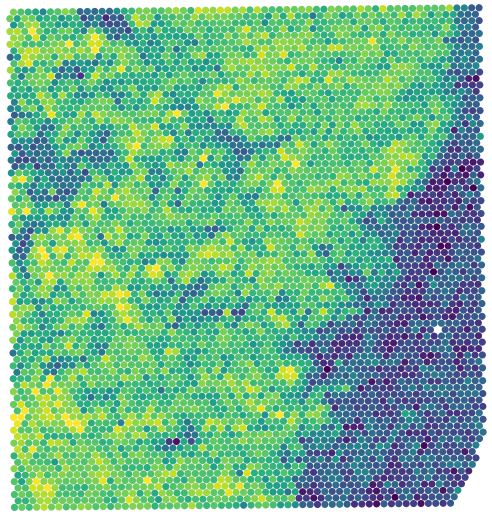
Tgf Beta Signaling



0 0.232

UCell score

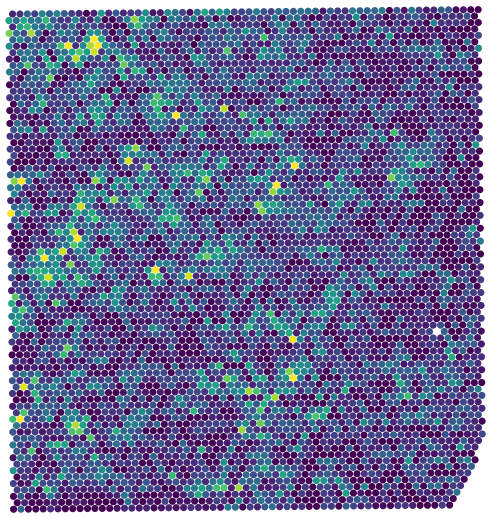
Cholesterol Homeostasis



0 0.21

UCell score

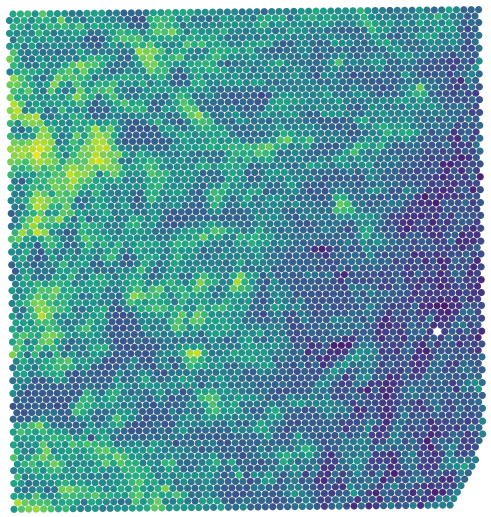
Cholesterol Biosynthesis



0 0.166

UCell score

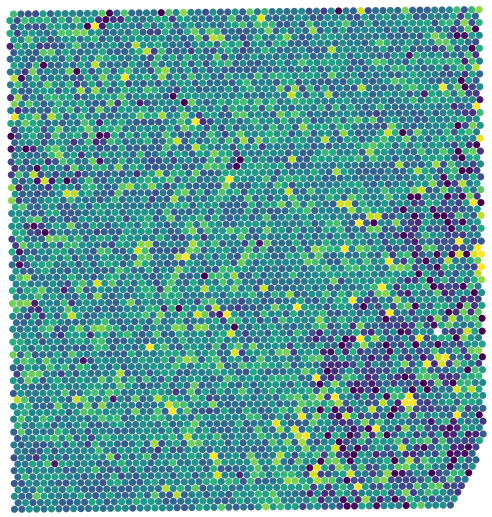
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



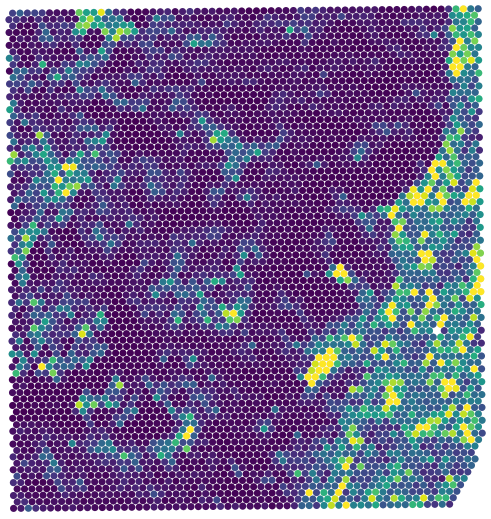
0 0.151

UCell score

Inhousedata_T4413900 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

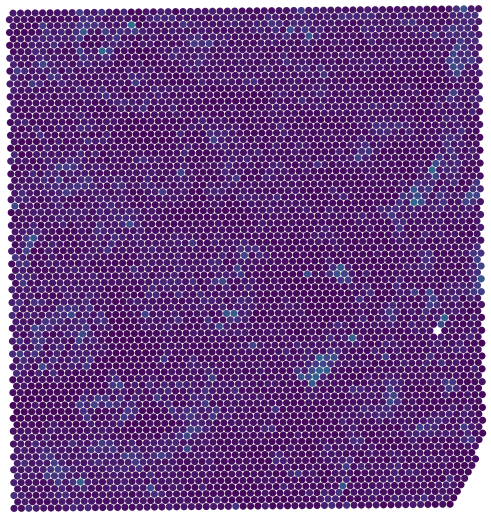
Treg composition



0 0.102

Fraction of all 30

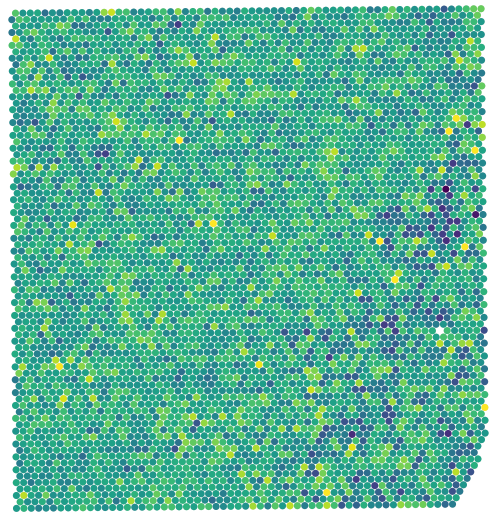
Treg estimated abundance



0 2.85

Posterior mean estimate

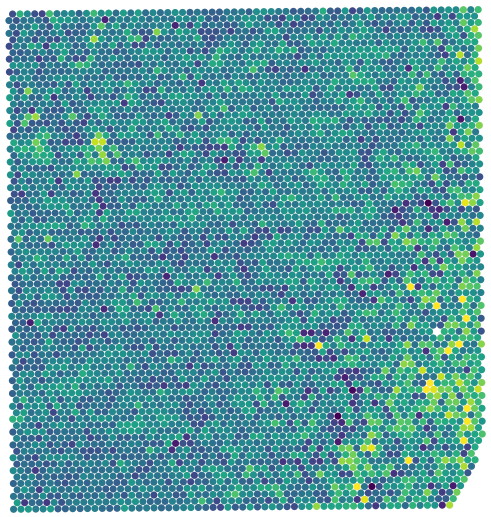
Treg Resting Vs Activated Dn



0 0.0778

UCell score

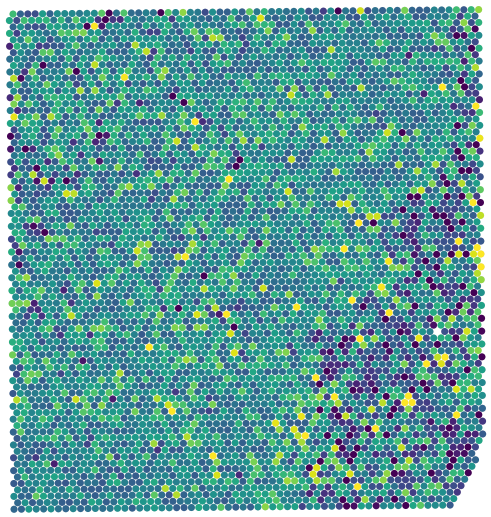
Treg Resting Vs Activated Up



0 0.0609

UCell score

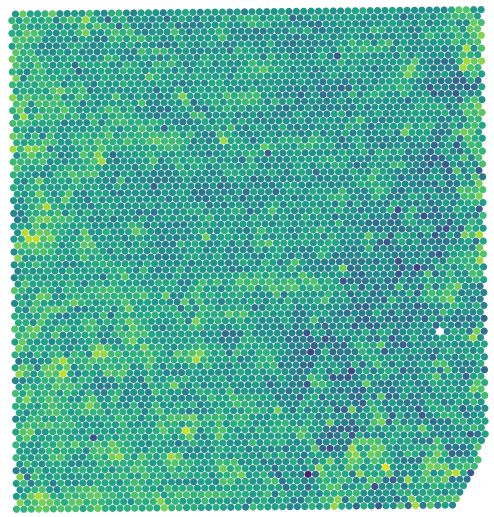
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

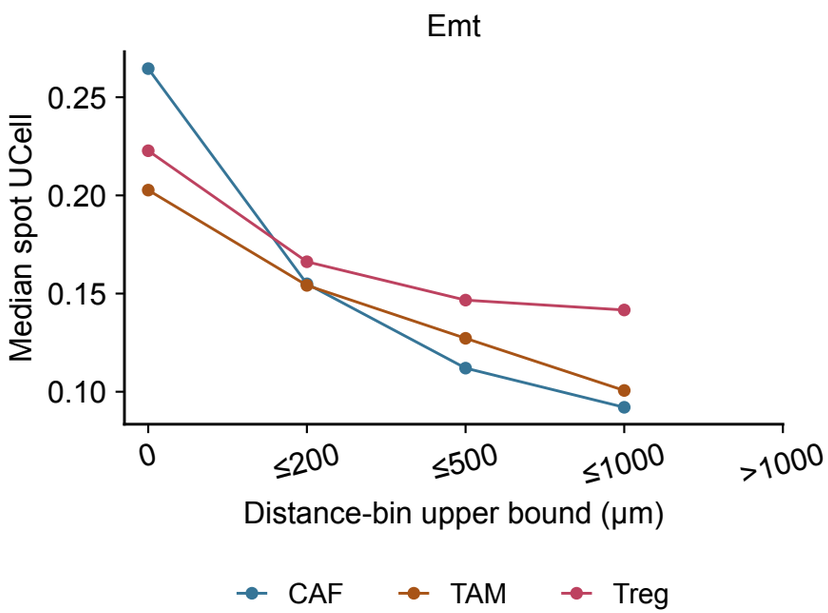


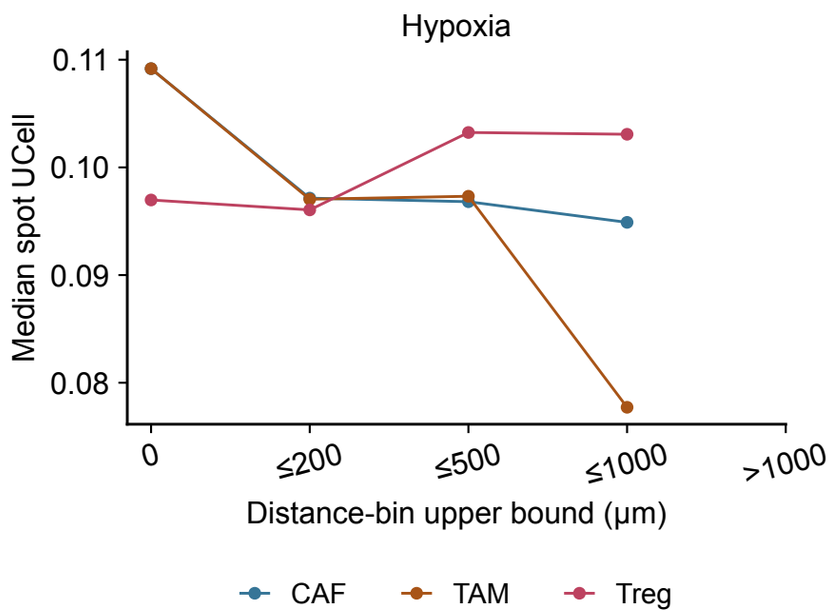
0 0.144

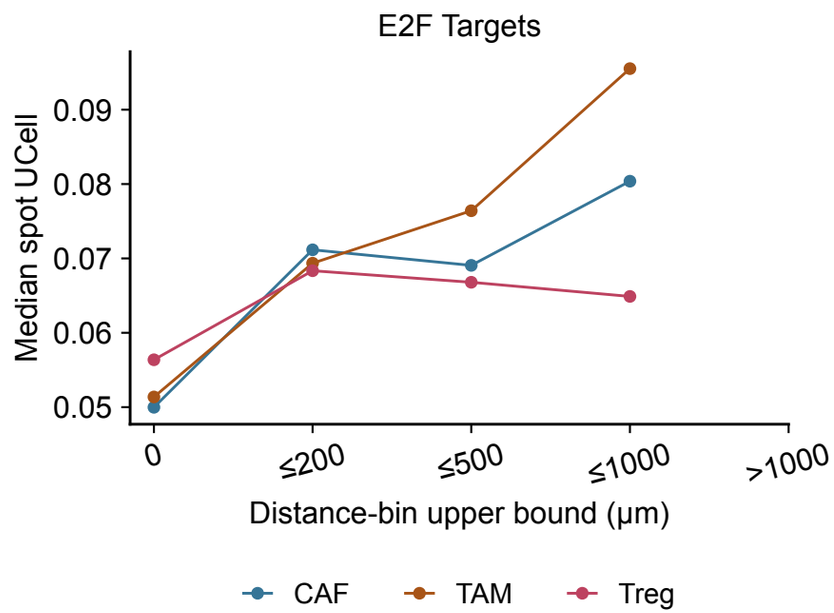
UCell score

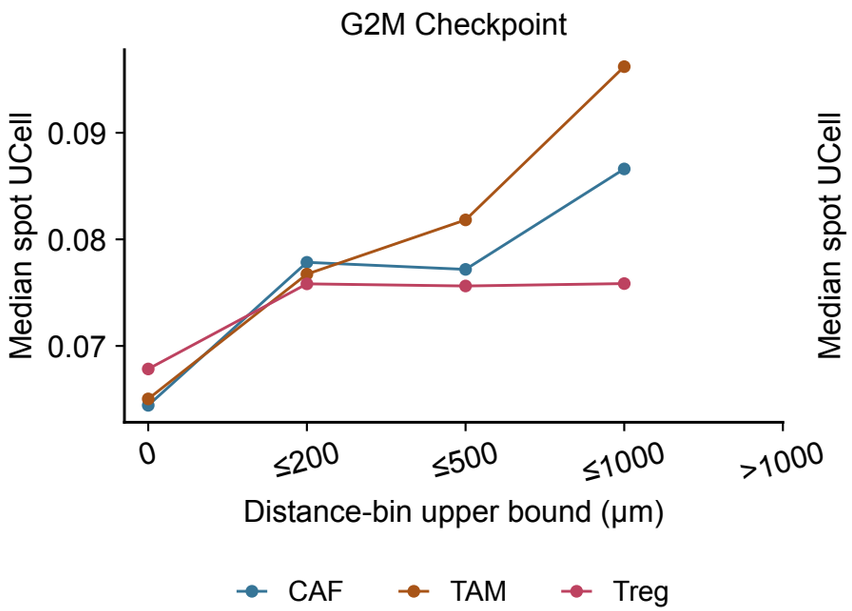
Inhousedata_T4413900 | Programs by target distance

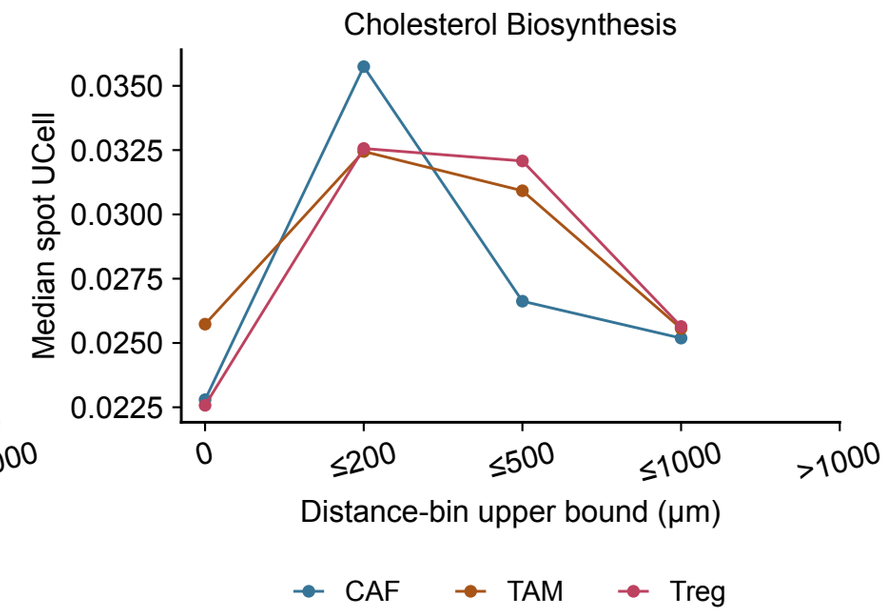
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

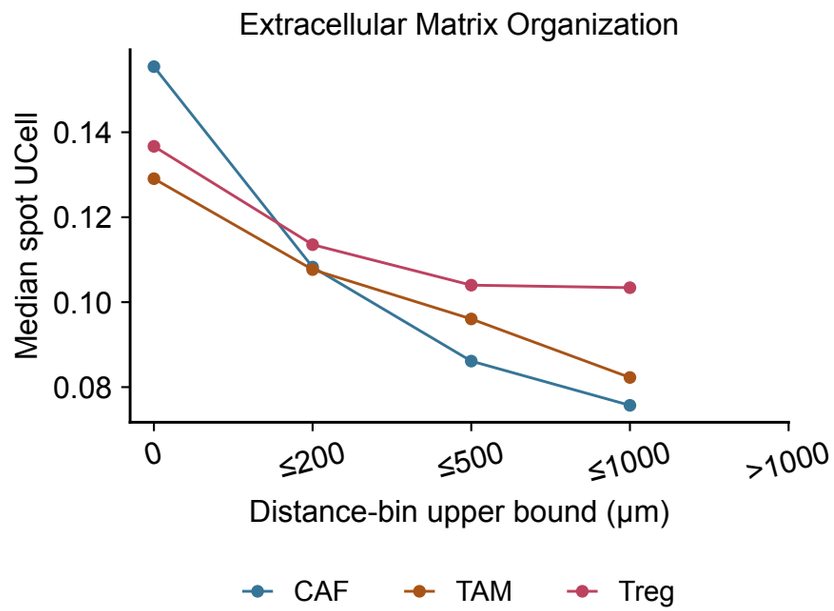






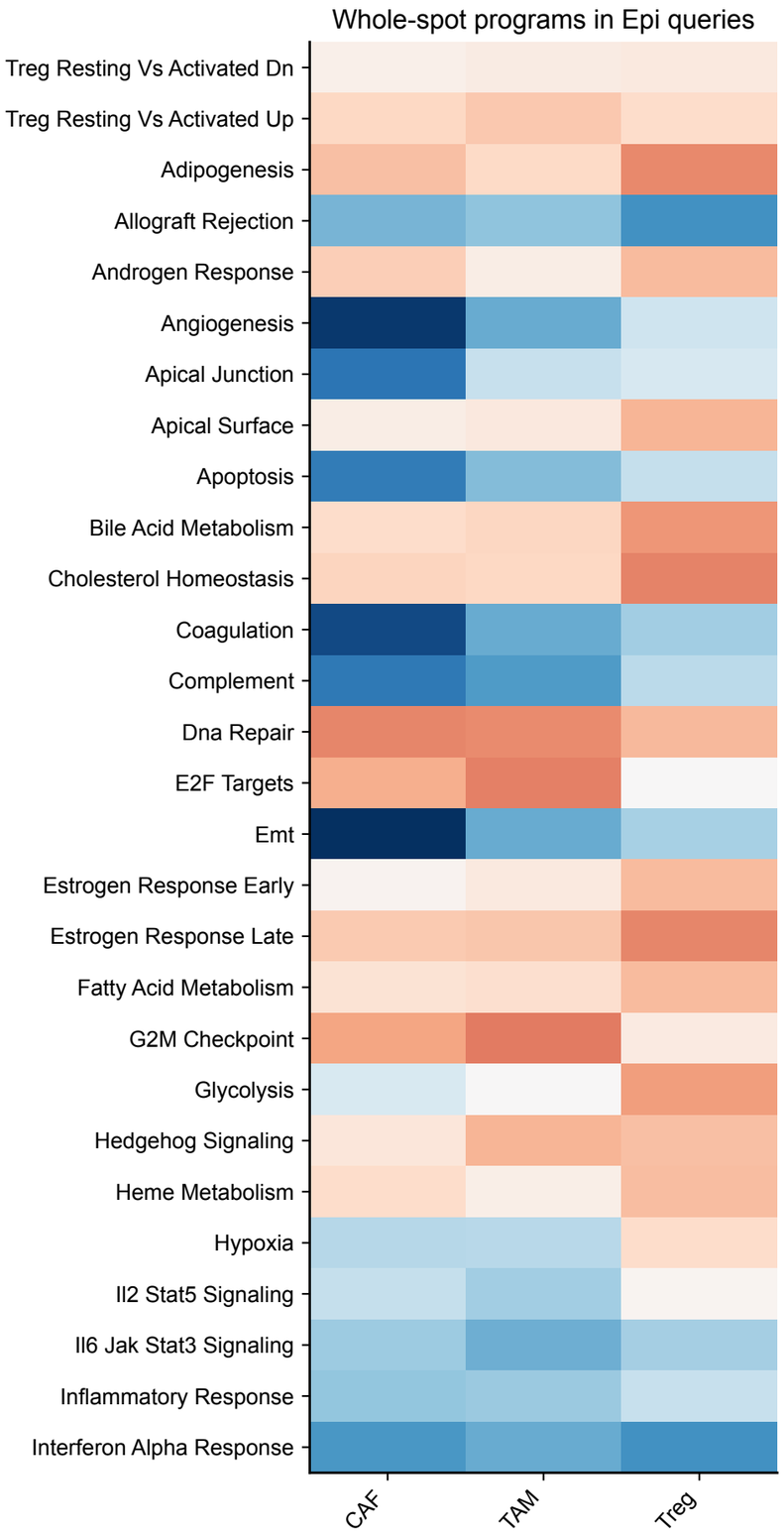


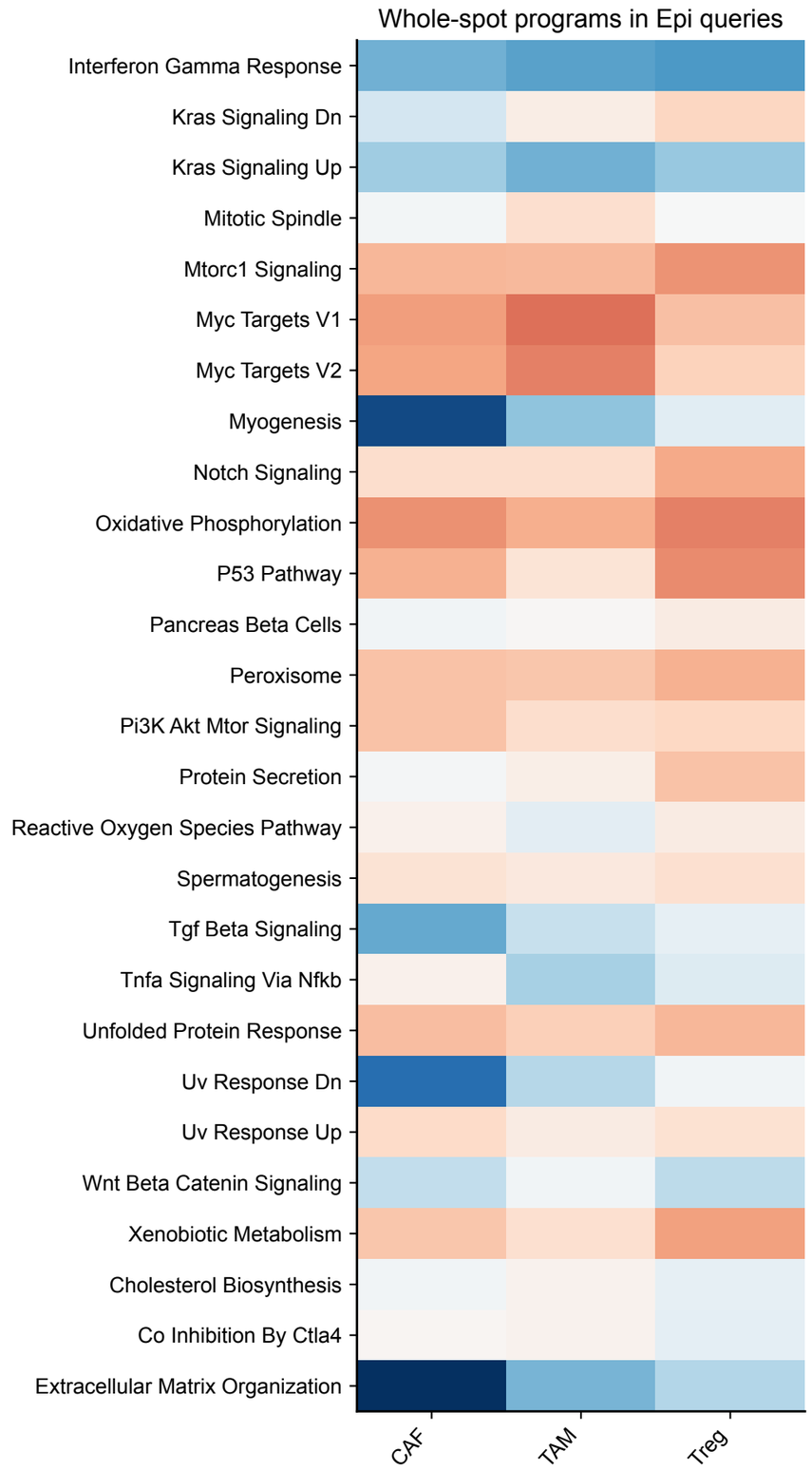




Inhousedata_T4413900 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

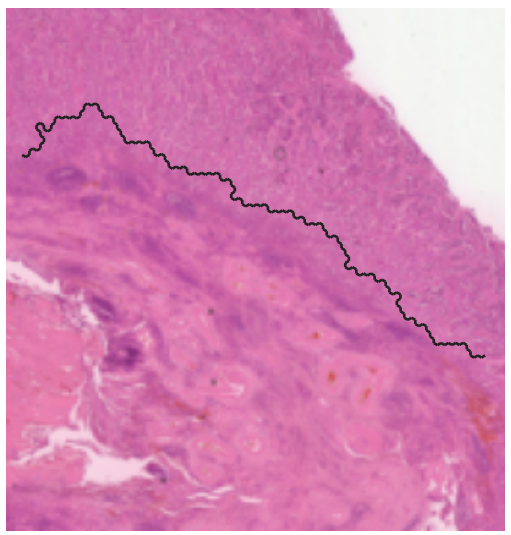




Inhousedata_T4463209 | Tissue context

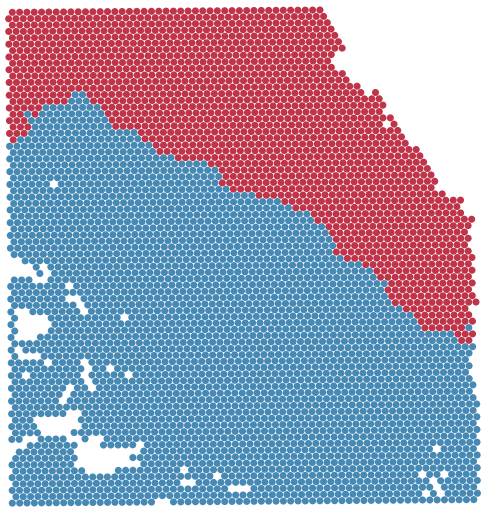
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



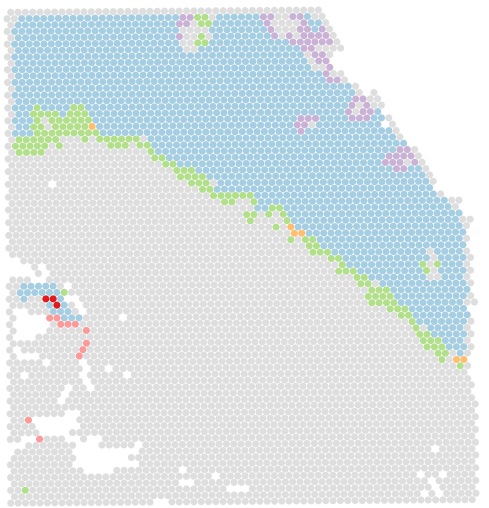
Black line: observed candidate outer edge

Final candidate location



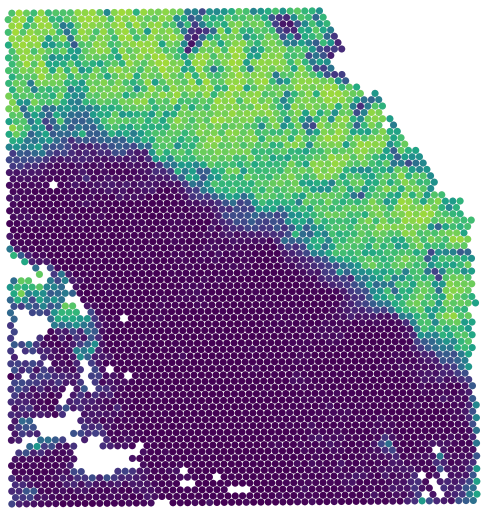
inside outside

Accepted R12 (original 0–8)



0 4 6 7 8 2

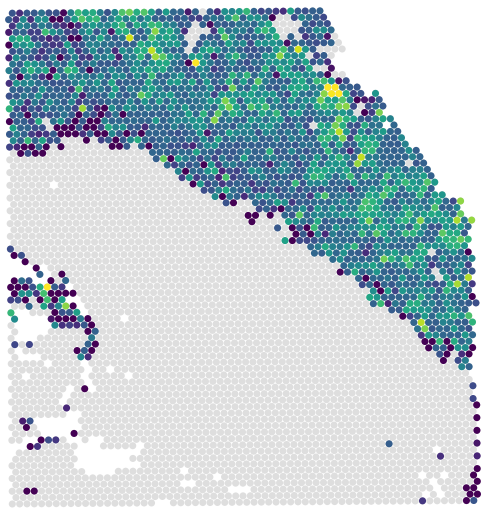
Epithelial support



0 1

Fraction of all 30

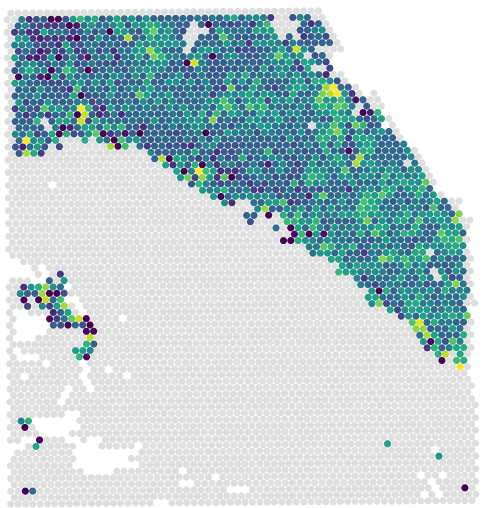
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



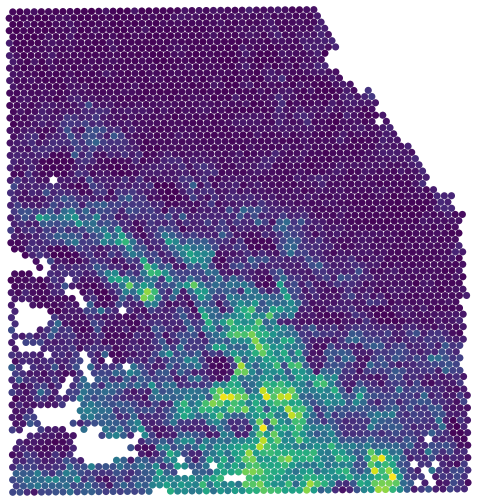
0 0.14

Matched-model AUCell

Inhousedata_T4463209 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

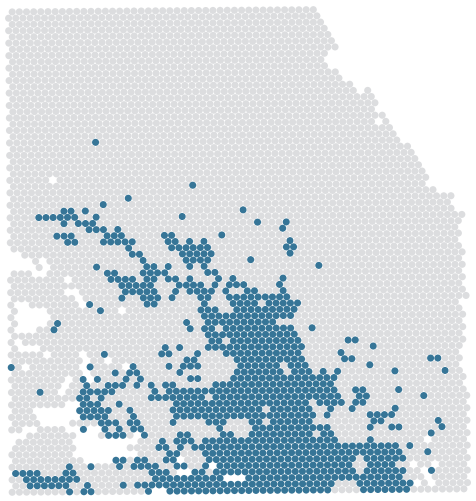
CAF composition



0 0.418

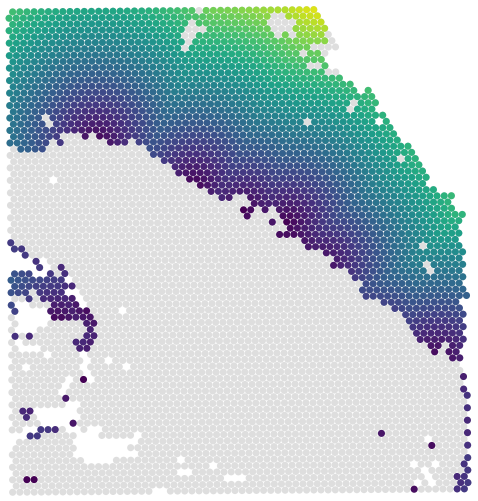
Fraction of all 30

CAF P80 targets



False True

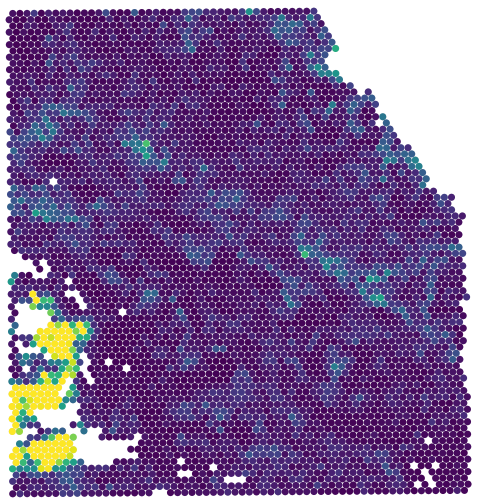
CAF distance



0 3.1e+03

Nearest spot center (µm)

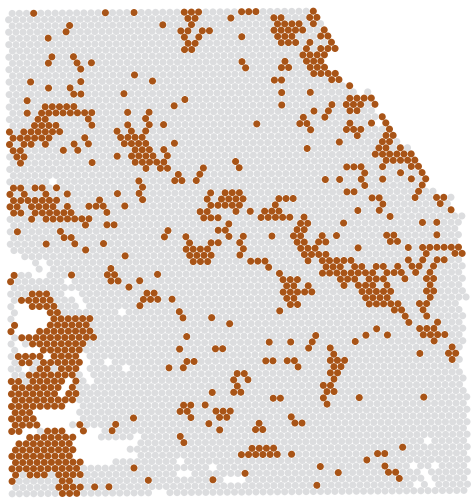
TAM composition



0 0.291

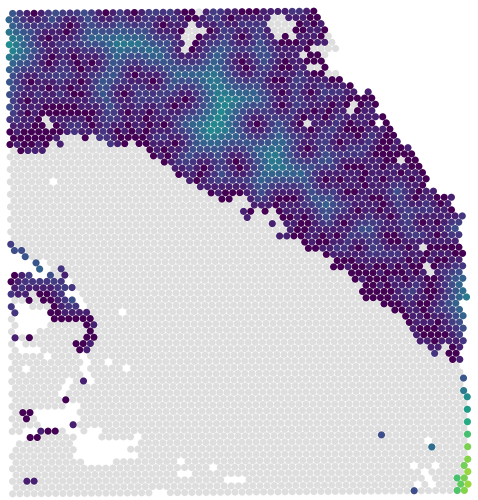
Fraction of all 30

TAM P80 targets



False True

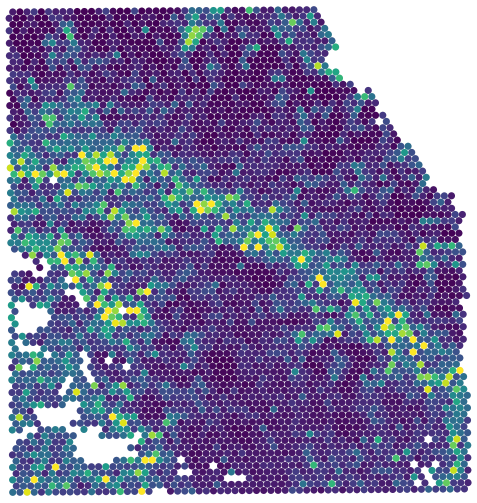
TAM distance



0 1.13e+03

Nearest spot center (µm)

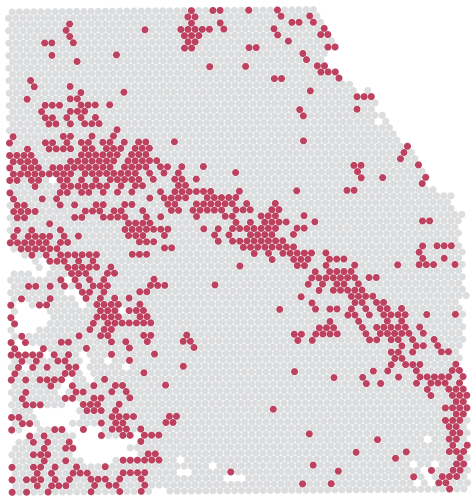
Treg composition



0 0.102

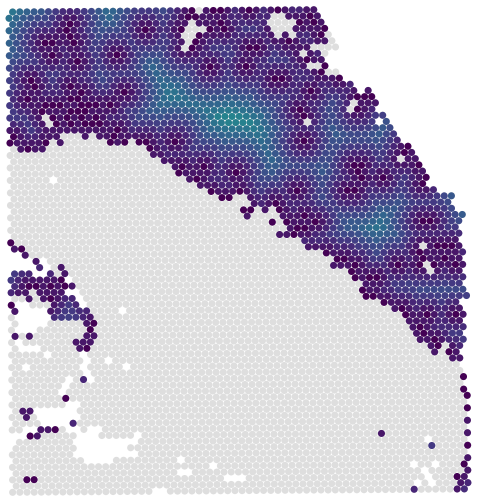
Fraction of all 30

Treg P80 targets



False True

Treg distance

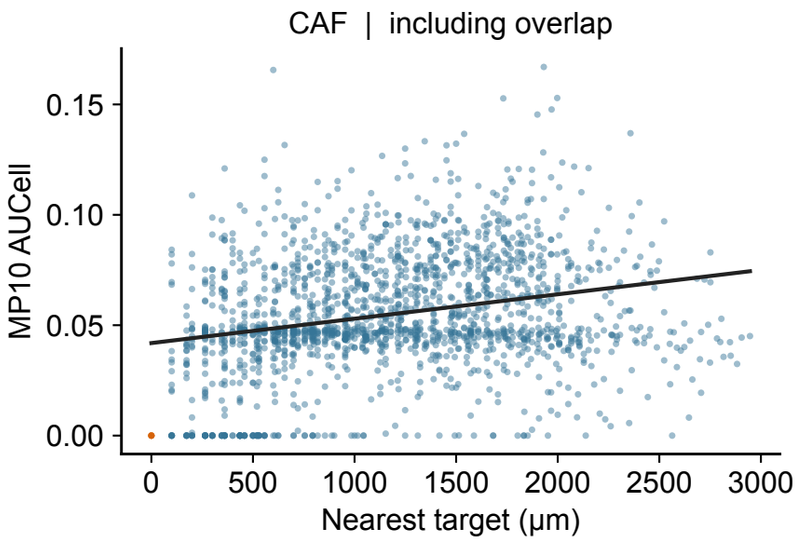


0 1.61e+03

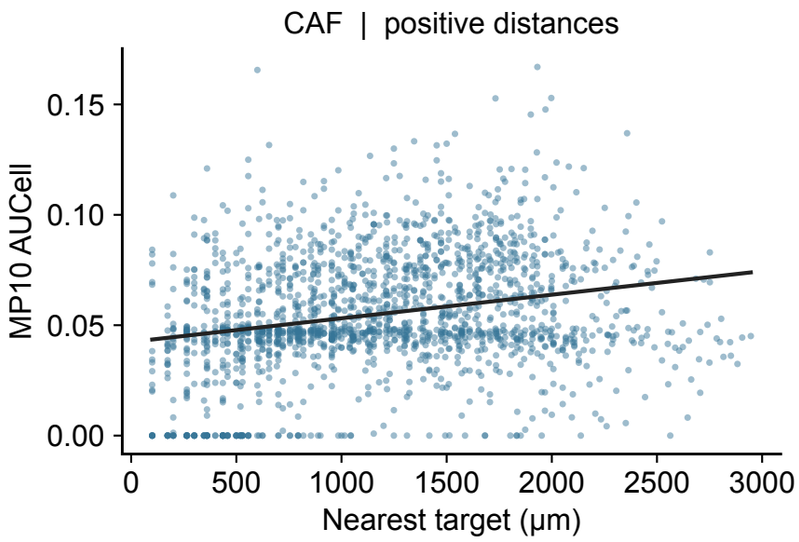
Nearest spot center (µm)

Inhousedata_T4463209 | MP10 versus nearest target

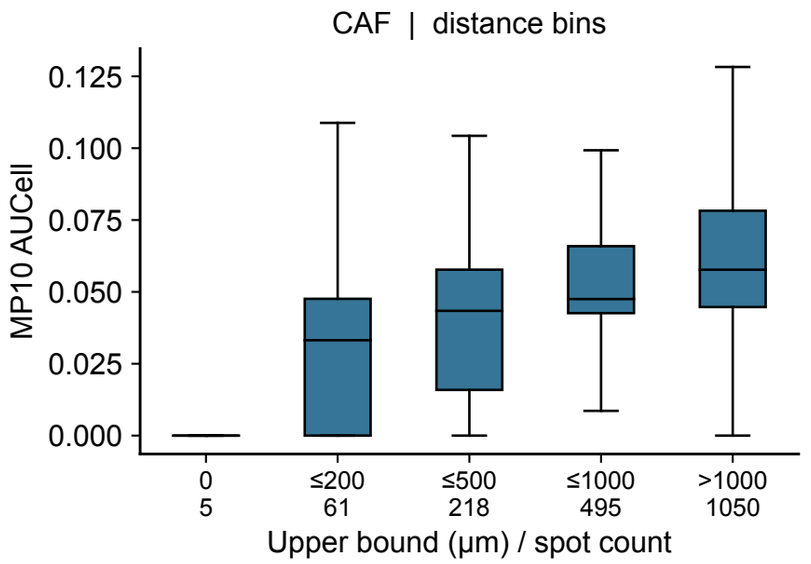
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



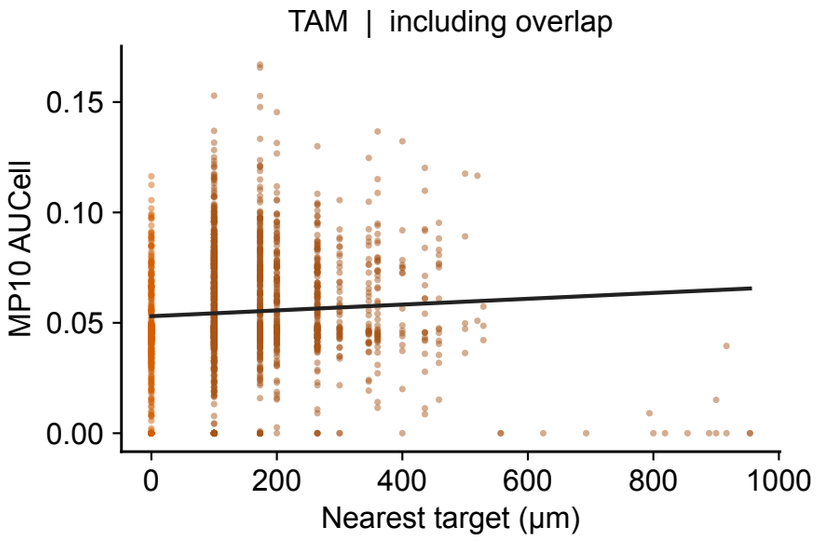
r +0.25 ρ +0.24
n 1829/1829 queries • 889 targets



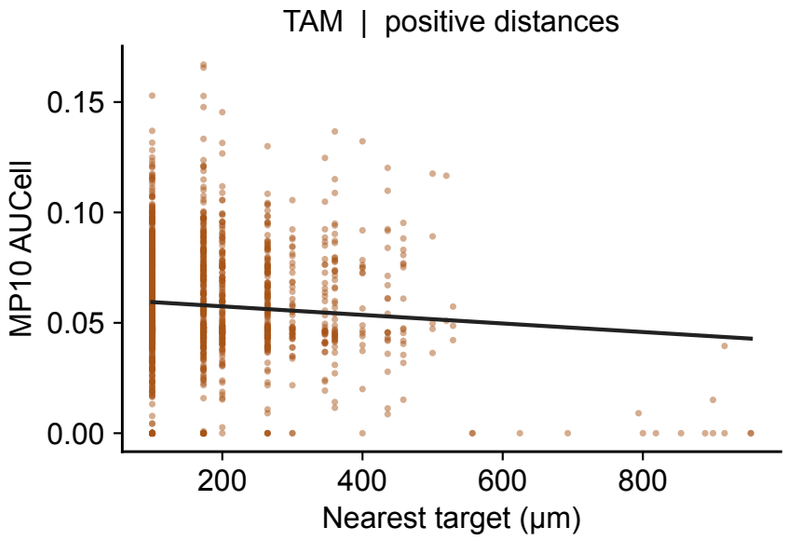
r +0.24 ρ +0.24
n 1824/1829 queries • 889 targets



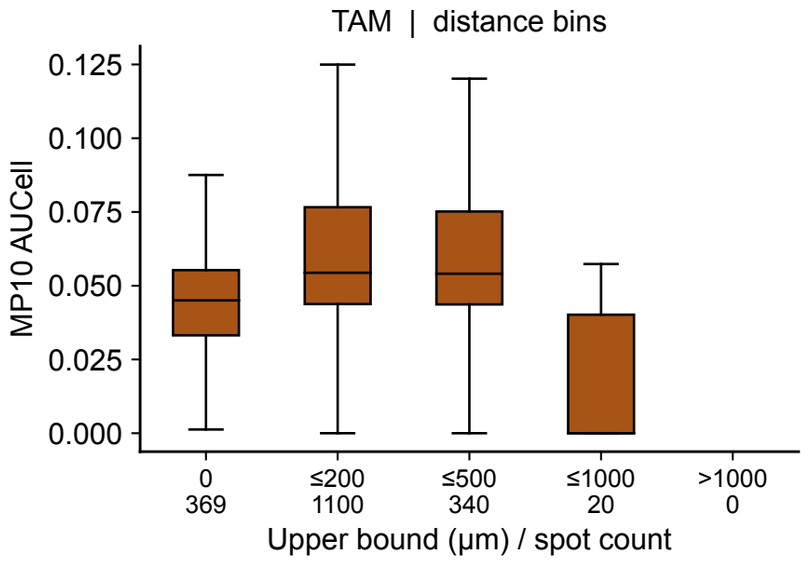
Orange = zero distance
Line = within-section linear fit



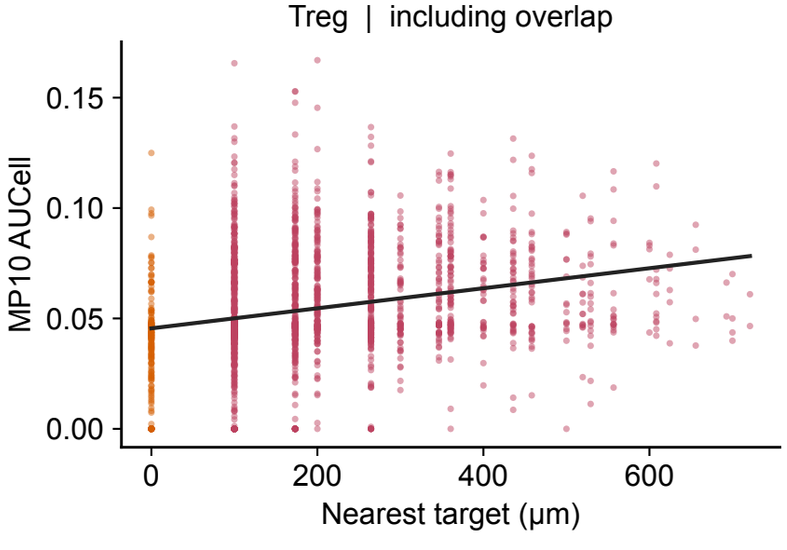
r +0.06 ρ +0.14
n 1829/1829 queries • 889 targets



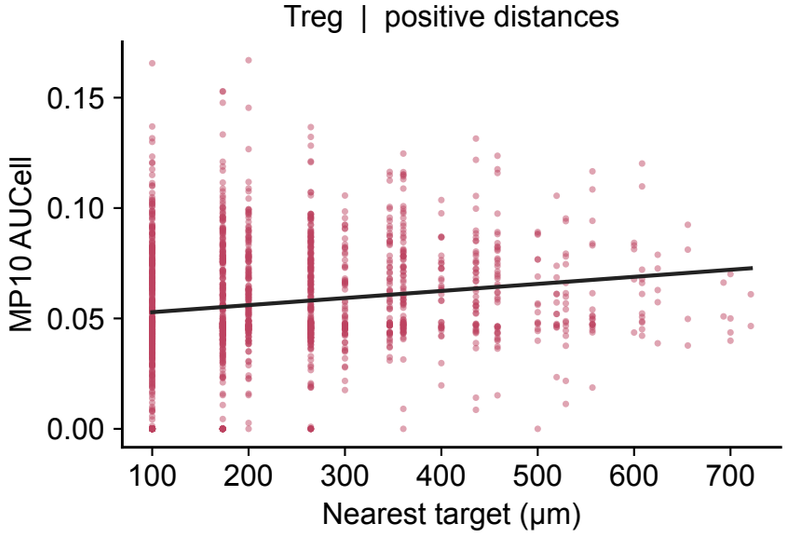
r -0.08 ρ -0.02
n 1460/1829 queries • 889 targets



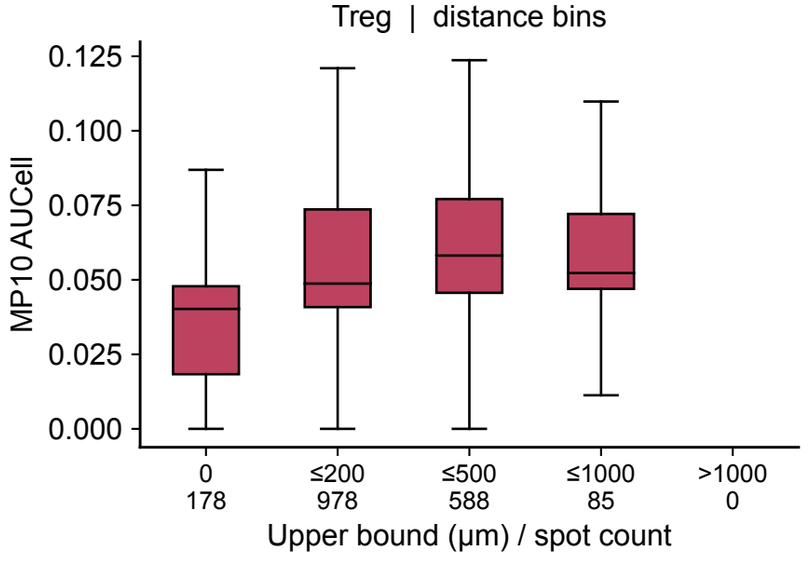
Orange = zero distance
Line = within-section linear fit



r +0.24 ρ +0.28
n 1829/1829 queries • 889 targets



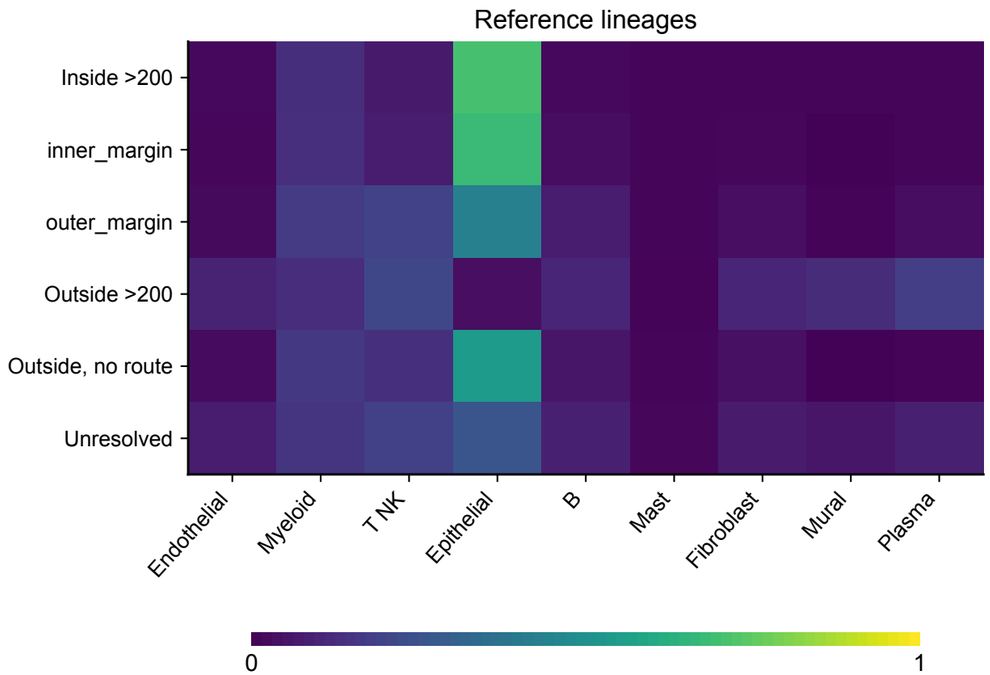
r +0.16 ρ +0.20
n 1651/1829 queries • 889 targets

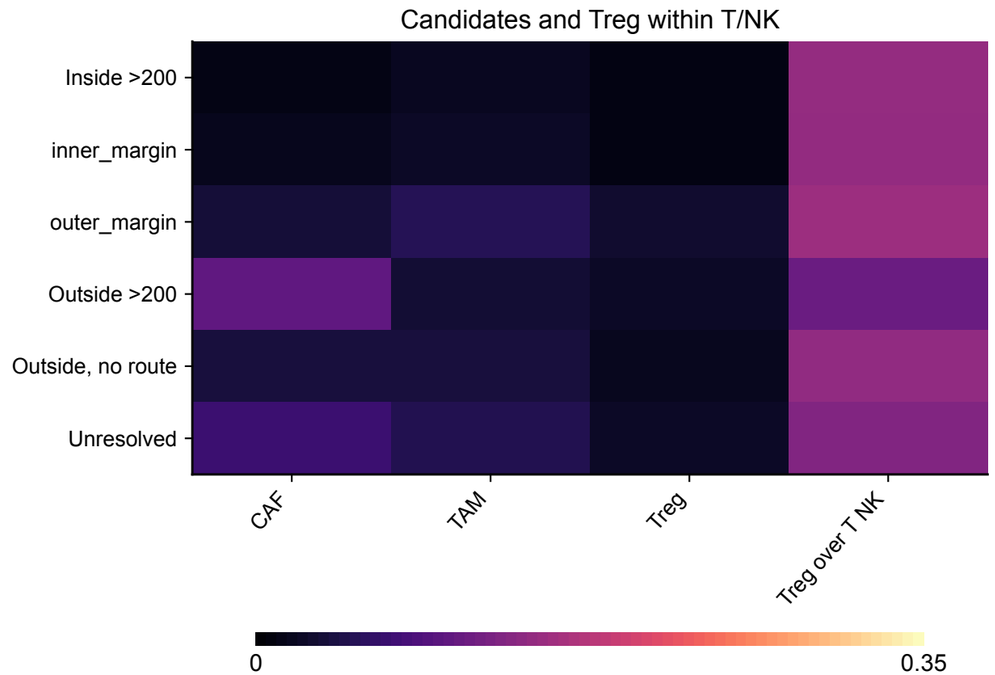


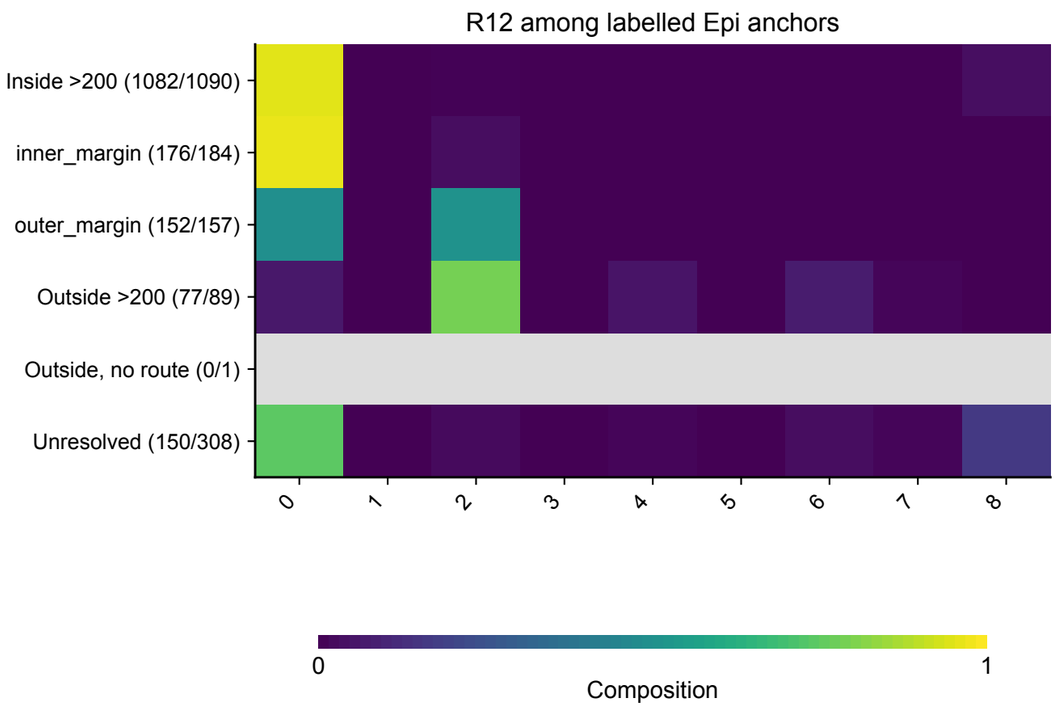
Orange = zero distance
Line = within-section linear fit

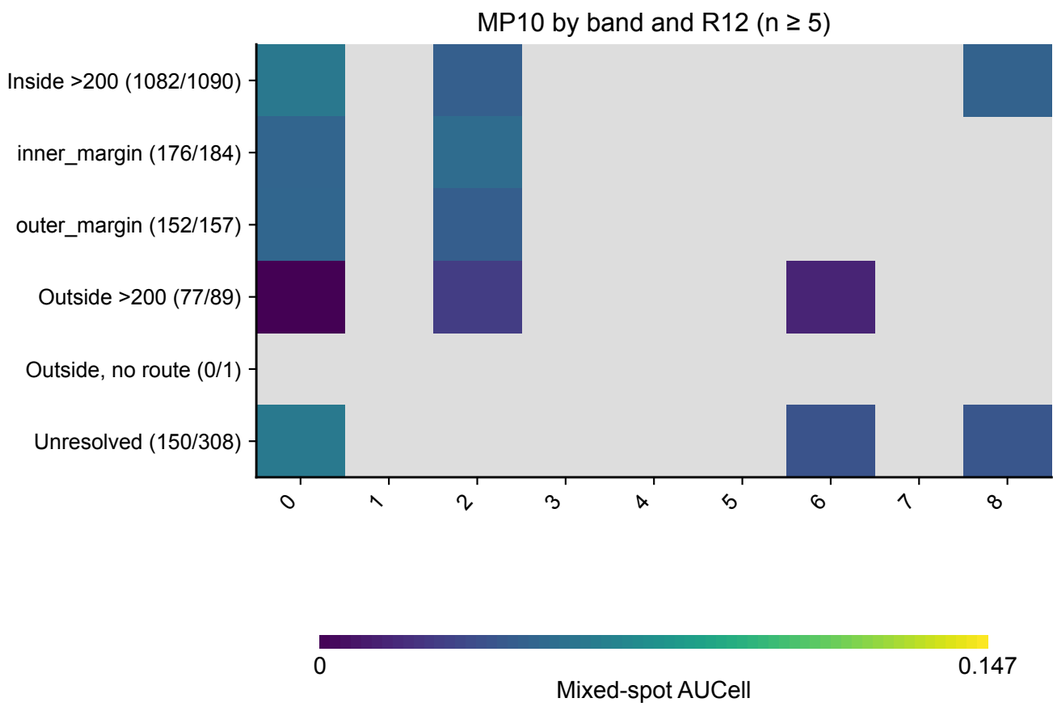
Inhousedata_T4463209 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





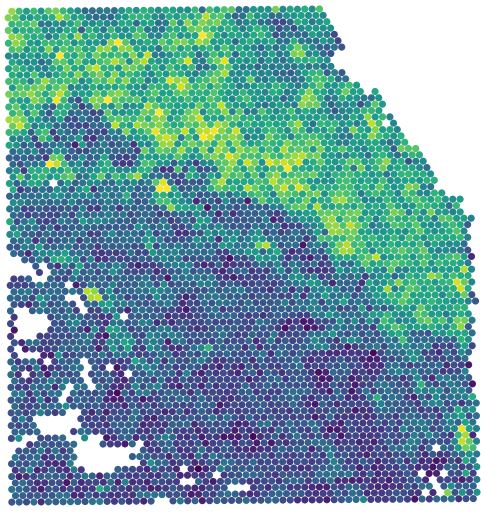




Inhousedata_T4463209 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

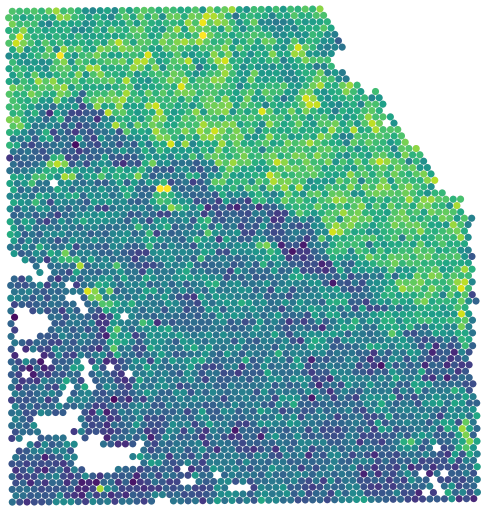
E2F Targets



0 0.152

UCell score

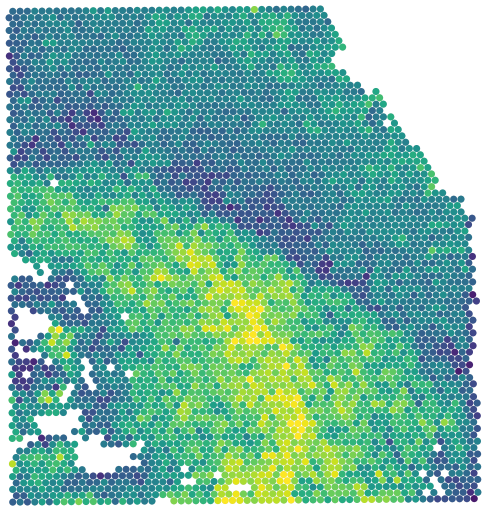
G2M Checkpoint



0 0.134

UCell score

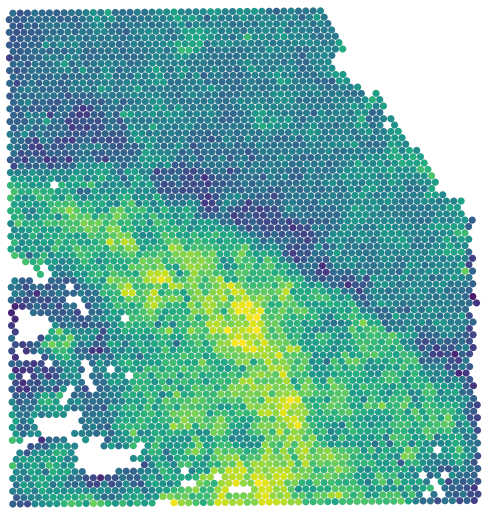
Hypoxia



0 0.211

UCell score

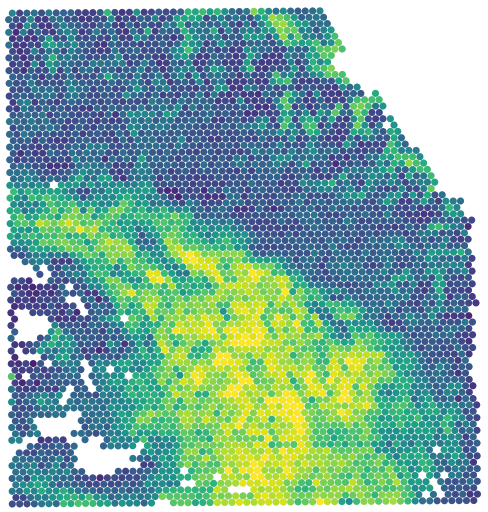
Tnfa Signaling Via Nfkb



0 0.275

UCell score

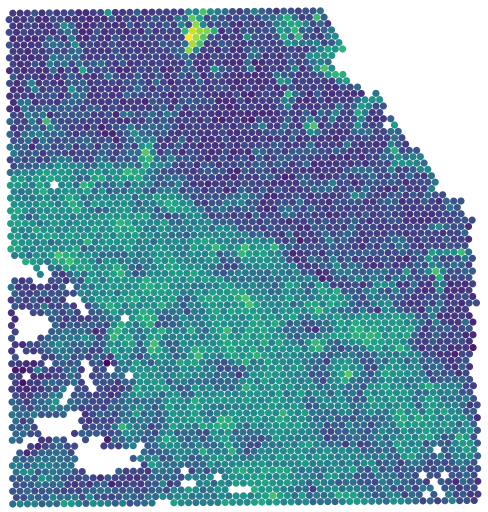
Emt



0 0.386

UCell score

Interferon Gamma Response



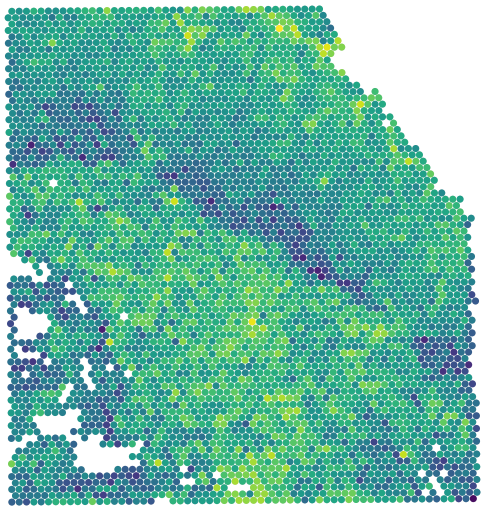
0 0.275

UCell score

Inhousedata_T4463209 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

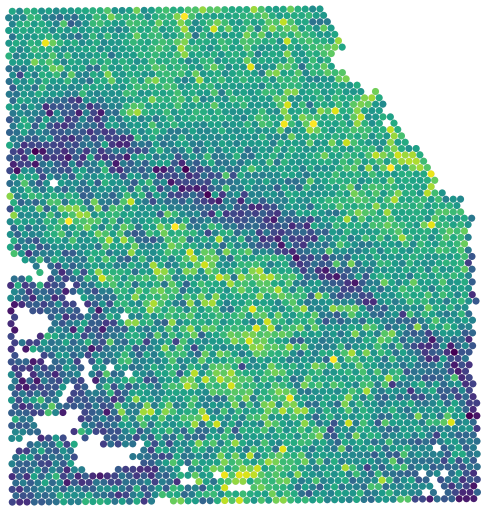
Il2 Stat5 Signaling



0 0.144

UCell score

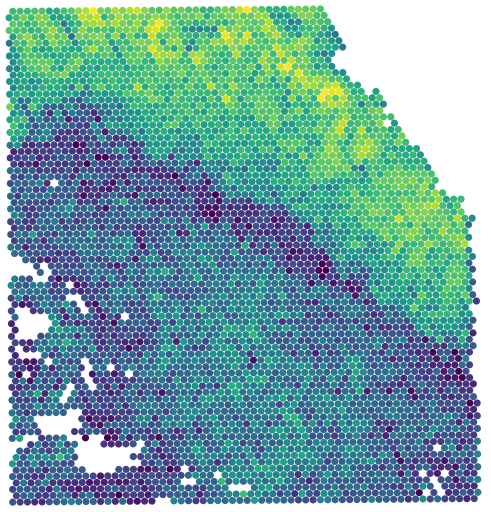
Tgf Beta Signaling



0 0.232

UCell score

Cholesterol Homeostasis



0 0.21

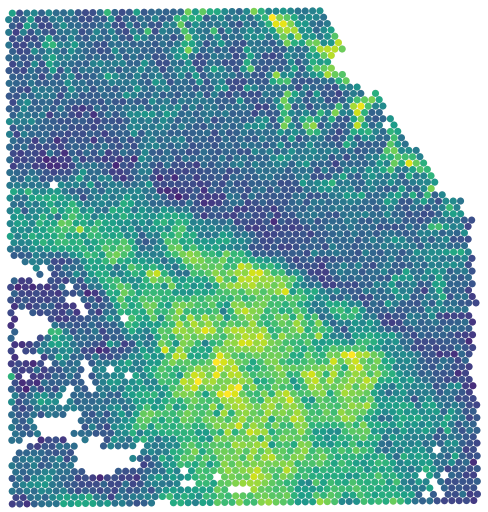
UCell score

Cholesterol Biosynthesis

0 0.166

UCell score

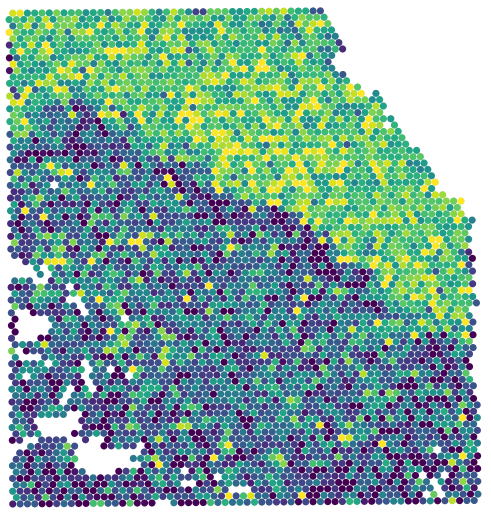
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



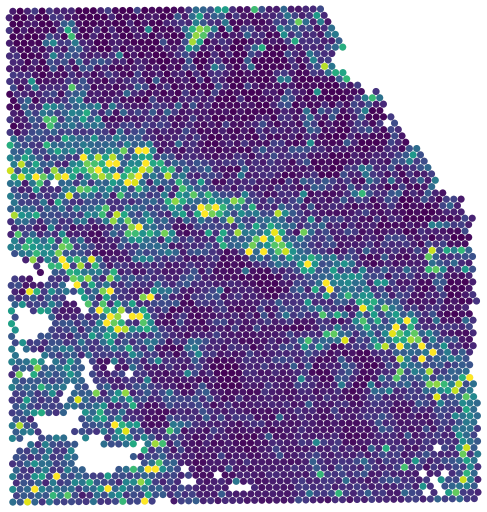
0 0.151

UCell score

Inhousedata_T4463209 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

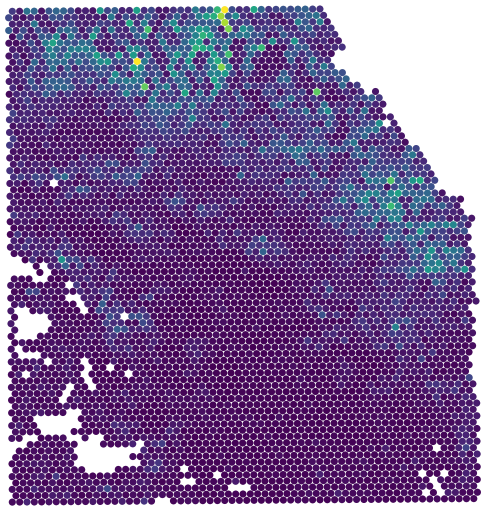
Treg composition



0 0.102

Fraction of all 30

Treg estimated abundance



0 2.85

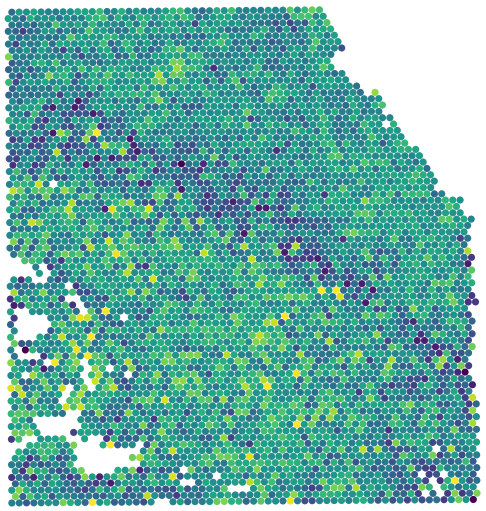
Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

UCell score

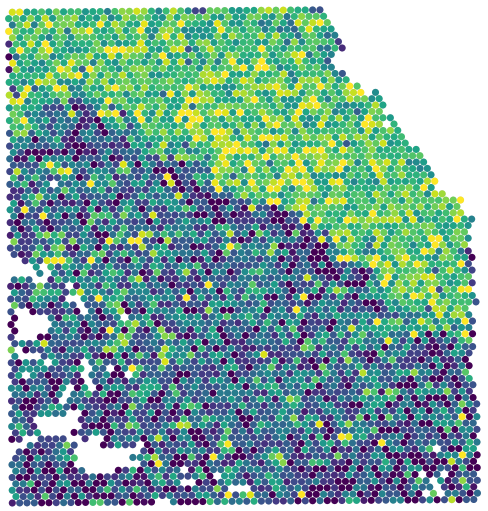
Treg Resting Vs Activated Up



0 0.0609

UCell score

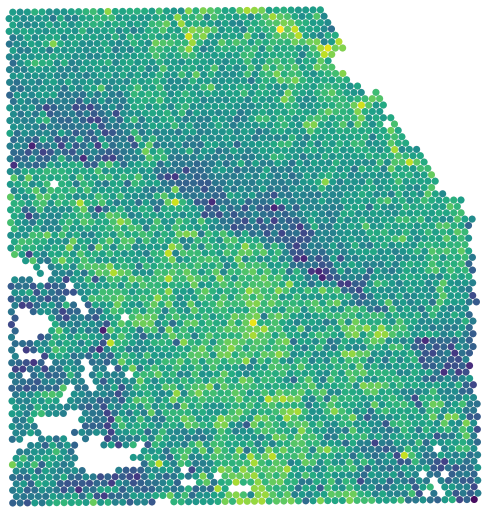
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

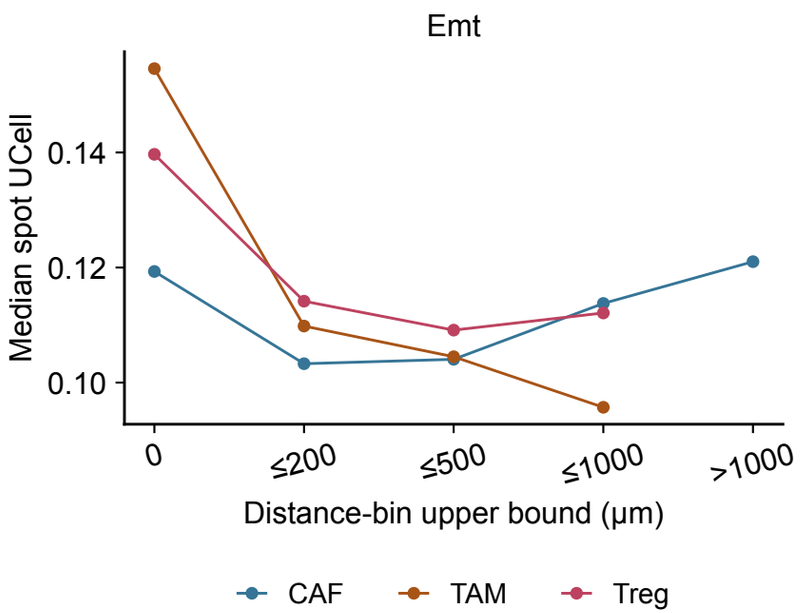


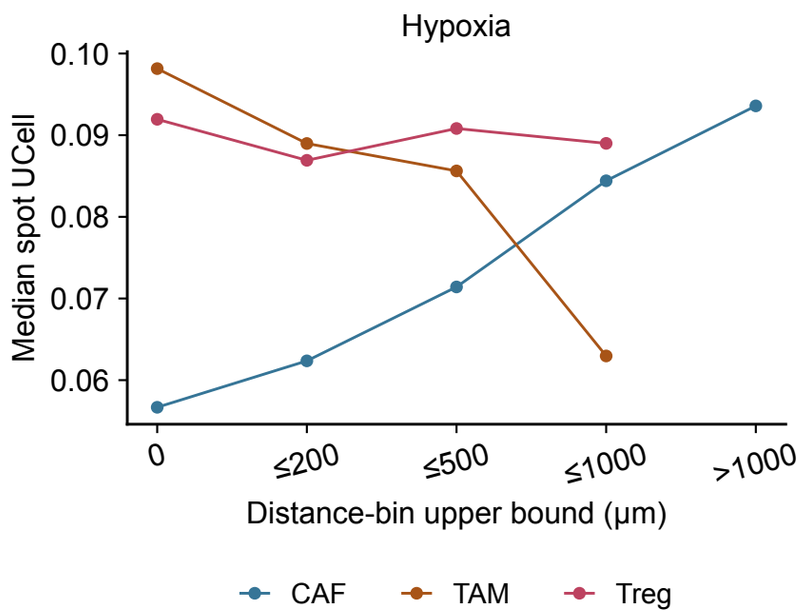
0 0.144

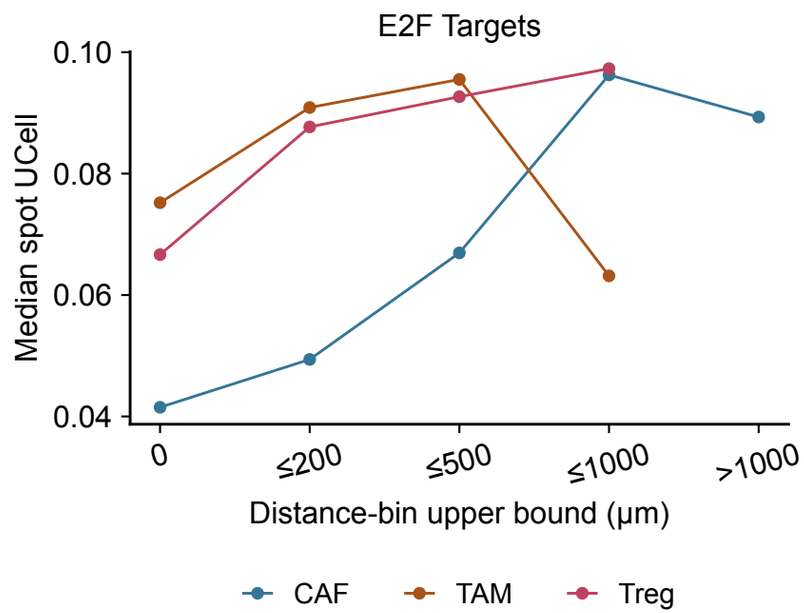
UCell score

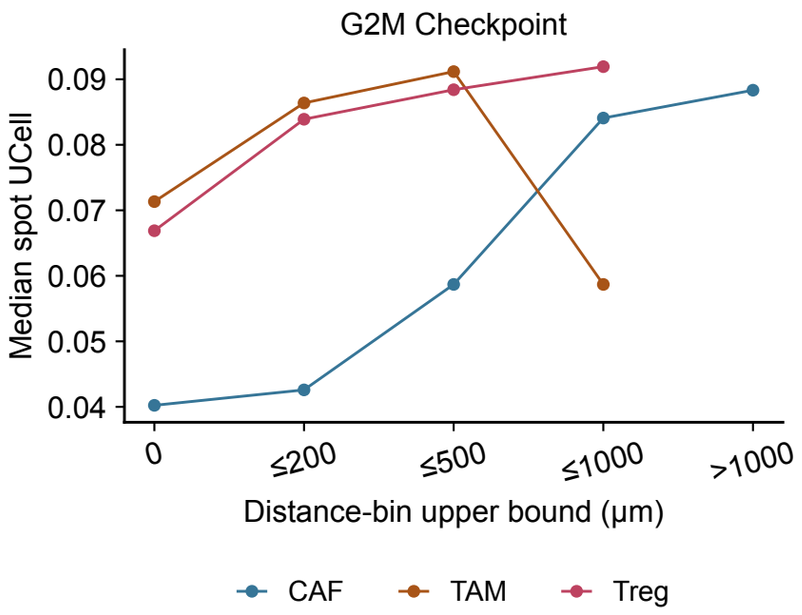
Inhousedata_T4463209 | Programs by target distance

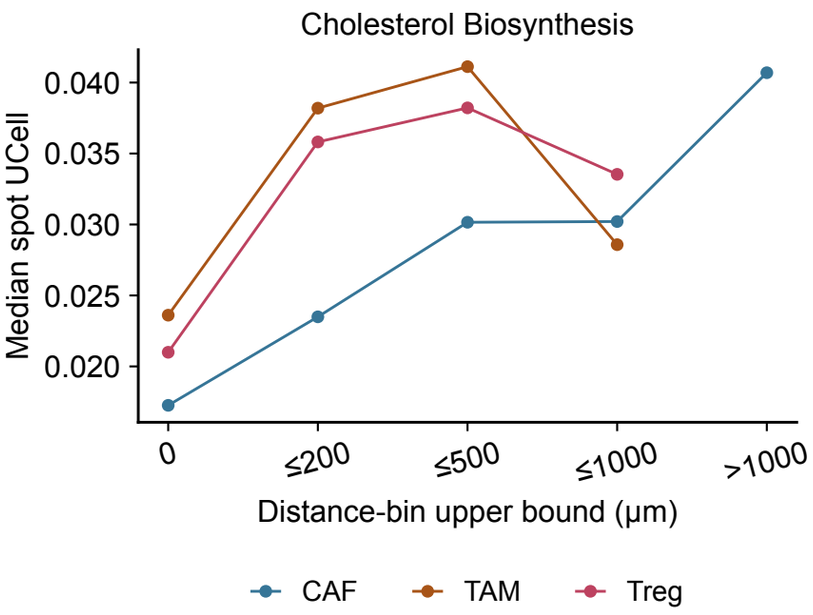
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

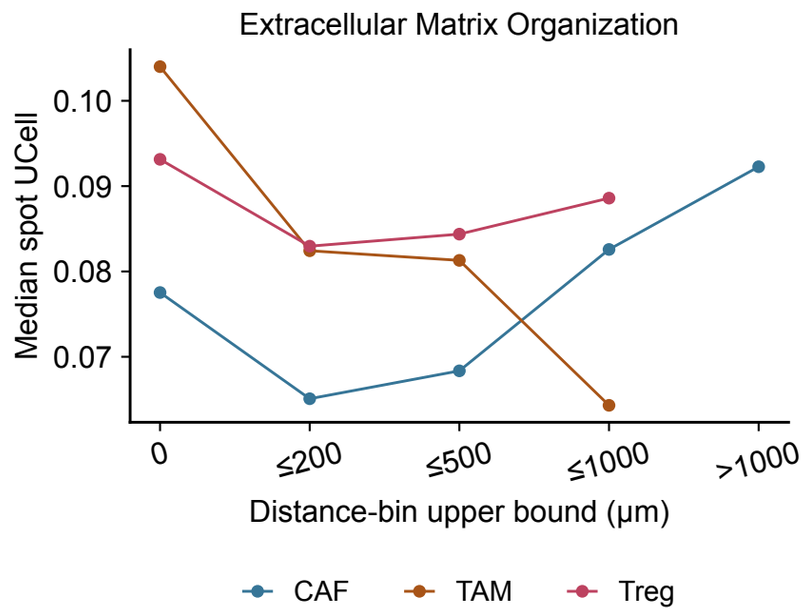






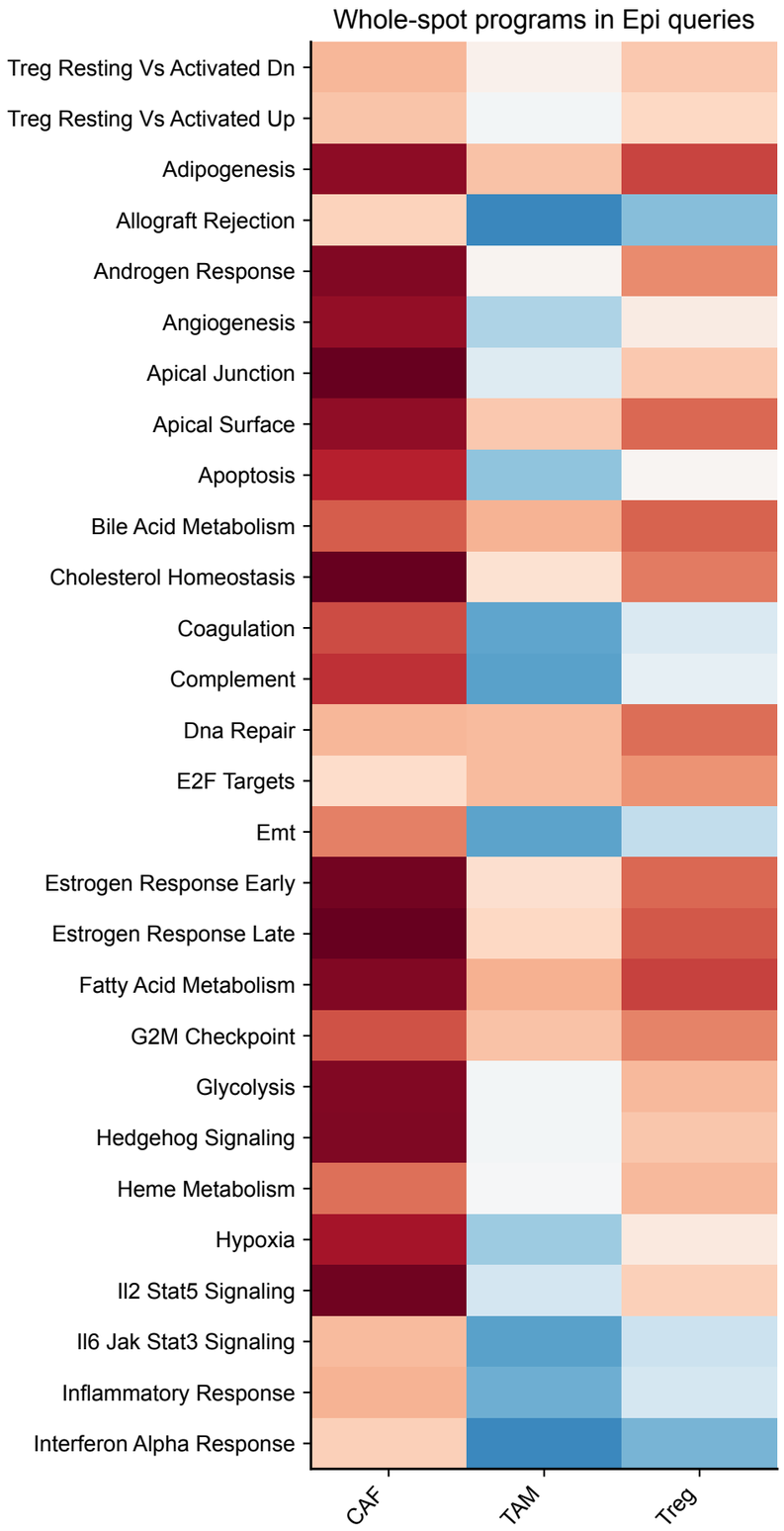


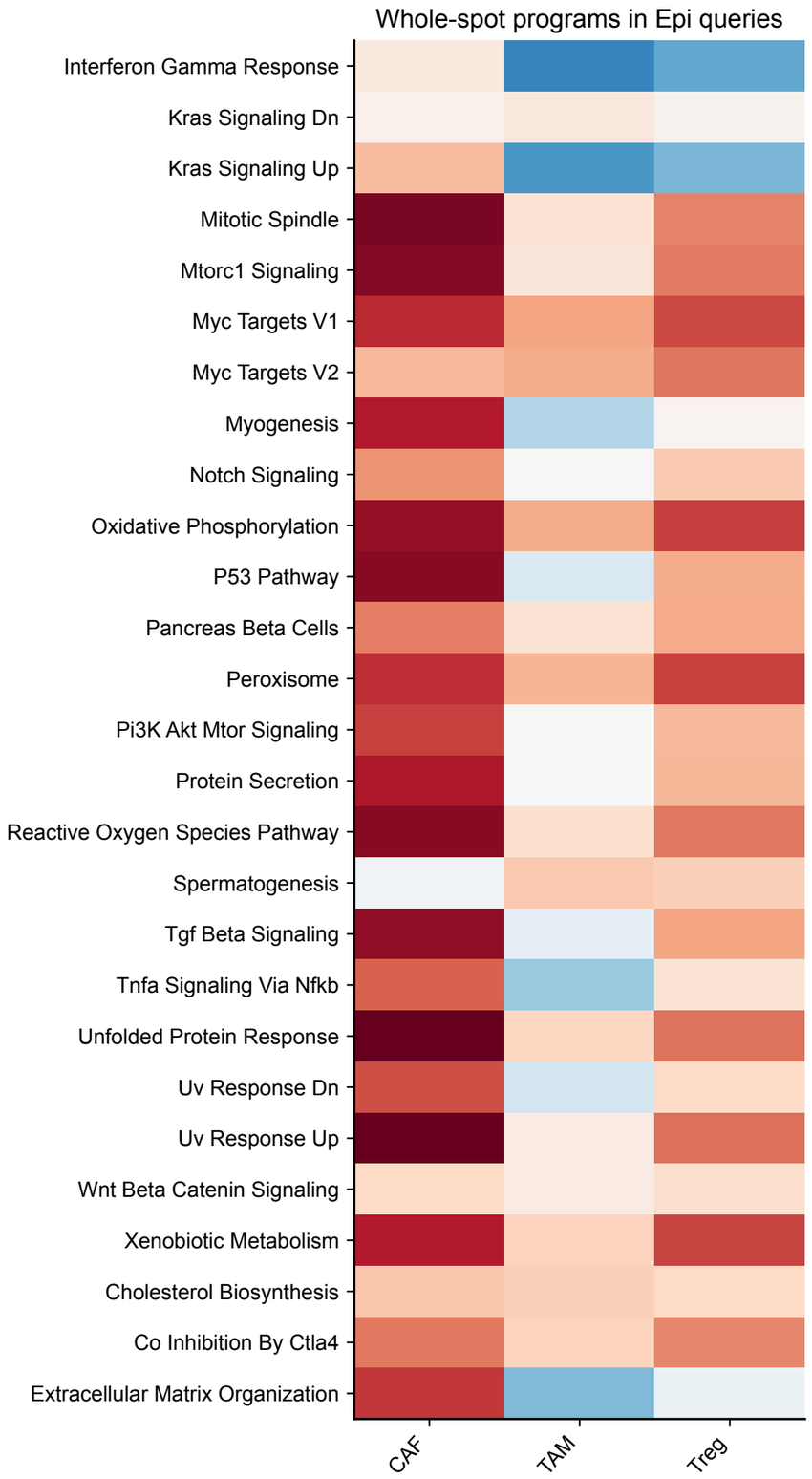




Inhousedata_T4463209 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

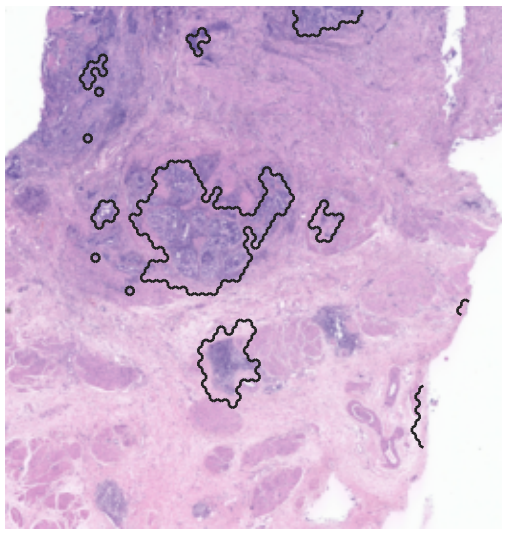




Inhousedata_T4465935 | Tissue context

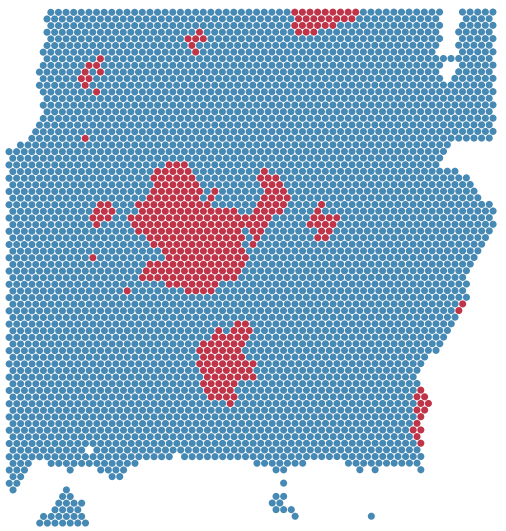
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



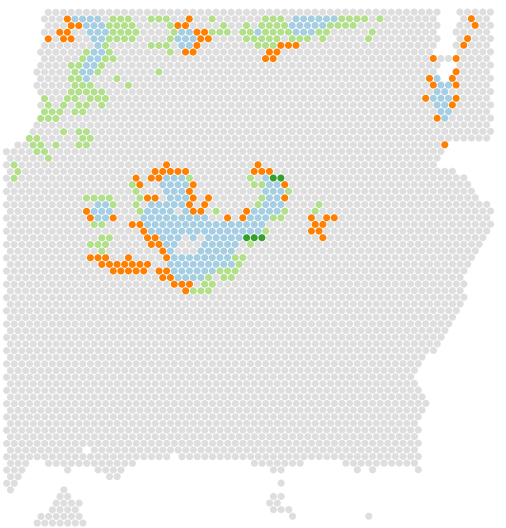
Black line: observed candidate outer edge

Final candidate location



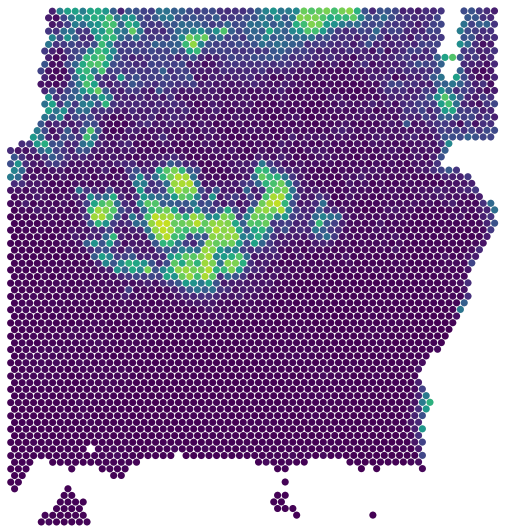
inside outside

Accepted R12 (original 0–8)



0 2 3 5

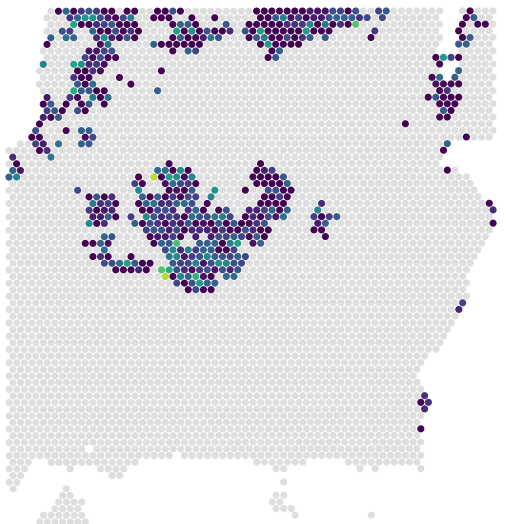
Epithelial support



0 1

Fraction of all 30

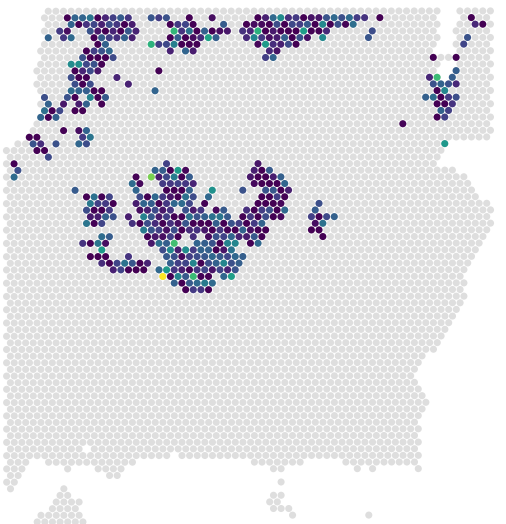
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



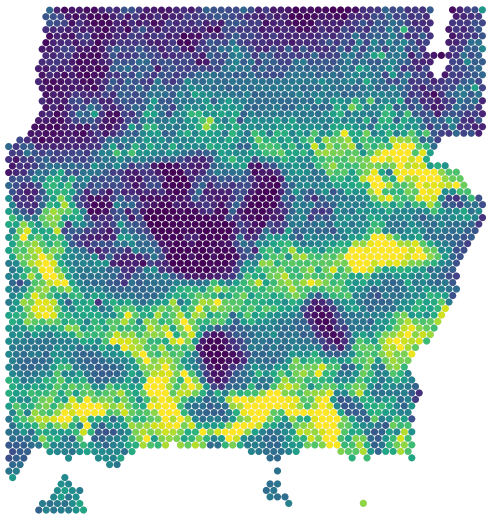
0 0.14

Matched-model AUCell

Inhousedata_T4465935 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

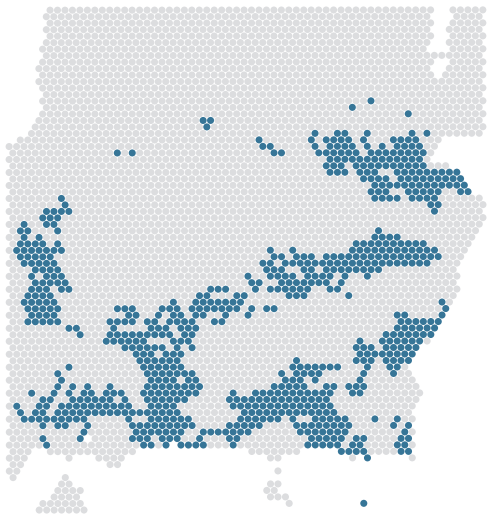
CAF composition



0 0.418

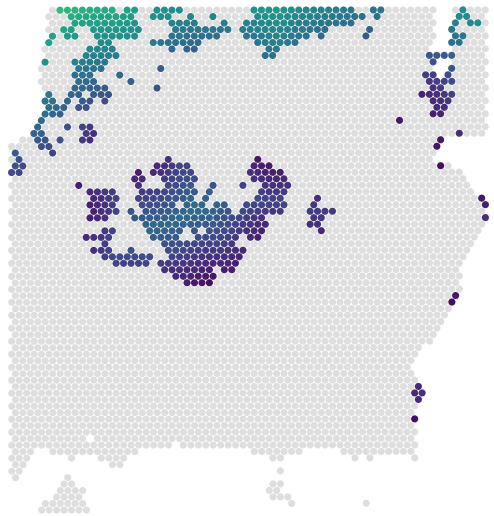
Fraction of all 30

CAF P80 targets



False True

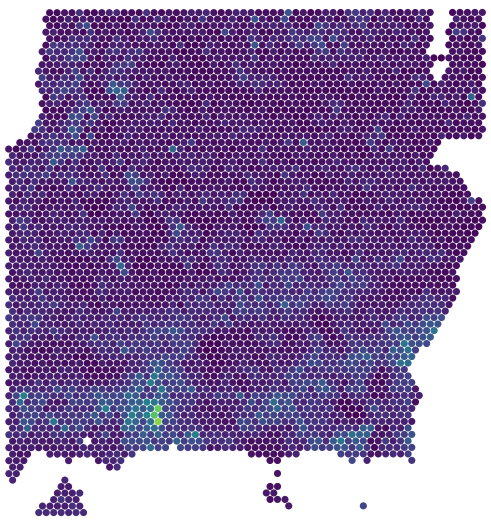
CAF distance



0 3.1e+03

Nearest spot center (μm)

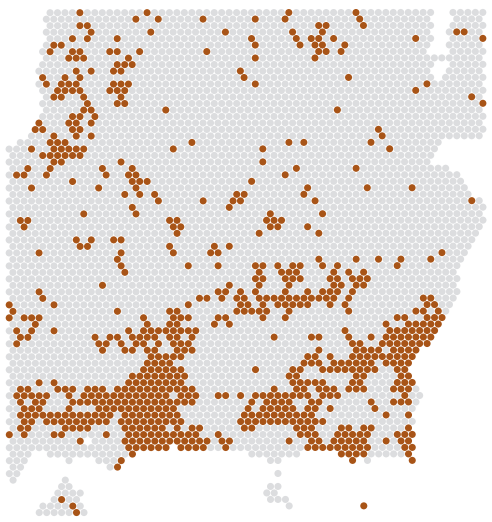
TAM composition



0 0.291

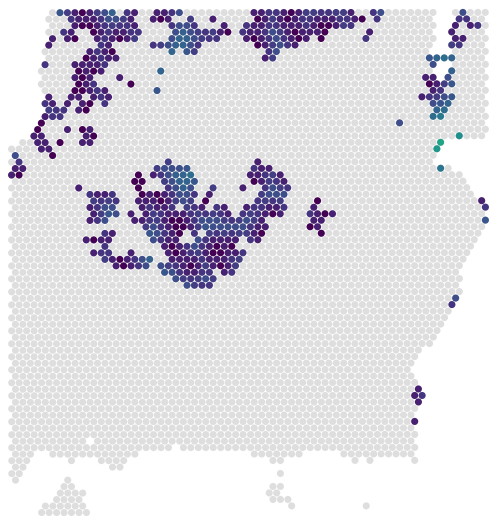
Fraction of all 30

TAM P80 targets



False True

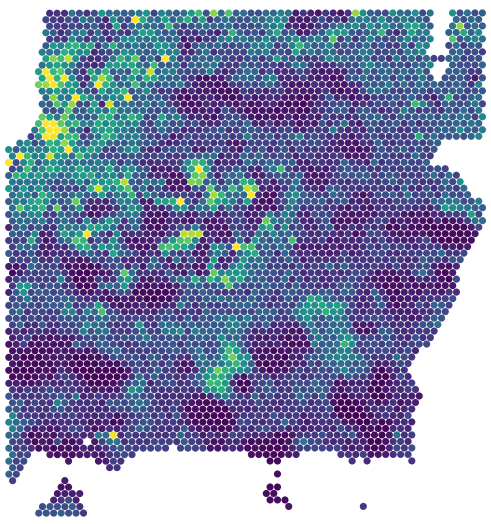
TAM distance



0 1.13e+03

Nearest spot center (μm)

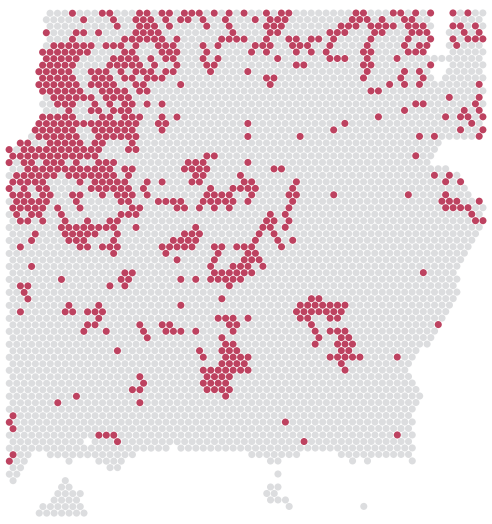
Treg composition



0 0.102

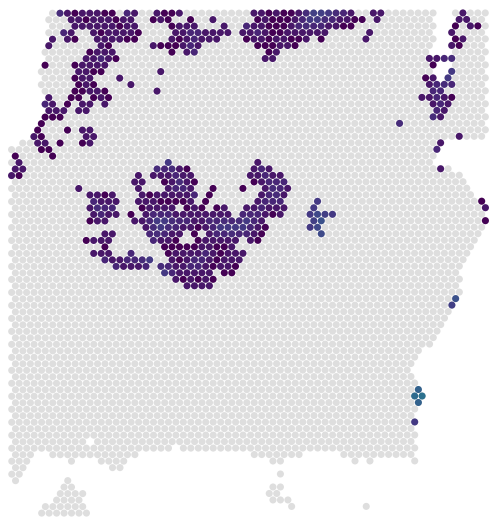
Fraction of all 30

Treg P80 targets



False True

Treg distance

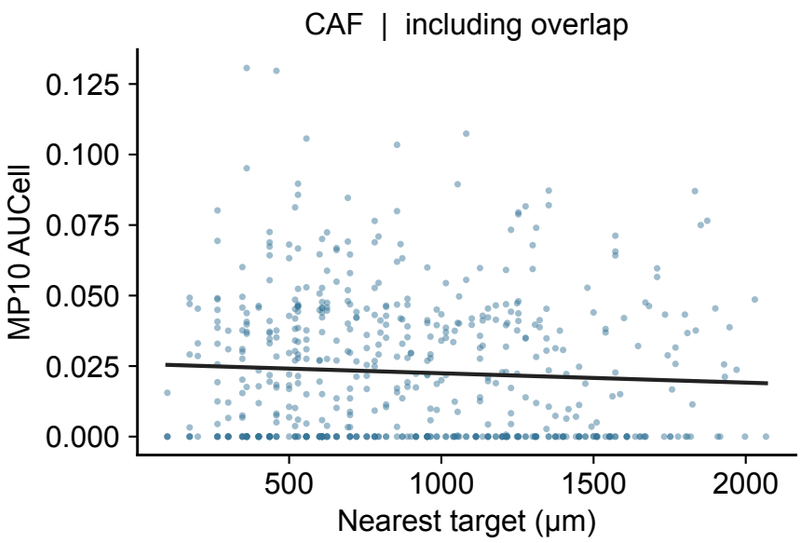


0 1.61e+03

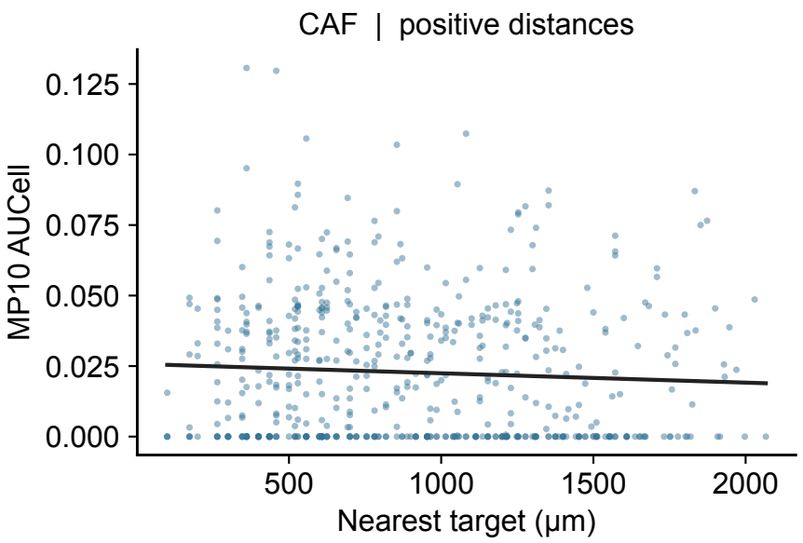
Nearest spot center (μm)

Inhousedata_T4465935 | MP10 versus nearest target

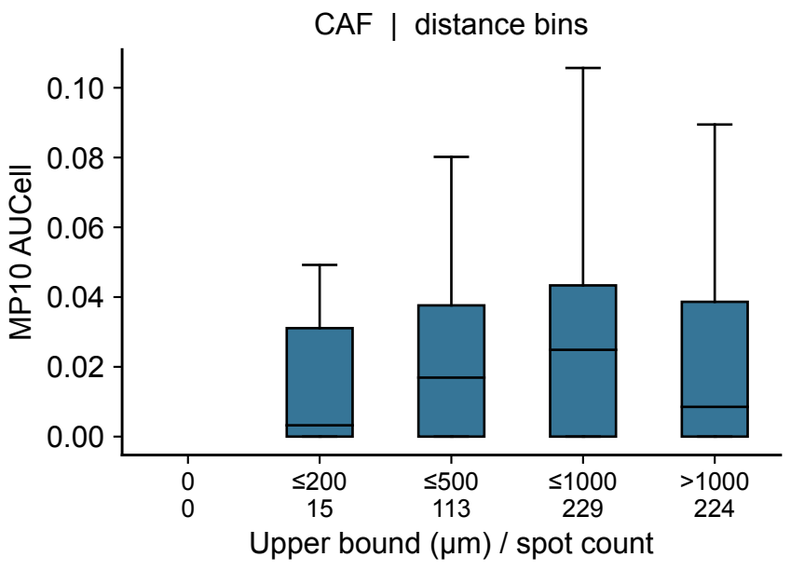
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



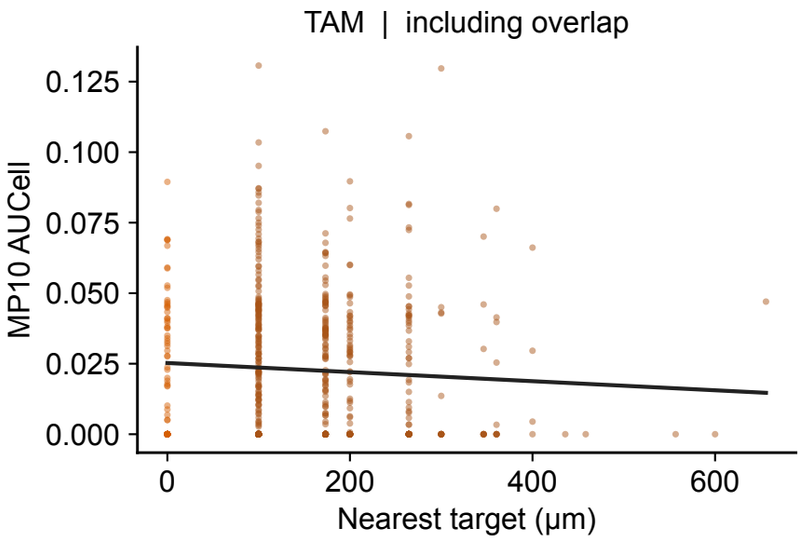
r -0.06 ρ -0.07
n 581/581 queries • 814 targets



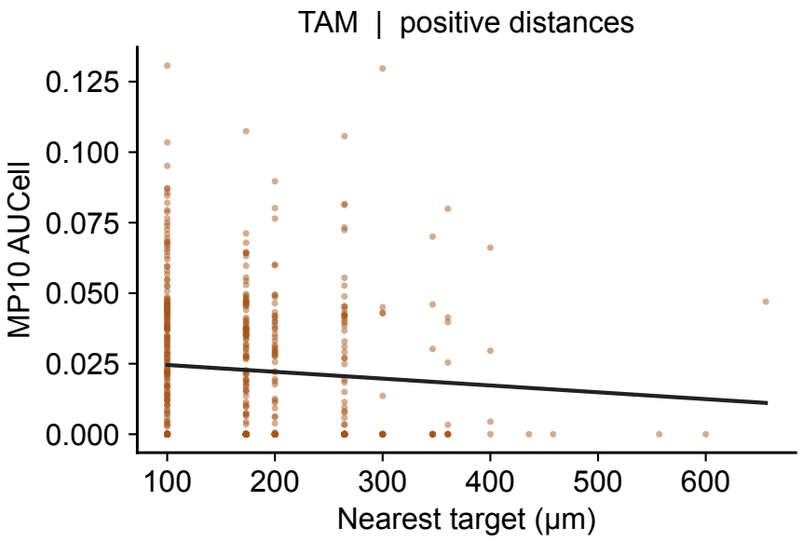
r -0.06 ρ -0.07
n 581/581 queries • 814 targets



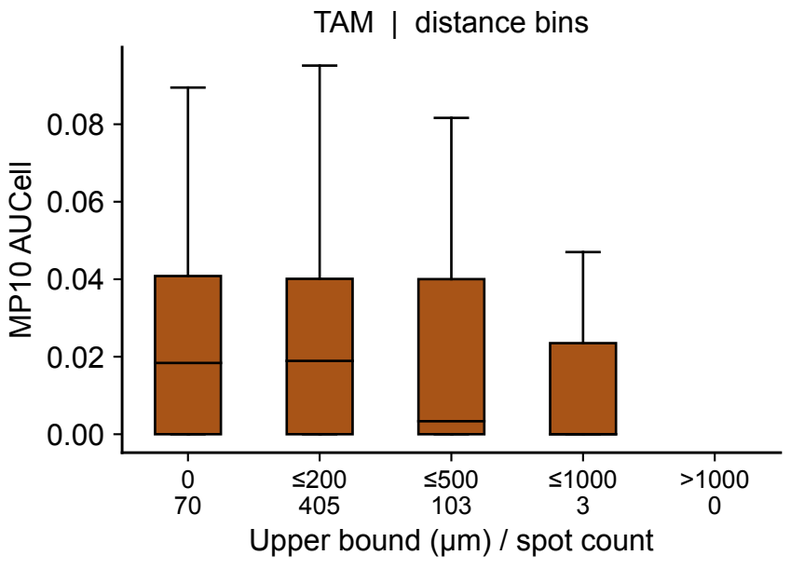
Orange = zero distance
Line = within-section linear fit



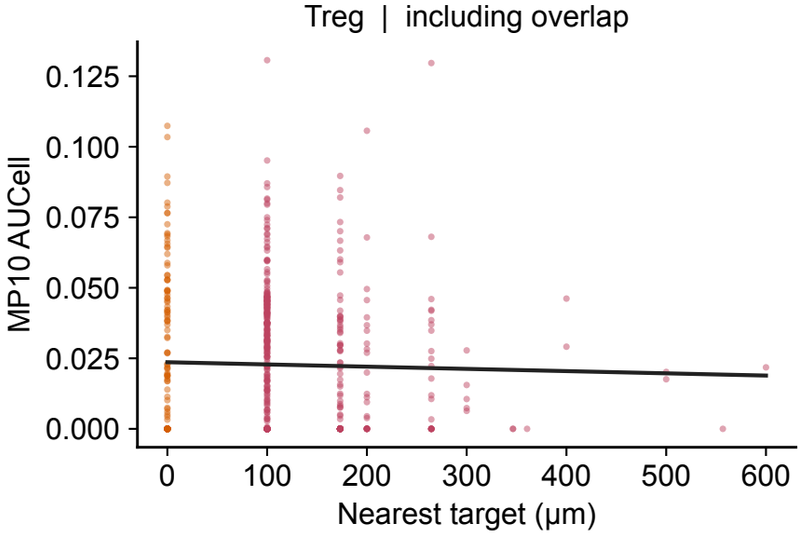
r -0.06 ρ -0.07
n 581/581 queries • 814 targets



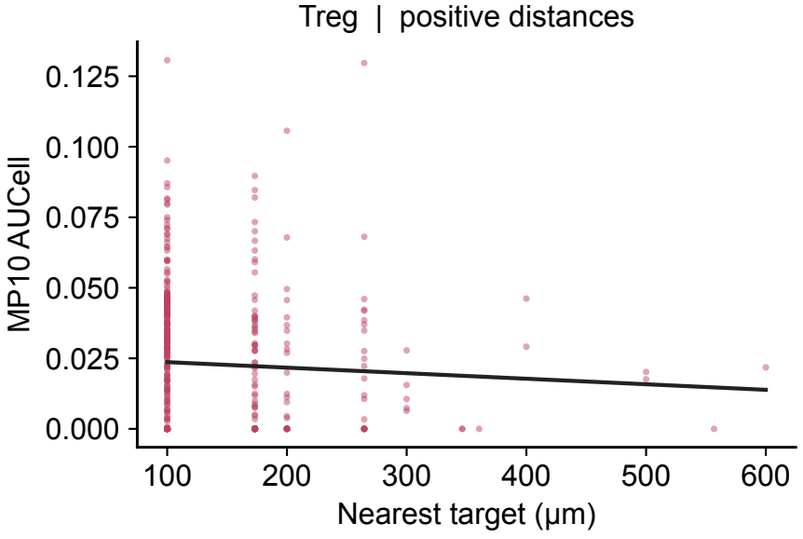
r -0.08 ρ -0.08
n 511/581 queries • 814 targets



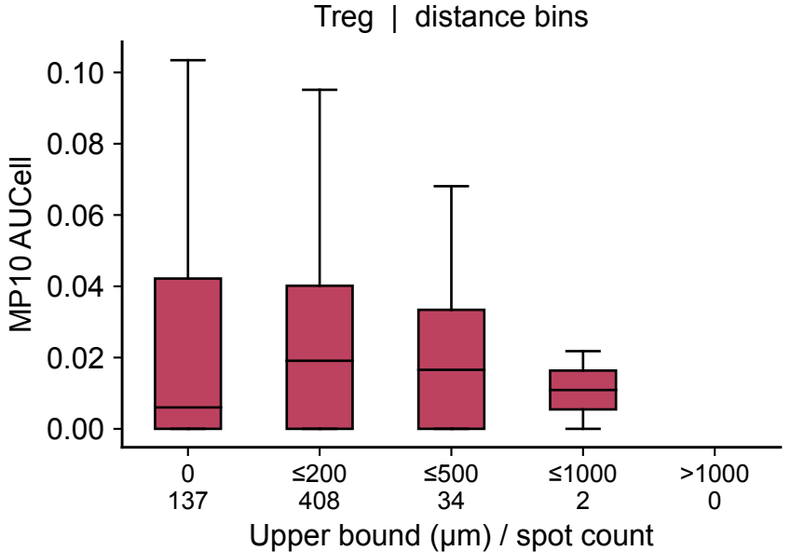
Orange = zero distance
Line = within-section linear fit



r -0.03 ρ +0.00
n 581/581 queries • 814 targets



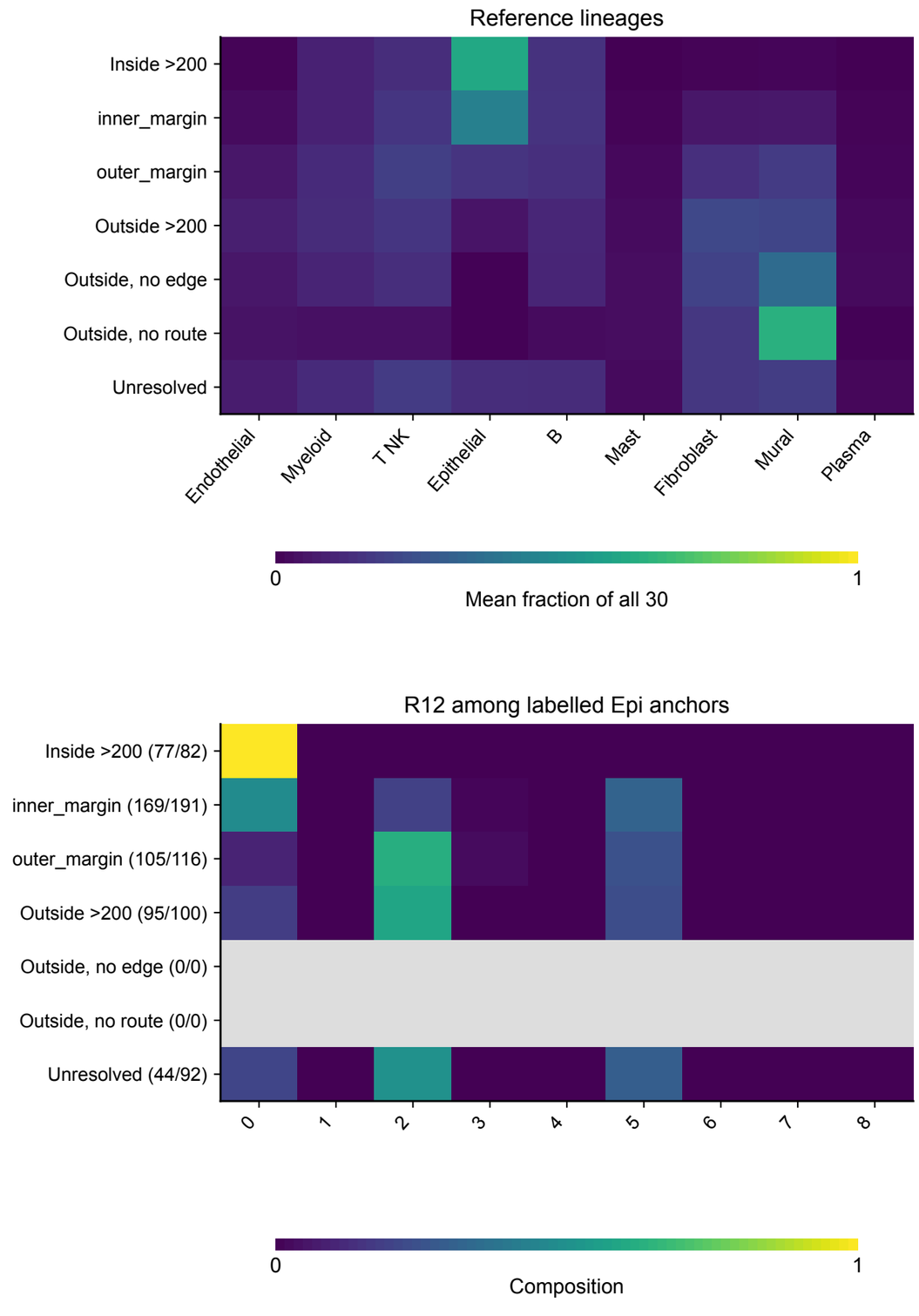
r -0.06 ρ -0.06
n 444/581 queries • 814 targets

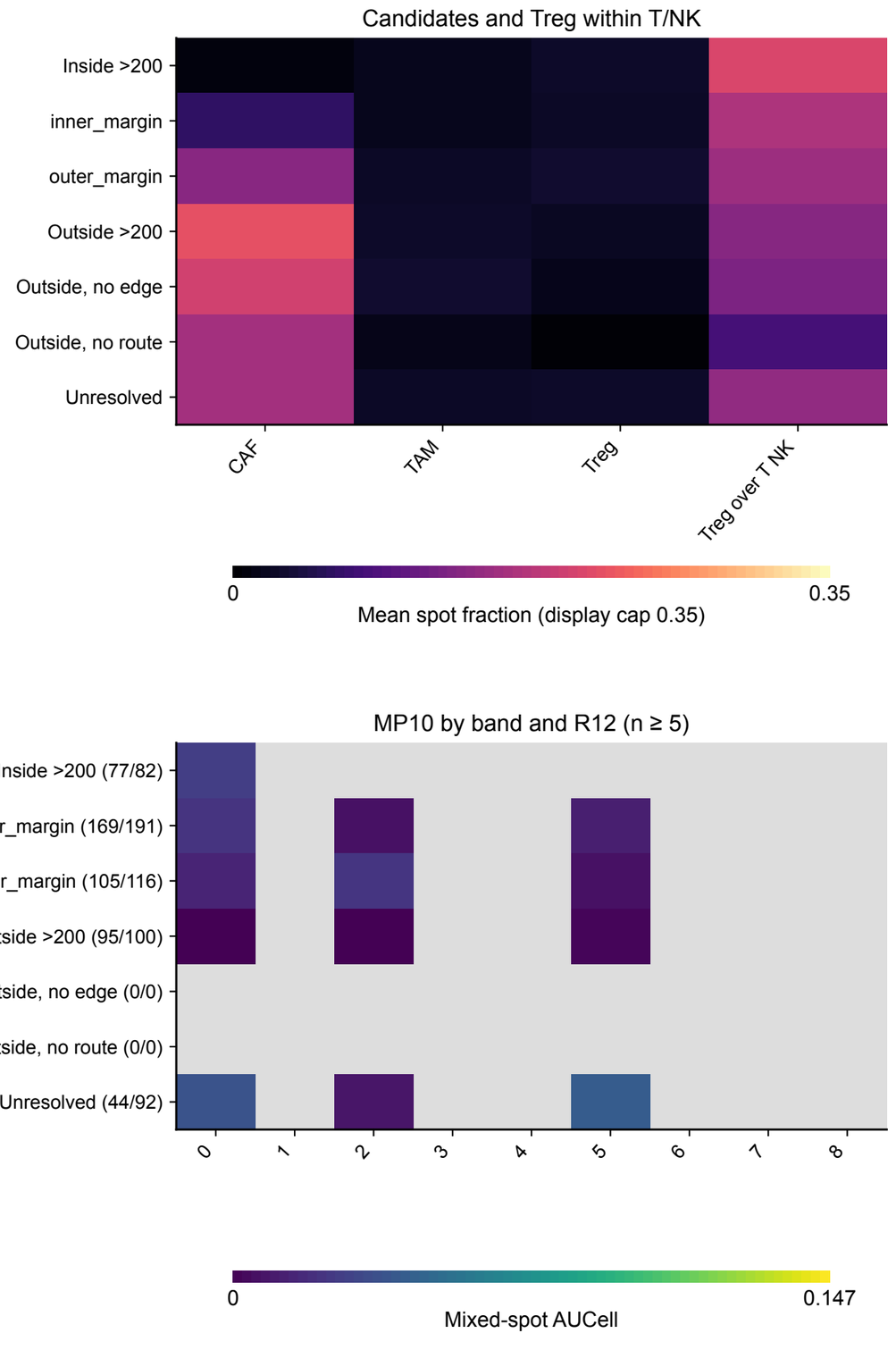


Orange = zero distance
Line = within-section linear fit

Inhousedata_T4465935 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable

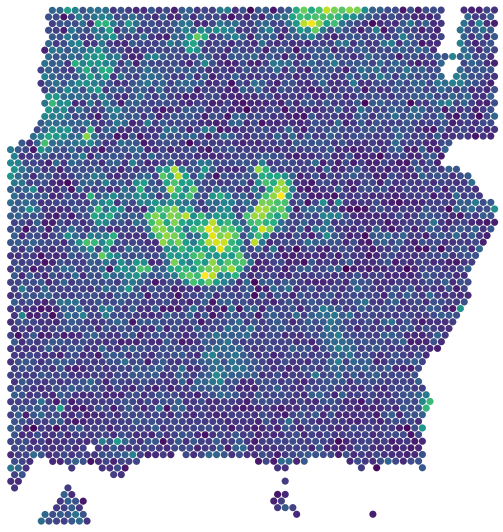




Inhousedata_T4465935 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

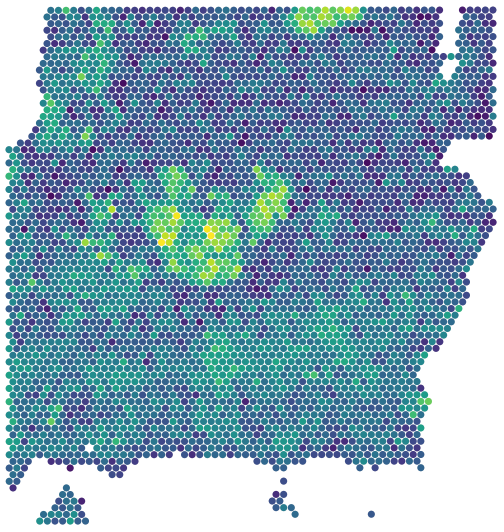
E2F Targets



0 0.152

UCell score

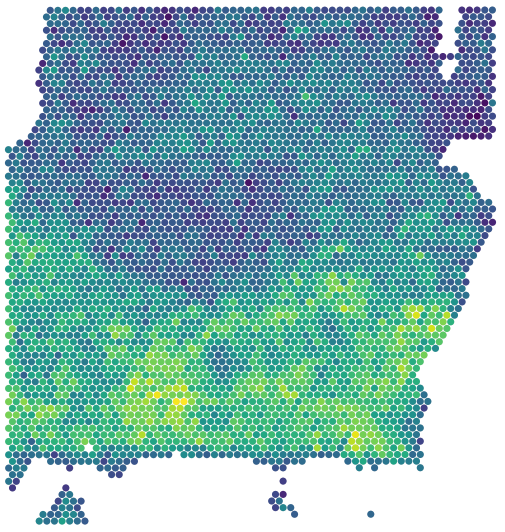
G2M Checkpoint



0 0.134

UCell score

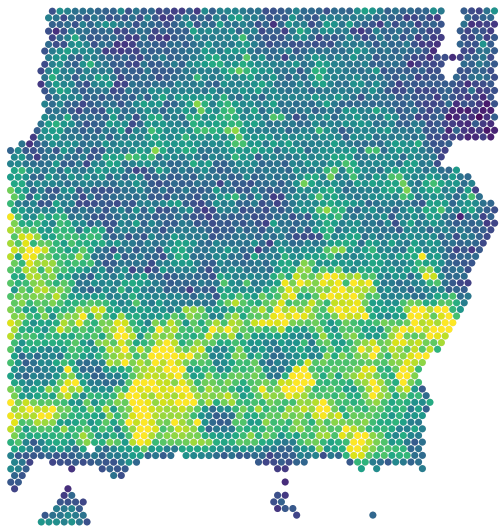
Hypoxia



0 0.211

UCell score

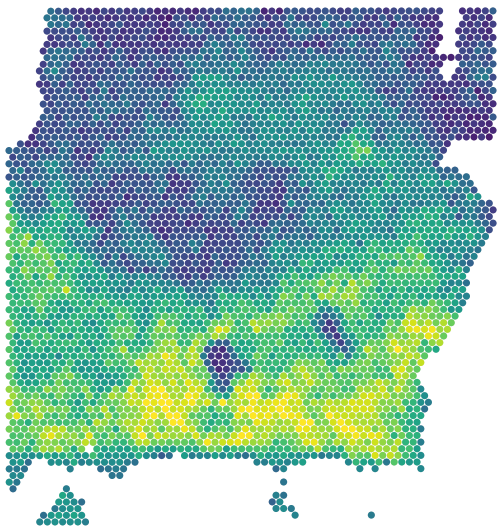
Tnfa Signaling Via Nfkb



0 0.275

UCell score

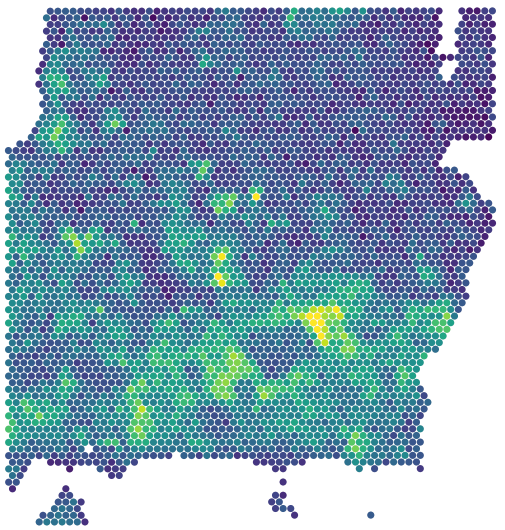
Emt



0 0.386

UCell score

Interferon Gamma Response



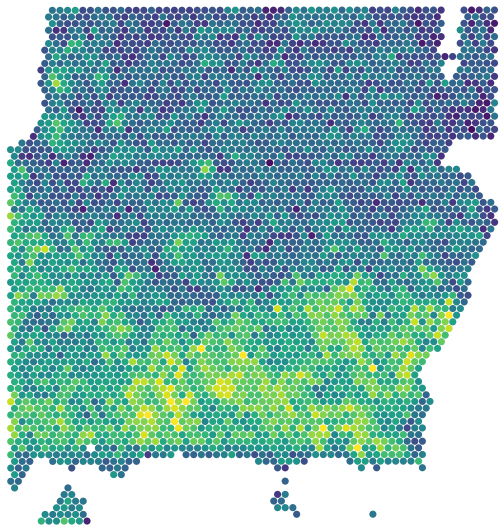
0 0.275

UCell score

Inhousedata_T4465935 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

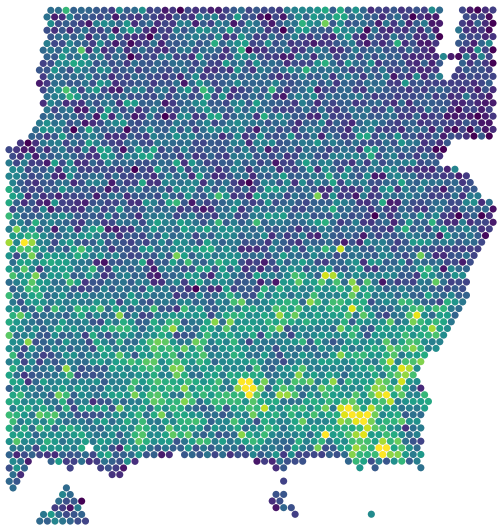
Il2 Stat5 Signaling



0 0.144

UCell score

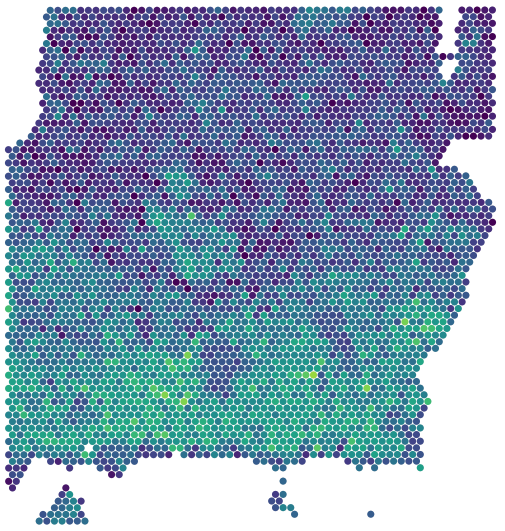
Tgf Beta Signaling



0 0.232

UCell score

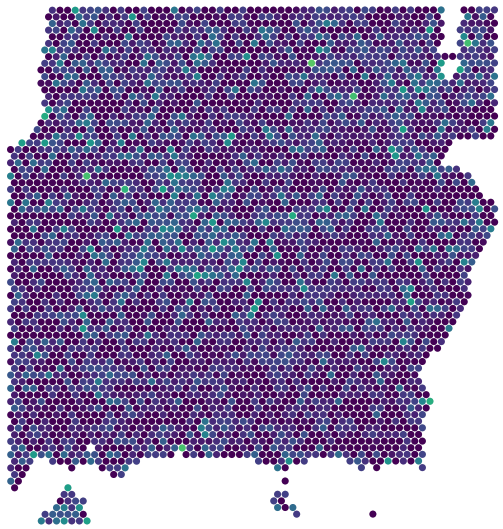
Cholesterol Homeostasis



0 0.21

UCell score

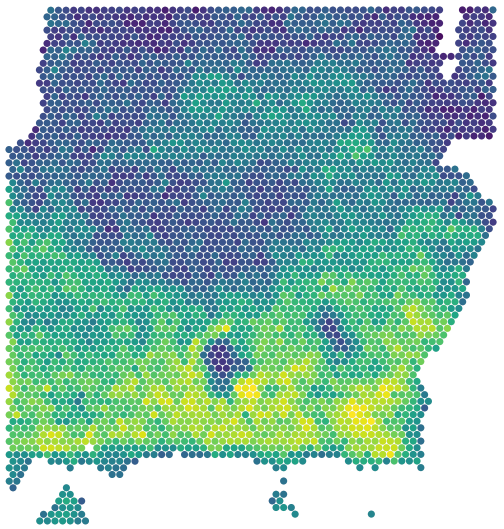
Cholesterol Biosynthesis



0 0.166

UCell score

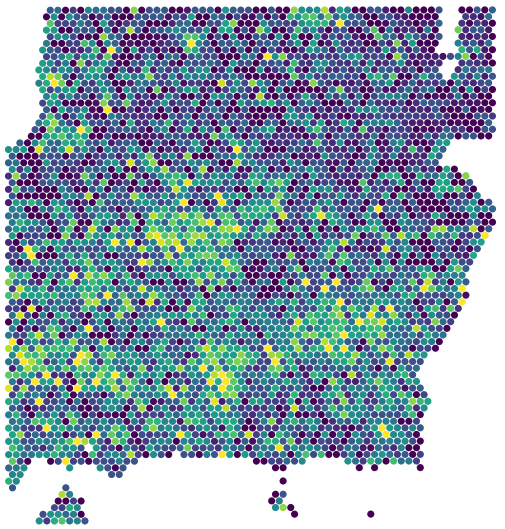
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



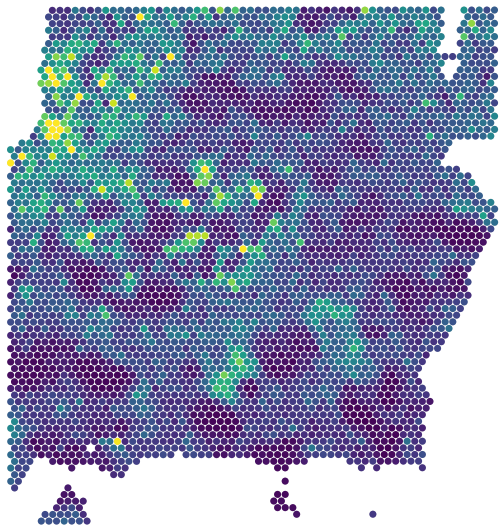
0 0.151

UCell score

Inhousedata_T4465935 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

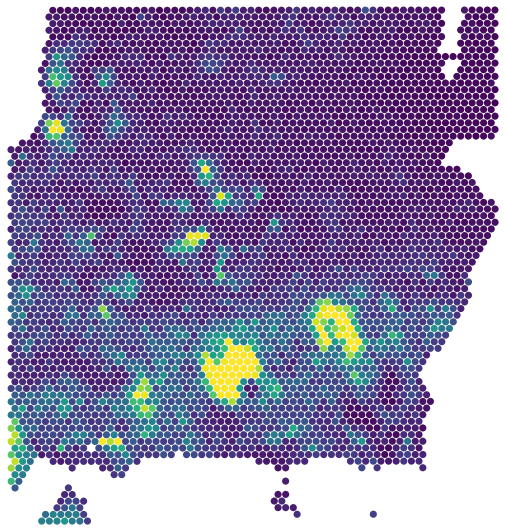
Treg composition



0 0.102

Fraction of all 30

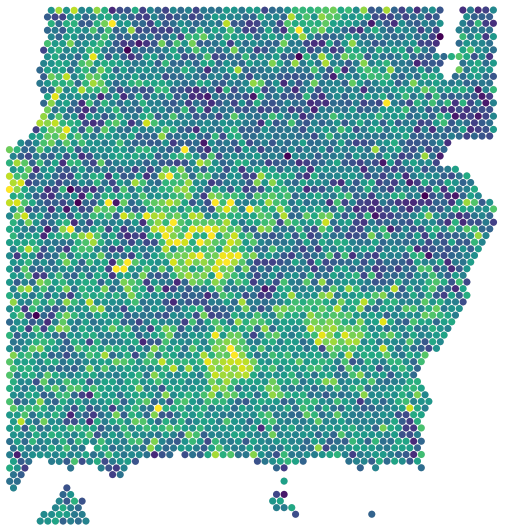
Treg estimated abundance



0 2.85

Posterior mean estimate

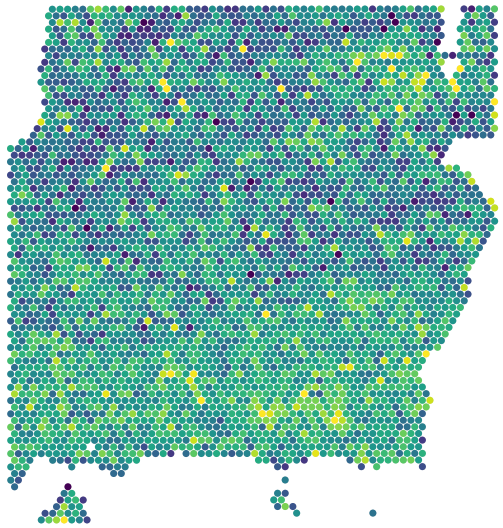
Treg Resting Vs Activated Dn



0 0.0778

UCell score

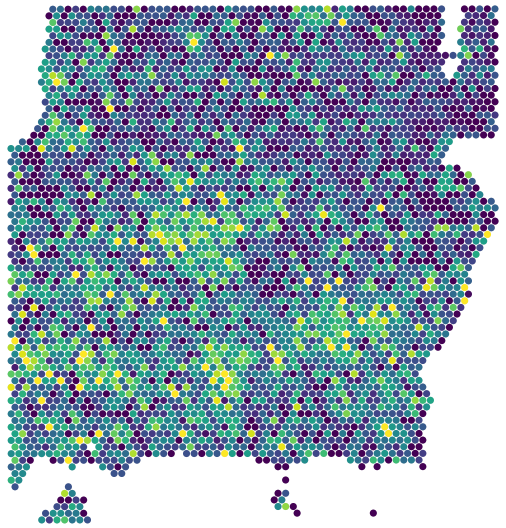
Treg Resting Vs Activated Up



0 0.0609

UCell score

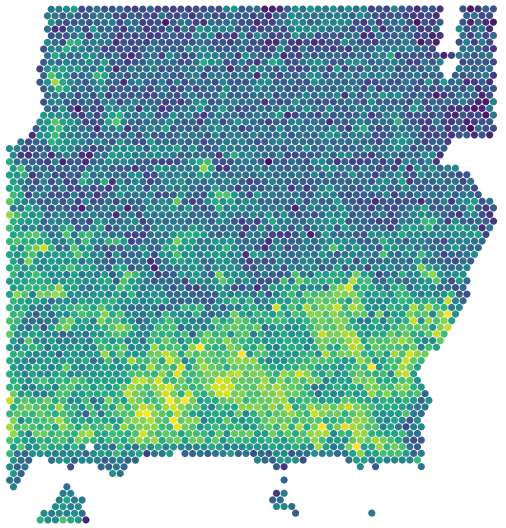
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

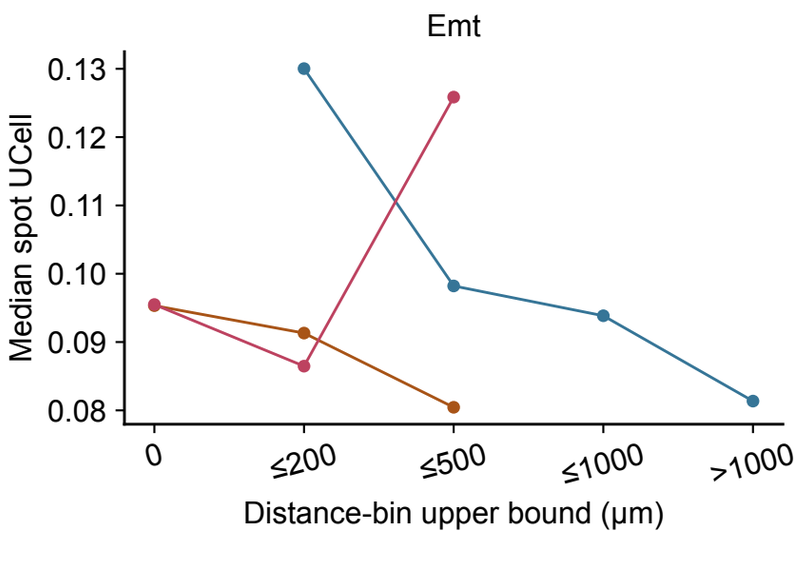


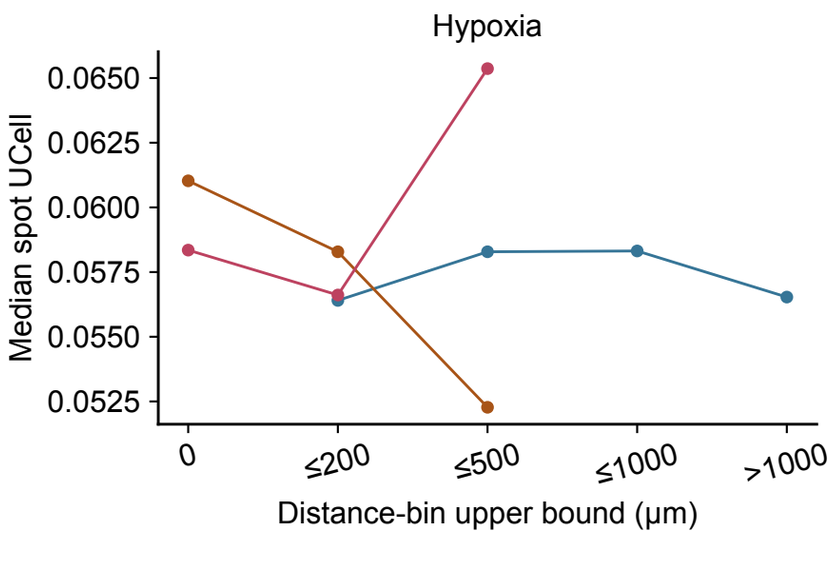
0 0.144

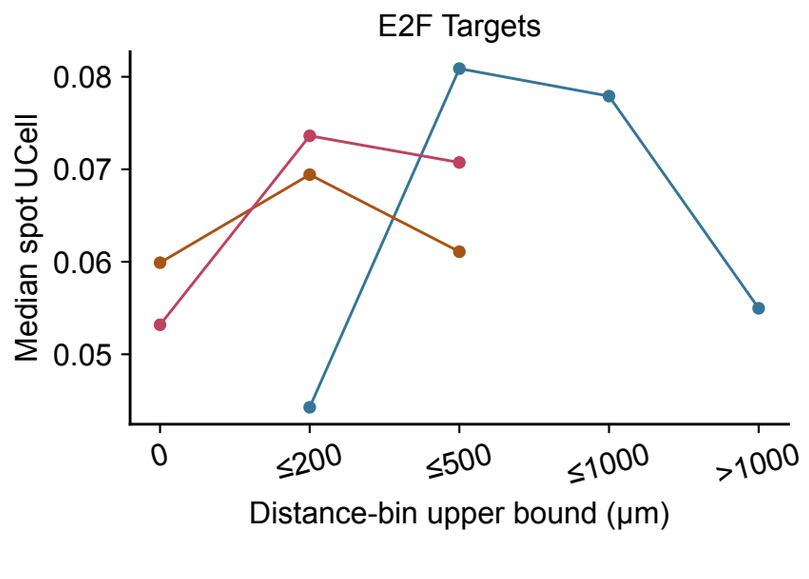
UCell score

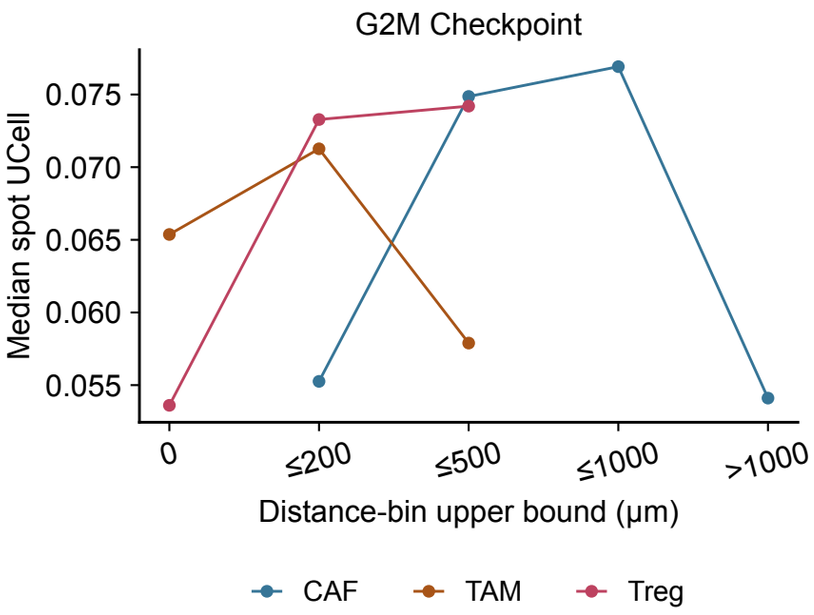
Inhousedata_T4465935 | Programs by target distance

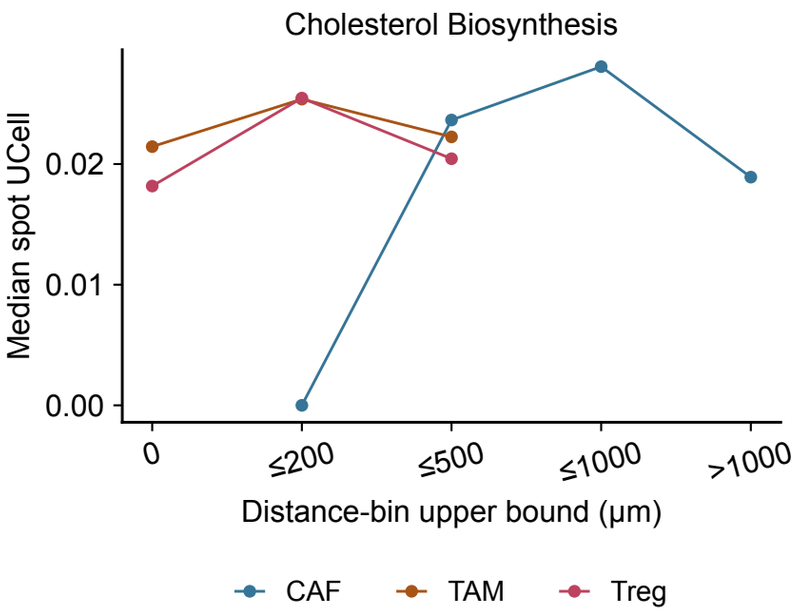
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

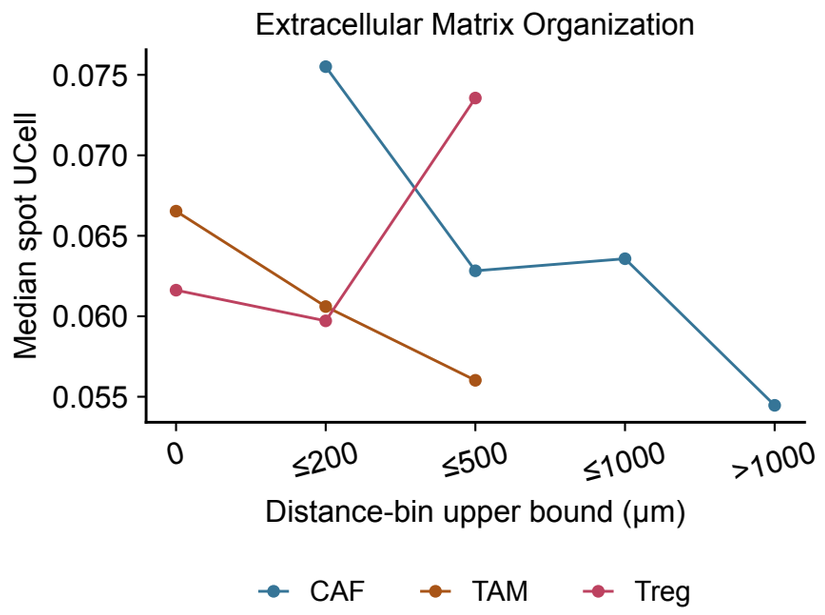






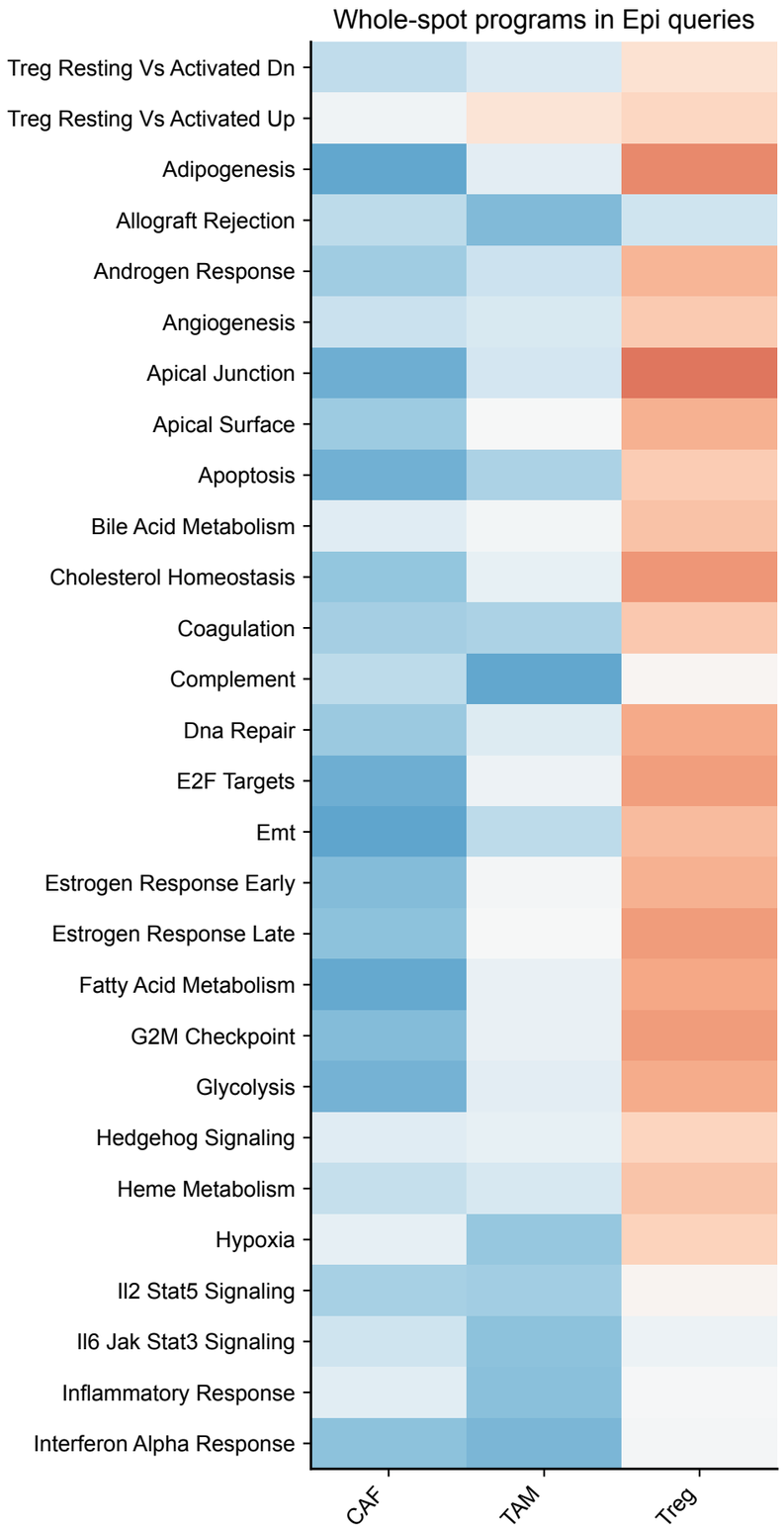


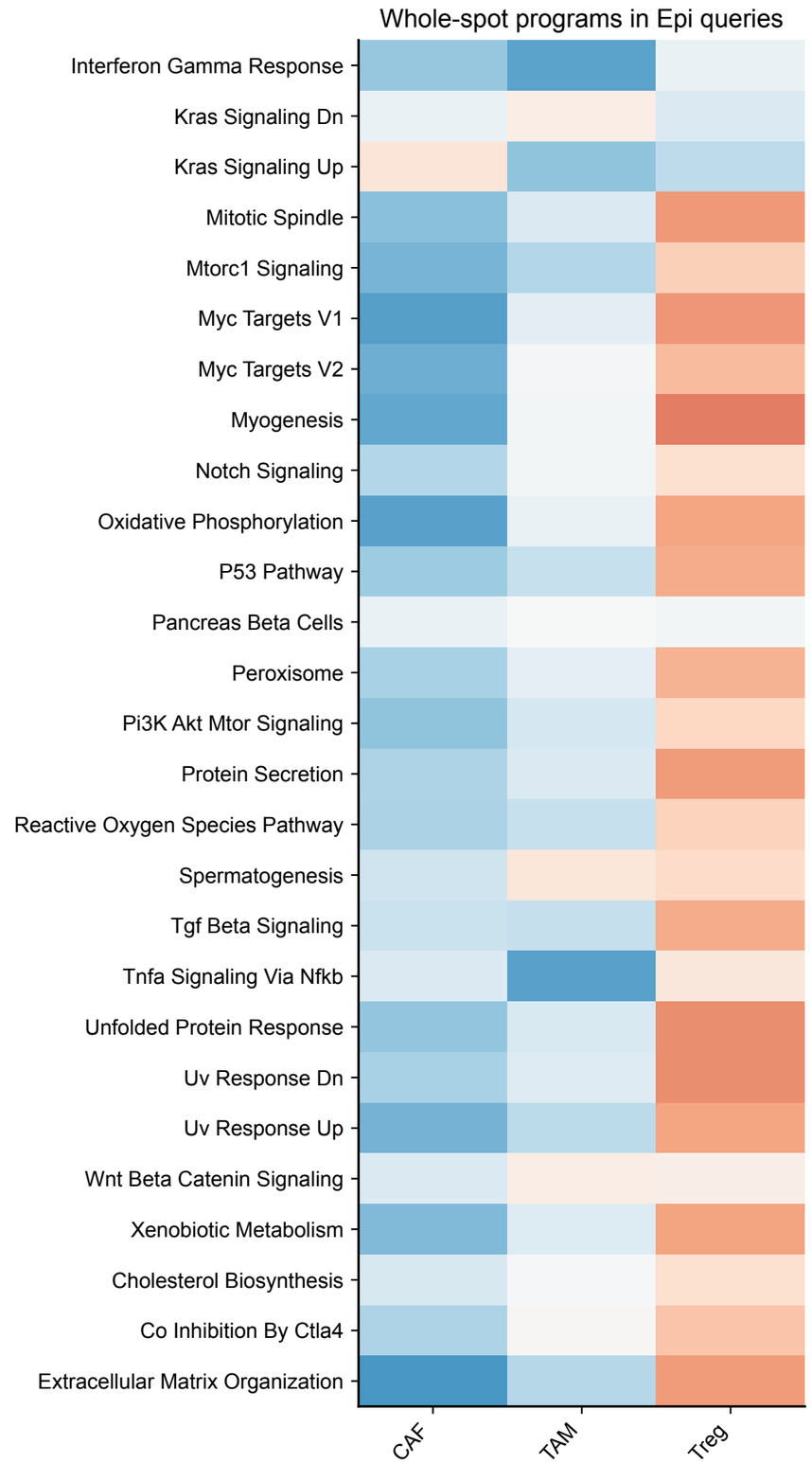




Inhousedata_T4465935 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

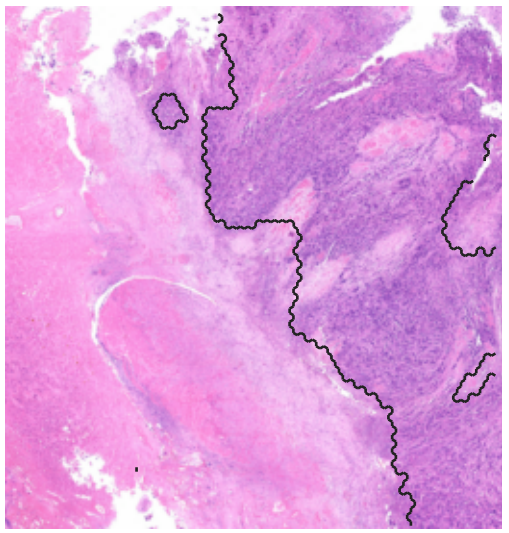




Inhousedata_T4469479 | Tissue context

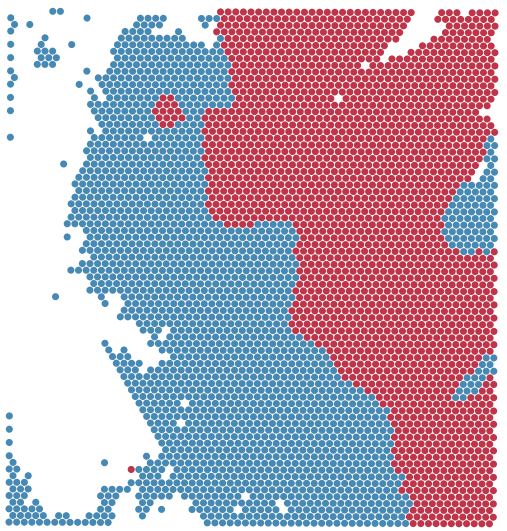
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



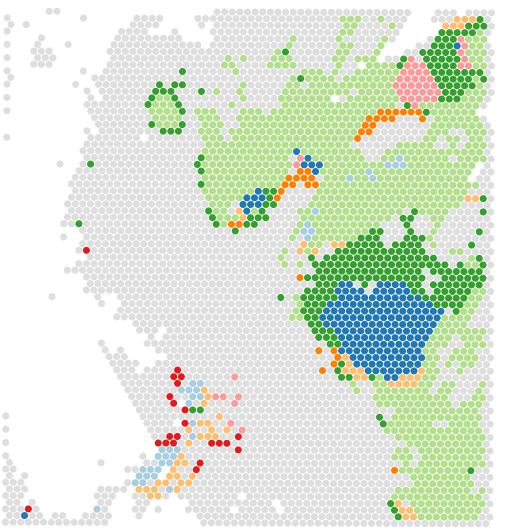
Black line: observed candidate outer edge

Final candidate location



inside outside

Accepted R12 (original 0–8)



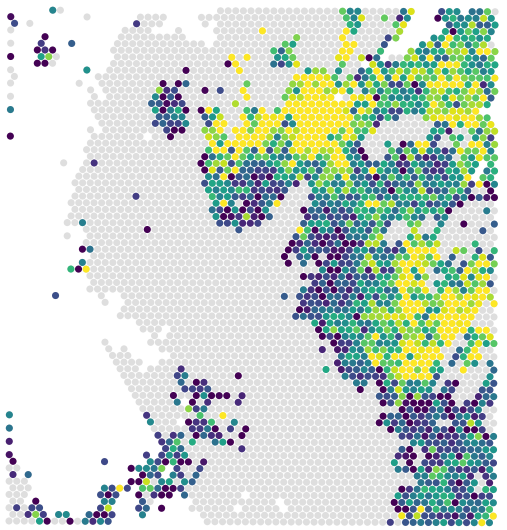
0 2 4 6 7
1 3 5

Epithelial support

0 1

Fraction of all 30

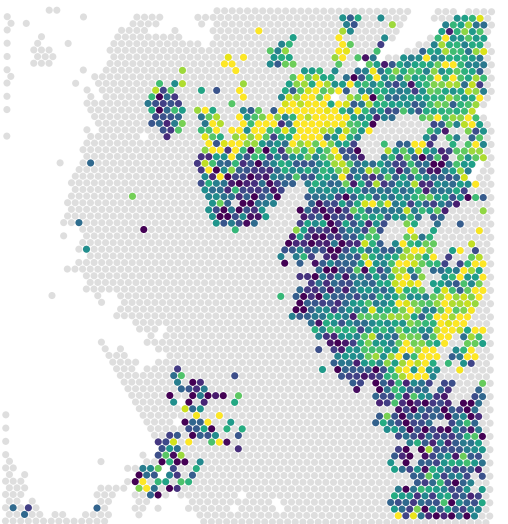
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



0 0.14

Matched-model AUCell

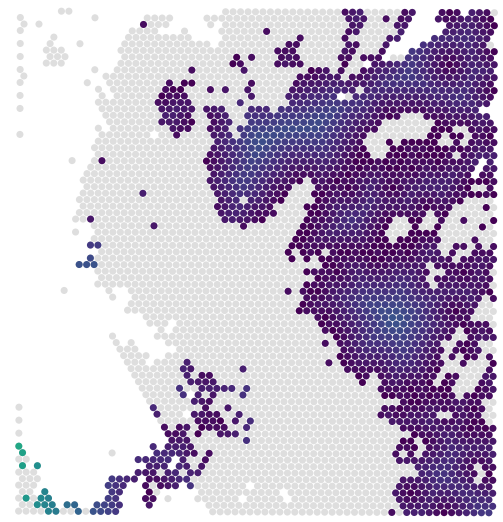
Inhousedata_T4469479 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition

CAF P80 targets

CAF distance

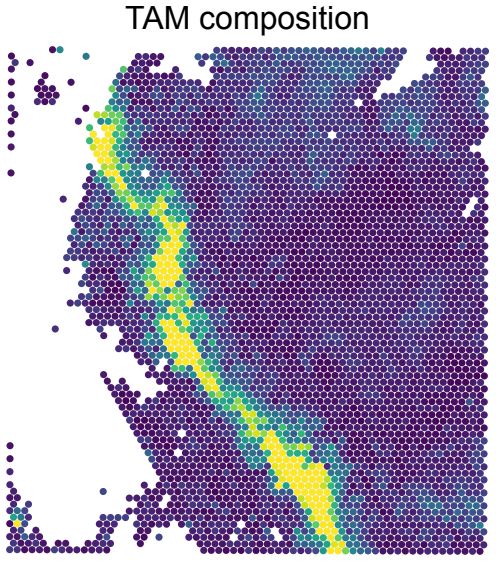


0 0.418
Fraction of all 30

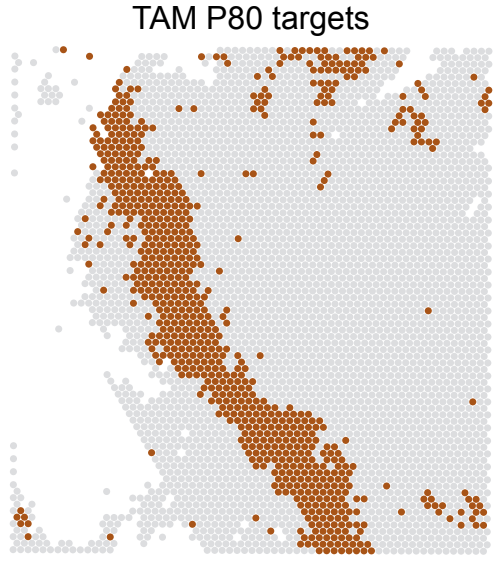
False True

0 3.1e+03
Nearest spot center (μm)

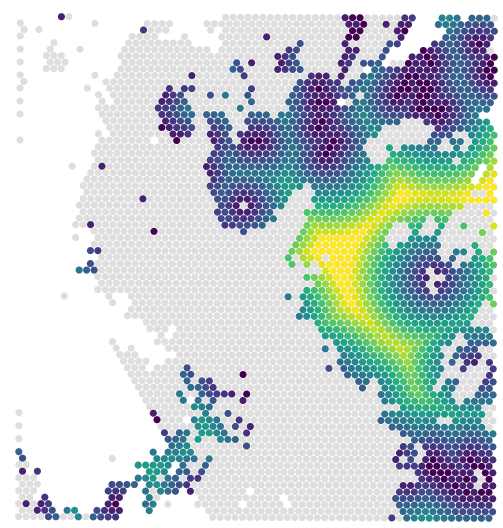
TAM composition



TAM P80 targets



TAM distance



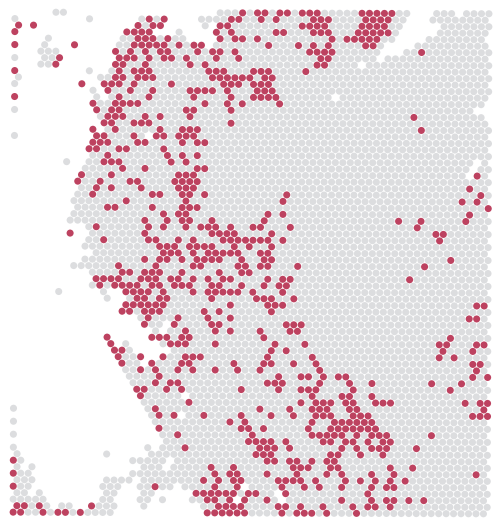
0 0.291
Fraction of all 30

False True

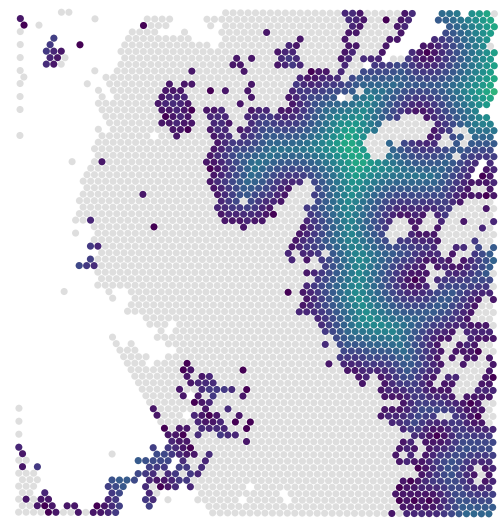
0 1.13e+03
Nearest spot center (μm)

Treg composition

Treg P80 targets



Treg distance



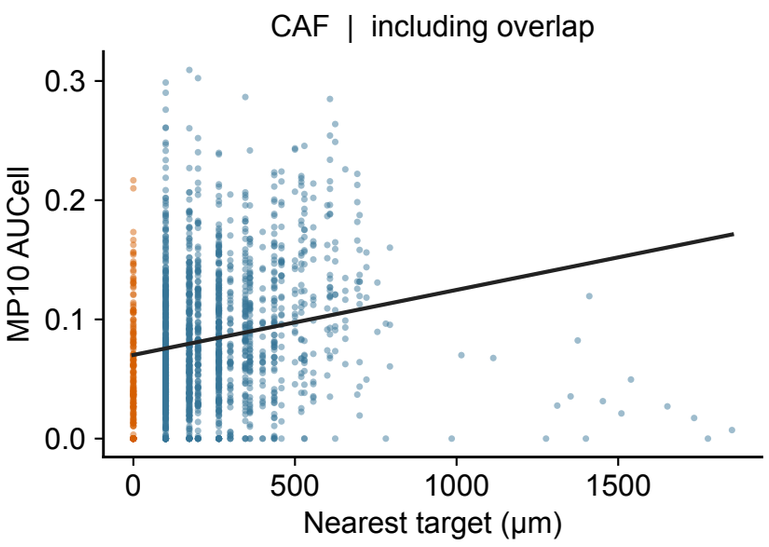
0 0.102
Fraction of all 30

False True

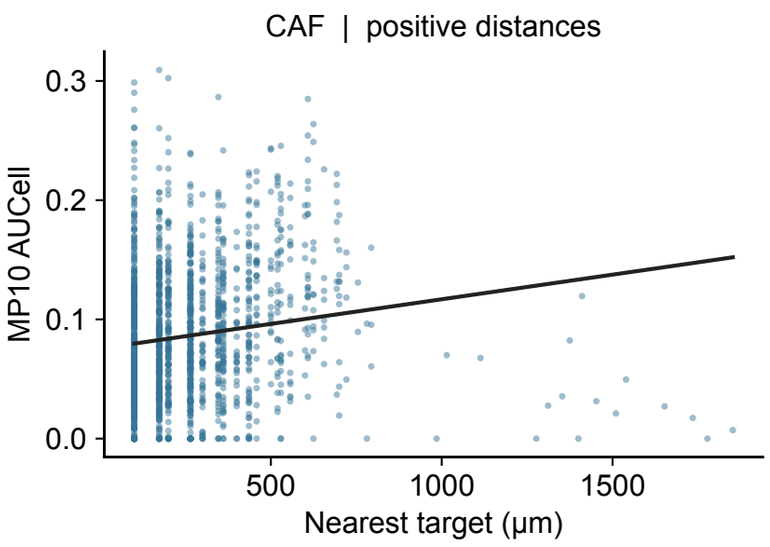
0 1.61e+03
Nearest spot center (μm)

Inhousedata_T4469479 | MP10 versus nearest target

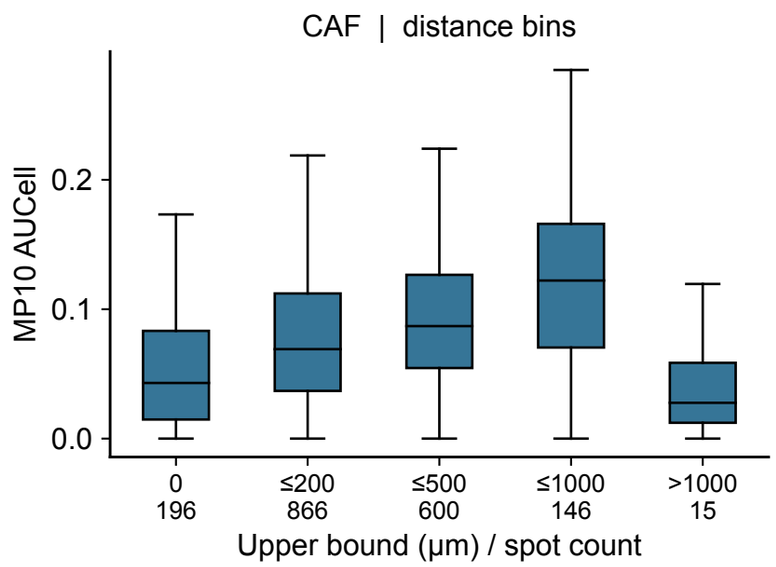
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



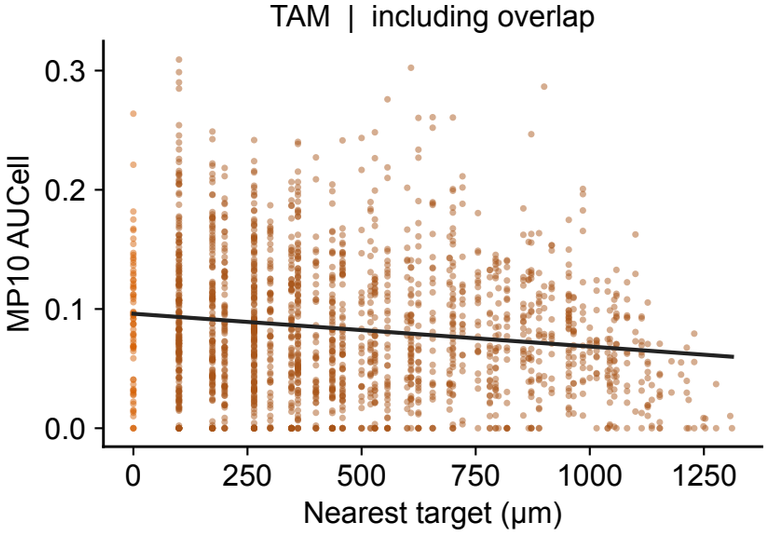
r +0.19 ρ +0.27
n 1823/1846 queries • 784 targets



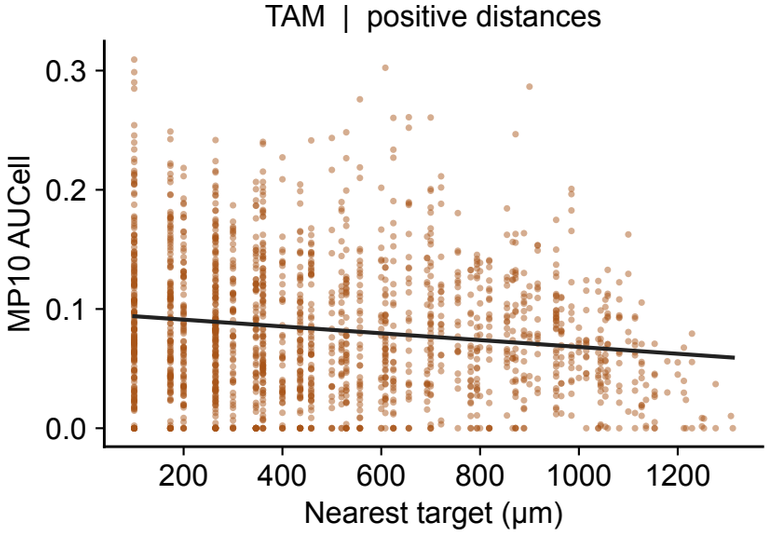
r +0.14 ρ +0.22
n 1627/1846 queries • 784 targets



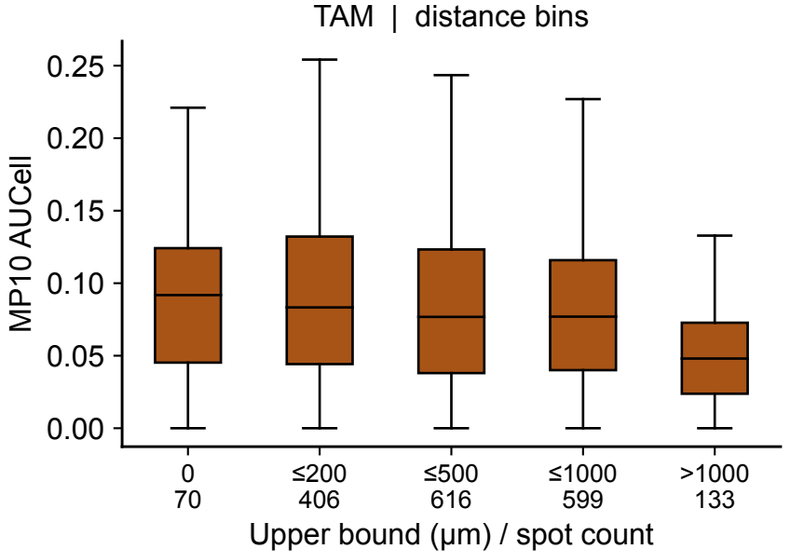
Orange = zero distance
Line = within-section linear fit



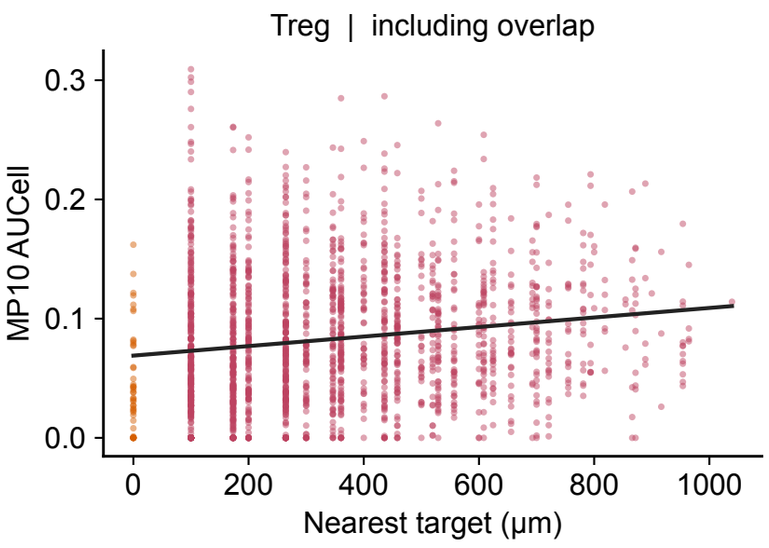
r -0.15 ρ -0.13
n 1824/1846 queries • 784 targets



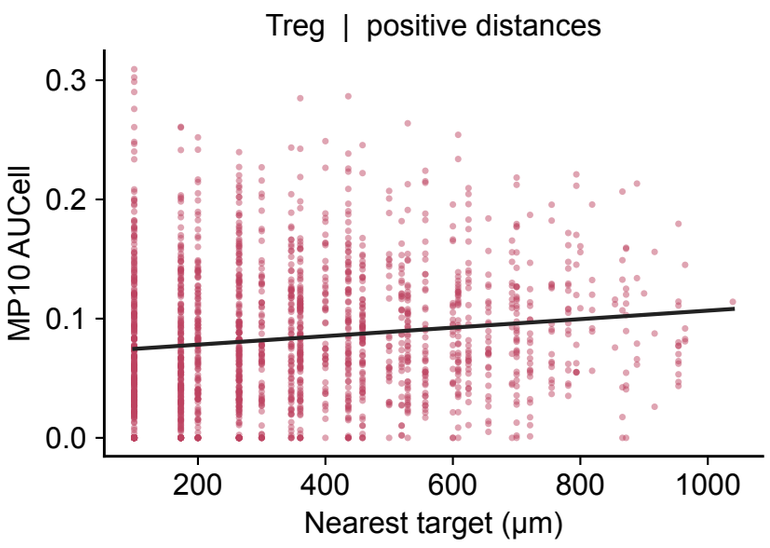
r -0.15 ρ -0.13
n 1754/1846 queries • 784 targets



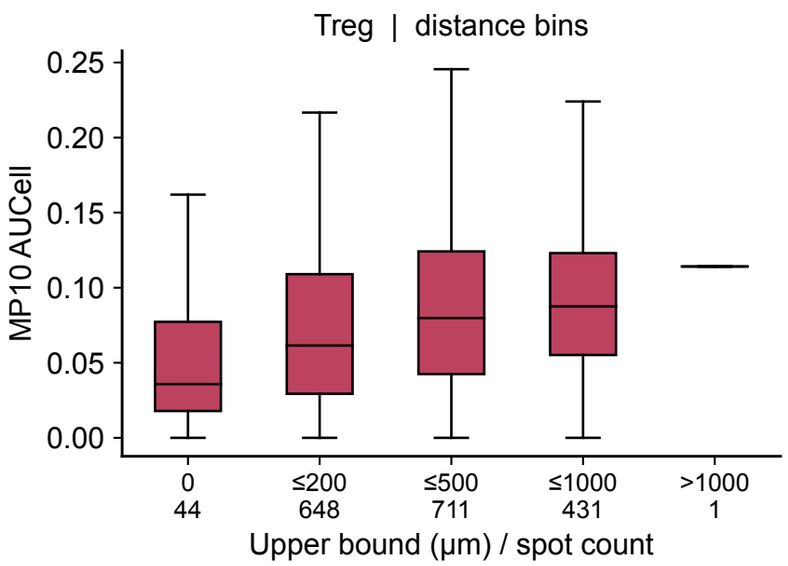
Orange = zero distance
Line = within-section linear fit



r +0.15 ρ +0.20
n 1835/1846 queries • 784 targets



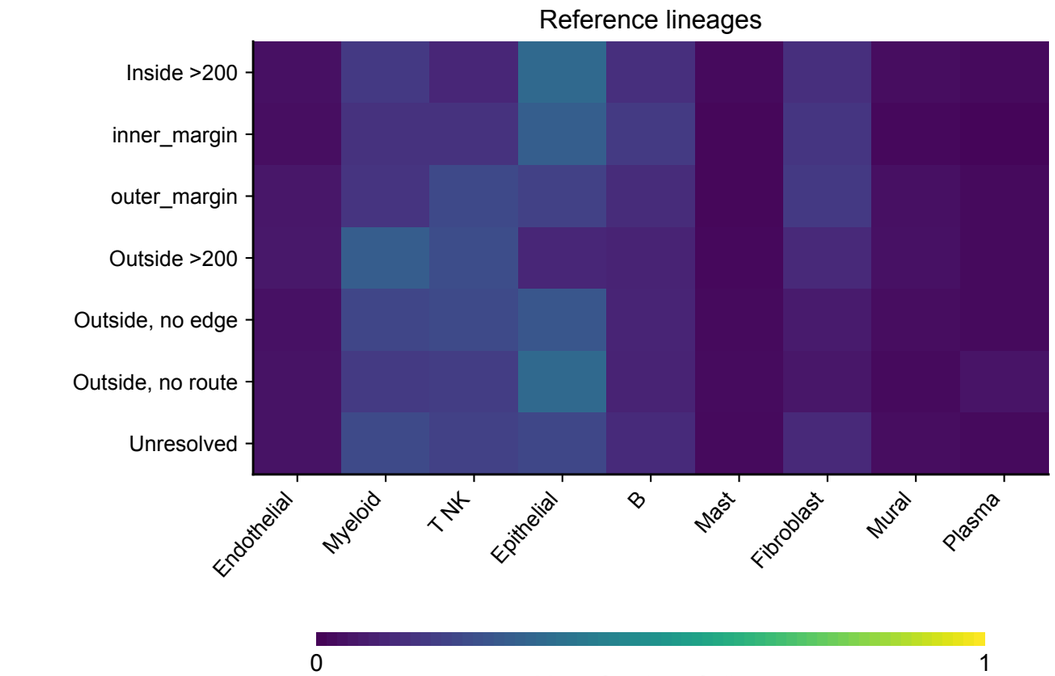
r +0.13 ρ +0.18
n 1791/1846 queries • 784 targets

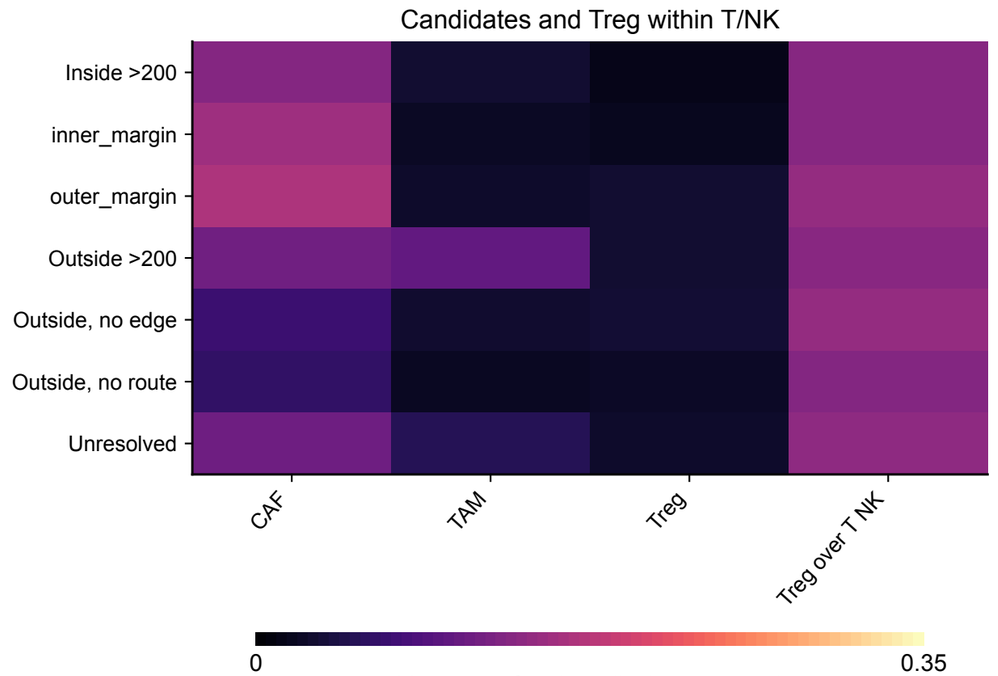


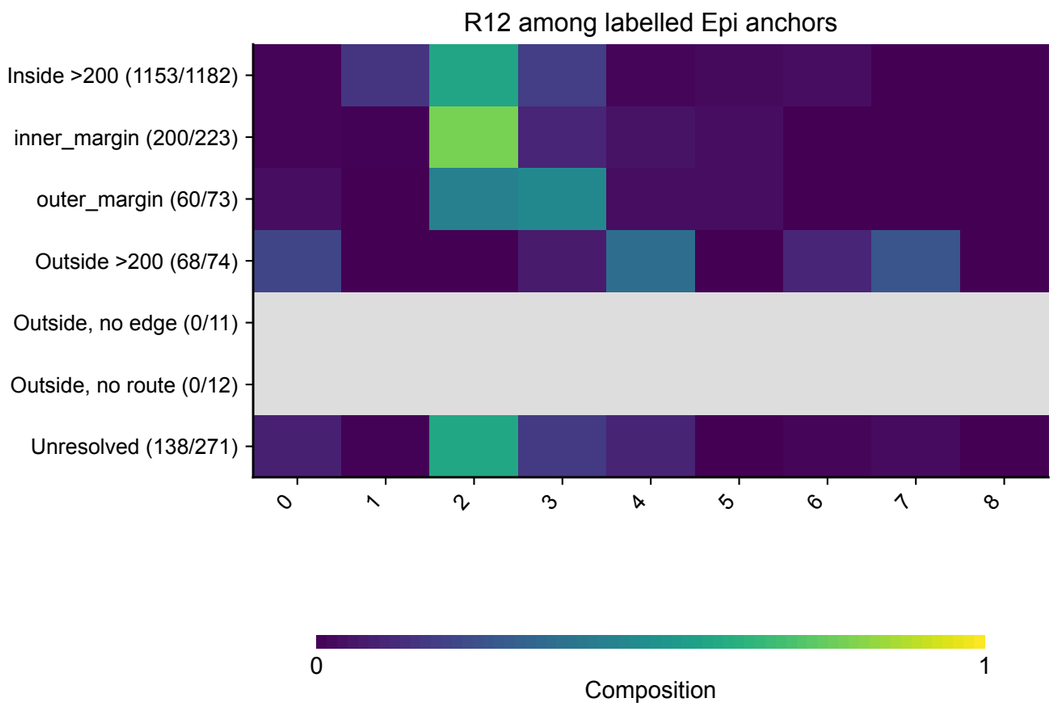
Orange = zero distance
Line = within-section linear fit

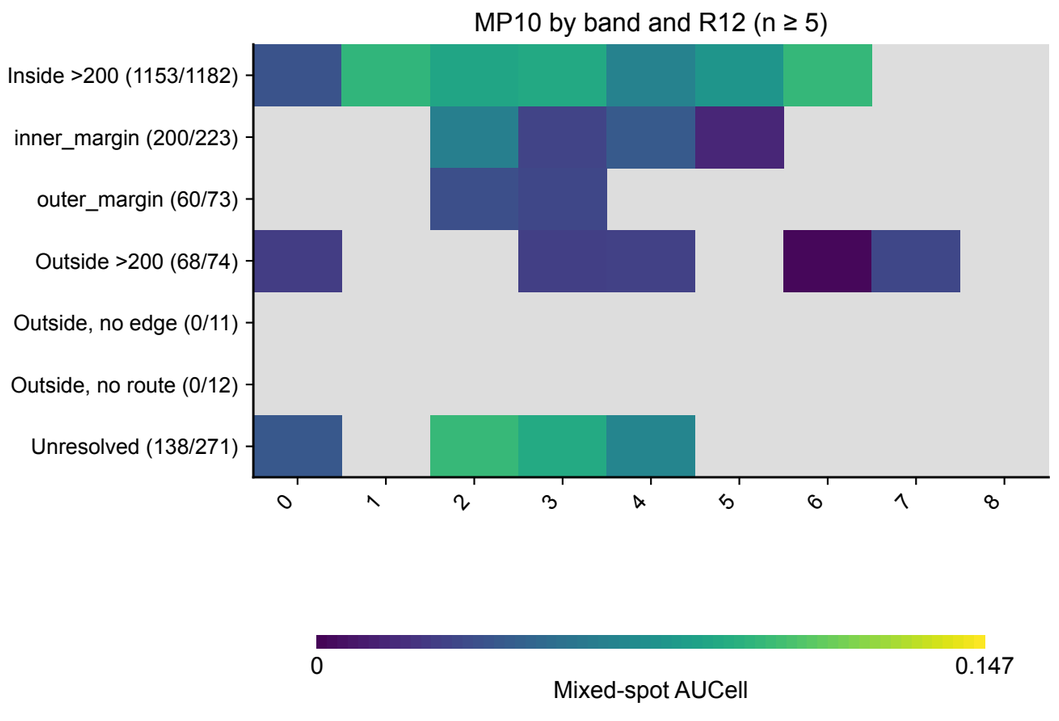
Inhousedata_T4469479 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable









Inhousedata_T4469479 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

E2F Targets

0 0.152

UCell score

G2M Checkpoint

0 0.134

UCell score

Hypoxia

0 0.211

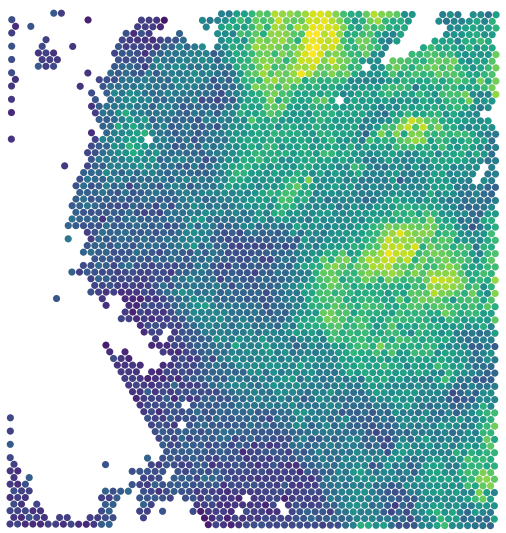
UCell score

Tnfa Signaling Via Nfkb

0 0.275

UCell score

Emt



0 0.386

UCell score

Interferon Gamma Response

0 0.275

UCell score

Inhousedata_T4469479 | Immune, lipid and matrix programs

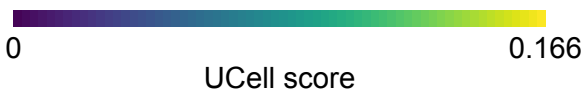
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling

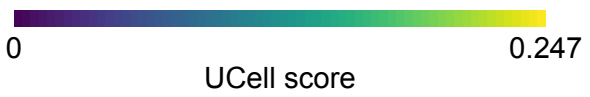
Tgf Beta Signaling

Cholesterol Homeostasis

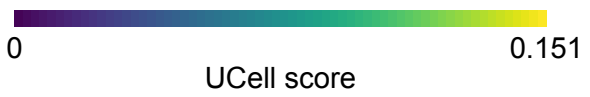
Cholesterol Biosynthesis



Extracellular Matrix Organization



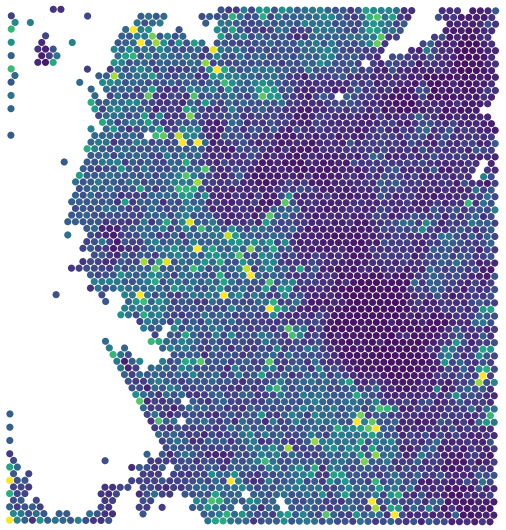
Co Inhibition By Ctla4



Inhousedata_T4469479 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition



0 0.102

Fraction of all 30

Treg estimated abundance

0 2.85

Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

UCell score

Treg Resting Vs Activated Up

0 0.0609

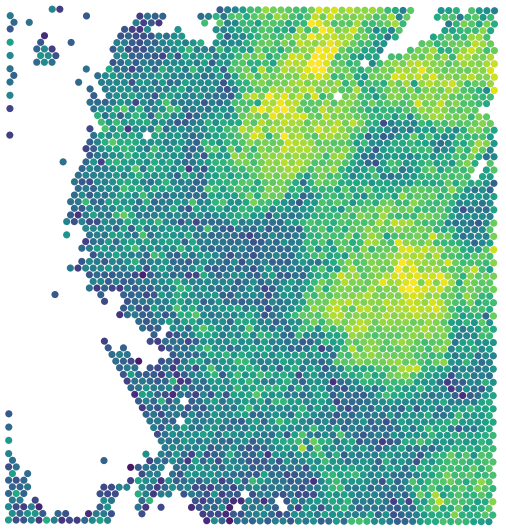
UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

Il2 Stat5 Signaling

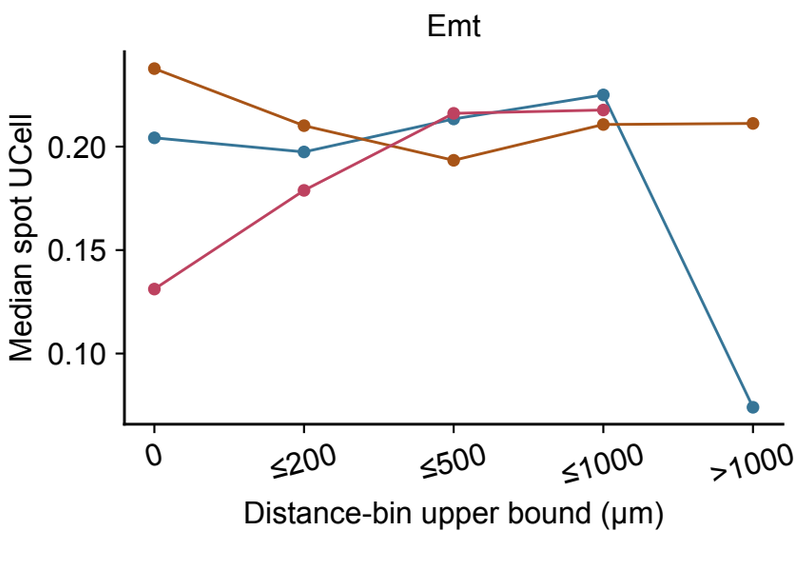


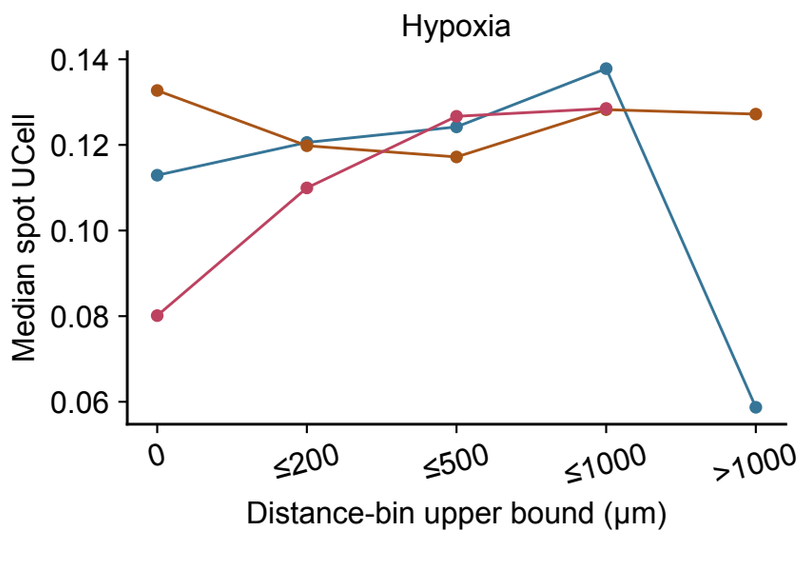
0 0.144

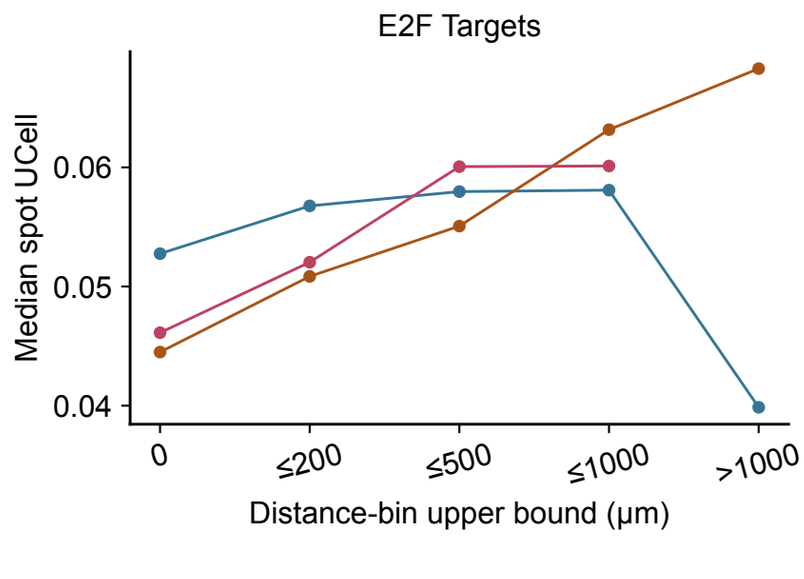
UCell score

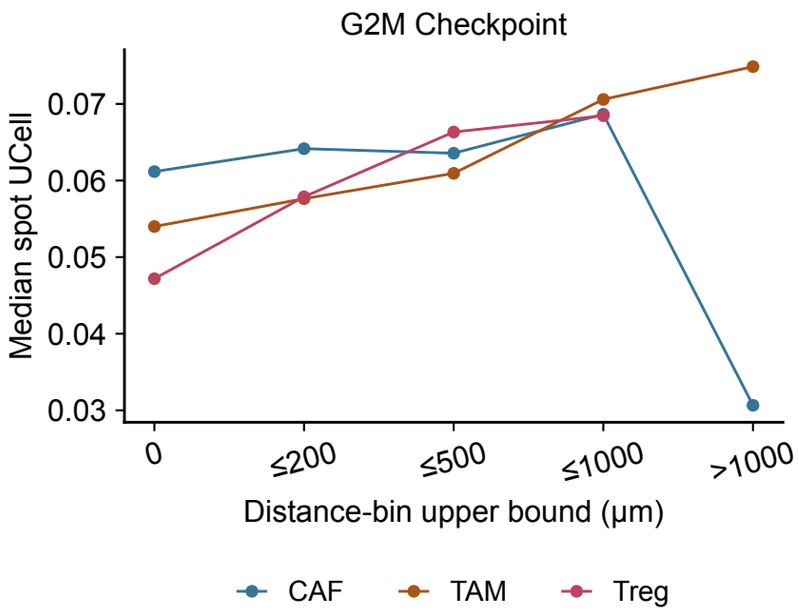
Inhousedata_T4469479 | Programs by target distance

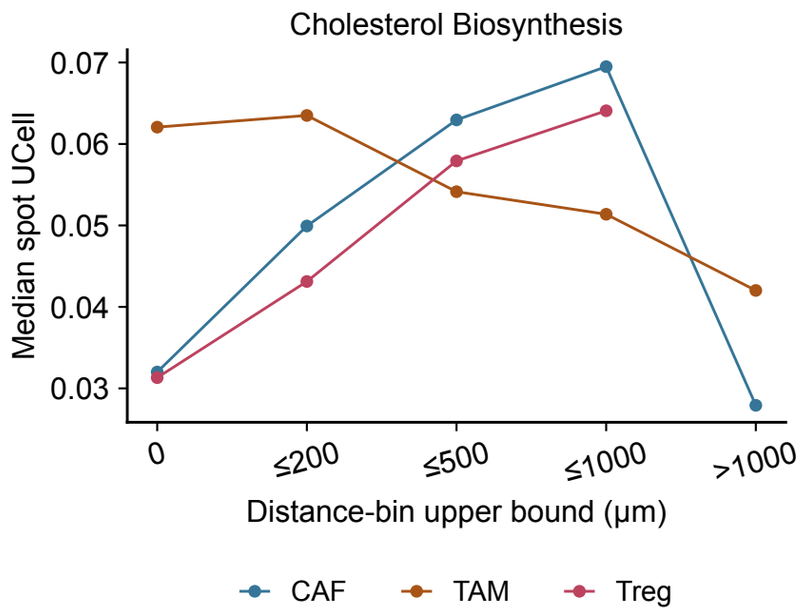
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

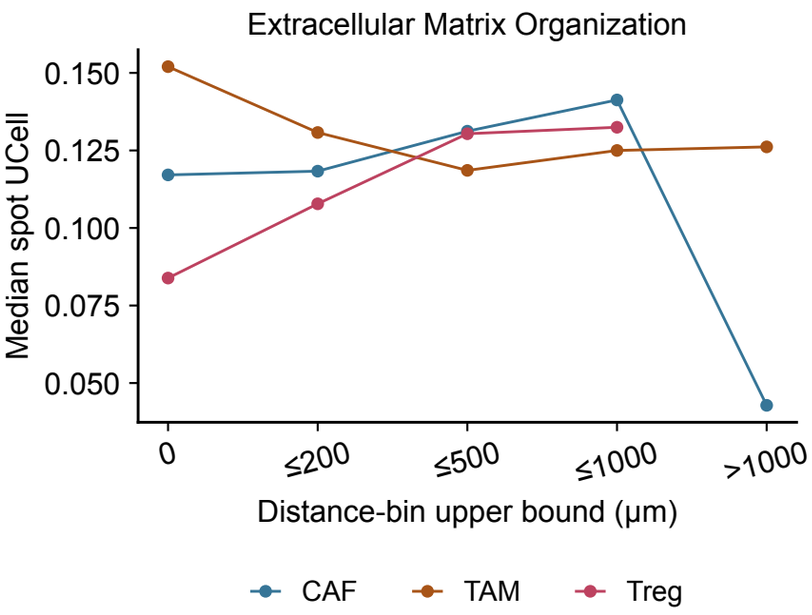






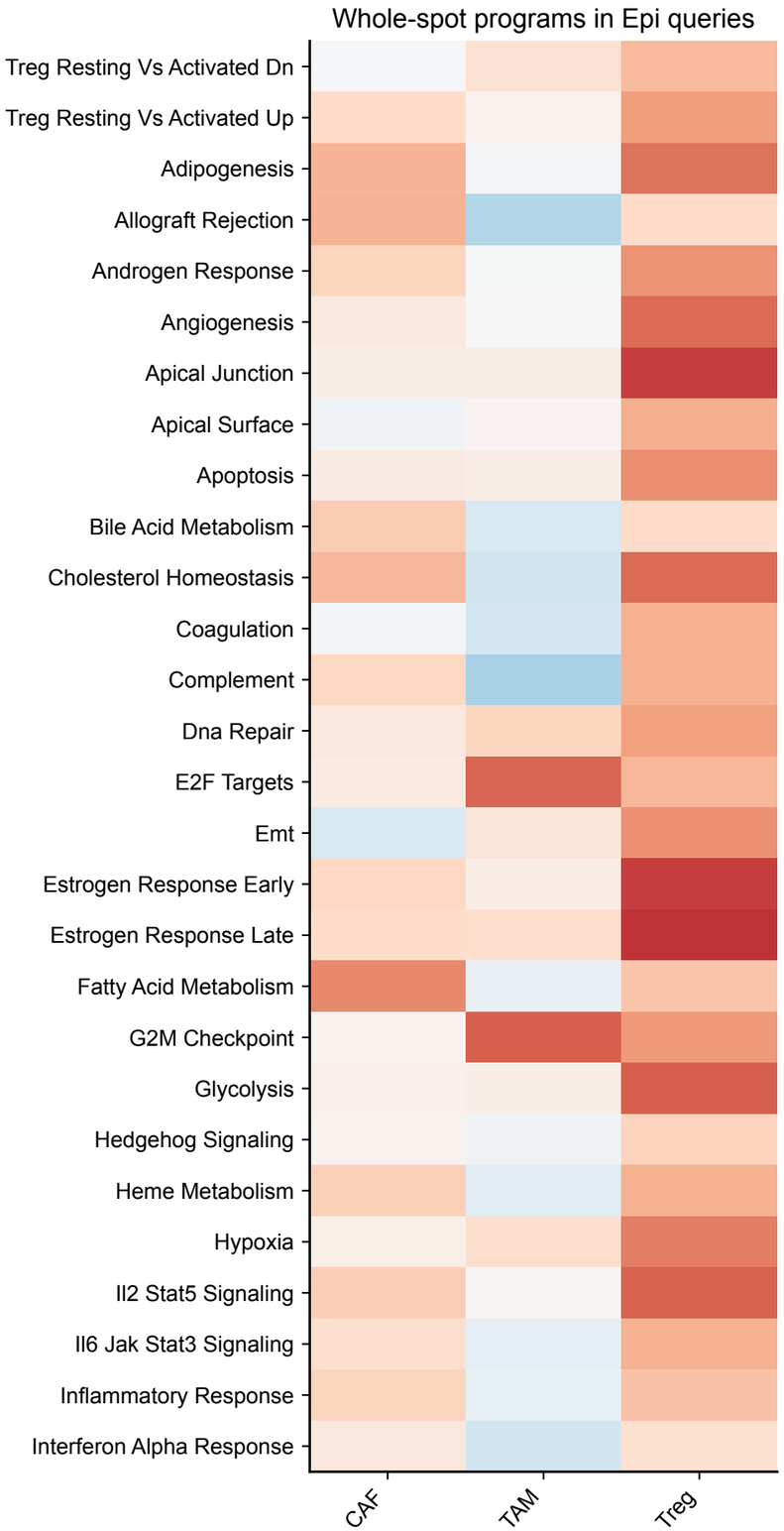


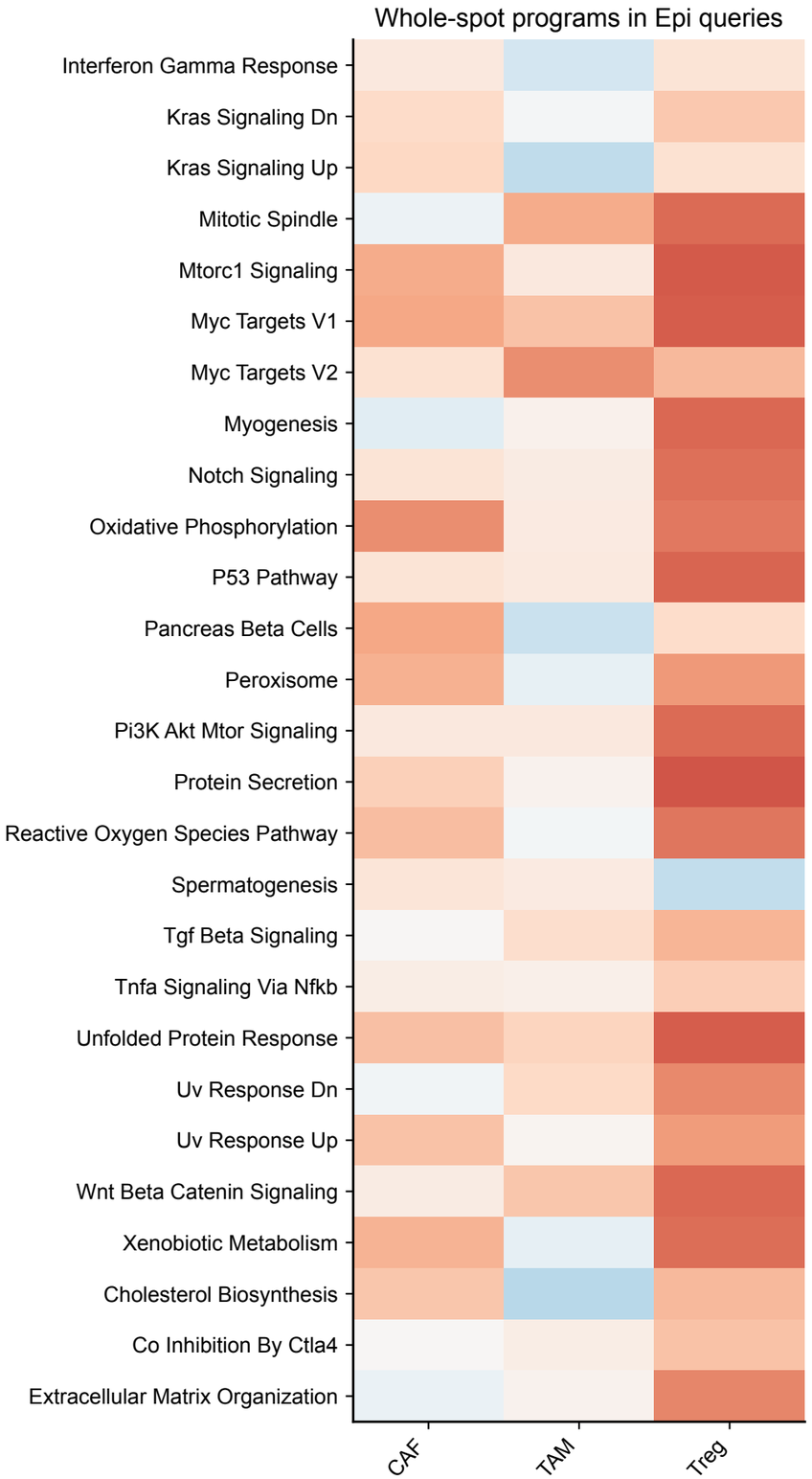




Inhousedata_T4469479 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

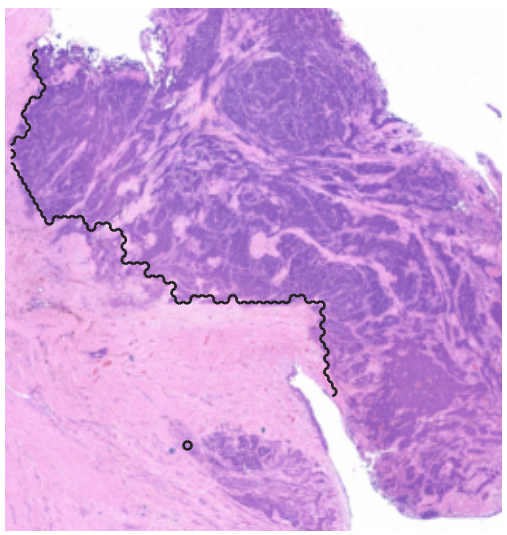




Inhousedata_T4471438 | Tissue context

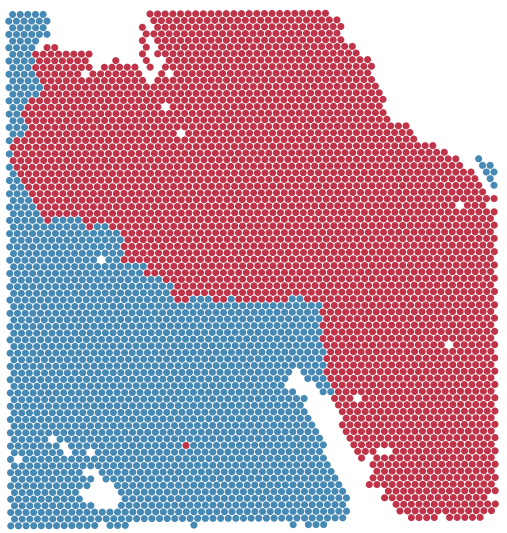
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



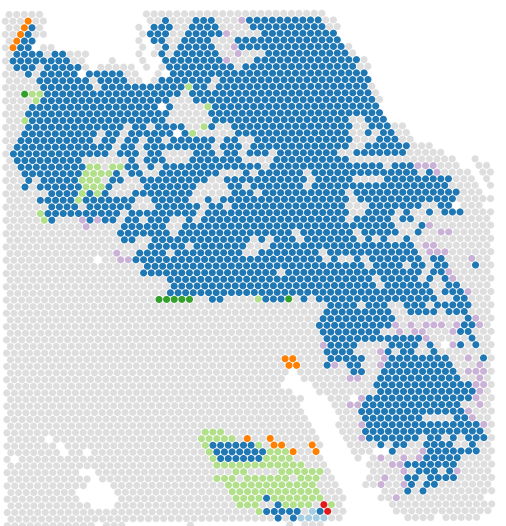
Black line: observed candidate outer edge

Final candidate location



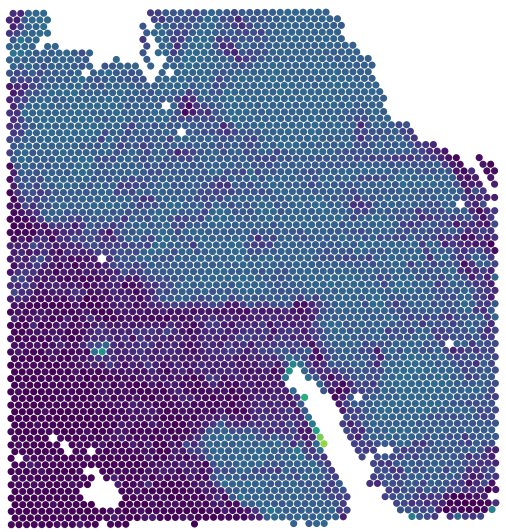
inside outside

Accepted R12 (original 0–8)



0 2 5 7 8
1 3

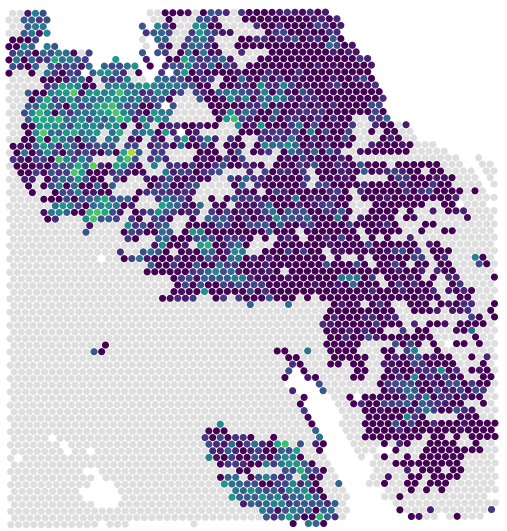
Epithelial support



0 1

Fraction of all 30

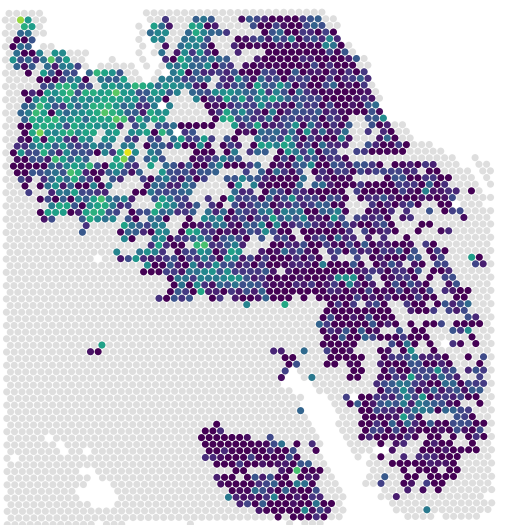
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



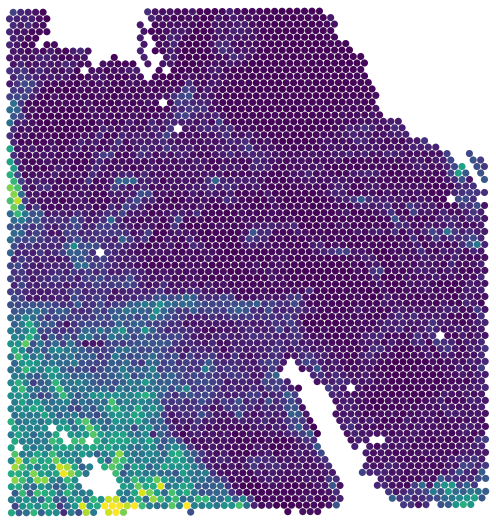
0 0.14

Matched-model AUCell

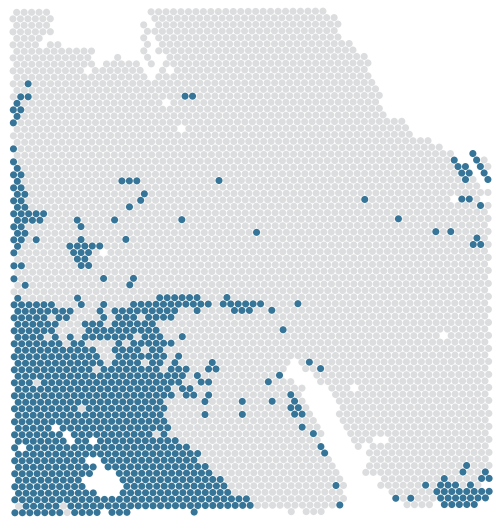
Inhousedata_T4471438 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

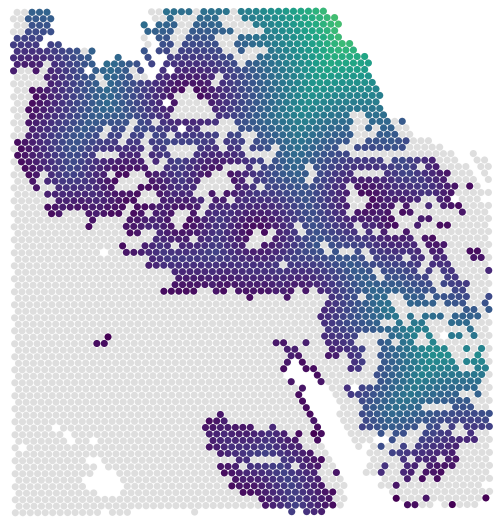
CAF composition

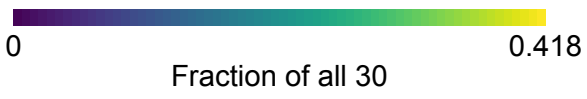


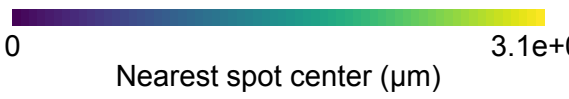
CAF P80 targets



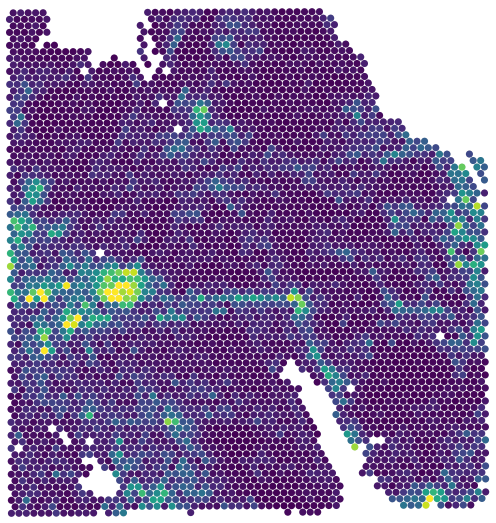
CAF distance



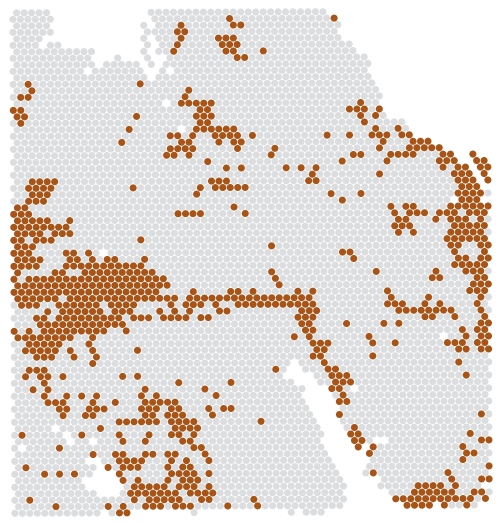




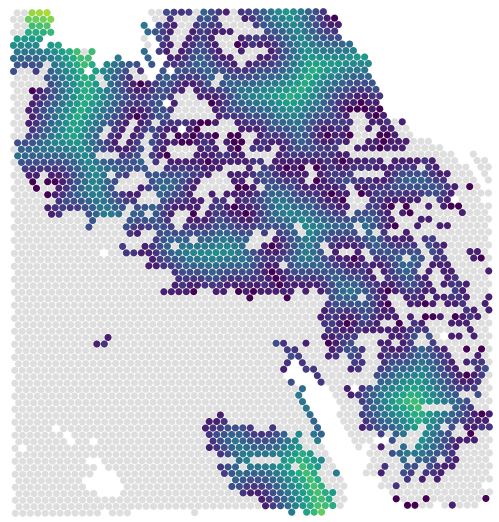
TAM composition

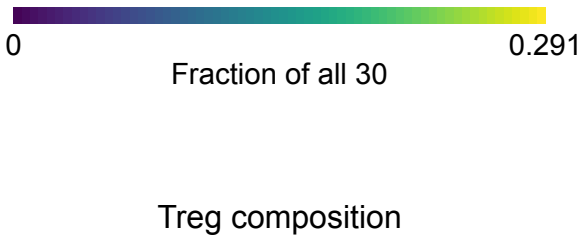


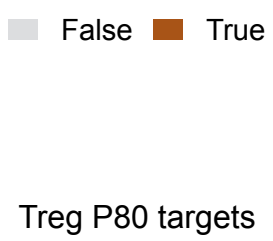
TAM P80 targets

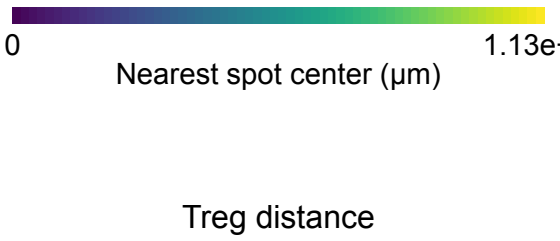


TAM distance

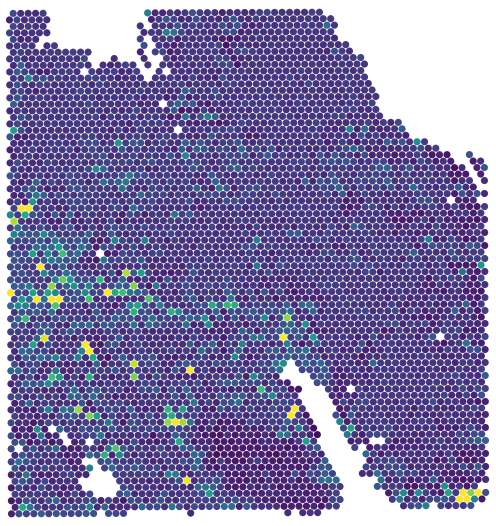




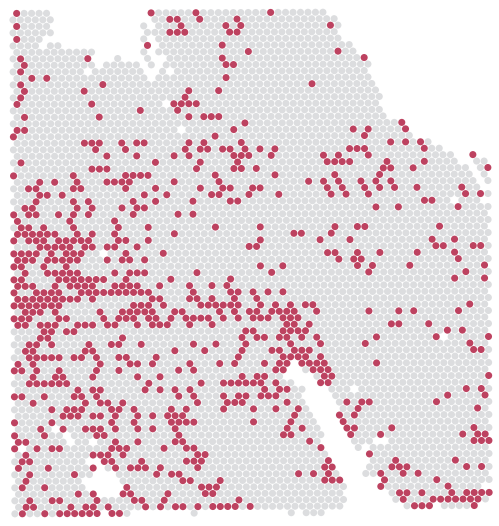




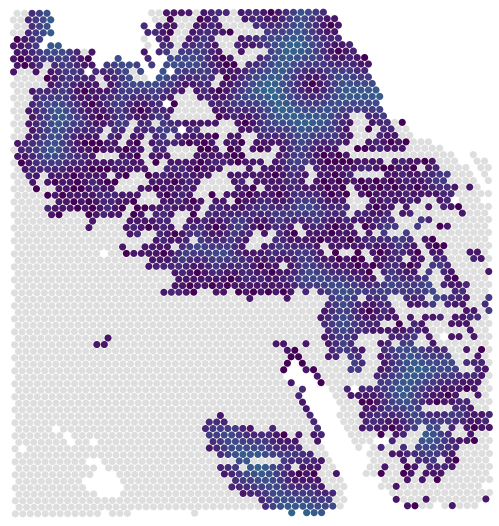
Treg composition

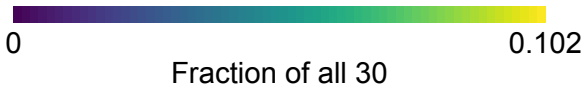


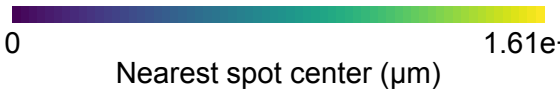
Treg P80 targets



Treg distance

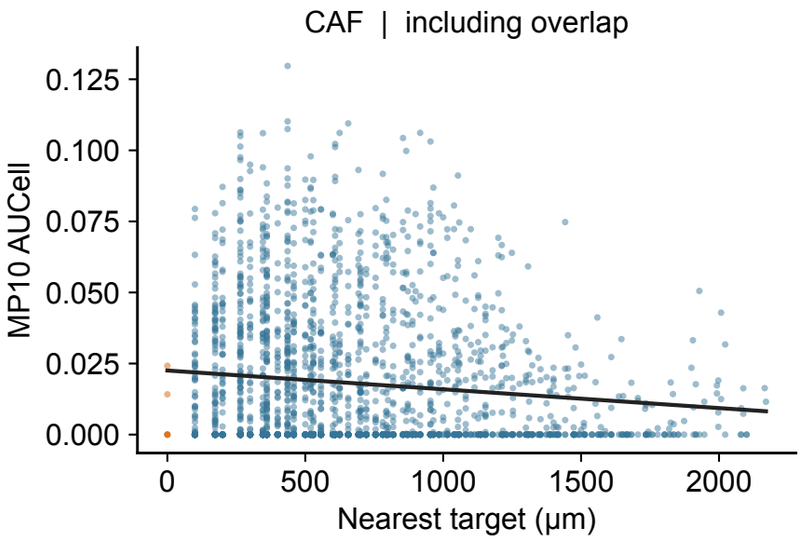




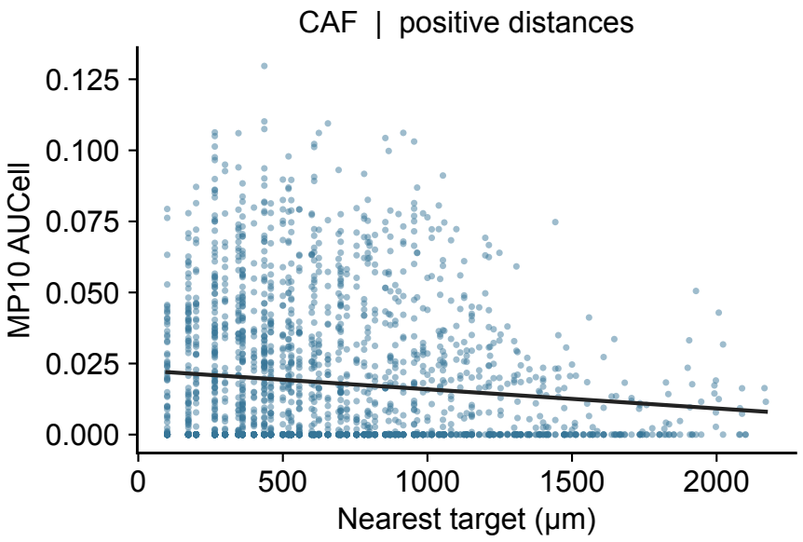


Inhousedata_T4471438 | MP10 versus nearest target

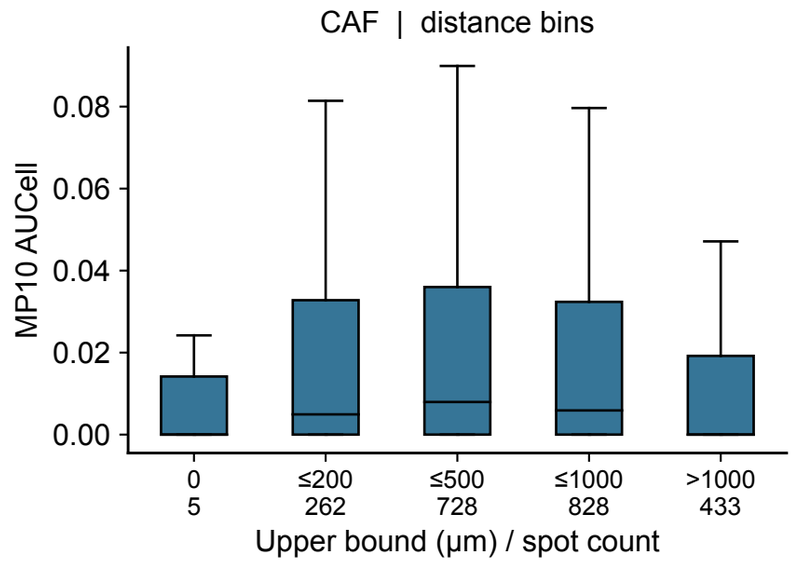
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



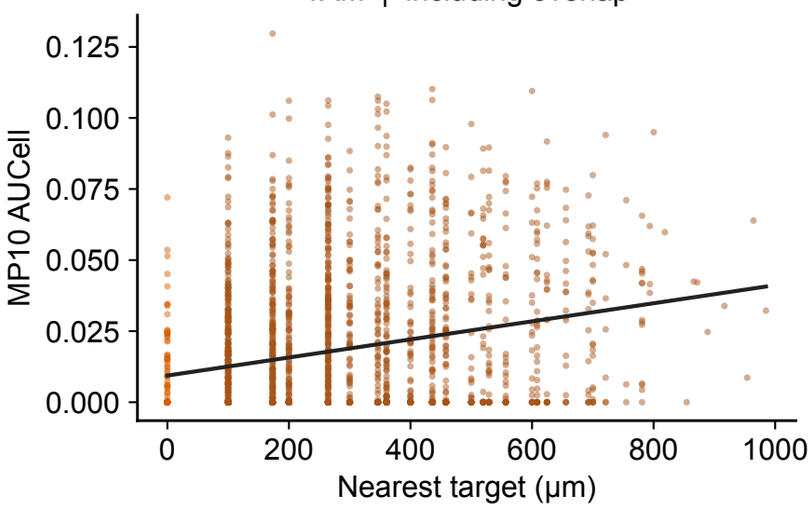
r -0.11 ρ -0.07
n 2256/2256 queries • 875 targets



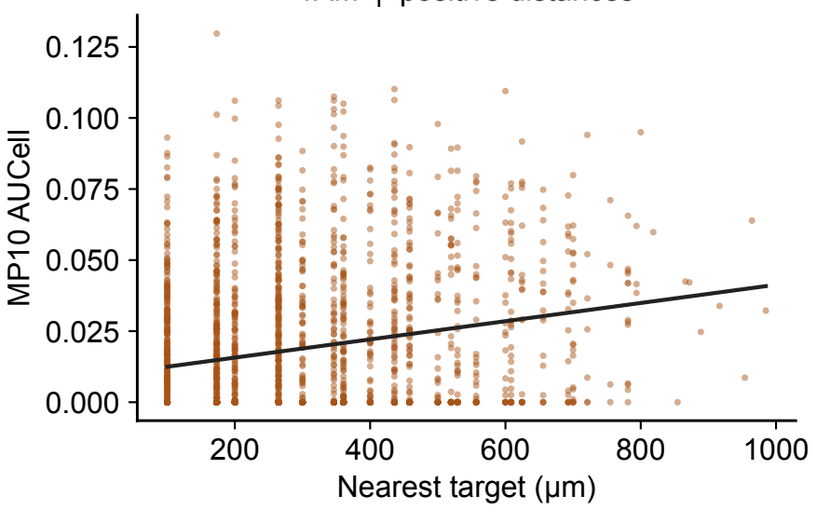
r -0.12 ρ -0.07
n 2251/2256 queries • 875 targets



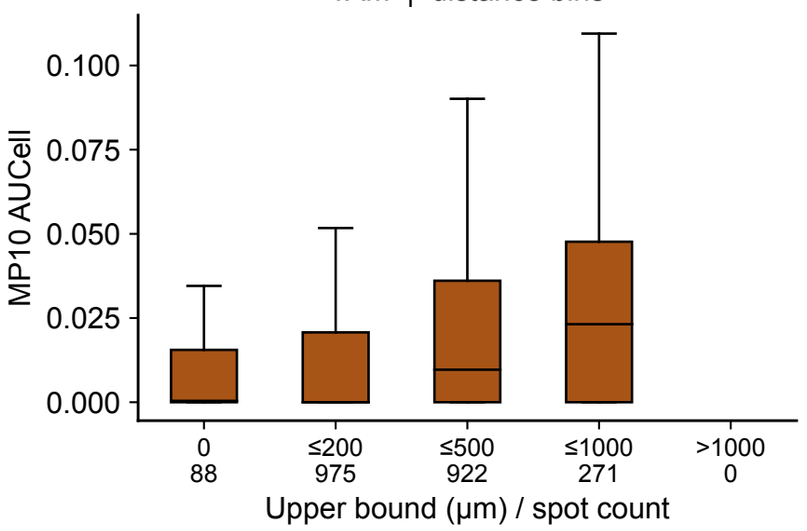
Orange = zero distance
Line = within-section linear fit



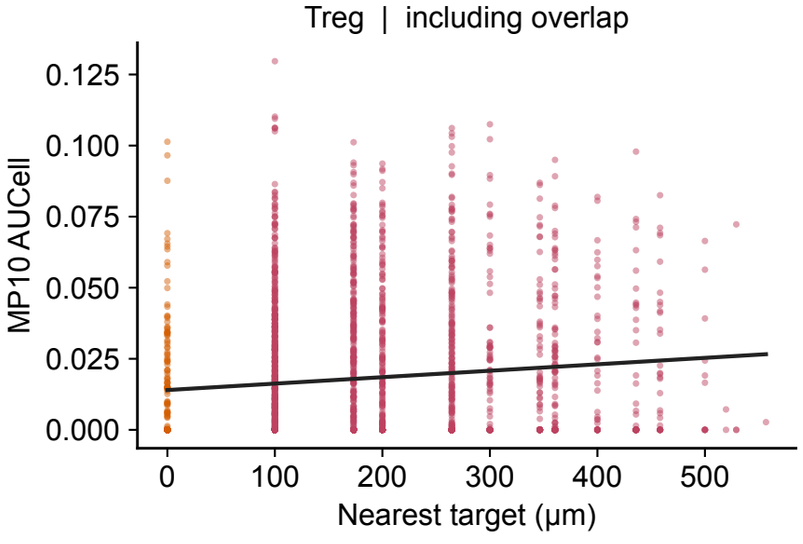
r +0.23 ρ +0.22
n 2256/2256 queries • 875 targets



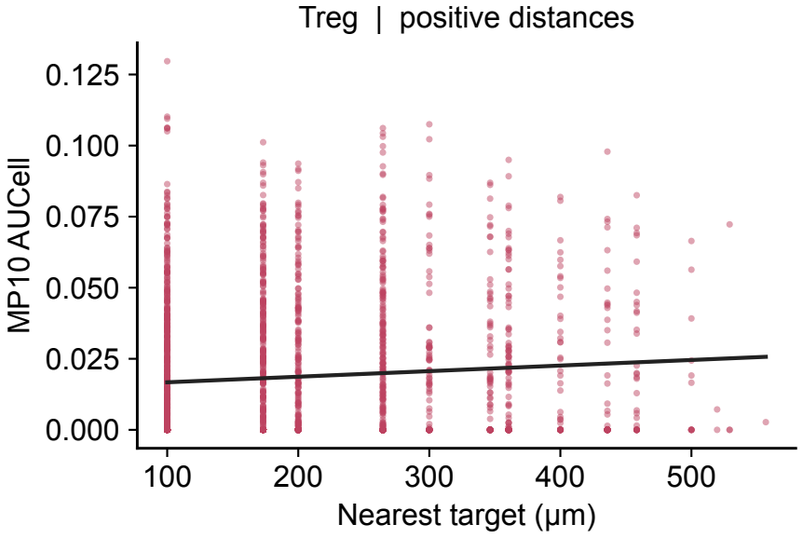
r +0.22 ρ +0.22
n 2168/2256 queries • 875 targets



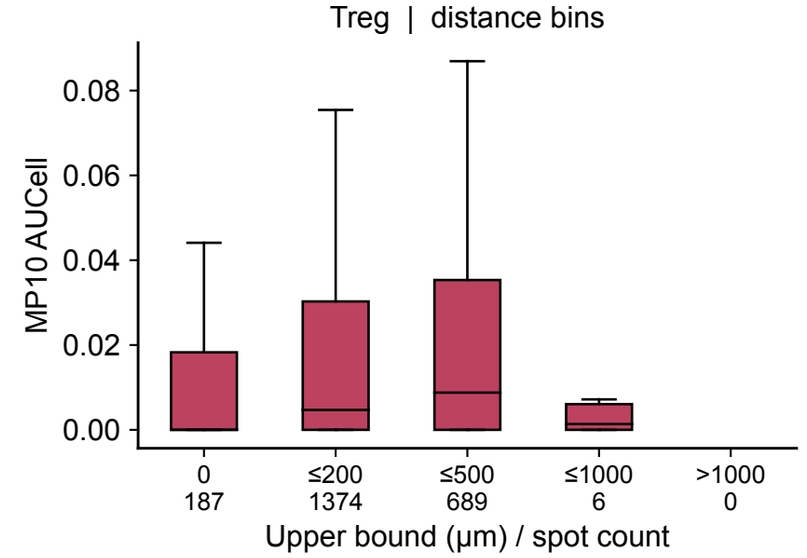
Orange = zero distance
Line = within-section linear fit



r +0.11 ρ +0.11
n 2256/2256 queries • 875 targets



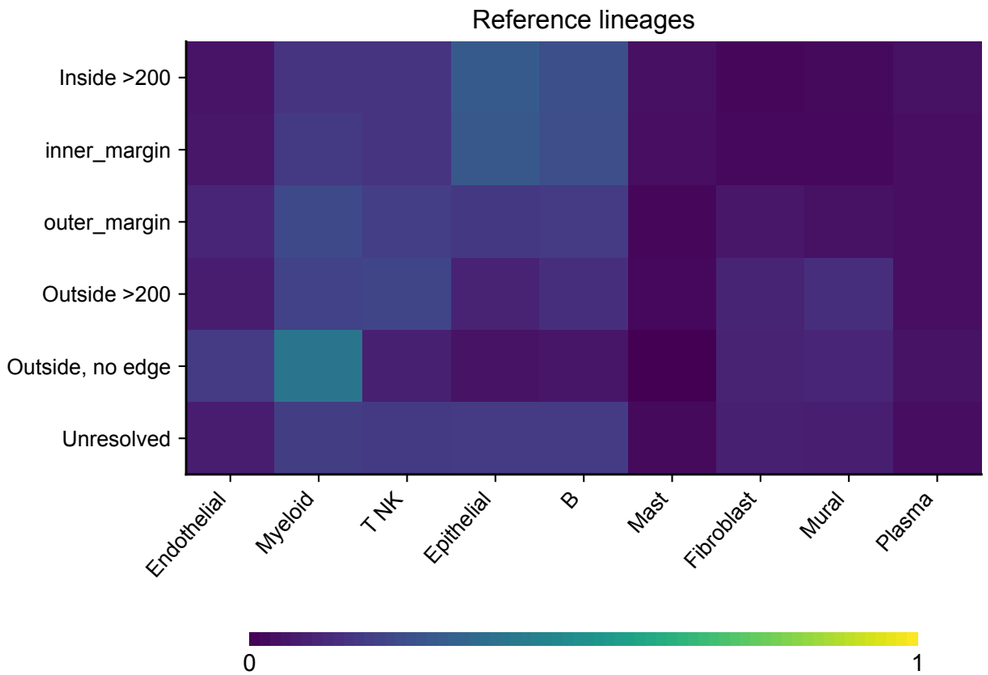
r +0.08 ρ +0.09
n 2069/2256 queries • 875 targets

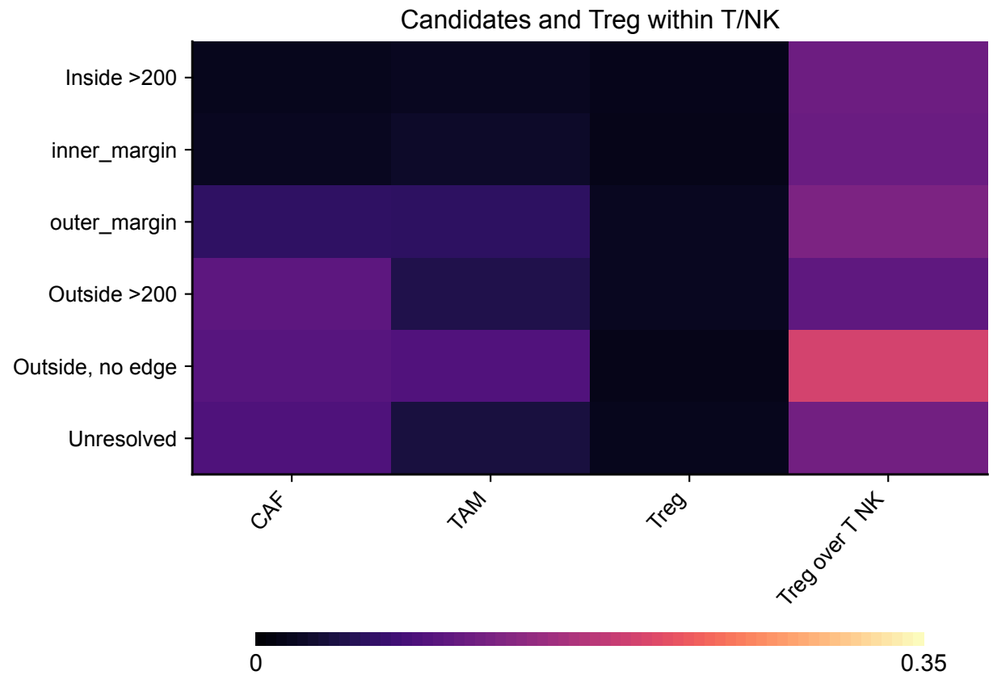


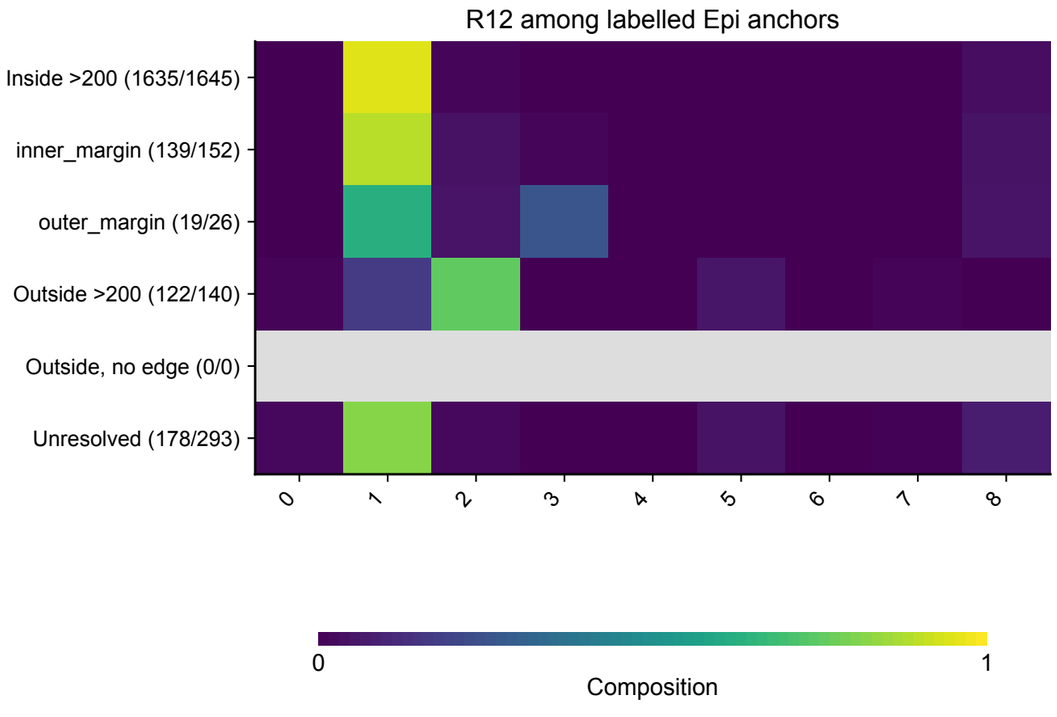
Orange = zero distance
Line = within-section linear fit

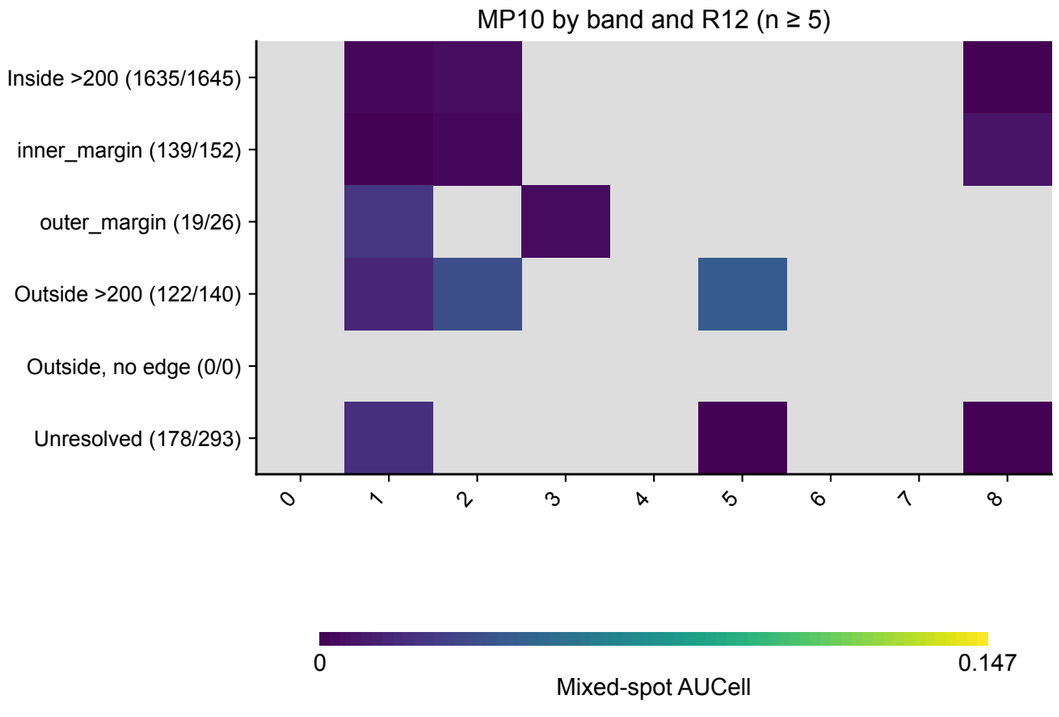
Inhousedata_T4471438 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





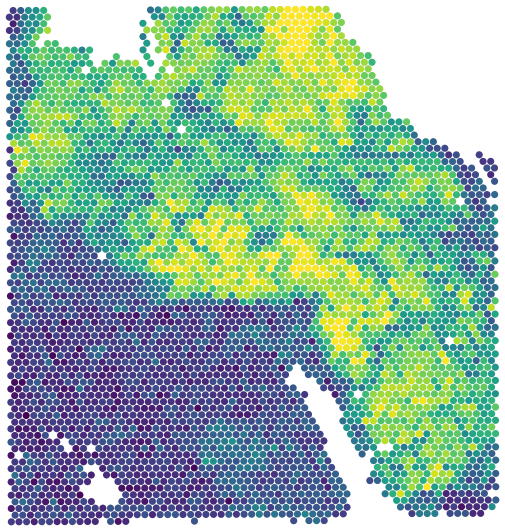




Inhousedata_T4471438 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

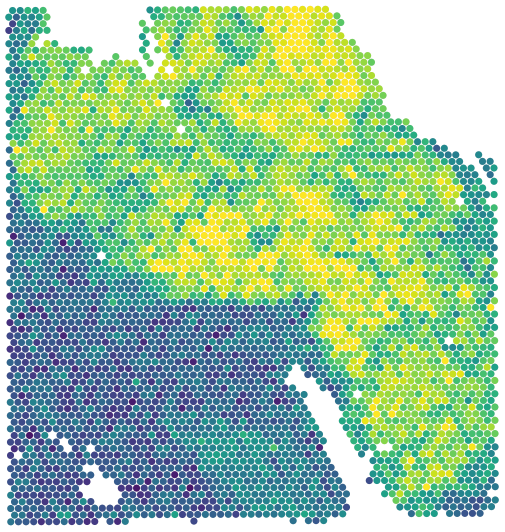
E2F Targets



0 0.152

UCell score

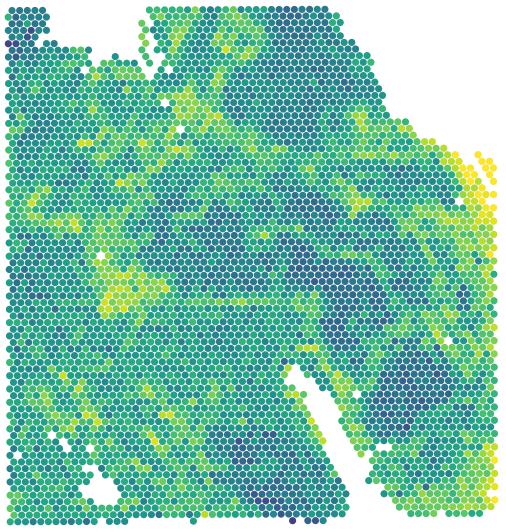
G2M Checkpoint



0 0.134

UCell score

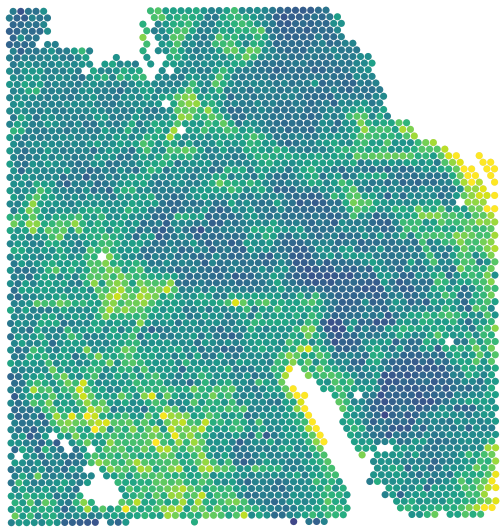
Hypoxia



0 0.211

UCell score

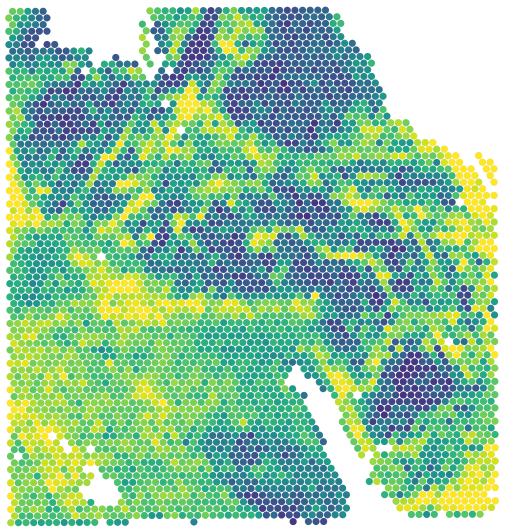
Tnfa Signaling Via Nfkb



0 0.275

UCell score

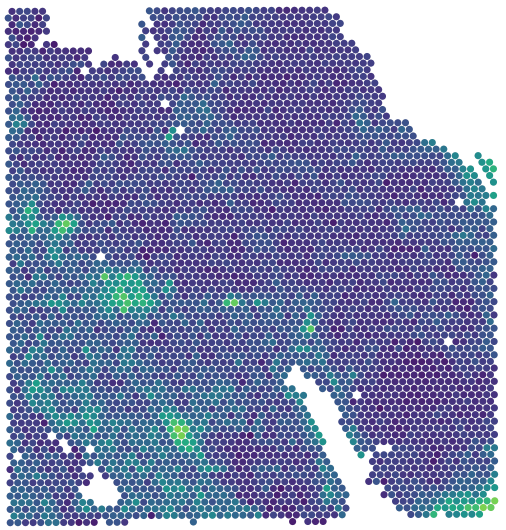
Emt



0 0.386

UCell score

Interferon Gamma Response



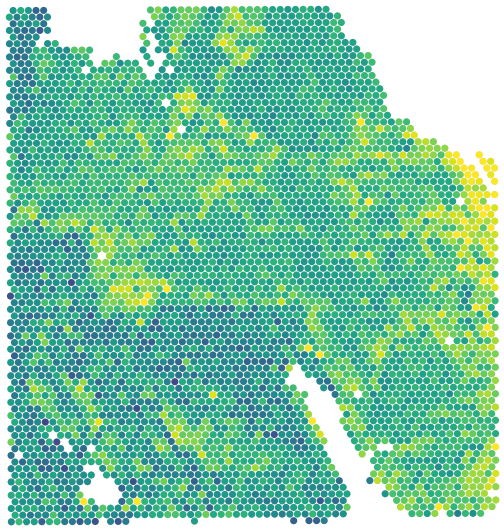
0 0.275

UCell score

Inhousedata_T4471438 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

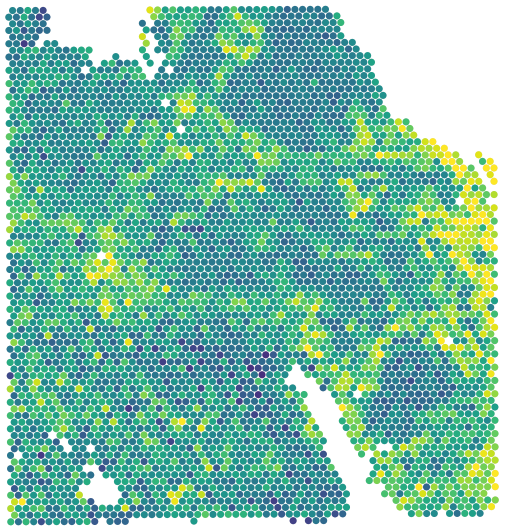
Il2 Stat5 Signaling



0 0.144

UCell score

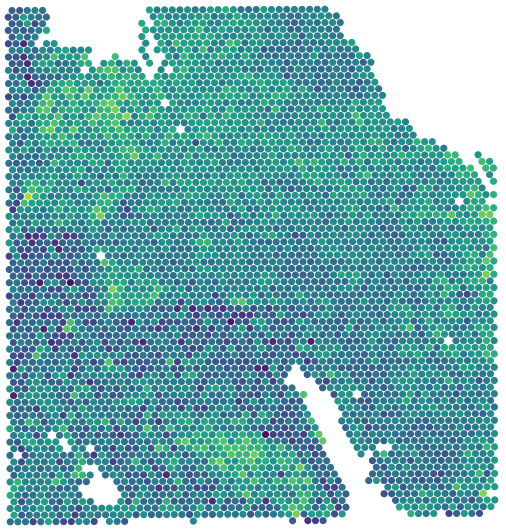
Tgf Beta Signaling



0 0.232

UCell score

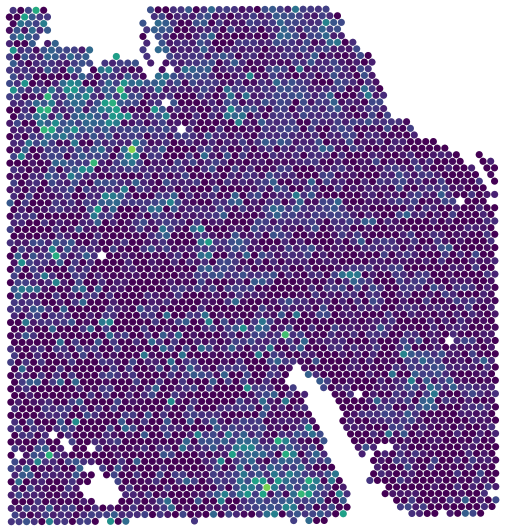
Cholesterol Homeostasis



0 0.21

UCell score

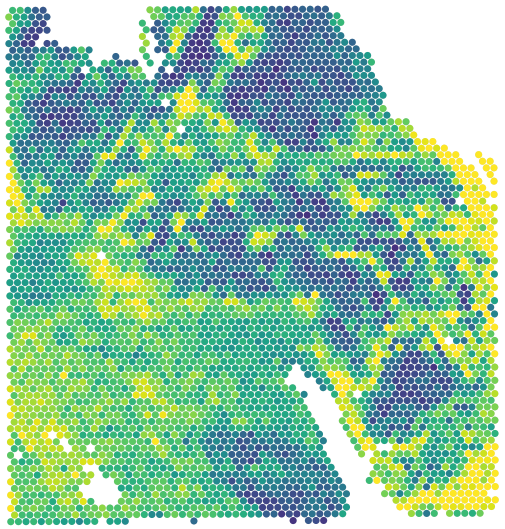
Cholesterol Biosynthesis



0 0.166

UCell score

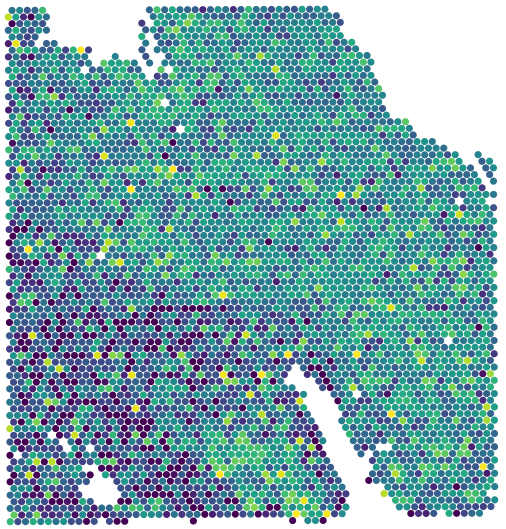
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



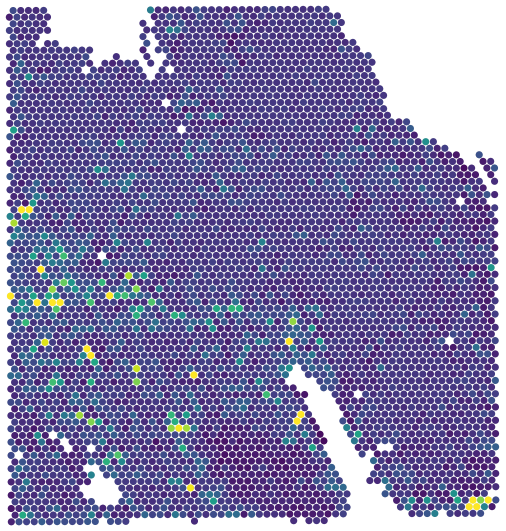
0 0.151

UCell score

Inhousedata_T4471438 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

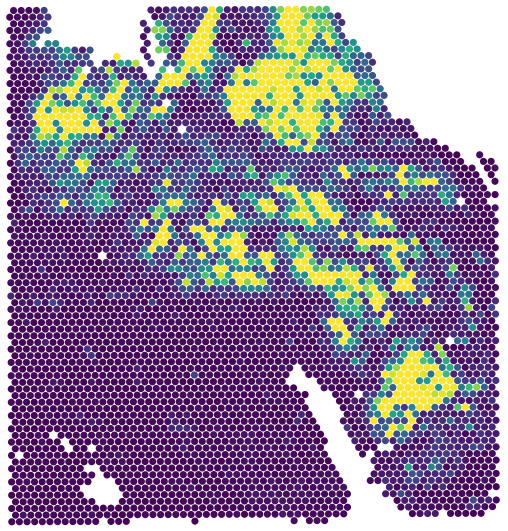
Treg composition



0 0.102

Fraction of all 30

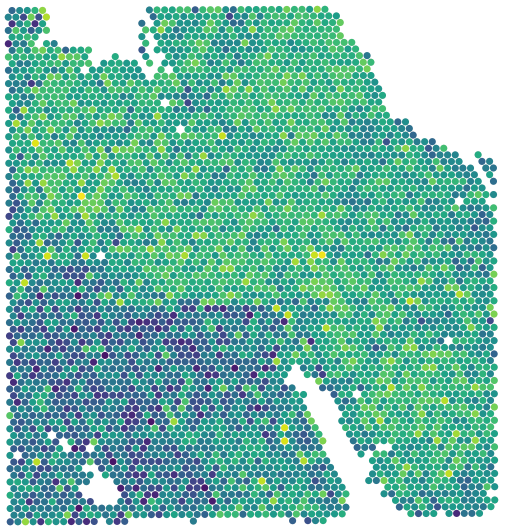
Treg estimated abundance



0 2.85

Posterior mean estimate

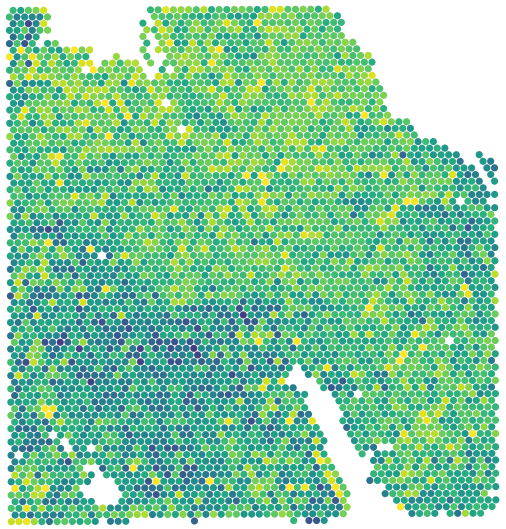
Treg Resting Vs Activated Dn



0 0.0778

UCell score

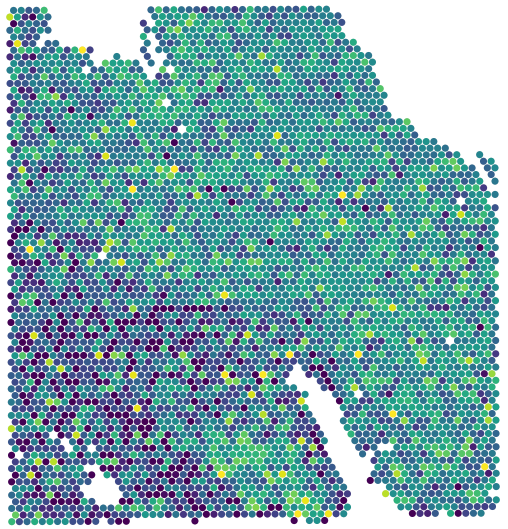
Treg Resting Vs Activated Up



0 0.0609

UCell score

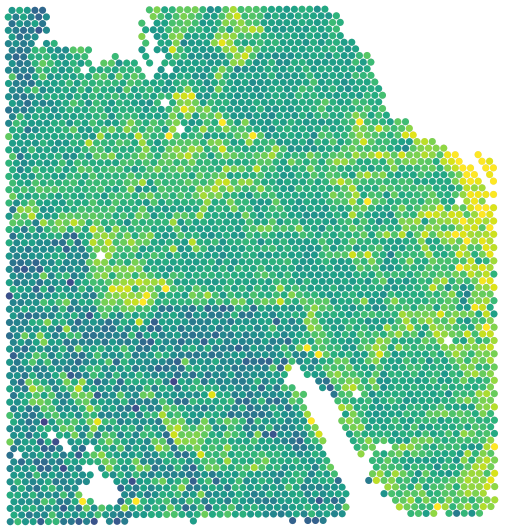
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

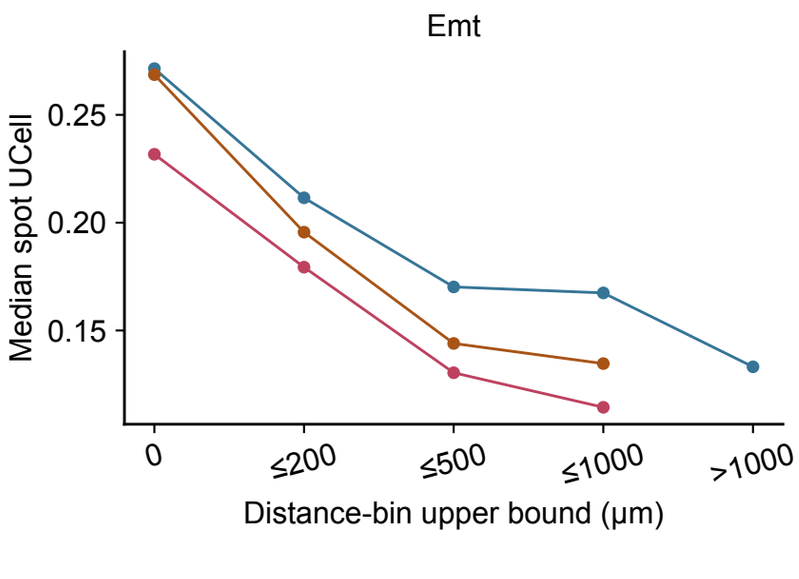


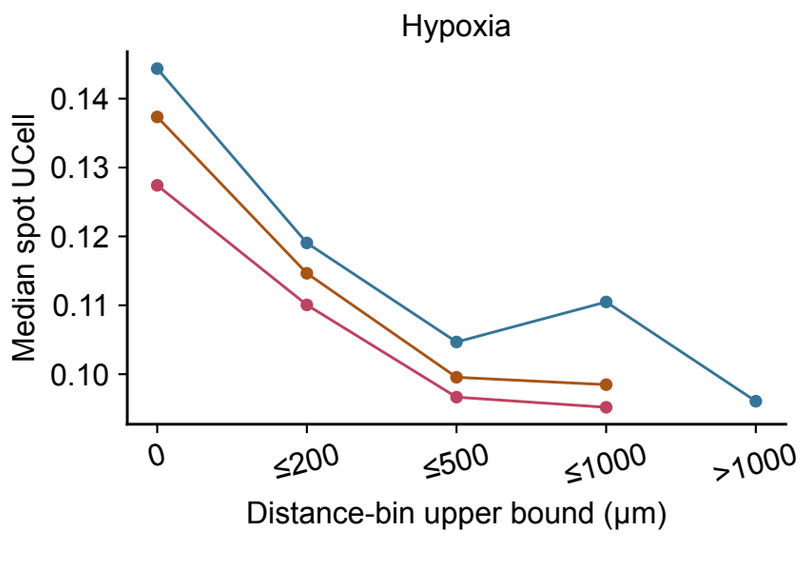
0 0.144

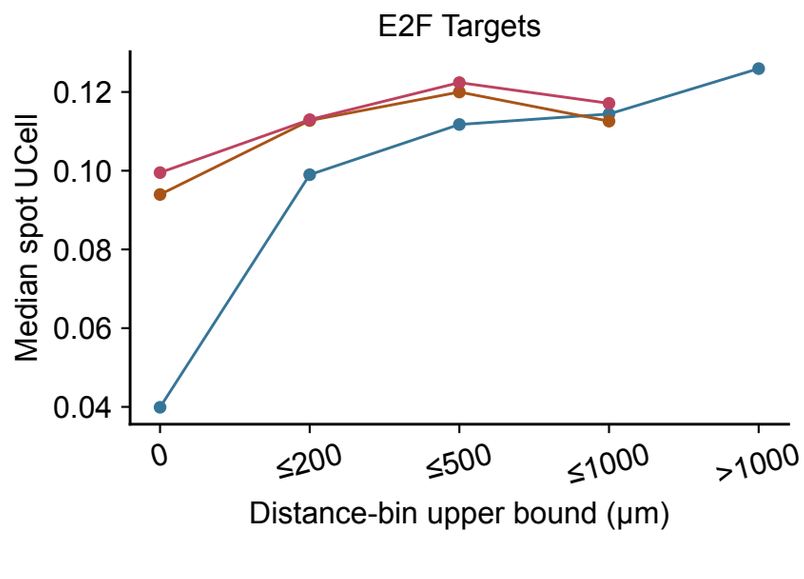
UCell score

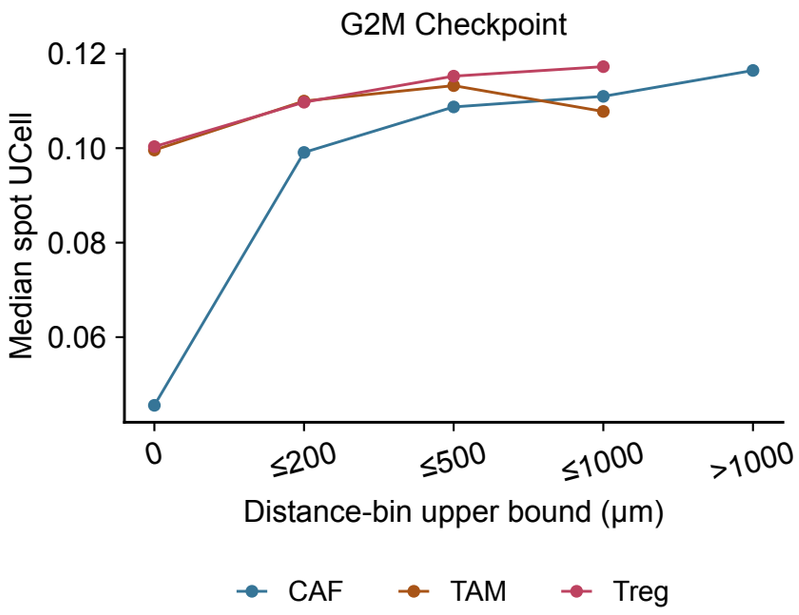
Inhousedata_T4471438 | Programs by target distance

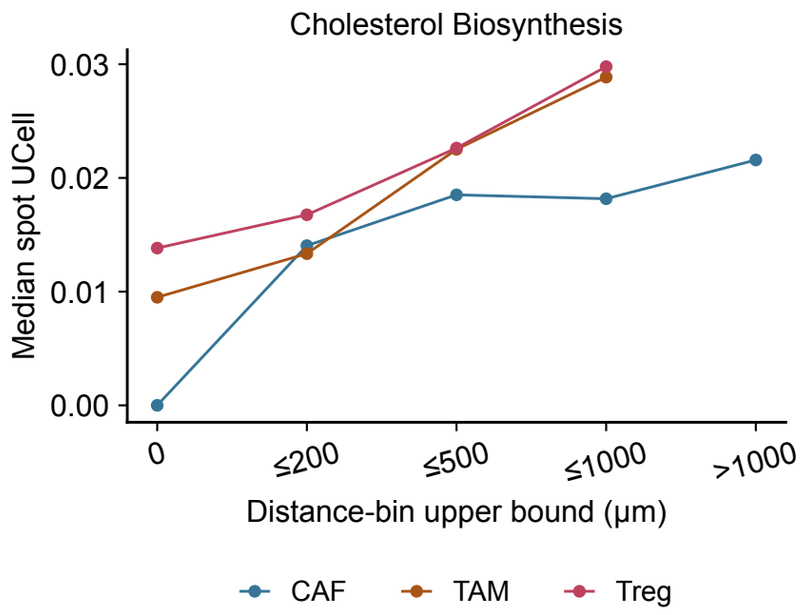
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

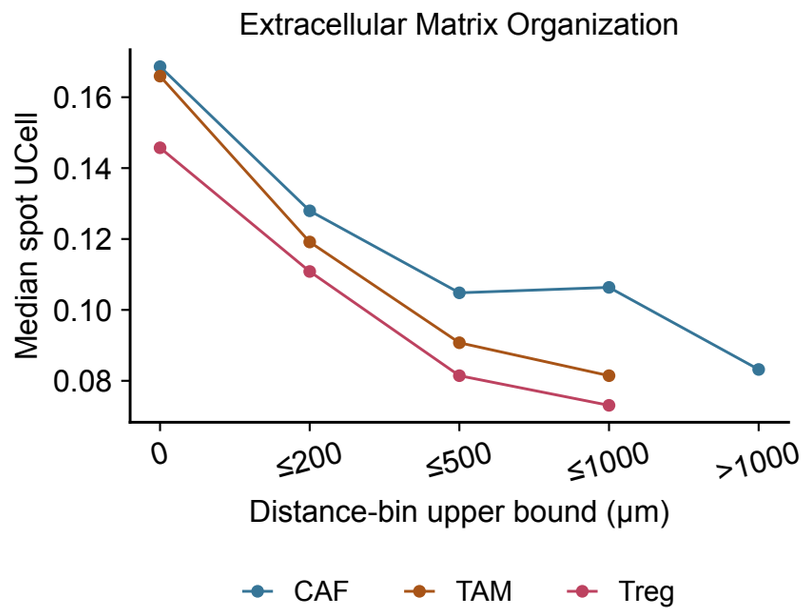












Inhousedata_T4471438 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6